# Mathematical Olympiad Treasures

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# Contents


**I Problems** **1**

**1 Algebra** **3**
- 1.1 An Algebraic Identity . . . 3
- 1.2 Cauchy-Schwartz revisited . . . 7
- 1.3 Easy ways through absolute values . . . 11
- 1.4 Parameters . . . 15
- 1.5 Take the conjugate! . . . 17
- 1.6 Inequalities with convex functions . . . 21
- 1.7 Induction at work . . . 25
- 1.8 Roots and coefficients . . . 28

**2 Geometry and Trigonometry** **35**
- 2.1 Geometric inequalities . . . 35
- 2.2 An interesting locus . . . 39
- 2.3 Cyclic Quads . . . 44
- 2.4 Equiangular polygons . . . 51
- 2.5 More on equilateral triangles . . . 55
- 2.6 The "carpets" theorem . . . 60
- 2.7 Quadrilaterals with an inscribed circle . . . 64
- 2.8 Dr. Trig learns complex numbers . . . 69

**3 Number Theory and Combinatorics** **73**
- 3.1 Arrays of numbers . . . 73
- 3.2 Functions defined on sets of points . . . 76
- 3.3 Count twice! . . . 80
- 3.4 Sequences of integers . . . 82
- 3.5 Equations with infinitely many solutions . . . 86
- 3.6 Equations with no solutions . . . 88

3.7 Powers of 2 . . . . . 90
3.8 Progressions . . . . . 93

**II Solutions** **97**

**4 Algebra** **99**
4.1 An Algebraic Identity . . . . . 99
4.2 Cauchy-Schwartz revisited . . . . . 105
4.3 Easy ways through absolute values . . . . . 109
4.4 Parameters . . . . . 114
4.5 Take the conjugate! . . . . . 118
4.6 Inequalities with convex functions . . . . . 124
4.7 Induction at work . . . . . 127
4.8 Roots and coef cients . . . . . 130

**5 Geometry and Trigonometry** **137**
5.1 Geometric inequalities . . . . . 137
5.2 An interesting locus . . . . . 145
5.3 Cyclic Quads . . . . . 153
5.4 Equiangular polygons . . . . . 161
5.5 More on equilateral triangles . . . . . 171
5.6 The ìcarpetsî theorem . . . . . 178
5.7 Quadrilaterals with an inscribed circle . . . . . 184
5.8 Dr. Trig learns complex numbers . . . . . 190

**6 Number Theory and Combinatorics** **195**
6.1 Arrays of numbers . . . . . 195
6.2 Functions de ned on sets of points . . . . . 199
6.3 Count twice! . . . . . 204
6.4 Sequences of integers . . . . . 209
6.5 Equations with in nitely many solutions . . . . . 212
6.6 Equations with no solutions . . . . . 216
6.7 Powers of 2 . . . . . 219
6.8 Progressions . . . . . 227

**Glossary** **233**

**Index of notations** **235**

**Index to the problems** **237**

# *Mathematical Olympiad Treasures*

# Part I

# Problems

# Chapter 1

# Algebra

## 1.1 An algebraic identity

A very useful algebraic identity is derived by considering the following problem.

**Problem 1.** Factor $a^3 + b^3 + c^3 - 3abc$.

**Solution.** Let $P$ denote the polynomial with roots $a, b, c$:

$$P(X) = X^3 - (a+b+c)X^2 + (ab+bc+ca)X - abc.$$

Because $a, b, c$ satisfy the equation $P(x) = 0$, we obtain

$$\begin{aligned} a^3 - (a+b+c)a^2 + (ab+bc+ca)a - abc &= 0, \\ b^3 - (a+b+c)b^2 + (ab+bc+ca)b - abc &= 0, \\ c^3 - (a+b+c)c^2 + (ab+bc+ca)c - abc &= 0. \end{aligned}$$

Adding up these three equalities yields

$$\begin{aligned} a^3 + b^3 + c^3 - (a+b+c)(a^2+b^2+c^2) \\ + (ab+bc+ca)(a+b+c) - 3abc = 0. \end{aligned}$$

Hence

$$a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca). \quad (1)$$

Note that the above identity leads to the following result: if $a+b+c=0$, then $a^3+b^3+c^3 = 3abc$.

Another way to obtain the identity (1) is to consider the determinant:

$$D = \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}.$$

Expanding $D$, we obtain

$$D = a^3 + b^3 + c^3 - 3abc.$$

On the other hand, adding all columns to the first one gives

$$D = \begin{vmatrix} a+b+c & b & c \\ a+b+c & a & b \\ a+b+c & c & a \end{vmatrix} = (a+b+c)\begin{vmatrix} 1 & b & c \\ 1 & a & b \\ 1 & c & a \end{vmatrix}$$

$$= (a+b+c)(a^2+b^2+c^2-ab-bc-ca).$$

Notice that the expression

$$a^2+b^2+c^2-ab-bc-ca$$

can also be written as

$$\tfrac{1}{2}\left[(a-b)^2+(b-c)^2+(c-a)^2\right].$$

We obtain another version of the identity (1):

$$a^3+b^3+c^3-3abc$$
$$= \tfrac{1}{2}(a+b+c)\left[(a-b)^2+(b-c)^2+(c-a)^2\right]. \quad (2)$$

This form leads to a short proof of the $AM-GM$ inequality for three variables. Indeed, from (2) it is clear that if $a, b, c$ are positive, then $a^3+b^3+c^3 \geq 3abc$. Now, if $x, y, z$ are positive numbers, taking $a = \sqrt[3]{x}, b = \sqrt[3]{y}$ and $c = \sqrt[3]{z}$ yields

$$\frac{x+y+z}{3} \geq \sqrt[3]{xyz},$$

with equality if and only if $x = y = z$.

Finally, let us regard

$$a^2+b^2+c^2-ab-bc-ca$$

as a quadratic in $a$, with parameters $b$, $c$. This quadratic has discriminant

$$\Delta = (b+c)^2 - 4(b^2+c^2-bc) = -3(b-c)^2.$$

Hence its roots are

$$a_1 = \frac{b+c-i(b-c)\sqrt{3}}{2} = b\frac{1-i\sqrt{3}}{2} + c\frac{1+i\sqrt{3}}{2}$$

and

$$a_2 = \frac{b+c+i(b-c)\sqrt{3}}{2} = b\frac{1+i\sqrt{3}}{2} + c\frac{1-i\sqrt{3}}{2}.$$

Setting $\omega = \frac{-1+i\sqrt{3}}{2}$, one of the primitive cubic roots of the unity, we have $\omega^2 = \frac{-1-i\sqrt{3}}{2}$, hence $a_1 = -b\omega - c\omega^2$ and $a_2 = -b\omega^2 - c\omega$.

This gives the factorization

$$a^2+b^2+c^2-ab-bc-ca = (a+b\omega+c\omega^2)(a+b\omega^2+c\omega),$$

which leads to the identity

$$a^3+b^3+c^3-3abc = (a+b+c)(a+b\omega+c\omega^2)(a+b\omega^2+c\omega). \quad (3)$$

Here are some problems that show how useful the above identities can be.

**Problem 2.** Factor $(x-y)^3+(y-z)^3+(z-x)^3$.

**Solution.** Observe that if $a+b+c=0$, then it follows from (1) that $a^3+b^3+c^3 = 3abc$. Because

$$(x-y)+(y-z)+(z-x) = 0,$$

we obtain the factorization

$$(x-y)^3+(y-z)^3+(z-x)^3 = 3(x-y)(y-z)(z-x).$$

**Problem 3.** Prove that $\sqrt[3]{2+\sqrt{5}}+\sqrt[3]{2-\sqrt{5}}$ is a rational number.

**Solution.** Let $x = \sqrt[3]{2+\sqrt{5}}+\sqrt[3]{2-\sqrt{5}}$. We then have

$$x - \sqrt[3]{2+\sqrt{5}} - \sqrt[3]{2-\sqrt{5}} = 0.$$

As we have seen, $a+b+c=0$ implies $a^3+b^3+c^3 = 3abc$, so we obtain

$$x^3 - (2+\sqrt{5}) - (2-\sqrt{5}) = 3x\sqrt[3]{(2+\sqrt{5})(2-\sqrt{5})},$$

or

$$x^3 + 3x - 4 = 0.$$

Clearly, one of the roots of this equation is $x = 1$ and the other two roots satisfy the equation $x^2 + x + 4 = 0$, which has no real solutions. Since $\sqrt[3]{2+\sqrt{5}} + \sqrt[3]{2-\sqrt{5}}$ is a real number, it follows that

$$\sqrt[3]{2+\sqrt{5}} + \sqrt[3]{2-\sqrt{5}} = 1,$$

which is a rational number.

Next come some proposed problems.

**Problem 4.** Factor $(a + 2b - 3c)^3 + (b + 2c - 3a)^3 + (c + 2a - 3b)^3$.

**Problem 5.** Let $x, y, z$ be integers such that

$$(x - y)^2 + (y - z)^2 + (z - x)^2 = xyz.$$

Prove that $x^3 + y^3 + z^3$ is divisible by $x + y + z + 6$.

**Problem 6.** Let $a, b, c$ be distinct real numbers. Prove that the following equality cannot hold:

$$\sqrt[3]{a-b} + \sqrt[3]{b-c} + \sqrt[3]{c-a} = 0.$$

**Problem 7.** Prove that the number

$$\sqrt[3]{45 + 29\sqrt{2}} + \sqrt[3]{45 - 29\sqrt{2}}$$

is an rational number.

**Problem 8.** Let $a, b, c$ be rational numbers such that

$$a + b\sqrt[3]{2} + c\sqrt[3]{4} = 0.$$

Prove that $a = b = c = 0$.

**Problem 9.** Let $r$ be a real number such that

$$\sqrt[3]{r} + \frac{1}{\sqrt[3]{r}} = 3.$$

Determine the value of

$$r^3 + \frac{1}{r^3}.$$

**Problem 10.** Find the locus of points $(x, y)$ for which

$$x^3 + y^3 + 3xy = 1.$$

**Problem 11.** Let $n$ be a positive integer. Factor

$$3^{3^n}\left(3^{3^n} + 1\right) + 3^{3^n+1} - 1.$$

**Problem 12.** Let $S$ be the set of integers $x$ such that $x = a^3 + b^3 + c^3 - 3abc$, for some integers $a, b, c$. Prove that if $x, y \in S$, then $xy \in S$.

**Problem 13.** Let $a, b, c$ be distinct positive integers and let $k$ be a positive integer such that

$$ab + bc + ca \geq 3k^2 - 1.$$

Prove that

$$\frac{a^3 + b^3 + c^3}{3} - abc \geq 3k.$$

**Problem 14.** Let $a, b, c$ be the side lengths of a triangle. Prove that

$$\sqrt[3]{\frac{a^3 + b^3 + c^3 + 3abc}{2}} \geq \max(a, b, c).$$

## 1.2 Cauchy–Schwartz revisited

Let $a_1, a_2, \ldots, a_n, b_1, b_2, \ldots, b_n$ be nonzero real numbers. Then

$$\left(a_1^2 + a_2^2 + \cdots + a_n^2\right)\left(b_1^2 + b_2^2 + \cdots + b_n^2\right) \geq (a_1b_1 + a_2b_2 + \cdots + a_nb_n)^2 \quad (*)$$

with equality if and only if

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \cdots = \frac{a_n}{b_n}.$$

This is the well-known Cauchy–Schwartz inequality. The standard elementary proof uses the properties of the quadratic function: consider the function

$$f(x) = (a_1 - b_1x)^2 + (a_2 - b_2x)^2 + \cdots + (a_n - b_nx)^2.$$

Clearly, $f(x) \geq 0$ for all real $x$, therefore, being a quadratic function, its discriminant $\Delta$ must be negative or zero. The inequality follows by observing that

$$\Delta = 4(a_1b_1 + a_2b_2 + \cdots + a_nb_n)^2 - 4(a_1^2 + a_2^2 + \cdots + a_n^2)(b_1^2 + b_2^2 + \cdots + b_n^2).$$

If we have equality in (*), then $\Delta = 0$ and the equation $f(x) = 0$ has a real root $x_0$. But then

$$(a_1 - b_1x_0)^2 + (a_2 - b_2x_0)^2 + \cdots + (a_n - b_nx_0)^2 = f(x_0) = 0,$$

so that $a_1 - b_1x_0 = a_2 - b_2x_0 = \cdots = a_n - b_nx_0 = 0$ and

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \cdots = \frac{a_n}{b_n} = x_0.$$

Conversely, if

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \cdots = \frac{a_n}{b_n},$$

then the equation $f(x) = 0$ has a real root, so $\Delta \geq 0$. Since $\Delta$ cannot be positive, it follows that $\Delta = 0$ and we have equality in (*).

Another proof uses a simple lemma which can also be helpful in proving a large number of algebraic inequalities:

*Lemma.* If $a, b, x, y$ are real numbers and $x, y > 0$, then the following inequality holds:

$$\frac{a^2}{x} + \frac{b^2}{y} \geq \frac{(a+b)^2}{x+y}.$$

The proof is straightforward. Clearing out denominators yields

$$a^2y(x+y)+b^2x(x+y)\geq(a+b)^2xy,$$

which readily simplifies to the obvious $(ay-bx)^2\geq 0$. We see that the equality holds if and only if $ay=bx$, that is, if

$$\frac{a}{x}=\frac{b}{y}.$$

Applying the lemma twice, we can extend the inequality to three pairs of numbers. Indeed,

$$\frac{a^2}{x}+\frac{b^2}{y}+\frac{c^2}{z}\geq\frac{(a+b)^2}{x+y}+\frac{c^2}{z}\geq\frac{(a+b+c)^2}{x+y+z},$$

and a simple inductive argument shows that

$$\frac{a_1^2}{x_1}+\frac{a_2^2}{x_2}+\cdots+\frac{a_n^2}{x_n}\geq\frac{(a_1+a_2+\cdots+a_n)^2}{x_1+x_2+\cdots+x_n}$$

for all real numbers $a_1, a_2, \ldots, a_n$ and $x_1, x_2, \ldots, x_n > 0$, with equality if and only if

$$\frac{a_1}{x_1}=\frac{a_2}{x_2}=\cdots=\frac{a_n}{x_n}.$$

Returning to Cauchy–Schwartz, let us use the lemma in the general case:

$$\begin{aligned}&a_1^2+a_2^2+\cdots+a_n^2\\&\qquad=\frac{a_1^2b_1^2}{b_1^2}+\frac{a_2^2b_2^2}{b_2^2}+\cdots+\frac{a_n^2b_n^2}{b_n^2}\geq\frac{(a_1b_1+a_2b_2+\cdots+a_nb_n)^2}{b_1^2+b_2^2+\cdots+b_n^2}.\end{aligned}$$

This yields

$$(a_1^2+a_2^2+\cdots+a_n^2)\,(b_1^2+b_2^2+\cdots+b_n^2)\geq(a_1b_1+a_2b_2+\cdots+a_nb_n)^2$$

and the equality holds if and only if

$$\frac{a_1}{b_1}=\frac{a_2}{b_2}=\cdots=\frac{a_n}{b_n}.$$

Let's see our lemma at work!

**Problem 1.** Let $a, b, c$ be positive real numbers. Prove that

$$\frac{a}{b+c}+\frac{b}{c+a}+\frac{c}{a+b} \geq \frac{3}{2}.$$

**Solution.** Observe that

$$\begin{aligned}\frac{a}{b+c}+\frac{b}{c+a}+\frac{c}{a+b} &= \frac{a^2}{ab+ac}+\frac{b^2}{bc+ba}+\frac{c^2}{ca+cb} \geq \frac{(a+b+c)^2}{2(ab+bc+ca)},\end{aligned}$$

so it suffices to prove

$$\frac{(a+b+c)^2}{2(ab+bc+ca)} \geq \frac{3}{2}.$$

A short computation shows that this is equivalent to

$$a^2+b^2+c^2 \geq ab+bc+ca,$$

which yields

$$(a-b)^2+(b-c)^2+(c-a)^2 \geq 0.$$

**Problem 2.** Let $a$ and $b$ be positive real numbers. Prove that

$$8(a^4+b^4) \geq (a+b)^4.$$

**Solution.** We apply the lemma twice:

$$a^4+b^4 = \frac{a^4}{1}+\frac{b^4}{1} \geq \frac{(a^2+b^2)^2}{2} \geq \frac{\left(\frac{(a+b)^2}{2}\right)^2}{2} = \frac{(a+b)^4}{8}.$$

Try to use the lemma to prove the following inequalities.

**Problem 3.** Let $x, y, z > 0$. Prove that

$$\frac{2}{x+y}+\frac{2}{y+z}+\frac{2}{z+x} \geq \frac{9}{x+y+z}.$$

**Problem 4.** Let $a, b, x, y, z$ be positive real numbers. Prove that

$$\frac{x}{ay+bz}+\frac{y}{az+bx}+\frac{z}{ax+by} \geq \frac{3}{a+b}.$$

**Problem 5.** Let $a, b, c > 0$. Prove that

$$\frac{a^2+b^2}{a+b}+\frac{b^2+c^2}{b+c}+\frac{a^2+c^2}{a+c}\geq a+b+c.$$

**Problem 6.** Let $a, b, c$ be positive numbers such that $abc = 1$. Prove that

$$\frac{1}{a^3(b+c)}+\frac{1}{b^3(a+c)}+\frac{1}{c^3(a+b)}\geq\frac{3}{2}.$$

**Problem 7.** Let $x, y, z > 0$. Prove that

$$\frac{x}{x+2y+3z}+\frac{y}{y+2z+3x}+\frac{z}{z+2x+3y}\geq\frac{1}{2}.$$

**Problem 8.** Let $x, y, z > 0$. Prove that

$$\frac{x^2}{(x+y)(x+z)}+\frac{y^2}{(y+z)(y+x)}+\frac{z^2}{(z+x)(z+y)}\geq\frac{3}{4}.$$

**Problem 9.** Let $a, b, c, d, e$ be positive real numbers. Prove that

$$\frac{a}{b+c}+\frac{b}{c+d}+\frac{c}{d+e}+\frac{d}{e+a}+\frac{e}{a+b}\geq\frac{5}{2}.$$

**Problem 10.** Let $a, b, c$ be positive real numbers such that

$$ab+bc+ca=\frac{1}{3}.$$

Prove that

$$\frac{a}{a^2-bc+1}+\frac{b}{b^2-ca+1}+\frac{c}{c^2-ab+1}\geq\frac{1}{a+b+c}.$$

## 1.3 Easy ways through absolute values

Everybody knows that sometimes solving equations or inequalities with absolute values can be boring. Most of the students facing such problems begin by writing the absolute values in an explicit manner. Let us consider, for instance, the following simple equation:

**Problem 1.** Solve the equation $|2x-1| = |x+3|$.

**Solution.** We have

$$|2x-1| = \begin{cases} -2x+1, x \le \frac{1}{2} \\ 2x-1, x > \frac{1}{2} \end{cases} \quad \text{and} \quad |x+3| = \begin{cases} -x-3, x \le -3 \\ x+3, x > -3 \end{cases}.$$

If $x \le -3$, the equation becomes $-2x+1 = -x-3$; hence $x = 4$. But $4 > -3$, so we have no solutions in this case. If $-3 < x \le \frac{1}{2}$, we obtain $-2x+1 = x+3$, so $x = -\frac{2}{3} \in (-3, \frac{1}{2}]$. Finally, if $x > \frac{1}{2}$, we obtain $x = 4$ again. We conclude that the solutions are $x = -\frac{2}{3}$ and $x = 4$.

Nevertheless, there is an easier way to solve the equation by noticing that $|a| = |b|$ if and only if $a = \pm b$, so that it is not necessary to write the absolute values in an explicit form. Here are some properties of the absolute values that might be useful in solving equations and inequalities:

$$|ab| = |a||b|,$$

$$\left|\frac{a}{b}\right| = \frac{|a|}{|b|},$$

$$|a+b| \le |a| + |b|,$$

with equality if and only if $ab \ge 0$,

$$|a-b| \le |a| + |b|,$$

with equality if and only if $ab \le 0$.

The last two inequalities can be written in a general form:

$$|\pm a_1 \pm a_2 \pm \cdots \pm a_n| \le |a_1| + |a_2| + \cdots + |a_n|.$$

**Problem 2.** Solve the equation $|x-1| + |x-4| = 2$.

**Solution.** Observe that

$$|x-1| + |x-4| \ge |(x-1) - (x-4)| = |3| = 3 > 2,$$

hence the equation has no solutions.

**Problem 3.** Solve the equation $|x-1| + |x| + |x+1| = x+2$.

**Solution.** We have

$$x+2 = |x-1| + |x| + |x+1| \ge |(x-1) - (x+1)| + |x| = |x| + 2.$$

On the other hand, clearly $x \le |x|$, thus all inequalities must be equalities: that is $x+2 = |x|+2$ and

$$|x-1| + |x| + |x+1| = |(x-1) - (x+1)| + |x|.$$

This implies that $x \ge 0$ and the expressions $x-1$ and $x+1$ are of different sign, so $x \in [-1, 1]$. Finally, the solution is $x \in [0, 1]$.

**Problem 4.** Find the minimum value of the expression:

$$E(x) = |x-1| + |x-2| + \cdots + |x-100|,$$

where $x$ is a real number.

**Solution.** Observe that for $k = 1, 2, \ldots, 50$, we have

$$|x-k| + |x-(101-k)| \geq 101 - 2k,$$

with equality if $x \in [k, 101-k]$. Adding these inequalities yields $E(x) \geq 2500$, and the equality holds for all $x$ such that

$$x \in \bigcap_{1 \leq k \leq 50} [k, 101-k] = [50, 51].$$

**Problem 5.** Find the values of $a$ for which the equation

$$(x-1)^2 = |x-a|$$

has exactly three solutions.

**Solution.** Observe that $(x-1)^2 = |x-a|$ if and only if

$$x - a = \pm(x-1)^2,$$

that is, if and only if

$$a = x \pm (x-1)^2.$$

The number of solutions of the equation is equal to the number of intersection points between the line $y = a$ and the graphs of the functions

$$f(x) = x + (x-1)^2 = x^2 - x + 1$$

and

$$g(x) = x - (x-1)^2 = -x^2 + 3x - 1.$$

The graph of $f$ is a parabola with vertex $B(\frac{1}{2}, \frac{3}{4})$ and the graph of $g$ a parabola with vertex $C(\frac{3}{2}, \frac{5}{4})$. Now, since the equation $f(x) = g(x)$ is a quadratic with one real root, it follows that the graphs are tangent to each other at point $A(1, 1)$. We deduce that the line $y = a$ intersects the two graphs at three points if and only if it passes through one of the points $A$, $B$, $C$; that is, if $a \in \{\frac{3}{4}, 1, \frac{5}{4}\}$.

Try to use some of the ideas above in solving the following problems:

**Problem 6.** Solve the equation $|x-3|+|x+1|=4$.

**Problem 7.** Show that the equation $|2x-3|+|x+1|+|5-x|=0.99$ has no solutions.

**Problem 8.** Let $a, b > 0$. Find the values of $m$ for which the equation

$$|x-a|+|x-b|+|x+a|+|x+b|=m(a+b)$$

has at least one real solution.

**Problem 9.** Find all possible values of the expression

$$E(x,y,z)=\frac{|x+y|}{|x|+|y|}+\frac{|y+z|}{|y|+|z|}+\frac{|z+x|}{|z|+|x|},$$

where $x$, $y$, $z$ are nonzero real numbers.

**Problem 10.** Find all positive real numbers $x, x_1, x_2, \ldots, x_n$ such that

$$\begin{aligned}|\log(xx_1)|+|\log(xx_2)|+\cdots+|\log(xx_n)|&\\ +\left|\log\left(\frac{x}{x_1}\right)\right|+\left|\log\left(\frac{x}{x_2}\right)\right|+\cdots+\left|\log\left(\frac{x}{x_n}\right)\right|&\\ =|\log x_1+\log x_2+\cdots+\log x_n|\,.&\end{aligned}$$

**Problem 11.** Prove that for all real numbers $a$, $b$, we have:

$$\frac{|a+b|}{1+|a+b|}\le\frac{|a|}{1+|a|}+\frac{|b|}{1+|b|}.$$

**Problem 12.** Let $n$ be an odd positive integer and let $x_1, x_2, \ldots, x_n$ be distinct real numbers. Find all one-to-one functions

$$f:\{x_1,x_2,\ldots,x_n\}\to\{x_1,x_2,\ldots,x_n\}$$

such that

$$|f(x_1)-x_1|=|f(x_2)-x_2|=\cdots=|f(x_n)-x_n|\,.$$

**Problem 13.** Suppose the sequence $a_1, a_2, \ldots, a_n$ satisfies the following conditions:

$$a_1=0,\ |a_2|=|a_1+1|\,,\ \ldots,\ |a_n|=|a_{n-1}+1|\,.$$

Prove that

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \geq -\tfrac{1}{2}.$$

**Problem 14.** Find real numbers $a, b, c$ such that

$$|ax + by + cz| + |bx + cy + az| + |cx + ay + bz| = |x| + |y| + |z|,$$

for all real numbers $x, y, z$.

## 1.4 Parameters

We start with the following problem.

**Problem 1.** Solve the equation:

$$x^3(x + 1) = 2(x + a)(x + 2a),$$

where $a$ is a real parameter.

**Solution**. The equation is equivalent to

$$x^4 + x^3 - 2x^2 - 6ax - 4a^2 = 0.$$

This fourth degree equation is difficult to solve. We might try to factor the left-hand side, but without some appropriate software, the process would get quite complicated. What if we think of $a$ as the unknown and $x$ as the parameter? In this case, the equation can be written as a quadratic:

$$4a^2 + 6xa - x^4 - x^3 + 2x^2 = 0,$$

whose discriminant is

$$36x^2 + 16\left(x^4 + x^3 - 2x^2\right) = 4x^2(2x + 1)^2,$$

fortunately a square. Solving for $a$, we obtain the solutions $a_1 = -\frac{1}{2}x^2 - x$ and $a_2 = \frac{1}{2}x^2 - \frac{1}{2}x$, yielding the factorization

$$4a^2 + 6ax - x^4 - x^3 + 2x^2 = 4(a + \tfrac{1}{2}x^2 + x)(a - \tfrac{1}{2}x^2 + \tfrac{1}{2}x)$$

$$= -(x^2 + 2x + 2a)(x^2 - x - 2a).$$

Finally, we obtain the solutions $x_{1,2} = -1 \pm \sqrt{1 - 2a}$, $x_{3,4} = \frac{1}{2} \pm \frac{1}{2}\sqrt{1 + 8a}$, which are all real if $a \in \left[-\frac{1}{8}, \frac{1}{2}\right]$.

**Problem 2.** Solve the equation $\sqrt{5-x} = 5 - x^2$.

**Solution.** There is no parameter! Note, however, that we must have $x \in [-\sqrt{5}, \sqrt{5}]$, since the left-hand side is nonnegative. Squaring both sides, we obtain the equation

$$x^4 - 10x^3 + x + 20 = 0$$

and, if we are lucky, we might observe the factorization

$$(x^2 + x - 5)(x^2 - x - 4) = 0.$$

What if we are not lucky? Well, let us introduce a parameter ourselves: replace 5 by $a$, where $a > 0$: $\sqrt{a-x} = a - x^2$. Squaring both sides yields the equation

$$x^4 - 2ax^2 + x + a^2 - a = 0.$$

Consider this as a quadratic in $a$ with $x$ as a parameter:

$$a^2 - (2x^2 + 1)a + x^4 + x = 0.$$

The discriminant of the quadratic is

$$\Delta = (2x - 1)^2$$

and thus the roots are $a_1 = x^2 + x$ and $a_2 = x^2 - x + 1$. It follows that

$$a^2 - (2x^2 + 1)a + x^4 + x = (a - x^2 - x)(a - x^2 + x - 1).$$

Returning to $a = 5$, we arrive at the desired factorization.

The equations $x^2 + x - 5 = 0$ and $x^2 - x - 4 = 0$ have the solutions $x_{1,2} = \frac{1}{2}\left(-1 \pm \sqrt{21}\right)$ and $x_{3,4} = \frac{1}{2}\left(1 \pm \sqrt{17}\right)$, respectively. Only two of them belong to the interval $[-\sqrt{5}, \sqrt{5}]$, and therefore the solutions of the initial equation are $\frac{1}{2}\left(-1 + \sqrt{21}\right)$ and $\frac{1}{2}\left(1 - \sqrt{17}\right)$.

Here are some suggested problems:

**Problem 3.** Solve the equation

$$x = \sqrt{a - \sqrt{a + x}},$$

where $a > 0$ is a parameter.

**Problem 4.** Let $a$ be a nonzero real number. Solve the equation

$$a^3x^4 + 2a^2x^2 + x + a + 1 = 0.$$

**Problem 5.** Let $a \in (0, \frac{1}{4})$. Solve the equation

$$x^2 + 2ax + \frac{1}{16} = -a + \sqrt{a^2 + x - \frac{1}{16}}.$$

**Problem 6.** Find the positive solutions of the following system of equations:

$$\begin{cases} \frac{a^2}{x^2} - \frac{b^2}{y^2} = 8(y^4 - x^4) \\ ax - by = x^4 - y^4 \end{cases},$$

where $a, b > 0$ are parameters.

**Problem 7.** Let $a, b, c > 0$. Solve the system of equations

$$\begin{cases} ax - by + \frac{1}{xy} = c \\ bz - cx + \frac{1}{zx} = a \\ cy - az + \frac{1}{yz} = b \end{cases}.$$

**Problem 8.** Solve the equation

$$x + a^3 = \sqrt[3]{a - x},$$

where $a$ is a real parameter.

## 1.5 Take the conjugate!

Let $a$ and $b$ be positive numbers. Recall that the $AM - GM$ states that $\frac{a+b}{2} \geq \sqrt{ab}$. Can we estimate the difference between the arithmetic and geometric means?

**Problem 1.** Prove that if $a \geq b > 0$, then

$$\frac{(a-b)^2}{8a} \leq \frac{a+b}{2} - \sqrt{ab} \leq \frac{(a-b)^2}{8b}.$$

**Solution.** By taking the conjugate of the difference above, we obtain

$$\frac{a+b}{2}-\sqrt{ab}=\frac{\left(\frac{a+b}{2}\right)^2-ab}{\frac{a+b}{2}+\sqrt{ab}}=\frac{(a-b)^2}{2(a+b+2\sqrt{ab})}.$$

Now, the conclusion follows by observing that $b \le \sqrt{ab} \le a$, and that

$$8b \le 2(a+b+2\sqrt{ab}) \le 8a.$$

**Problem 2.** Evaluate the integer part of the number

$$A=\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}+\cdots+\frac{1}{\sqrt{10000}}.$$

**Solution.** Observe that

$$\frac{1}{\sqrt{k}}=\frac{2}{\sqrt{k}+\sqrt{k}}<\frac{2}{\sqrt{k}+\sqrt{k-1}}.$$

Taking the conjugate, we obtain

$$\frac{1}{\sqrt{k}}<2\left(\sqrt{k}-\sqrt{k-1}\right).$$

Setting $k=2,3,\ldots,10000$ and adding up yields

$$A<2\left(\sqrt{10000}-1\right)=198.$$

Similarly,

$$\frac{1}{\sqrt{k}}=\frac{2}{\sqrt{k}+\sqrt{k}}>\frac{2}{\sqrt{k+1}+\sqrt{k}}=2\left(\sqrt{k+1}-\sqrt{k}\right),$$

hence

$$A>2\left(\sqrt{10001}-\sqrt{2}\right)>197.$$

It follows that the integer part of the number $A$ equals 197.

**Problem 3.** Let $n$ be a positive integer. Prove that $\left\lfloor \left(2+\sqrt{3}\right)^n \right\rfloor$ is an odd number.

**Solution.** Observe that the number

$$x_n = \left(2+\sqrt{3}\right)^n + \left(2-\sqrt{3}\right)^n$$

is an even integer. Indeed, $x_1 = 4$, $x_2 = 14$ and

$$x_{n+2} = 4x_{n+1} - x_n,$$

for all $n \geq 1$, so the assertion follows inductively. Since $0 < \left(2-\sqrt{3}\right)^n < 1$, we have

$$y_n = \left\lfloor \left(2+\sqrt{3}\right)^n \right\rfloor = x_n - 1.$$

Thus $y_n$ satisfies the recursive relation

$$y_{n+2} = 4y_{n+1} - y_n + 2$$

and since $y_1 = 3$ is odd, all $y_n$ are odd .

**Problem 4.** Solve the equation

$$\sqrt{1+mx} = x + \sqrt{1-mx},$$

where $m$ is a real parameter.

**Solution.** We can try squaring both terms, but since it is difficult to control the sign of the right-hand side, we prefer to write the equation as follows:

$$\sqrt{1+mx} - \sqrt{1-mx} = x.$$

Taking the conjugate yields the equivalent form

$$\frac{2mx}{\sqrt{1+mx} + \sqrt{1-mx}} = x.$$

We obtain $x = 0$ as solution and, for $x \neq 0$ :

$$2m = \sqrt{1+mx} + \sqrt{1-mx}.$$

Thus $m$ is necessarily positive. We now square and obtain:

$$2m^2 - 1 = \sqrt{1-m^2x^2}.$$

We obtain another condition for $m$, that is $2m^2 - 1 \geq 0$, and squaring again we finally obtain the solutions

$$x = \pm 2\sqrt{1 - m^2},$$

where $m \in \left[\frac{1}{\sqrt{2}}, 1\right]$.

Take the conjugate in the following problems:

**Problem 5.** Let $a$ and $b$ be distinct positive numbers and let $A = \frac{a+b}{2}$, $B = \sqrt{ab}$. Prove the inequality

$$B < \frac{(a-b)^2}{8(A-B)} < A.$$

**Problem 6.** Let $m, n$ be positive integers with $m < n$. Find a closed form for the sum

$$\frac{1}{\sqrt{m} + \sqrt{m+1}} + \frac{1}{\sqrt{m+1} + \sqrt{m+2}} + \cdots + \frac{1}{\sqrt{n-1} + \sqrt{n}}.$$

**Problem 7.** For any positive integer $n$, let

$$f(n) = \frac{4n + \sqrt{4n^2 - 1}}{\sqrt{2n+1} + \sqrt{2n-1}}.$$

Evaluate the sum $f(1) + f(2) + \cdots + f(40)$.

**Problem 8.** Let $a$ and $b$ be distinct real numbers. Solve the equation

$$\sqrt{x - b^2} - \sqrt{x - a^2} = a - b.$$

**Problem 9.** Solve the following equation, where $m$ is a real parameter:

$$\sqrt{x + \sqrt{x}} - \sqrt{x - \sqrt{x}} = m\sqrt{\frac{x}{x + \sqrt{x}}}.$$

**Problem 10.** Prove that for every positive integer $k$, there exists a positive integer $n_k$ such that

$$\left(\sqrt{3} - \sqrt{2}\right)^k = \sqrt{n_k} - \sqrt{n_k - 1}.$$

**Problem 11.** Let $a$ and $b$ be nonzero integers with $|a| \leq 100$, $|b| \leq 100$. Prove that

$$\left|a\sqrt{2}+b\sqrt{3}\right| \geq \frac{1}{350}.$$

**Problem 12.** Let $n$ be a positive integer. Prove that

$$\left[\left(\frac{1+\sqrt{5}}{2}\right)^{4n-2}\right]-1$$

is a perfect square.

## 1.6 Inequalities with convex functions

A real-valued function $f$ defined on an interval $I \subset \mathbf{R}$ is called *convex* if for all $x_A, x_B \in I$ and for any $\lambda \in [0, 1]$ the following inequality holds:

$$f(\lambda x_A + (1-\lambda)x_B) \leq \lambda f(x_A) + (1-\lambda) f(x_B).$$

Although the definition seems complicated, it has a very simple geometrical interpretation. Let us assume that $f$ is a continuous function. Then $f$ is convex on $I$ if and only if no matter how we choose two points on the function's graph, the segment joining these points lies above the graph (see Figure 1.1).

To see why, let $A(x_A, y_A)$ and $B(x_B, y_B)$ be two points on the graph of $f$, with $x_A < x_B$, and let $M(x_M, f(x_M))$ be an arbitrary point on the graph with $x_A < x_M < x_B$. If $N(x_M, y_N)$ is on the segment $AB$, it suffices to verify that $f(x_M) \leq y_N$. If we let

$$\lambda = \frac{x_B - x_M}{x_B - x_A},$$

then $\lambda \in [0, 1]$ and

$$x_M = \lambda x_A + (1-\lambda)x_B.$$

Now it is easy to see that

$$\lambda = \frac{BN}{BA} = \frac{y_B - y_N}{y_B - y_A},$$

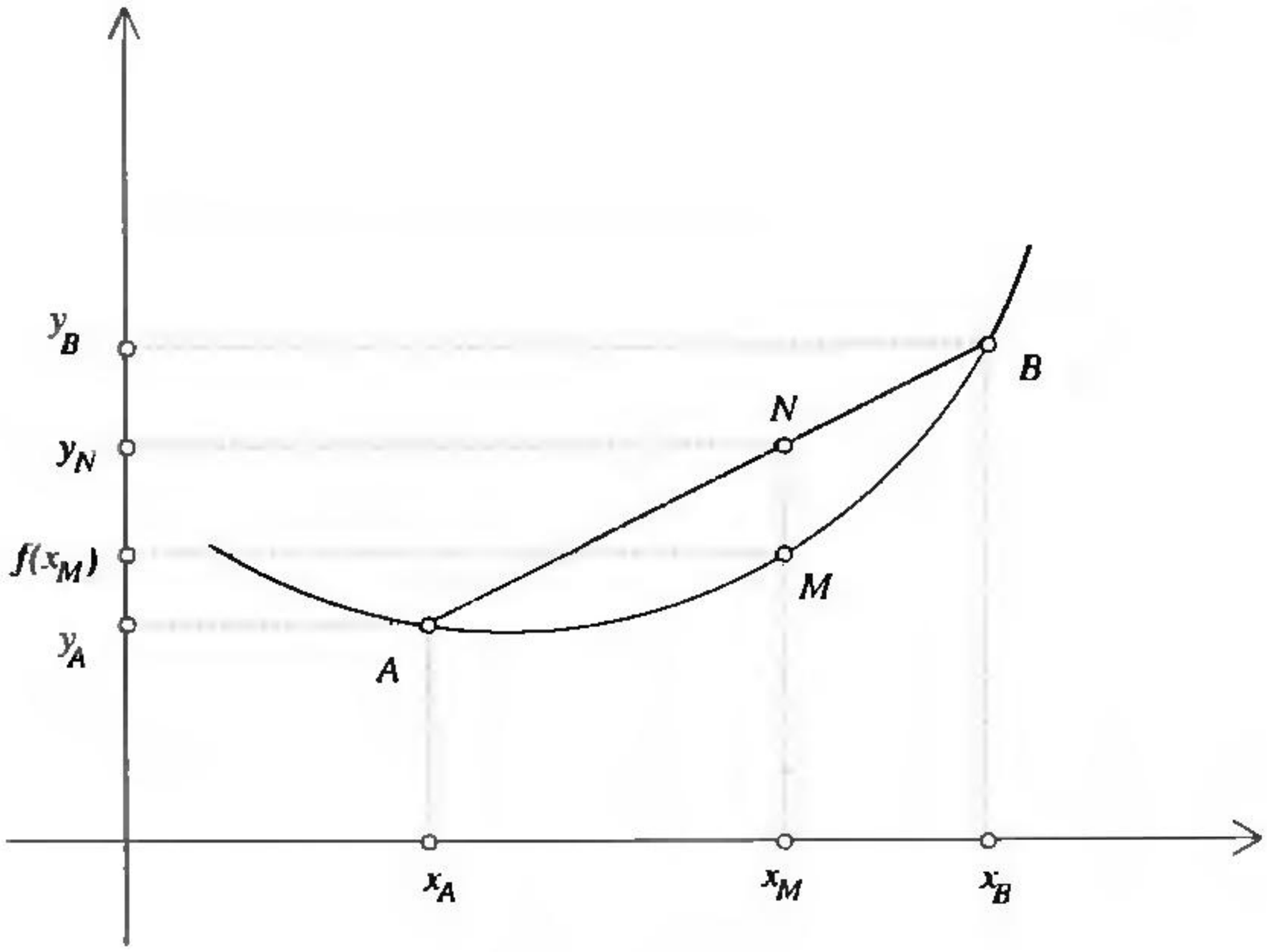


Figure 1.1

**that is,**

$$y_N = \lambda y_A + (1 - \lambda) y_B = \lambda f(x_A) + (1 - \lambda) f(x_B).$$

Thus the condition $f(x_M) \leq y_N$ is equivalent to the convexity condition

$$f(\lambda x_A + (1 - \lambda) x_B) \leq \lambda f(x_A) + (1 - \lambda) f(x_B).$$

It can be shown that if $f$ is a convex function on the interval $I$, then for any $x_1, x_2, \ldots, x_n \in I$ and any positive numbers $\lambda_1, \lambda_2, \ldots, \lambda_n$ such that $\lambda_1 + \lambda_2 + \cdots + \lambda_n = 1$, we have

$$f(\lambda_1 x_1 + \lambda_2 x_2 + \cdots + \lambda_n x_n) \leq \lambda_1 f(x_1) + \lambda_2 f(x_2) + \cdots + \lambda_n f(x_n).$$

A real-valued function $f$ defined on an interval $I \subset \mathbf{R}$ is called *convex* if for all $x_A, x_B \in I$ and for any $\lambda \in [0, 1]$ the following inequality holds:

$$f(\lambda x_A + (1 - \lambda) x_B) \geq \lambda f(x_A) + (1 - \lambda) f(x_B).$$

Similar inequalities are valid for concave functions, replacing everywhere $\leq$ by $\geq$.

It is known that if $f$ is twice differentiable, then $f$ is convex on $I$ if an only if $f'' \geq 0$ on $I$ (and concave if $f'' \leq 0$ on $I$ ).

**Problem 1.** Let $f$ be a convex function on the interval $I \subset R$. Prove that for any $a < b < c$ in $I$, we have

$$f(a-b+c) \leq f(a) - f(b) + f(c).$$

**Solution.** Since $b \in (a, c)$, there exists $\lambda$ such that $b = \lambda a + (1-\lambda)c$ (take $\lambda = \frac{c-b}{c-a}$ and notice that $\lambda \in [0, 1]$). Since $f$ is convex, we have

$$f(b) \leq \lambda f(a) + (1-\lambda) f(c).$$

On the other hand, we can see that

$$a - b + c = a - (\lambda a + (1-\lambda)c) + c = (1-\lambda)a + \lambda c.$$

Therefore, by convexity

$$f(a-b+c) \leq (1-\lambda) f(a) + \lambda f(c).$$

Adding up the two inequalities, we obtain

$$f(a-b+c) + f(b) \leq f(a) + f(c),$$

the desired result.

A simple induction argument shows that if $f$ is convex and

$$a_1 < a_2 < \cdots < a_{2n+1},$$

then

$$\begin{aligned} f(a_1 - a_2 + a_3 - \cdots - a_{2n} + a_{2n+1}) \\ \leq f(a_1) - f(a_2) + f(a_3) - \cdots - f(a_{2n}) + f(a_{2n+1}). \end{aligned}$$

If $f$ is concave, then the inequality is reversed.

**Problem 2.** Let $a, b, c > 0$. Prove the inequality

$$\frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} \geq \frac{3}{2}.$$

**Solution.** This is a classical inequality. Usual solutions use Cauchy–Schwartz inequality or simple algebraic inequalities obtained by denoting $b + c = x$, etc. Another proof is given in the section "Cauchy–Schwartz revisited" of this book.

Set $s = a + b + c$. Then the inequality becomes

$$\frac{a}{s-a} + \frac{b}{s-b} + \frac{c}{s-c} \geq \frac{3}{2}.$$

Consider the function $f(x) = \frac{x}{s-x}$, defined on $(0, s)$. We have

$$f''(x) = \frac{2s}{(s-x)^3} > 0,$$

so that $f$ is convex on $(0, s)$. We then have

$$f\left(\frac{a+b+c}{3}\right) \leq \frac{f(a) + f(b) + f(c)}{3},$$

which means that

$$\frac{a}{s-a} + \frac{b}{s-b} + \frac{c}{s-c} \geq 3f\left(\frac{s}{3}\right) = 3\frac{\frac{s}{3}}{s-\frac{s}{3}} = \frac{3}{2},$$

as desired.

**Problem 3.** Let $f : [1, 13] \to R$ be a convex and integrable function. Prove that

$$\int_1^3 f(x)dx + \int_{11}^{13} f(x)\,dx \geq \int_5^9 f(x)\,dx.$$

**Solution.** We have seen that if $a < b < c$,

$$f(a-b+c) + f(b) \leq f(a) + f(c).$$

Let $c = a + 10$ and $b = a + 4$. Then

$$f(a+6) + f(a+4) \leq f(a) + f(a+10).$$

If we integrate both sides for $a \in [1, 3]$, and notice that

$$\int_1^3 f(a+6)\,da = \int_7^9 f(x)\,dx, \int_1^3 f(a+4)\,da = \int_5^7 f(x)\,dx,$$

and

$$\int_1^3 f(a+10)\,da = \int_{11}^{13} f(x)\,dx,$$

the result follows.

Proposed problems:

**Problem 4.** Let $a, b > 0$ and let $n$ be a positive integer. Prove the inequality

$$\frac{a^n + b^n}{2} \geq \left(\frac{a+b}{2}\right)^n.$$

**Problem 5.** Prove that

$$\sqrt[3]{3 - \sqrt[3]{3}} + \sqrt[3]{3 + \sqrt[3]{3}} < 2\sqrt[3]{3}.$$

**Problem 6.** Prove the $AM - GM$ inequality

$$\frac{x_1 + x_2 + \cdots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \ldots x_n},$$

for all $x_1, x_2, \ldots, x_n > 0$.

**Problem 7.** Let $a_1 < a_2 < \cdots < a_{2n+1}$ be positive real numbers. Prove the inequality

$$\sqrt[n]{a_1 - a_2 + a_3 - \cdots - a_{2n} + a_{2n+1}} \geq \sqrt[n]{a_1} - \sqrt[n]{a_2} + \cdots + \sqrt[n]{a_{2n+1}}.$$

**Problem 8.** Let $x, y, z > 0$. Prove that

$$\frac{x}{2x+y+z} + \frac{y}{x+2y+z} + \frac{z}{x+y+2z} \leq \frac{3}{4}.$$

**Problem 9.** Prove that if $a, b, c, d > 0$ and $a \leq 1, a+b \leq 5, a+b+c \leq 14, a+b+c+d \leq 30$, then

$$\sqrt{a} + \sqrt{b} + \sqrt{c} + \sqrt{d} \leq 10.$$

## 1.7 Induction at work

A large number of identities, inequalities etc. can be proved by induction. Sometimes induction is helpful in less obvious situations. We will examine several examples.

**Problem 1.** Let $n$ be a positive integer. Find the roots of the polynomial

$$P_n(X) = 1 + \frac{X}{1!} + \frac{X(X+1)}{2!} + \cdots + \frac{X(X+1)\ldots(X+n-1)}{n!}.$$

**Solution.** For $n = 1$, the polynomial $P_1(X) = 1 + X$ has the root $-1$. For $n = 2$, $P_2(X) = 1 + \frac{X}{1!} + \frac{X(X+1)}{2!} = \frac{1}{2}(X+2)(X+1)$ has roots $-1$ and $-2$. It is natural to presume that the roots of $P_n$ are $-1, -2, \ldots, -n$. We prove this assertion by induction on $n$. Suppose it holds true for $n$. Then $P_n$ factors as

$$P_n(X) = c(X+1)(X+2)\cdots(X+n),$$

the number $c$ being the coefficient of $X^n$. Checking again the definition of $P_n$, we see that $c = \frac{1}{n!}$, hence

$$P_n(X) = \frac{1}{n!}(X+1)(X+2)\cdots(X+n).$$

Now, observe that

$$\begin{aligned} P_{n+1}(X) &= P_n(X) + \frac{X(X+1)\cdots(X+n)}{(n+1)!} \\ &= \frac{1}{n!}(X+1)(X+2)\cdots(X+n) + \frac{X(X+1)\cdots(X+n)}{(n+1)!} \\ &= \frac{1}{(n+1)!}(X+1)(X+2)\cdots(X+n)(n+1+X), \end{aligned}$$

hence the roots of $P_{n+1}$ are $-1, -2, \ldots, -n$ and $-(n+1)$.

**Problem 2.** Let $n$ be a positive integer. Prove the inequality

$$\left(1+\frac{1}{1^3}\right)\left(1+\frac{1}{2^3}\right)\left(1+\frac{1}{3^3}\right)\cdots\left(1+\frac{1}{n^3}\right) < 3.$$

**Solution.** Apparently induction does not work here, since when passing from $n$ to $n+1$, the left-hand side increases while the right-hand side is constant. We can make induction work by proving a stronger result:

$$\left(1+\frac{1}{1^3}\right)\left(1+\frac{1}{2^3}\right)\left(1+\frac{1}{3^3}\right)\cdots\left(1+\frac{1}{n^3}\right) < 3 - \frac{1}{n}.$$

The assertion is clearly true for $n = 1$. Suppose it is true for $n$ and multiply the above inequality by $1 + \frac{1}{(n+1)^3}$. It suffices to prove that

$$\left(3-\frac{1}{n}\right)\left(1+\frac{1}{(n+1)^3}\right) \le 3 - \frac{1}{n+1}.$$

The difference

$$\left(3-\frac{1}{n}\right)\left(1+\frac{1}{(n+1)^3}\right)-3+\frac{1}{n+1}$$

factors as

$$-\frac{n^2-n+2}{n\,(n+1)^3},$$

hence it is negative. The claim is proved.

**Problem 3.** Let $p$ be a prime number. Prove that for any positive integer $a$ the number $a^p - a$ is divisible by $p$ (Fermat's little theorem).

**Solution.** We proceed by induction on $a$. The statement is true for $a = 1$, so suppose it holds true for some $a > 1$. We want to prove that $(a+1)^p - (a+1)$ is divisible by $p$. The binomial theorem gives

$$(a+1)^p = \sum_{k=0}^{p}\binom{p}{k}a^{p-k},$$

hence

$$(a+1)^p-(a+1) = a^p - a + \sum_{k=1}^{p-1}\binom{p}{k}a^{p-k}.$$

The conclusion follows from the induction hypothesis and from the fact that the binomial coefficients $\binom{p}{k}$ for $k = 1, 2, \ldots, p-1$ are divisible by $p$.

Use induction in solving the following problems.

**Problem 4.** Let $n$ be a positive integer. Prove the inequality

$$\left(1+\tfrac{1}{2}\right)\left(1+\frac{1}{2^2}\right)\left(1+\frac{1}{2^3}\right)\cdots\left(1+\frac{1}{2^n}\right) < \frac{5}{2}.$$

**Problem 5.** Let $n$ be a positive integer. Prove that the number

$$2^{2^n}-1$$

has at least $n$ distinct prime divisors.

**Problem 6.** Let $a$ and $n$ be positive integers such that $a < n!$. Prove that $a$ can be represented as a sum of at most $n$ distinct divisors of $n!$.

**Problem 7.** Let $x_1, x_2, \ldots, x_m, y_1, y_2, \ldots, y_n$ be positive integers such that the sums $x_1 + x_2 + \cdots + x_m$ and $y_1 + y_2 + \cdots + y_n$ are equal and less than $mn$. Prove that in the equality

$$x_1 + x_2 + \cdots + x_m = y_1 + y_2 + \cdots + y_n$$

one can cancel some terms and obtain another equality.

**Problem 8.** The sequence $(x_n)_{n\geq 1}$ is defined by $x_1 = 1$, $x_{2n} = 1 + x_n$ and $x_{2n+1} = \frac{1}{x_{2n}}$ for all $n \geq 1$. Prove that for any positive rational number $r$ there exists a unique $n$ such that $r = x_n$.

## 1.8 Roots and coefficients

Let $P(X) = a_0 + a_1X + \cdots + a_nX^n$ be a polynomial and $x_1, x_2, \ldots, x_n$ its roots (real or complex). It is known that the following equalities hold:

$$x_1 + x_2 + \cdots + x_n = -\frac{a_{n-1}}{a_n},$$

$$x_1x_2 + x_1x_3 + \cdots + x_{n-1}x_n = \frac{a_{n-2}}{a_n},$$

$$x_1x_2x_3 + x_1x_2x_4 + \cdots + x_{n-2}x_{n-1}x_n = -\frac{a_{n-3}}{a_n},$$

$$\vdots$$

$$x_1x_2 \ldots x_n = (-1)^n\frac{a_0}{a_n}.$$

These are usually called Viète's relations.

For instance, for a third degree polynomial

$$P(X) = a_0 + a_1X + a_2X^2 + a_3X^3,$$

we have

$$x_1 + x_2 + x_3 = -\frac{a_2}{a_3},$$

$$x_1x_2 + x_1x_3 + x_2x_3 = \frac{a_1}{a_3},$$

$$x_1x_2x_3 = -\frac{a_0}{a_3}.$$

Viète's relations can be very useful in solving problems not necessarily involving polynomials.

**Problem 1.** Let $a, b, c$ be nonzero real numbers such that

$$(ab+bc+ca)^3 = abc(a+b+c)^3.$$

Prove that $a, b, c$ are terms of a geometric sequence.

**Solution.** Consider the monic polynomial

$$P(X) = X^3 + mX^2 + nX + p,$$

with roots $a, b, c$. Then, by Viète's relations, we have

$$a+b+c = -m,$$
$$ab+bc+ca = n,$$
$$abc = -p.$$

The given equality yields $n^3 = m^3 p$, hence, if $m \neq 0$, the equation $P(x) = 0$ can be written as

$$x^3 + mx^2 + nx + \frac{n^3}{m^3} = 0,$$

or

$$m^3x^3 + m^4x^2 + nm^3x + n^3 = 0.$$

It is not difficult to factor the left-hand side:

$$(mx+n)\left(m^2x^2 - mnx + n^2\right) + m^3x\,(mx+n)$$

$$= (mx+n)\left(m^2x^2 + \left(m^3 - mn\right)x + n^2\right).$$

It follows that one of the roots of $P$ is $x_1 = -\frac{n}{m}$ and the other two satisfy the condition $x_2x_3 = \frac{n^2}{m^2}$ (Viète's relations for the second degree polynomial $m^2X^2 + \left(m^3 - mn\right)X + n^2$ ). We obtained $x_1^2 = x_2x_3$, thus the roots are the terms of a geometric sequence. If $m = 0$, then $n = 0$ but in this case, the polynomial $X^3 + p$ cannot have three real roots.

*Observation.* Using appropriate software, one can obtain the factorization

$$(ab+bc+ca)^3 - abc(a+b+c)^3 = (a^2-bc)(b^2-ac)(c^2-ab),$$

and the conclusion follows.

**Problem 2.** Solve in real numbers the system of equations

$$\begin{cases} x+y+z=4 \\ x^2+y^2+z^2=14 \\ x^3+y^3+z^3=34 \end{cases}.$$

**Solution.** Consider the monic polynomial

$$P(t)=t^3+at^2+bt+c,$$

with roots $x, y, z$.

Because $x+y+z=4$, it follows that $a=-4$, hence

$$P(t)=t^3-4t^2+bt+c.$$

We have

$$x^2+y^2+z^2=(x+y+z)^2-2(xy+xz+yz).$$

It follows that

$$b=xy+xz+yz=1.$$

The numbers $x, y, z$ are the roots of $P$, thus

$$\begin{aligned} x^3-4x^2+x+c=0, \\ y^3-4y^2+y+c=0, \\ z^3-4z^2+z+c=0. \end{aligned}$$

Adding these equalities and using the equations of the system, we obtain $c=6$, hence

$$P(t)=t^3-4t^2+t+6.$$

We observe that $t_1=-1$ is a root, so $P$ factors as

$$P(t)=(t+1)(t^2-5t+6),$$

the other two roots being $t_2=2$ and $t_3=3$. It follows that the solutions of the system are the triple $(-1.\,2.\,3)$ and all of its permutations.

**Problem 3.** Let $a$ and $b$ be two of the roots of the polynomial $X^4+X^3-1$. Prove that $ab$ is a root of the polynomial $X^6 + X^4 + X^3 - X^2 - 1$.

**Solution.** Let $c$ and $d$ be the other two roots of $X^4 + X^3 - 1$. Viète's relations yield

$$a+b+c+d=-1,$$
$$ab+ac+ad+bc+bd+cd=0,$$
$$abc+abd+acd+bcd=0,$$
$$abcd=-1.$$

Write these equalities in terms of $s=a+b$, $s'=c+d$, $p=ab$ and $p'=cd$ (this is often useful) to obtain

$$s+s'=-1,$$
$$p+p'+ss'=0,$$
$$ps'+p's=0,$$
$$pp'=-1.$$

Substituting $p'=-\frac{1}{p}$ and $s'=-1-s$ in the second and in the third equalities yields

$$p-\frac{1}{p}-s^2-s=0$$

and

$$p(-1-s)-\frac{s}{p}=0.$$

It follows from the second equality that

$$s=-\frac{p^2}{p^2+1}.$$

Plugging this into the first equality gives

$$p-\frac{1}{p}-\frac{p^4}{(p^2+1)^2}+\frac{p^2}{p^2+1}=0.$$

A short computation shows that this is equivalent to

$$p^6+p^4+p^3-p^2-1=0,$$

hence $p = ab$ is a root of the polynomial

$$X^6 + X^4 + X^3 - X^2 - 1.$$

Here are some suggested problems.

**Problem 4.** Let $a, b, c$ be nonzero real numbers such that $a + b + c \neq 0$ and

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{a+b+c}.$$

Prove that for all odd integers $n$

$$\frac{1}{a^n} + \frac{1}{b^n} + \frac{1}{c^n} = \frac{1}{a^n + b^n + c^n}.$$

**Problem 5.** Let $a \leq b \leq c$ be real numbers such that

$$a + b + c = 2$$

and

$$ab + bc + ca = 1.$$

Prove that

$$0 \leq a \leq \tfrac{1}{3} \leq b \leq 1 \leq c \leq \frac{4}{3}.$$

**Problem 6.** Prove that two of the four roots of the polynomial $X^4 + 12X - 5$ add up to 2.

**Problem 7.** Find $m$ and solve the following equation, knowing that its roots form a geometrical sequence:

$$X^4 - 15X^3 + 70X^2 - 120X + m = 0.$$

**Problem 8.** Let $x_1, x_2, \ldots, x_n$ be the roots of the polynomial $X^n + X^{n-1} + \cdots + X + 1$. Prove that

$$\frac{1}{1-x_1} + \frac{1}{1-x_2} + \cdots + \frac{1}{1-x_n} = \frac{n}{2}.$$

**Problem 9.** Let $a, b, c$ be rational numbers and let $x_1, x_2, x_3$ be the roots of the polynomial $P(X) = X^3 + aX^2 + bX + c$. Prove that if $\frac{x_1}{x_2}$ is a rational number, different from 0 and $-1$, then $x_1, x_2, x_3$ are rational numbers.

**Problem 10.** Solve in real numbers the system of equations

$$\begin{cases} x+y+z=0 \\ x^3+y^3+z^3=18 \\ x^7+y^7+z^7=2058 \end{cases}$$

**Problem 11.** Solve in real numbers the system of equations

$$\begin{cases} a+b=8 \\ ab+c+d=23 \\ ad+bc=28 \\ cd=12 \end{cases}$$

# Chapter 2

# Geometry and Trigonometry

## 2.1 Geometric inequalities

One of the most basic geometric inequalities is the triangle inequality: in every triangle, the length of one side is less than the sum of the two other sides' lengths. More generally, for any three points $A$, $B$, $C$ one has

$$AC + BC \geq AB$$

with equality if and only if $C$ lies on the line segment $AB$. Many interesting problems can be solved using this simple idea.

**Problem 1.** Let $ABCD$ be a convex quadrilateral and let $M$, $N$ be the midpoints of $AD$ and $BC$, respectively. Prove that

$$MN = \frac{AB + CD}{2}$$

if and only if $AB$ is parallel to $CD$.

**Solution.** Let $P$ be the midpoint of the diagonal $AC$ (see Figure 2.1). Then $MP$ and $PN$ are parallel to $CD$ and $AB$, respectively. Moreover, we have $MP = \frac{CD}{2}$ and $PN = \frac{AB}{2}$. Applying the triangle inequality to $\Delta MNP$ gives

$$\frac{AB}{2} + \frac{CD}{2} = PN + MP \geq MN.$$

The equality occurs if and only if $P$ lies on the line segment $MN$; that is, if the lines $MP$, $PN$ and $MN$ coincide. Since $MP||CD$ and $NP||AB$, the latter holds true if and only if $AB$ and $CD$ are parallel.

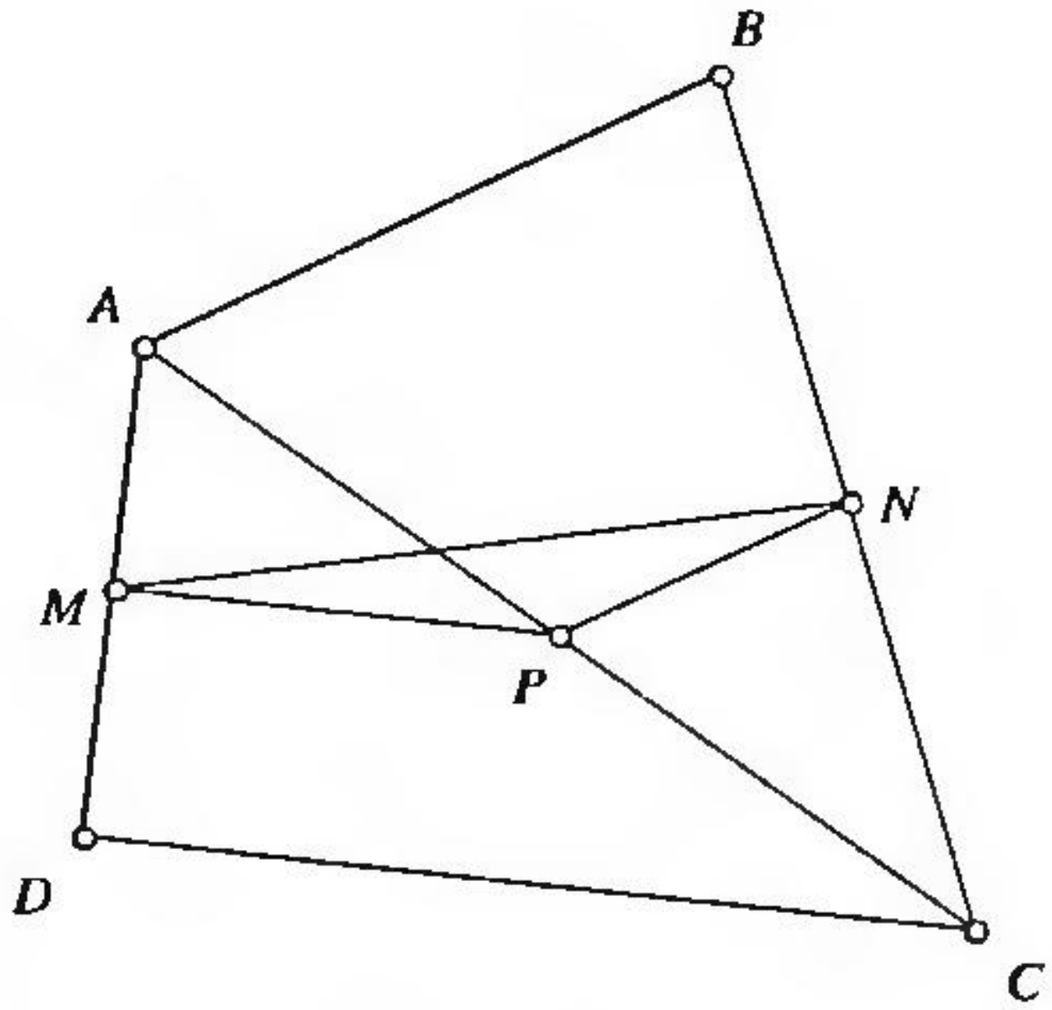


Figure 2.1

**Problem 2.** Let $D$ be the midpoint of the side $BC$ of triangle $ABC$. Prove that

$$AD < \frac{AB + AC}{2}.$$

**Solution.** Let the point $A'$ be such that $D$ is the midpoint of $AA'$. Thus, the quadrilateral $ABA'C$ is a parallelogram and $AD = \frac{AA'}{2}$. In triangle $ABA'$, we have $AA' < AB + BA'$. The conclusion follows observing that $BA' = AC$.

**Problem 3.** Let $M$ be a point inside the triangle $ABC$. Prove that

$$AB + AC > MB + MC.$$

**Solution.** Let $N$ be the point at which $BM$ intersects $AC$. Then we successively have:

$$\begin{aligned} AB + AC &= AB + AN + NC \\ &> BN + NC = BM + MN + NC > BM + MC. \end{aligned}$$

**Problem 4.** Let $A, B, C$ and $D$ be four points in space, not in the same plane. Prove that

$$AC \cdot BD < AB \cdot CD + AD \cdot BC.$$

**Solution.** Consider a sphere which passes through the points $B, C$ and $D$ and intersects the segments $AB$, $AC$ and $AD$ at the points $B', C'$ and

$D'$. The intersection of the sphere and the plane $(ABC)$ is a circle, thus the quadrilateral $BB'C'C$ is cyclic. It follows that triangles $ABC$ and $AC'B'$ are similar; hence

$$\frac{AB'}{AC} = \frac{AC'}{AB} = \frac{B'C'}{BC}.$$

We obtain

$$B'C' = \frac{BC \cdot AB'}{AC}.$$

Analogously, triangles $ABD$ and $AD'B'$ are similar, so that

$$B'D' = \frac{BD \cdot AB'}{AD}.$$

We deduce that

$$\frac{B'C'}{B'D'} = \frac{BC \cdot AD}{AC \cdot BD}.$$

In a similar way, we obtain

$$\frac{C'D'}{B'D'} = \frac{AB \cdot CD}{AC \cdot BD}.$$

But in triangle $A'B'C'$ we have $B'C' + C'D' > B'D'$, and therefore it follows that

$$\frac{BC \cdot AD}{AC \cdot BD} + \frac{AB \cdot CD}{AC \cdot BD} > 1,$$

so that $AC \cdot BD < AB \cdot CD + AD \cdot BC$.

*Observation.* If $A$, $B$, $C$, $D$ are coplanar points, then one can prove that

$$AC \cdot BD \leq AB \cdot CD + AD \cdot BC,$$

with equality if and only if $ABCD$ is a cyclic quadrilateral. The proof is similar, but instead of constructing the sphere, one uses inversion.

Here are some proposed problems.

**Problem 5.** Let $ABCD$ be a convex quadrilateral. Prove that

$$max(AB + CD, AD + BC) < AC + BD < AB + BC + CD + DA.$$

**Problem 6.** Let $M$ be the midpoint of segment $AB$. Prove that if $O$ is an arbitrary point, then

$$|OA - OB| \leq 2OM.$$

**Problem 7.** Prove that in an arbitrary triangle, the sum of the lengths of the altitudes is less than the triangle's perimeter.

**Problem 8.** Denote by $P$ the perimeter of triangle $ABC$. If $M$ is a point in the interior of the triangle, prove that

$$\frac{1}{2}P < MA + MB + MC < P.$$

**Problem 9.** Prove that if $A'$, $B'$ and $C'$ are the midpoints of the sides $BC$, $CA$, and $AB$, respectively, then:

$$\frac{3}{4}P < AA' + BB' + CC' < P.$$

**Problem 10.** In the convex quadrilateral $ABCD$, we have

$$AB + BD \leq AC + CD.$$

Prove that $AB < AC$.

**Problem 11.** Consider $n$ red and $n$ blue points in the plane, no three of them being collinear. Prove that one can connect each red point to a blue one with a segment such that no two segments intersect.

**Problem 12.** Let $n$ be an odd positive integer. On some field, $n$ gunmen are placed such that all pairwise distances between them are different. At a signal, every gunman takes out his gun and shoots the closest gunman. Prove that:

(a) at least one gunman survives;

(b) no gunman is shot more than five times;

(c) the trajectories of the bullets do not intersect.

**Problem 13.** Prove that the medians of a given triangle can form a triangle.

**Problem 14.** Let $A$ and $B$ be two points situated on the same side of a line $XY$. Find the position of a point $M$ on the line such that the sum $AM + MB$ is minimal.

**Problem 15.** Let $ABC$ be an acute triangle. Find the positions of the points $M$, $N$, $P$ on the sides $BC$, $CA$, $AB$, respectively, such that the perimeter of the triangle $MNP$ is minimal.

**Problem 16.** Seven real numbers are given in the interval (1,13). Prove that at least three of them are the lengths of a triangle's sides.

## 2.2 An interesting locus

Let us consider a triangle $ABC$ and let $D$ be the midpoint of segment $BC$. It is not difficult to see that

$$[ABD] = [ACD].$$

Moreover, for each point $M$ on the line $AD$ we have

$$[ABM] = [ACM].$$

We want to determine the locus of points $M$ such that $[ABM] = [ACM]$. The triangles $ABM$ and $ACM$ have the common side $AM$; hence their areas are equal if and only if the distances from $B$ and $C$ to $AM$ are equal.

It is not difficult to see that this happens in two situations: $AM$ passes through the midpoint of $BC$ or $AM$ is parallel to $BC$ (see Figure 2.2). We

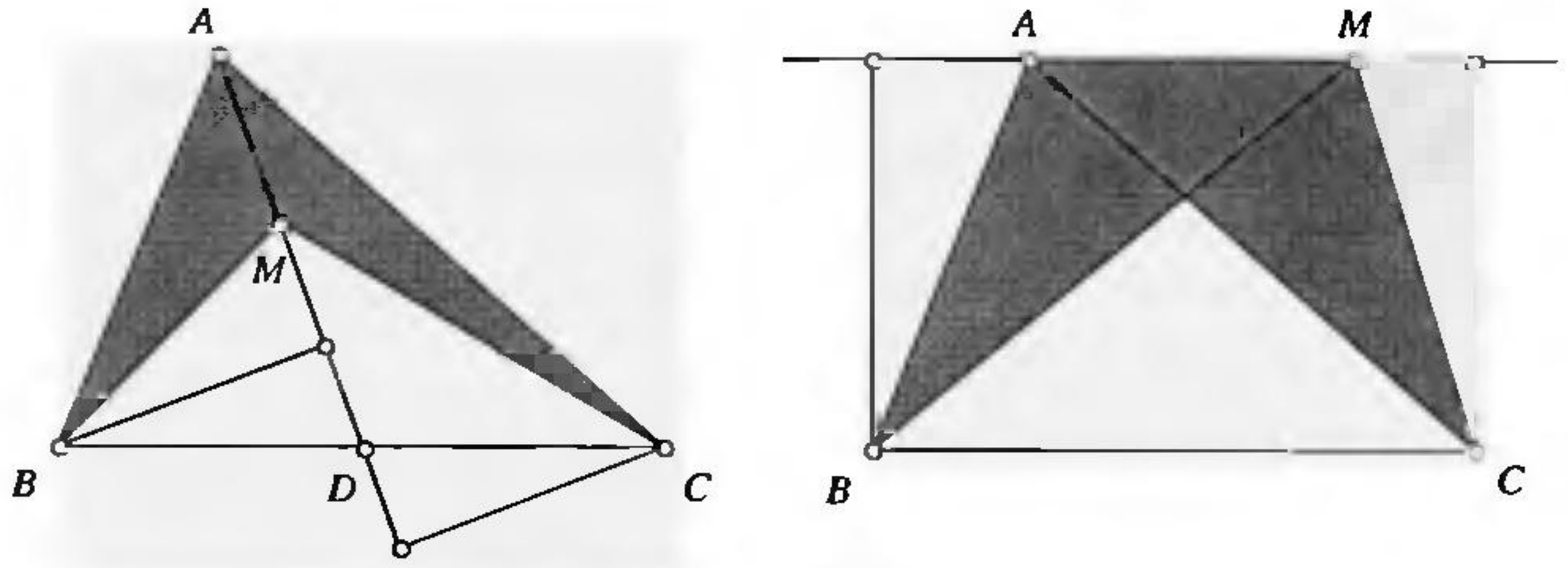


Figure 2.2

derive that the locus of points $M$ in the interior of triangle $ABC$ for which $[ABM] = [ACM]$ is the median $AD$.

Now, let us translate the sides $AB$ and $AC$ to $A'B$ and $A''C$ and ask a similar question. Find the locus of points $M$ such that

$$[A'BM] = [A''CM]$$

(see Figure 2.3).

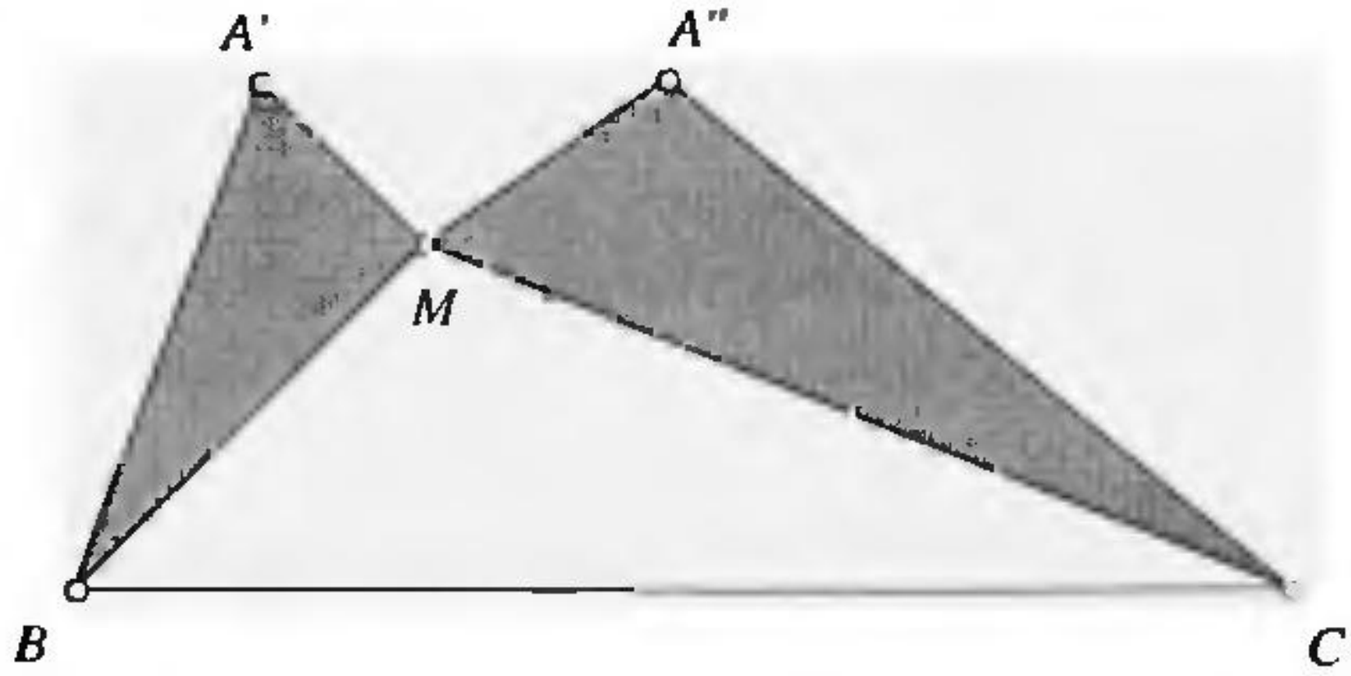


Figure 2.3

Let us rephrase this:

**Problem 1.** Given the quadrilateral $ABCD$ such that $AB$ and $CD$ are not parallel, find the locus of points $M$ inside $ABCD$ for which $[ABM] = [CDM]$.

**Solution.** We apply other translations. Denote by $T$ the point of intersection of the lines $AB$ and $CD$. We translate the segment $AB$ to $TX$ and the segment $CD$ to $TY$. It is not difficult to see that $[ABM] = [TXM]$ and $[CDM] = [TYM]$, hence $M$ lies on the median of triangle $TXY$.

We deduce that the desired locus is a segment: the part of the median of $TXY$ lying inside the quadrilateral $ABCD$ (see Figure 2.4.).

**Problem 2.** Prove that in a trapezoid, the midpoints of parallel sides, the point of intersection of the diagonals, and the point of intersection of the nonparallel sides are collinear.

**Solution.** This is an immediate application of the above property. We have (with the notation in the Figure 2.5):

$$[ABP] = [CDP], [ABR] = [CDR], [ABS] = [CDS] = 0,$$

so all we have to do is to prove that $[ABQ] = [CDQ]$.

But

$$[ABQ] + [BQC] = [ABC] = [DBC] = [CDQ] + [BQC],$$

and we are done.

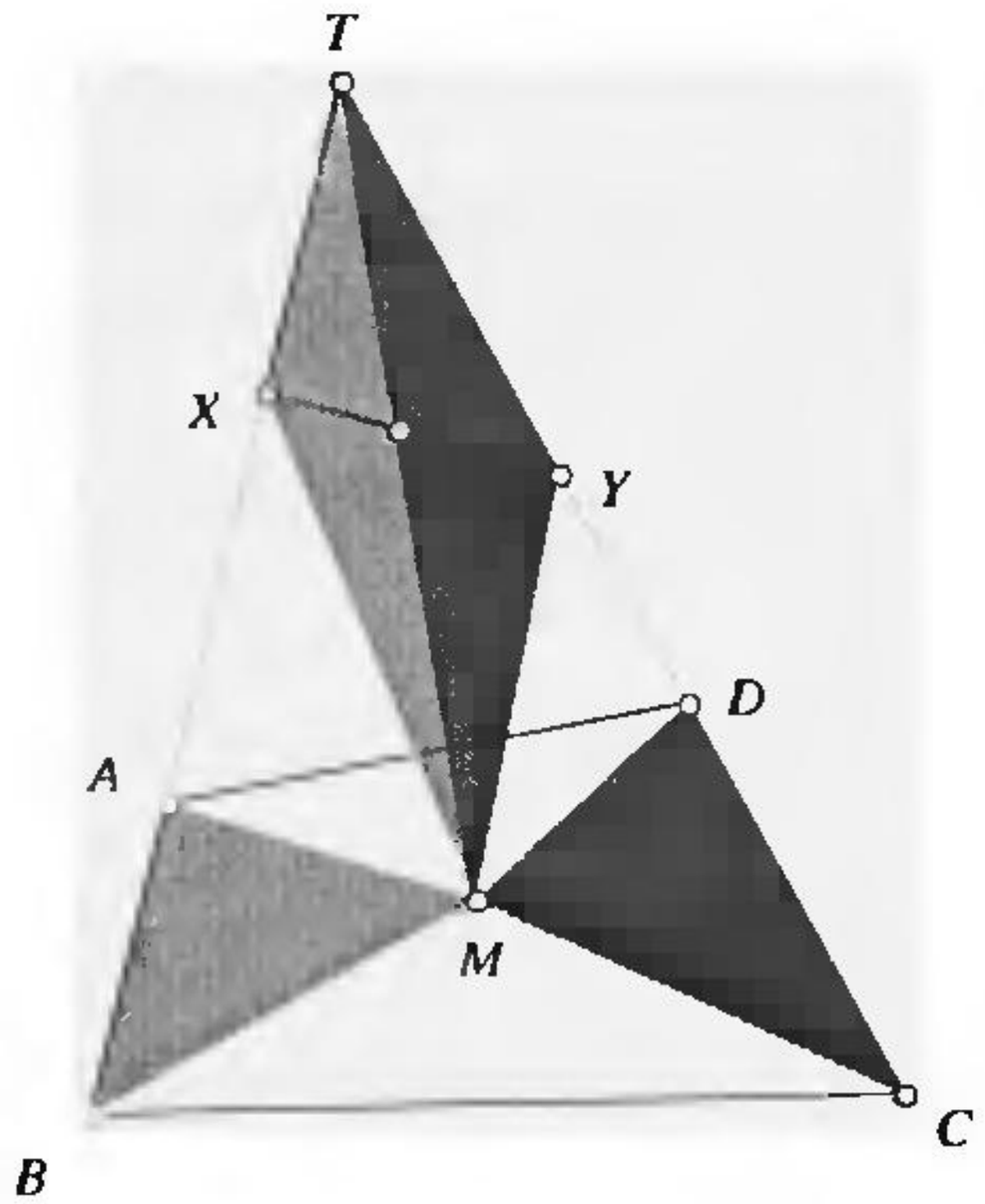


Figure 2.4

**Problem 3.** Suppose we are given a positive number $k$ and a quadrilateral $ABCD$ in which $AB$ and $CD$ are not parallel. Find the locus of points $M$ inside $ABCD$ for which $[ABM] + [CDM] = k$.

**Solution.** Using the construction from Problem 1, we obtain that

$$[ABM] + [CDM] = [TXMY] = [TXY] + [XMY].$$

Since $X$ and $Y$ are fixed points, it follows that $[XMY]$ is a constant; therefore the point $M$ lies on a parallel to the line $XY$. According to the value of $k$, the locus is either a segment (or a point) or it is the empty set.

*Observation.* If $M$ lies on this parallel, but in the exterior of the quadrilateral, it is not difficult to see that in this case we either must have

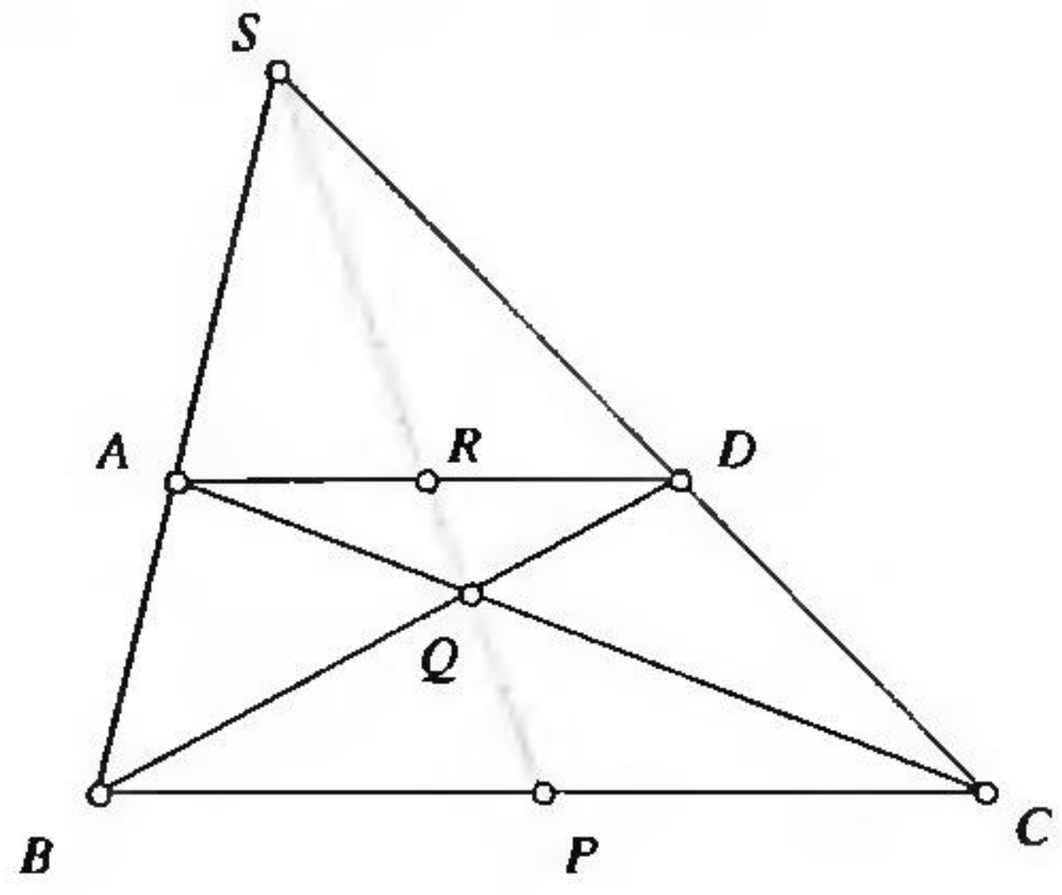


Figure 2.5

$[ABM] - [CDM] = k$, or $[CDM] - [ABM] = k$.

**Problem 4.** A circle with center $O$ is inscribed in the convex quadrilateral $ABCD$. If $M$ and $N$ are the midpoints of the diagonals $AC$ and $BD$, prove that points $O$, $M$, and $N$ are collinear.

**Solution.** Since a circle is inscribed in quadrilateral $ABCD$ (see Figure 2.6), we know that

$$AB + CD = AD + BC.$$

Multiplying the equality by $r/2$, where $r$ is the radius of the circle, we obtain

$$[OAB] + [OCD] = [OAD] + [OBC].$$

Thus

$$[OAB] + [OCD] = \tfrac{1}{2}[ABCD].$$

On the other hand,

$$[NAB] + [NCD] = \tfrac{1}{2}[ABD] + \tfrac{1}{2}[BCD] = \tfrac{1}{2}[ABCD].$$

Similarly,

$$[MAB] + [MCD] = \tfrac{1}{2}[ABCD].$$

It follows that $O$, $M$ and $N$ belong to the locus of the points $X$ such that

$$[XAB] + [XCD] = \tfrac{1}{2}[ABCD],$$

and they lie on the same segment if the opposite sides of $ABCD$ are not parallel. If $AB$ is parallel to $CD$ or $AD$ is parallel to $BC$, the result is trivial.

Let us investigate the following problems:

**Problem 5.** Find the locus of points $M$ in the plane of triangle $ABC$ such that $[ABM] = 2[ACM]$.

**Problem 6.** Let $D$ be a point on the side $BC$ of triangle $ABC$ and let $M$ be a point on $AD$. Prove that

$$\frac{[ABM]}{[ACM]} = \frac{BD}{CD}.$$

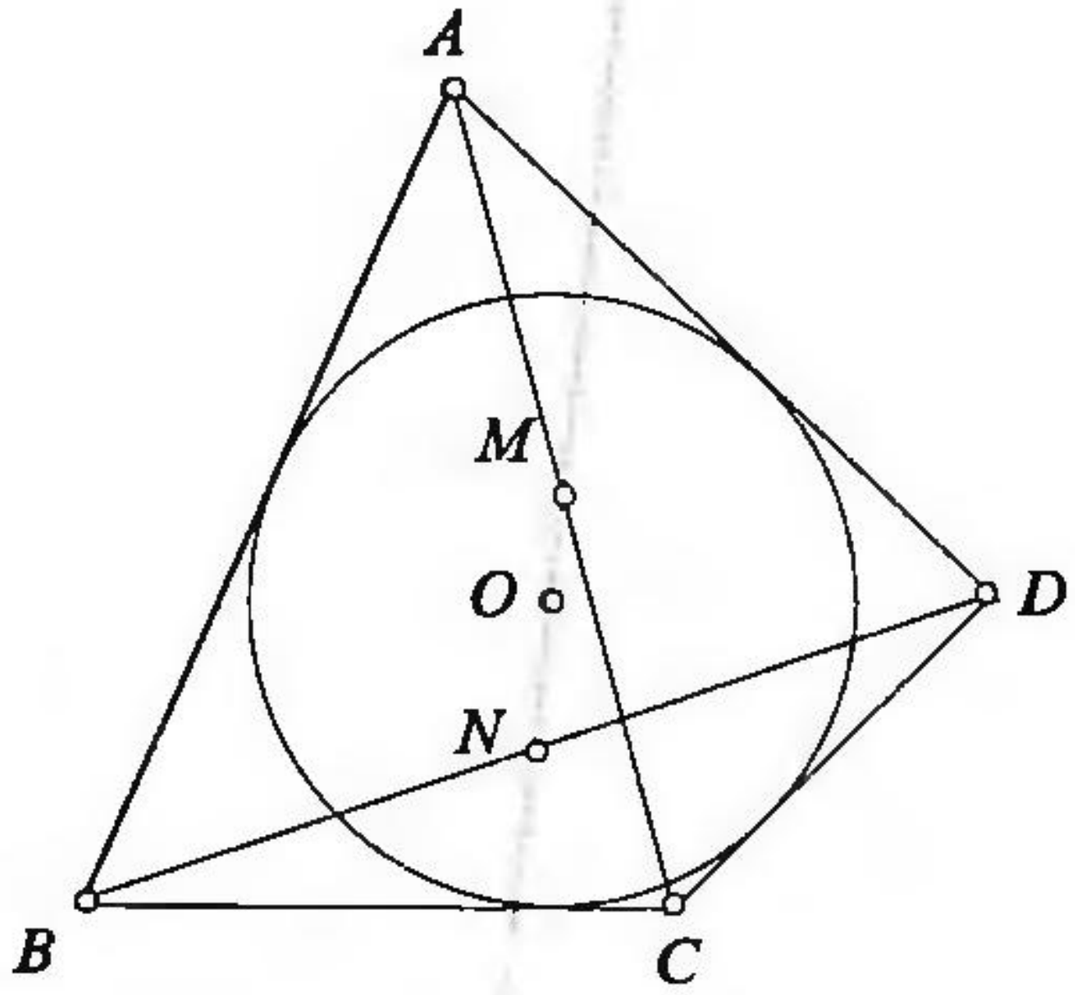


Figure 2.6

Deduce Ceva's theorem: If the segments $AD$, $BE$ and $CF$ are concurrent, then

$$\frac{BD}{CD} \cdot \frac{CE}{AE} \cdot \frac{AF}{BF} = 1.$$

**Problem 7.** Let $ABCD$ be a convex quadrilateral and let $M$ be a point in its interior such that

$$[MAB] = [MBC] = [MCD] = [MDA].$$

Prove that one of the diagonals of $ABCD$ passes through the midpoint of the other diagonal.

**Problem 8.** Let $ABCD$ be a convex quadrilateral. Find the locus of points $M$ in its interior such that

$$[MAB] = 2[MCD].$$

**Problem 9.** Let $ABCD$ be a convex quadrilateral and let $k > 0$ be a real number. Find the locus of points $M$ in its interior such that

$$[MAB] + 2[MCD] = k.$$

**Problem 10.** Let $d, d'$ be two nonparallel lines in the plane and let $k > 0$. Find the locus of points, the sum of whose distances to $d$ and $d'$ is equal to $k$.

**Problem 11.** Let $ABCD$ be a convex quadrilateral and let $E$ and $F$ be the points of intersections of the lines $AB$, $CD$ and $AD$, $BC$, respectively. Prove that the midpoints of the segments $AC$, $BD$, and $EF$ are collinear.

**Problem 12.** In the interior of a quadrilateral $ABCD$, consider a variable point $P$. Prove that if the sum of distances from $P$ to the sides is constant, then $ABCD$ is a parallelogram.

## 2.3 Cyclic quads

A convex quadrilateral is called *cyclic* if its vertices lie on a circle. It is not difficult to see that a necessary and sufficient condition for this is that the sum of the opposite angles of the quadrilateral be equal to 180°. (Figure 2.7.)

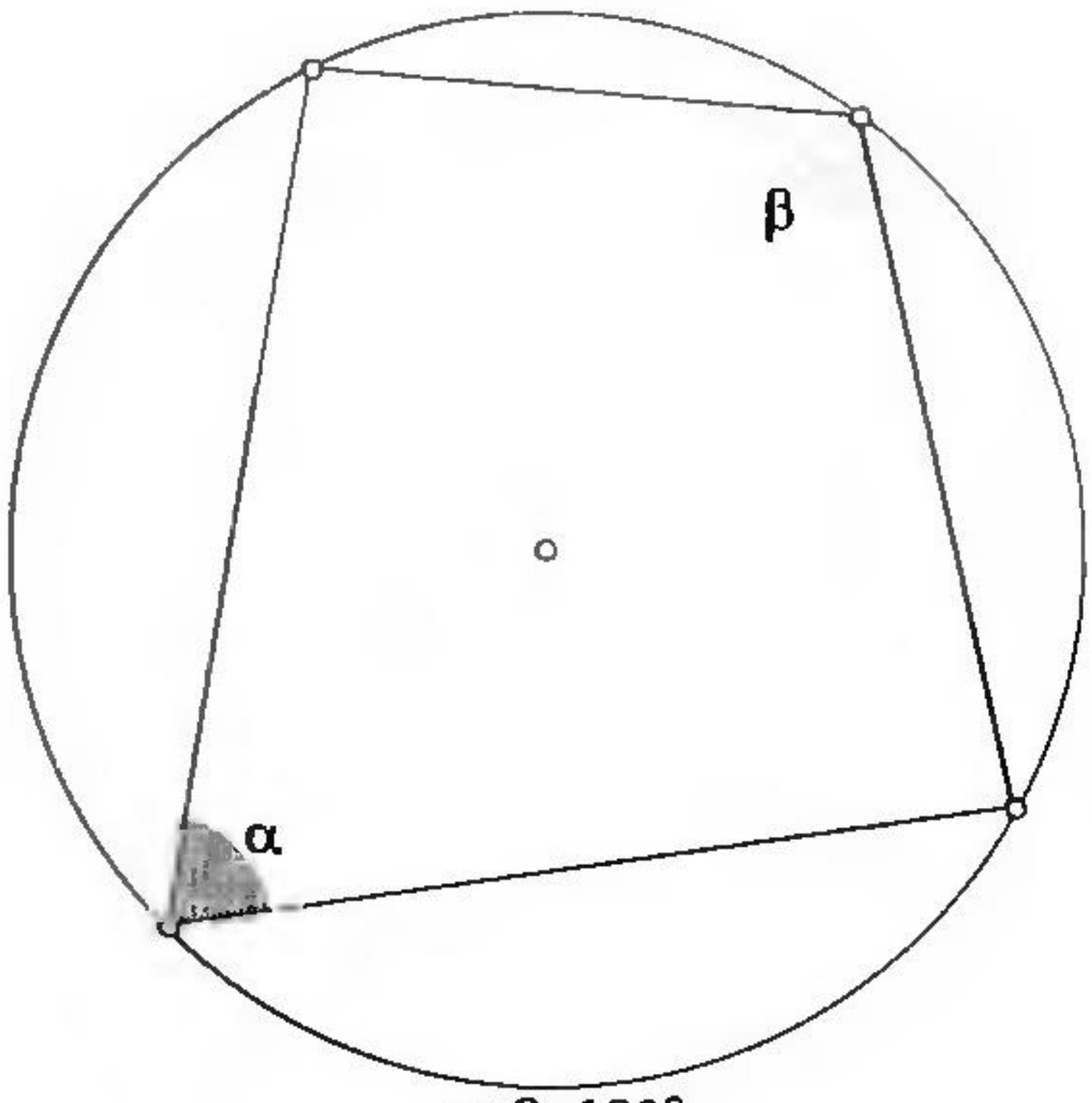


Figure 2.7

As a special case, if two opposite angles of the quadrilateral are right angles, then the quadrilateral is cyclic and one of its diagonals is a diameter of the circumscribed circle.

Another necessary and sufficient condition is that the angle between one side and a diagonal be equal to the angle between the opposite side and the other diagonal (see Figure 2.8).

**Problem 1.** Let $ABCD$ be a cyclic quadrilateral. Recall that the incenter of a triangle is the intersection of the angles' bisectors. Prove that the incenters of triangles $ABC$, $BCD$, $CDA$ and $DAB$ are the vertices of a rectangle.

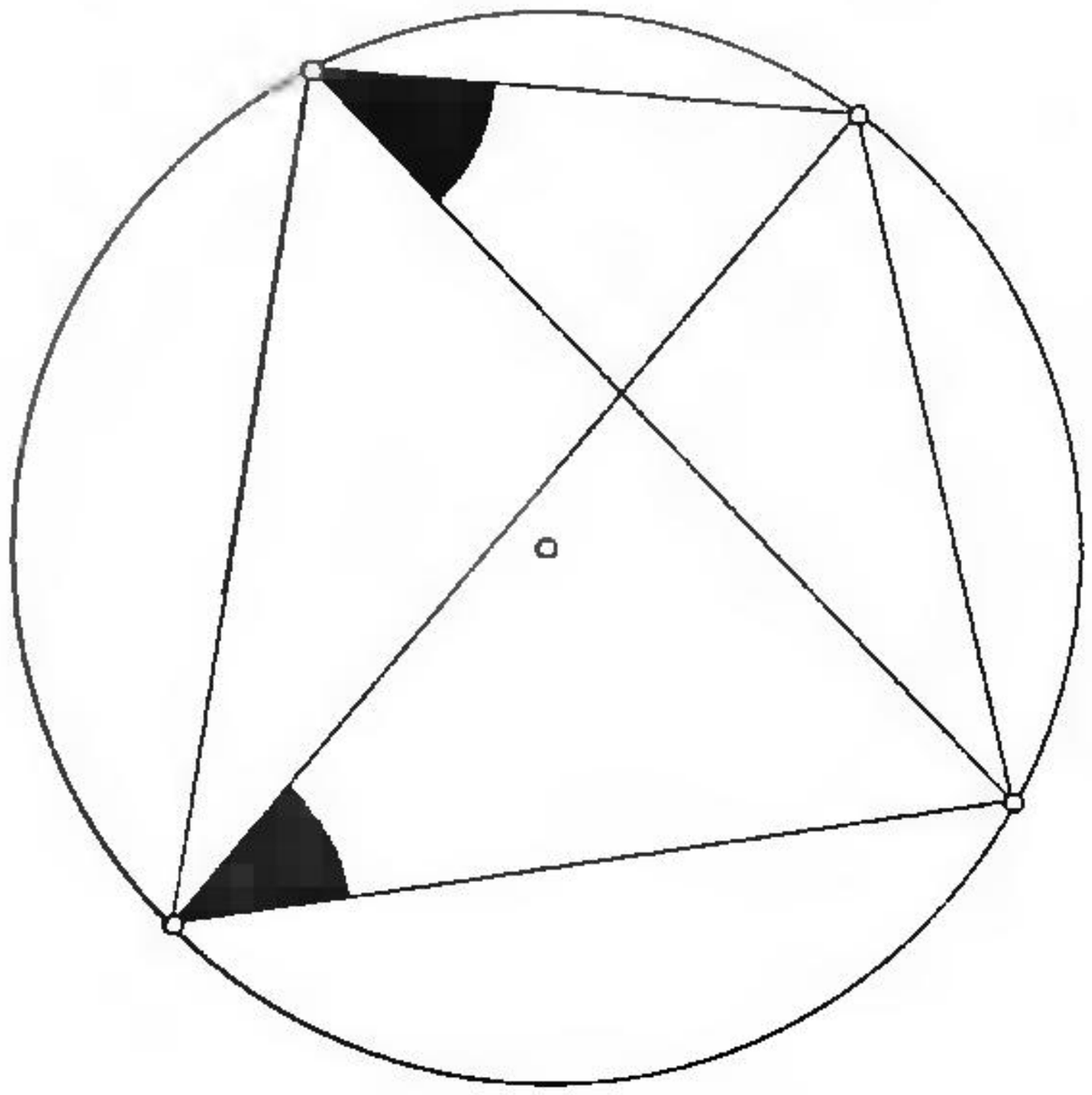

Figure 2.8

**Solution.** The incenter of a triangle is the intersection of the angle's bisectors. Let $K, L, M$ and $N$ be the four incenters (Figure 2.9.). We then have

$$\angle AKB = 180^\circ - \angle BAC/2 - \angle ABC/2 = 90^\circ + \angle ACB/2.$$

Similarly,

$$\angle ANB = 90^\circ + \angle ADB/2.$$

But since $ABCD$ is cyclic, we have $\angle ACB = \angle ADB$, hence $\angle AKB = \angle ANB$. This means that the quadrilateral $ANKB$ is also cyclic, so

$$\angle NKB = 180^\circ - \angle BAN = 180^\circ - \angle A/2.$$

In the same manner we obtain that

$$\angle BKL = 180^\circ - \angle C/2,$$

hence

$$\angle NKL = 360^\circ - (\angle NKB + \angle BKL) = (\angle A + \angle C)/2 = 180^\circ/2 = 90^\circ.$$

Analogously, we prove that the other three angles of $KLMN$ are right angles.

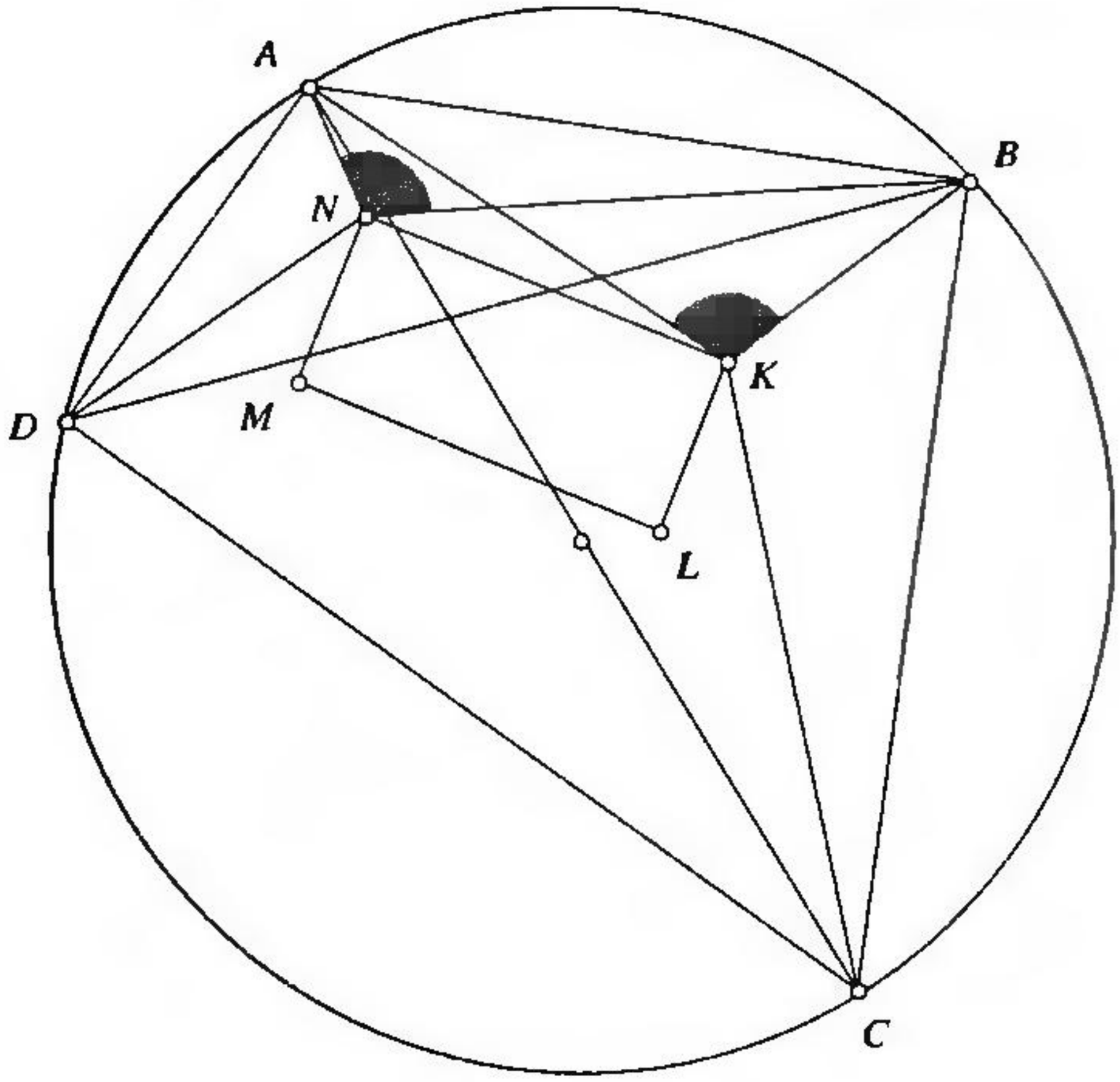


**Figure 2.9**

**Problem 2.** In the triangle $ABC$, the altitude, angle bisector and median from $C$ divide the angle $\angle C$ into four equal angles. Find the angles of the triangle.

**Solution.** No cyclic quads in the problem! (Figure 2.10.) However, a solution using some trigonometry is at hand: with usual notation, from the angle bisector theorem, we have

$$\frac{AE}{EB} = \frac{AC}{CB} = \frac{b}{a},$$

so that

$$\frac{AE}{AB} = \frac{b}{a+b}$$

and

$$AE = \frac{bc}{a+b}.$$

Also,

$$\frac{FB}{EB} = \frac{BC}{CE} = \frac{a}{b}.$$

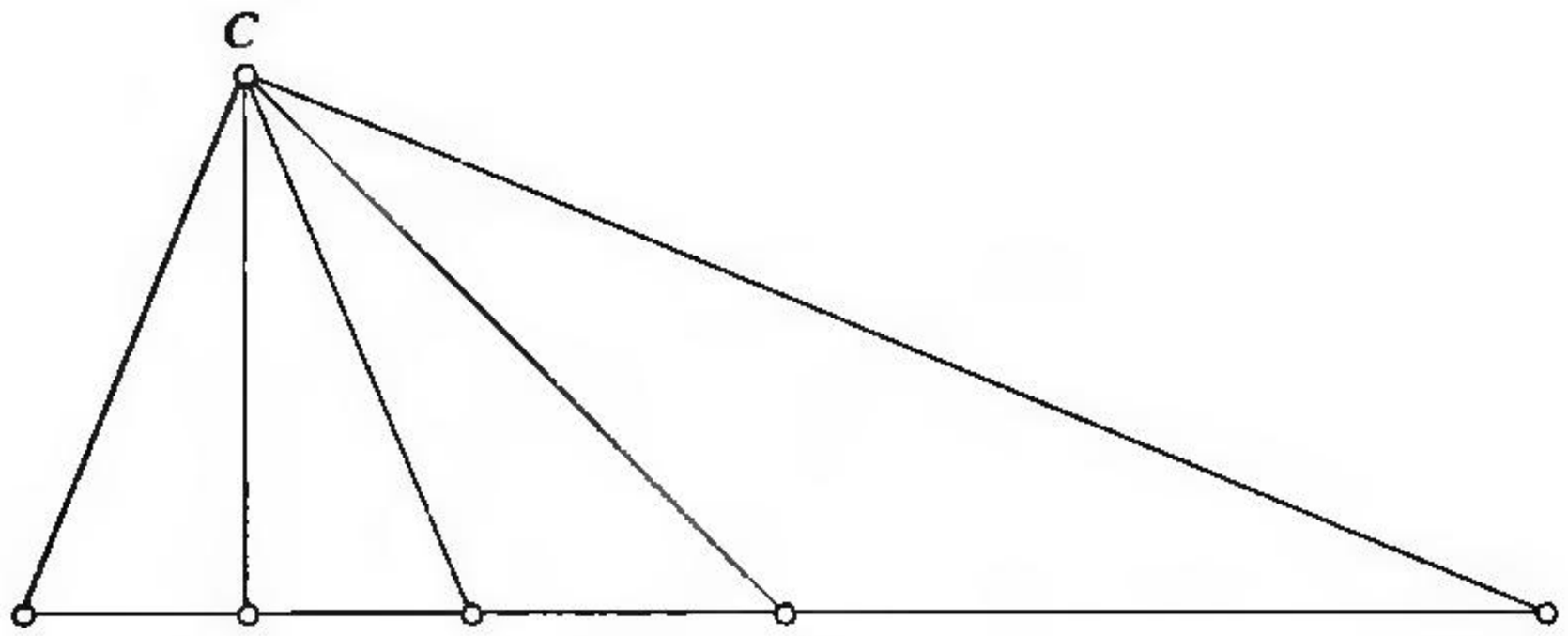


Figure 2.10

But

$$FB = \frac{c}{2}, FE = AF - AE = \frac{c}{2} - \frac{bc}{a+b} = \frac{c(a-b)}{2(a+b)},$$

so that we obtain

$$\frac{\frac{c}{2}}{\frac{c(a-b)}{2(a+b)}} = \frac{a}{b},$$

which is equivalent to

$$\frac{a+b}{a-b} = \frac{a}{b},$$

or $b^2 + 2ab - a^2 = 0$. We get the quadratic equation

$$\left(\frac{b}{a}\right)^2 + 2\frac{b}{a} - 1 = 0,$$

from which we deduce that

$$\frac{b}{a} = \sqrt{2} - 1.$$

On the other hand, we have

$$b = CE = \frac{2ab}{a+b}\cos\frac{\angle C}{2};$$

hence

$$\cos\frac{\angle C}{2} = \frac{a+b}{2a} = \tfrac{1}{2} + \tfrac{1}{2}\frac{b}{a} = \tfrac{1}{2} + \tfrac{1}{2}(\sqrt{2}-1) = \frac{\sqrt{2}}{2}.$$

We conclude that $\frac{\angle C}{2} = 45°$, so $\angle C = 90°$, $\angle A = 67.5°$ and $\angle B = 22.5°$.

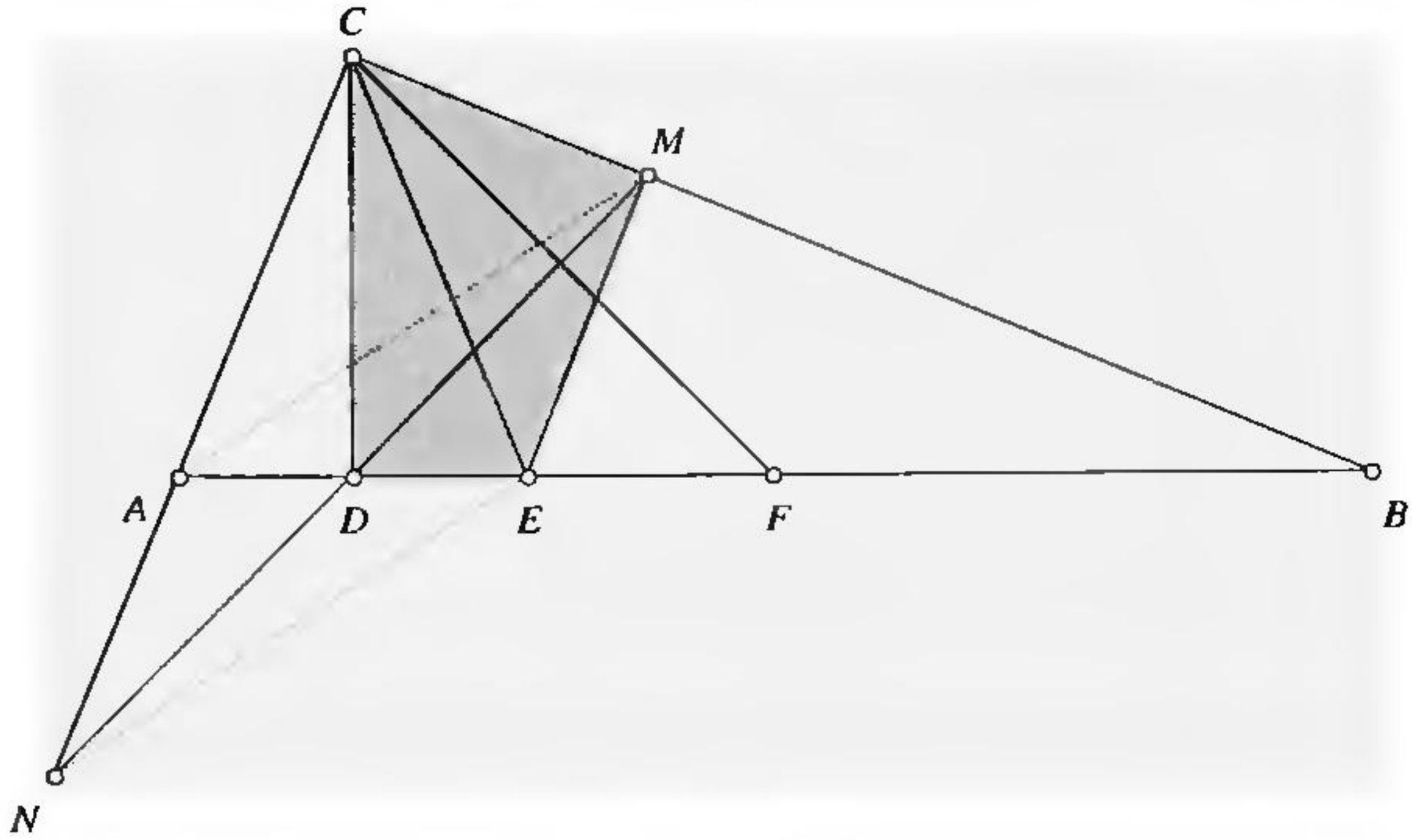


Figure 2.11

Now, let's try a purely geometric approach. Let $D$, $E$ and $F$ on $AB$ be the feet of the altitude, angle bisector and median. (Figure 2.11.)

Drop a perpendicular from $E$ to $BC$ in the point $M$ and let $N$ be the intersection between $MD$ and $AC$.

The quadrilateral $CDEM$ is cyclic, since

$$\angle CDE = \angle CME = 90^\circ.$$

Then

$$\angle CMD = \angle CED = \angle CAD.$$

It follows that triangles $CMN$ and $CAB$ are similar.

This implies that $CD$ is the median from $C$ in triangle $CMN$, hence $AMEN$ is a parallelogram ( $MN$ and $AE$ have the same midpoint).

Finally, $AN \parallel ME$ and since $ME \perp BC$, it follows that $AC \perp BC$.

We conclude that $\angle C = 90^\circ$, $\angle A = 67.5^\circ$ and $\angle B = 22.5^\circ$.

*Observation.* Another proof using cyclic quads follows from the lemma below.

*Lemma.* Let $ABC$ be a triangle inscribed in the circle centered at $O$ such that the angles $\angle B$ and $\angle C$ are acute. If $H$ is its orthocenter, then $\angle BAH = \angle CAO$. (Figure 2.12.)

Proof: Let $A'$ be the point on the circumcircle such that $AA'$ is a diameter. Then $ABA'C$ is cyclic, hence $\angle ABC = \angle AA'C$. It follows that their complementary angles $\angle BAH$ and $\angle CAA'$ are equal as well.

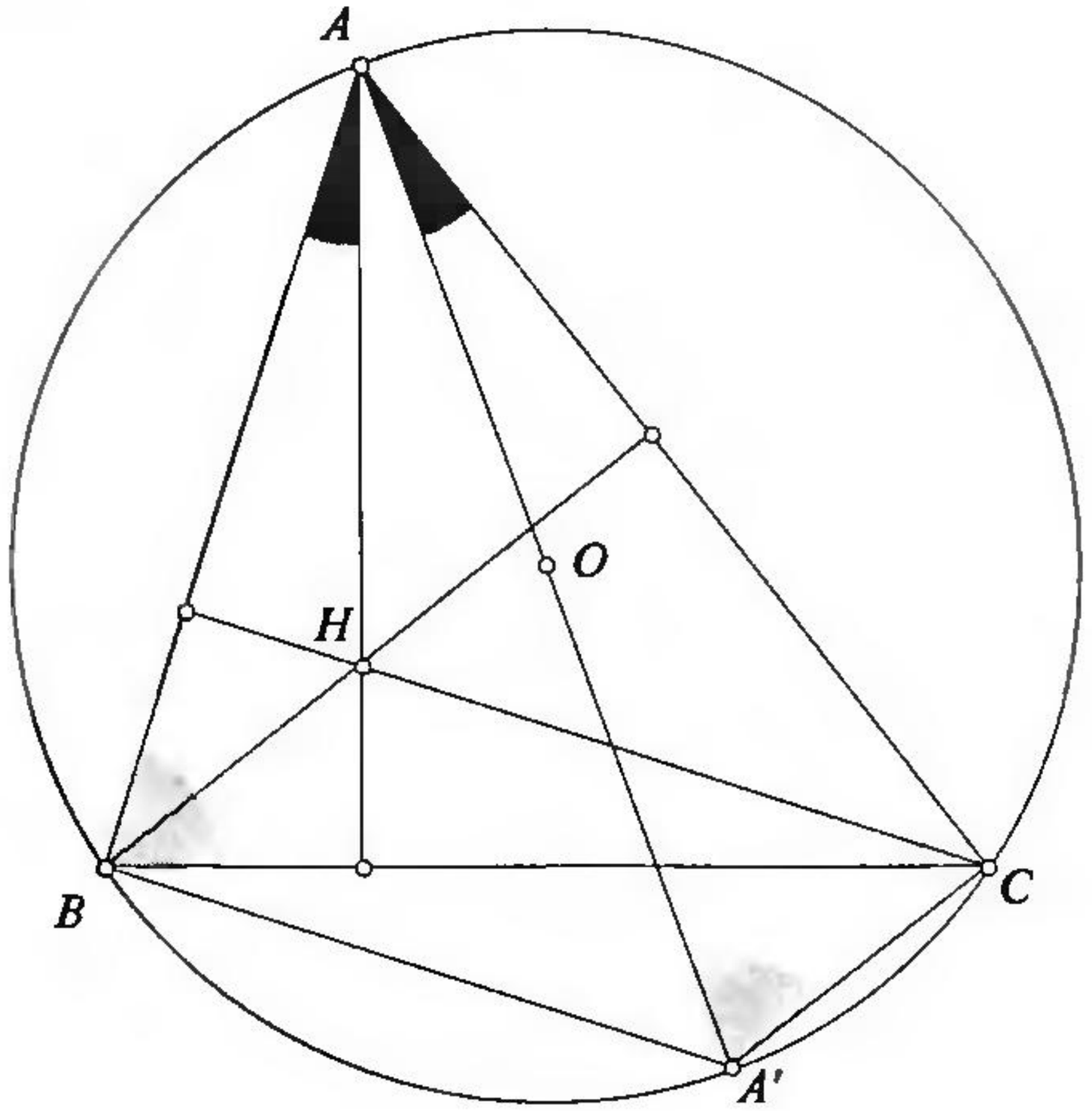


Figure 2.12

Returning to our problem, note that from the statement it follows that the angles $A$ and $B$ are acute. The orthocenter of the triangle $ABC$ lies on the altitude $CD$. Because $\angle ACD = \angle BCF$, it follows from the lemma that the circumcenter of the triangle $ABC$ lies on the line $CF$. On the other hand, the circumcenter lies on the perpendicular bisector of the segment $AB$. We deduce that the circumcenter is the point $F$, hence $\angle C = 90^\circ$.

**Problem 3.** Let $E$ and $F$ be two points on the sides $BC$ and $CD$ of the square $ABCD$, such that $\angle EAF = 45^\circ$. Let $M$ and $N$ be the intersections of the diagonal $BD$ with $AE$ and $AF$, respectively. Let $P$ be the intersection of $MF$ and $NE$. Prove that $AP$ is perpendicular to $EF$. (Figure 2.13.)

**Solution.** First, observe that $\angle NBE = 45^\circ$, so $ABEN$ is cyclic. This implies that

$$\angle AEN = \angle ABN = 45^\circ,$$

so that triangle $ANE$ is a right isosceles triangle. It follows that $EN$ is perpendicular to $AF$. Similarly, $FM$ is perpendicular to $AE$, so $P$ is the orthocenter of triangle $AEF$. The conclusion is obvious.

Look for the cyclic quads in the following problems.

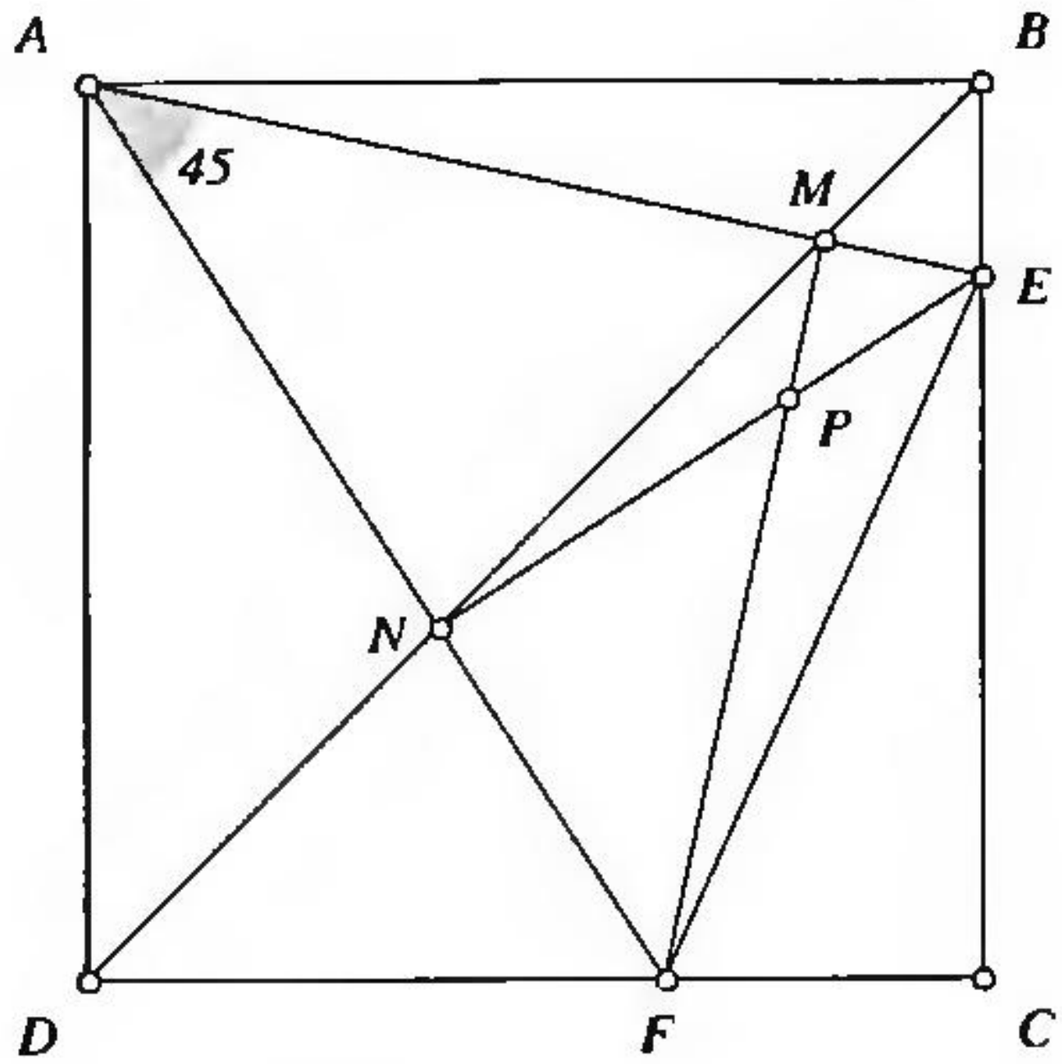


Figure 2.13

**Problem 4.** Let $D$, $E$, and $F$ be the feet of the altitudes of the triangle $ABC$. Prove that the altitudes of $ABC$ are the angle bisectors of the triangle $DEF$.

**Problem 5.** Let $ABCD$ be a convex quadrilateral. Prove that $AB \cdot CD + AD \cdot BC = AC \cdot BD$ if and only if $ABCD$ is cyclic (Ptolemy's theorem).

**Problem 6.** Let $A'$, $B'$ and $C'$ be points on the sides $BC$, $CA$, and $AB$ of the triangle $ABC$. Prove that the circumcircles of the triangles $AB'C'$, $BA'C'$, and $CA'B'$ have a common point.

**Problem 7.** Let $ABCD$ be a cyclic quadrilateral. Prove that the orthocenters of the triangles $ABC$, $BCD$, $CDA$, and $DAB$ are the vertices of a quadrilateral congruent to $ABCD$ and prove that the centroids of the same triangles are the vertices of a cyclic quadrilateral.

**Problem 8.** Let $K$, $L$, $M$, and $N$ be the midpoints of the sides $AB$, $BC$, $CD$, and $DA$, respectively, of a cyclic quadrilateral $ABCD$. Prove that the orthocenters of the triangles $AKN$, $BKL$, $CLM$, and $DMN$ are the vertices of a parallelogram.

**Problem 9.** Prove that the perpendiculars dropped from the midpoints of the sides of a convex quadrilateral to the opposite sides are concurrent.

**Problem 10.** In the convex quadrilateral $ABCD$ the diagonals $AC$ and $BD$ intersect at $O$ and are perpendicular. Prove that projections of $O$ on the quadrilateral's sides are the vertices of a cyclic quadrilateral.

## 2.4 Equiangular polygons

We call a convex polygon equiangular if its angles are congruent. Thus, an equiangular triangle is an equilateral one, an equiangular quadrilateral is a rectangle (or a square). One interesting property of the equiangular polygons is stated in the following problem.

**Problem 1.** Let $P$ be a variable point in the interior or on the sides of an equiangular polygon. Prove that the sum of distances from $P$ to the polygon's sides is constant.

**Solution.** It is not difficult to see that the statement is true for regular polygons. Indeed, connect $P$ to the polygon's vertices and write its area as a sum of areas of the triangles thus obtained (see Figure 2.14 in the case of a pentagon).

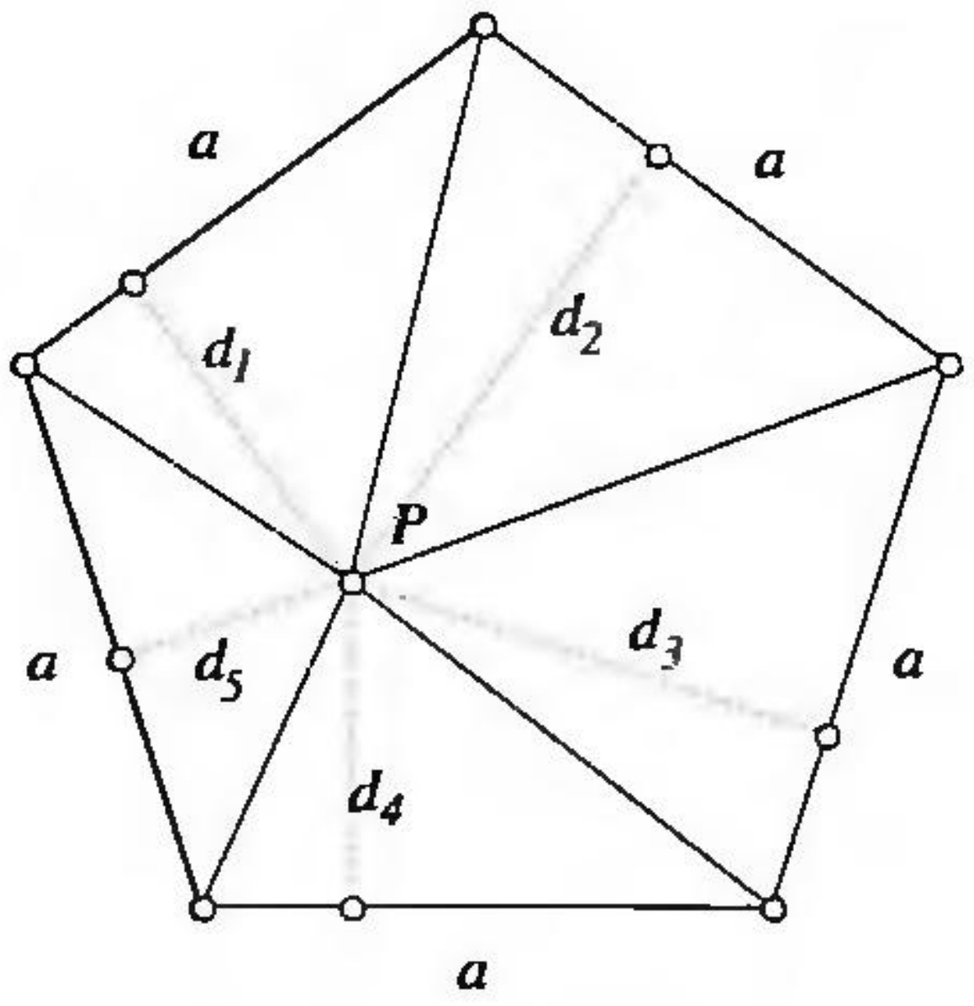


Figure 2.14

If the polygon's sides have length $a$ and the distances from $P$ to the sides are $d_1, d_2, \ldots, d_n$, respectively, then the area $A$ of the polygon equals $\frac{1}{2}\sum_{k=1}^{n} ad_k$ hence $d_1 + d_2 + \cdots + d_n = \frac{2A}{a}$ is constant.

Now, if the polygon is an arbitrary equiangular one, we can always "expand" it to a regular polygon, adding thus to the sum of distances a constant value (the sum of distances between parallel sides) as seen in Figure 2.15.

The converse of the above property is true in the case of a triangle. Indeed, if $P$ is a variable point in the interior or on the sides of triangle $ABC$ and the sum of distances from $P$ to the triangle's sides is constant, then $ABC$ is an equilateral triangle. We can see this if we move $P$ to $A$, $P$ to $B$ and $P$ to $C$

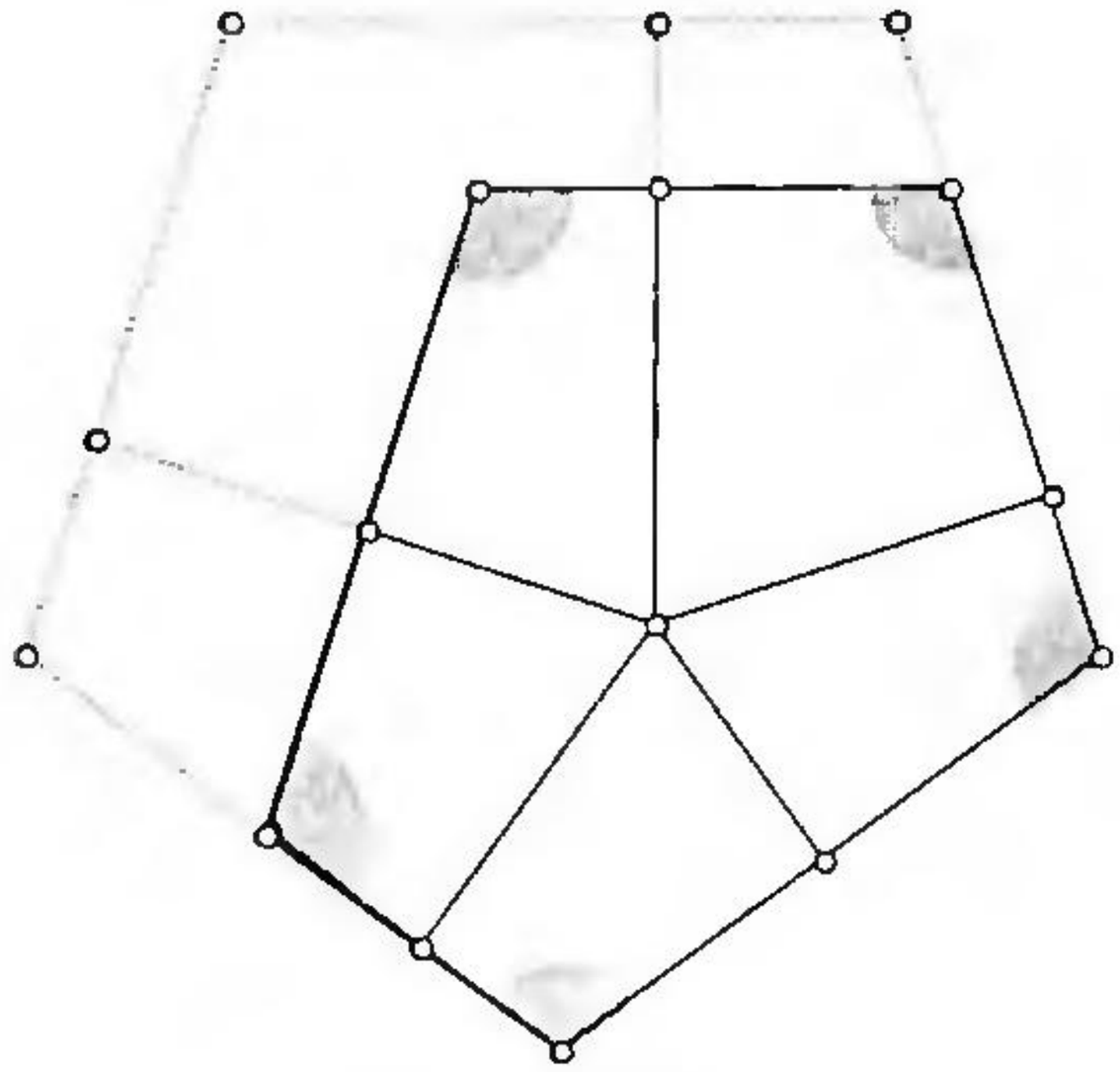

Figure 2.15

and observe that the triangle's altitudes must have the same length. It is not difficult to see that this happens if and only if $ABC$ is equilateral.

The converse is not true for a quadrilateral, because a parallelogram has this property and (unless it's a rectangle) it is not an equiangular polygon.

**Problem 2.** Let $a_1, a_2, \ldots a_n$ be positive real numbers and let $\varepsilon$ be a primitive $n$th root of unity, e.g., $\varepsilon = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$. Prove that if the sides of an equiangular polygon have lengths $a_1, a_2, \ldots, a_n$ (in counterclockwise order), then

$$a_1 + a_2\varepsilon + a_3\varepsilon^2 + \cdots + a_n\varepsilon^{n-1} = 0.$$

**Solution.** We draw a picture for the case $n = 6$ (the general case is essentially the same).

Consider the polygon's sides as vectors, oriented clockwise (see Figure 2.16). Then the sum of the vectors equals zero. Now, translate all vectors such that they have the same origin $O$. If we look at the complex numbers corresponding to their extremities, choosing $a_1$ on the positive real axis, we see that these are $a_1, a_2\varepsilon, a_3\varepsilon^2, \ldots,$ and $a_n\varepsilon^{n-1}$, respectively. Since the sum of the vectors equals zero, we deduce that $a_1 + a_2\varepsilon + a_3\varepsilon^2 + \cdots + a_n\varepsilon^{n-1} = 0$.

*Observation.* The converse of the statement is generally not true. For instance, if $a, b, c, d$ are the side lengths of a quadrilateral and $a + bi + ci^2 + di^3 = 0$, then $(a-c) + i(b-d) = 0$; this equality is fulfilled if the quadrilateral is a parallelogram (and not necessarily an equiangular quadrilateral, that is, a rectangle). However, from the solution we see that if $a_1, a_2, \ldots, a_n$ are

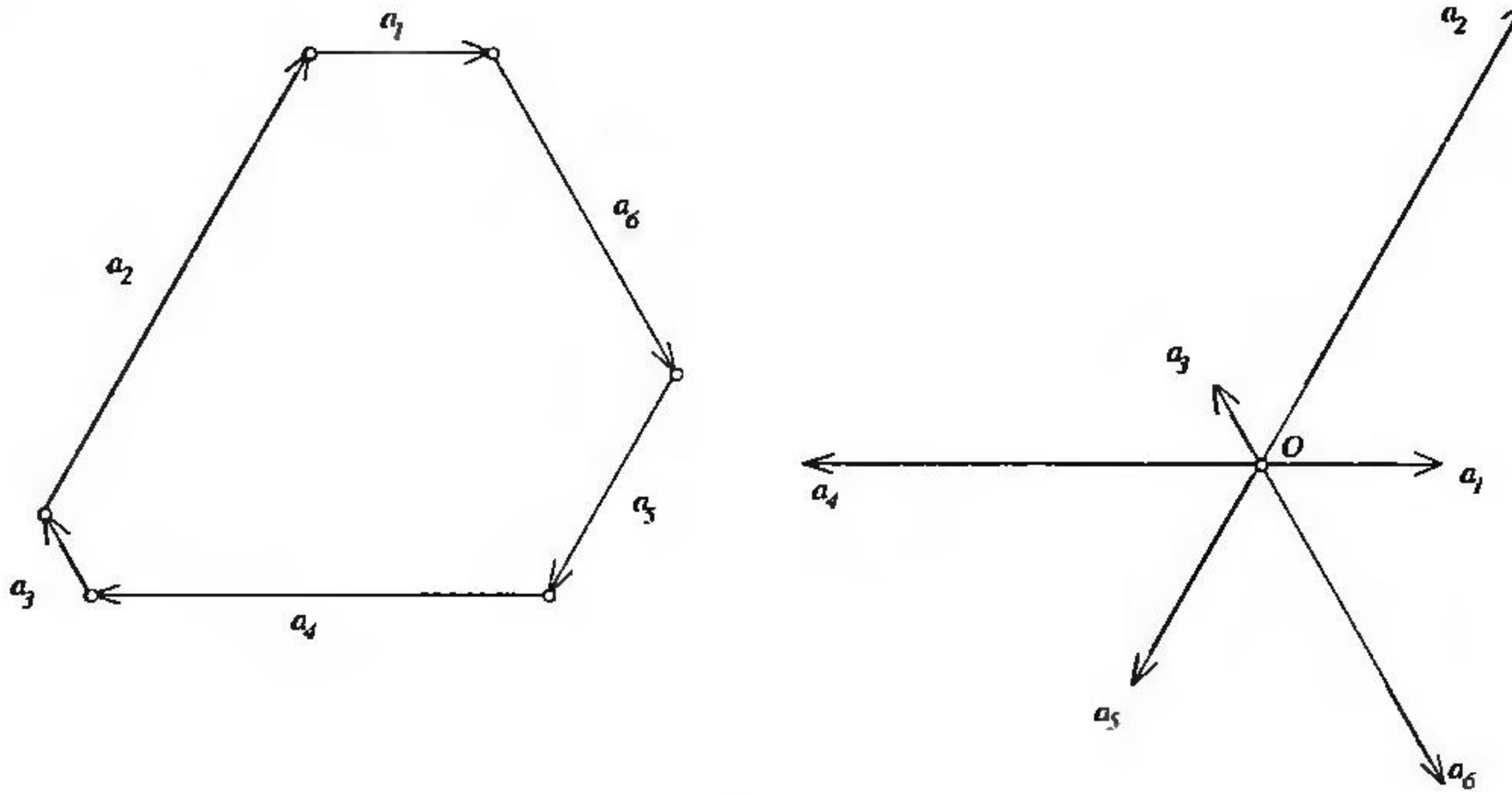


Figure 2.16

positive numbers and $a_1 + a_2\varepsilon + a_3\varepsilon^2 + \cdots + a_n\varepsilon^{n-1} = 0$, then there exists an equiangular polygon with sides of lengths $a_1, a_2, \ldots, a_n$.

It is worth mentioning that in the case $n$=3, there exists another necessary and sufficient condition: if $a, b, c$ are the complex numbers corresponding to the (distinct) points $A, B, C$, then the triangle $ABC$ is equilateral if and only if $a + b\varepsilon + c\varepsilon^2 = 0$, where $\varepsilon$ is one of the (nonreal) cubic roots of unity.

**Problem 3.** Prove that if an equiangular hexagon have side lengths $a_1$, $\ldots, a_6$ (in this order), then $a_1 - a_4 = a_5 - a_2 = a_3 - a_6$.

**Solution.** Choose three non-adjacent vertices of the hexagon and draw parallels to the sides through them, as in Figure 2.17. Suppose that the parallels mutually intersect at the points $K, L, M$.

Thus, the hexagon is partitioned into three parallelograms and a triangle. Since the hexagon is equiangular, triangle $KLM$ is equilateral. Finally, observe that $LM = a_1 - a_4$, $KL = a_5 - a_2$ and $MK = a_3 - a_6$.

Another solution is possible using the preceding problem. Let $\varepsilon = \cos\frac{2\pi}{6} + i\sin\frac{2\pi}{6}$ be a primitive sixth root of unity. Then

$$a_1 + a_2\varepsilon + a_3\varepsilon^2 + a_4\varepsilon^3 + a_5\varepsilon^4 + a_6\varepsilon^5 = 0.$$

But $\varepsilon^3 = \cos\pi + i\sin\pi = -1$, so $\varepsilon^4 = -\varepsilon$ and $\varepsilon^5 = -\varepsilon^2$. We deduce that

$$(a_1 - a_4) + (a_2 - a_5)\varepsilon + (a_3 - a_6)\varepsilon^2 = 0.$$

On the other hand, since $\varepsilon^3 = -1$ (and $\varepsilon \neq -1$) we see that $\varepsilon^2 - \varepsilon + 1 = 0$. Thus, $\varepsilon$ is a common root of the equations $(a_1-a_4)+(a_2-a_5)z+(a_3-a_6)z^2 =$

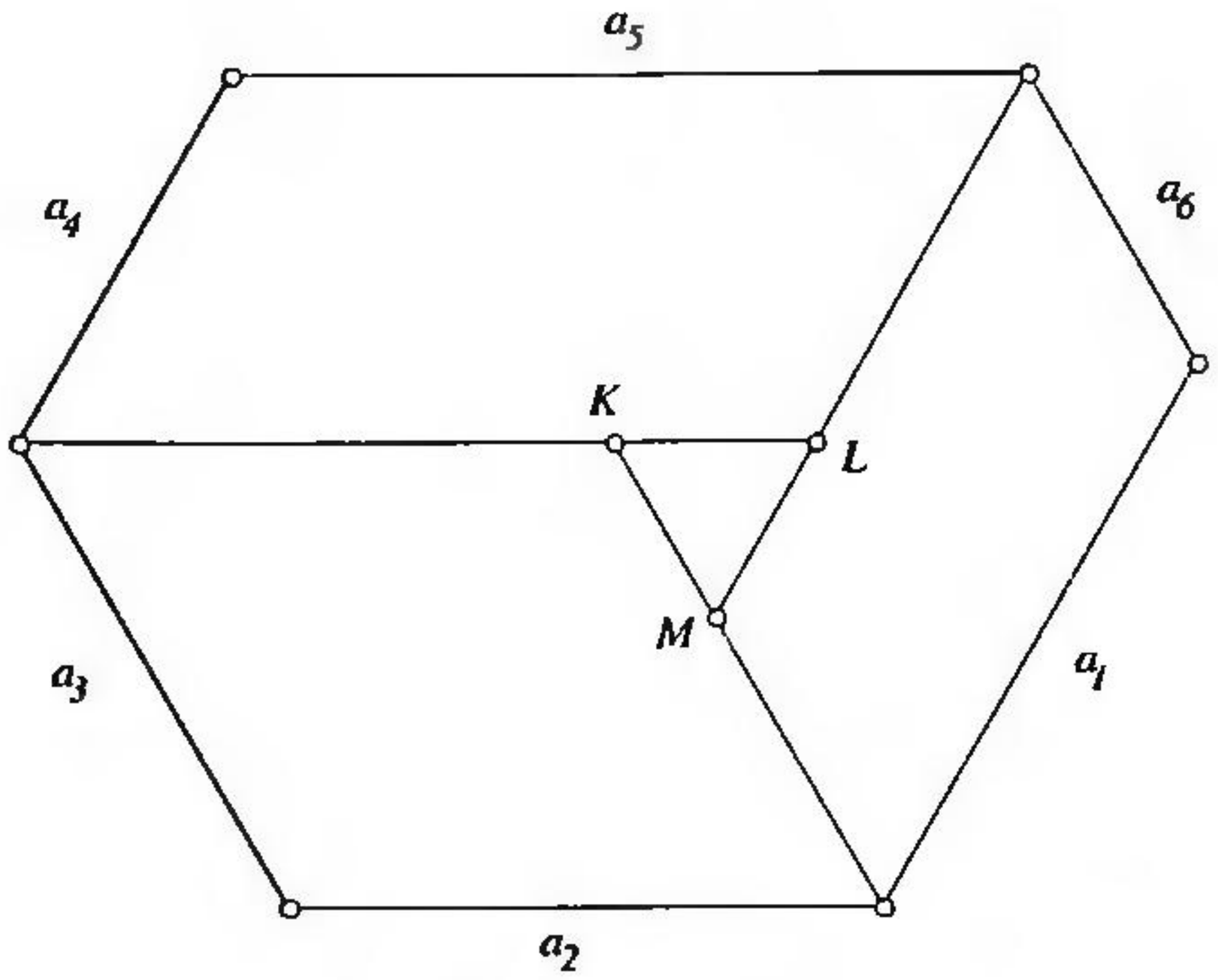


Figure 2.17

0 and $z^2 - z + 1 = 0$, both with real coefficients. Since $\varepsilon \notin \mathbf{R}$, it follows that the two equations share another common root $\overline{\varepsilon}$, so the coefficients of the two equations must be proportional; that is, $(a_1 - a_4) = -(a_2 - a_5) = (a_3 - a_6)$, as desired.

Try the following problems.

**Problem 4.** Let $ABCDE$ be an equiangular pentagon whose side lengths are rational numbers. Prove that the pentagon is regular.

**Problem 5.** Prove that $p$ is a prime number if and only if every equiangular polygon with $p$ sides of rational lengths is regular.

**Problem 6.** An equiangular polygon with an odd number of sides is inscribed in a circle. Prove that the polygon is regular.

**Problem 7.** Let $a_1, a_2, \ldots, a_n$ be the side lengths of an equiangular polygon. Prove that if $a_1 \geq a_2 \geq \cdots \geq a_n$, then the polygon is regular.

**Problem 8.** The side lengths of an equiangular octagon are rational numbers. Prove that the octagon has a symmetry center.

**Problem 9.** A convex hexagon is given in which any two opposite sides have the following property: the distance between their midpoints is $\sqrt{3}/2$ times the sum of their lengths. Prove that the hexagon is equiangular.

**Problem 10.** Prove that the segments joining the midpoints of the opposite sides of an equiangular hexagon are concurrent.

## 2.5 More on equilateral triangles

One of the most beautiful problems in geometry is the following one.

**Problem 1.** In the exterior of triangle $ABC$, three equilateral triangles $ABC'$, $BCA'$ and $CAB'$ are constructed. Prove that the centroids of these triangles are the vertices of an equilateral triangle. (Figure 2.18.)

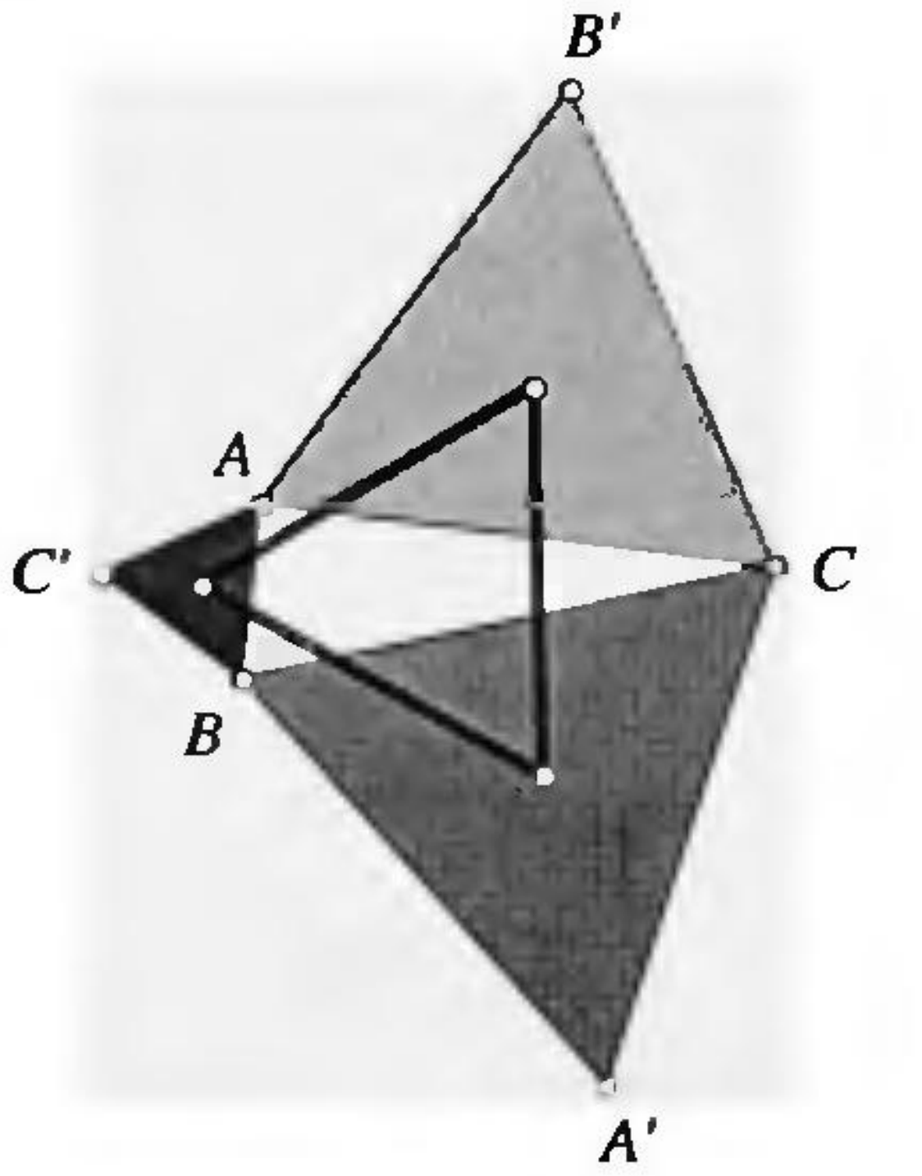


Figure 2.18

**Solution.** It is said that the problem was discovered by Napoleon. We do not know the proof that he gave to this statement, but surely it was different from the following one. We use complex numbers. It is known (see Equiangular polygons) that if triangle $ABC$ is equilateral and its vertices correspond to the complex numbers $a, b, c$, then

$$a + b\varepsilon + c\varepsilon^2 = 0, \qquad (*)$$

where $\varepsilon$ is one of the (nonreal) cubic roots of unity. The converse is also true: suppose (*) holds. Notice that $\varepsilon^2 + \varepsilon + 1 = 0$; then $\varepsilon^2 = -\varepsilon - 1$ and we obtain $\varepsilon(b - c) = c - a$, hence $|b - c| = |c - a|$. Analogously, we obtain $|c - a| = |a - b|$, and thus the side lengths of triangle $ABC$ are equal. Now, let us return to Napoleon's problem.

We have

$$a + b\varepsilon + c'\varepsilon^2 = 0, b + c\varepsilon + a'\varepsilon^2 = 0, c + a\varepsilon + b'\varepsilon^2 = 0, \qquad (**)$$

where $a'$, $b'$, $c'$ are the complex numbers corresponding to the points $A'$, $B'$, $C'$. The complex numbers corresponding to the centroids of triangles $A'BC$, $AB'C$ and $ABC'$ are

$$a'' = \tfrac{1}{3}(a' + b + c),$$

$$b'' = \tfrac{1}{3}(a + b' + c)$$

and

$$c'' = \tfrac{1}{3}(a + b + c'),$$

respectively.

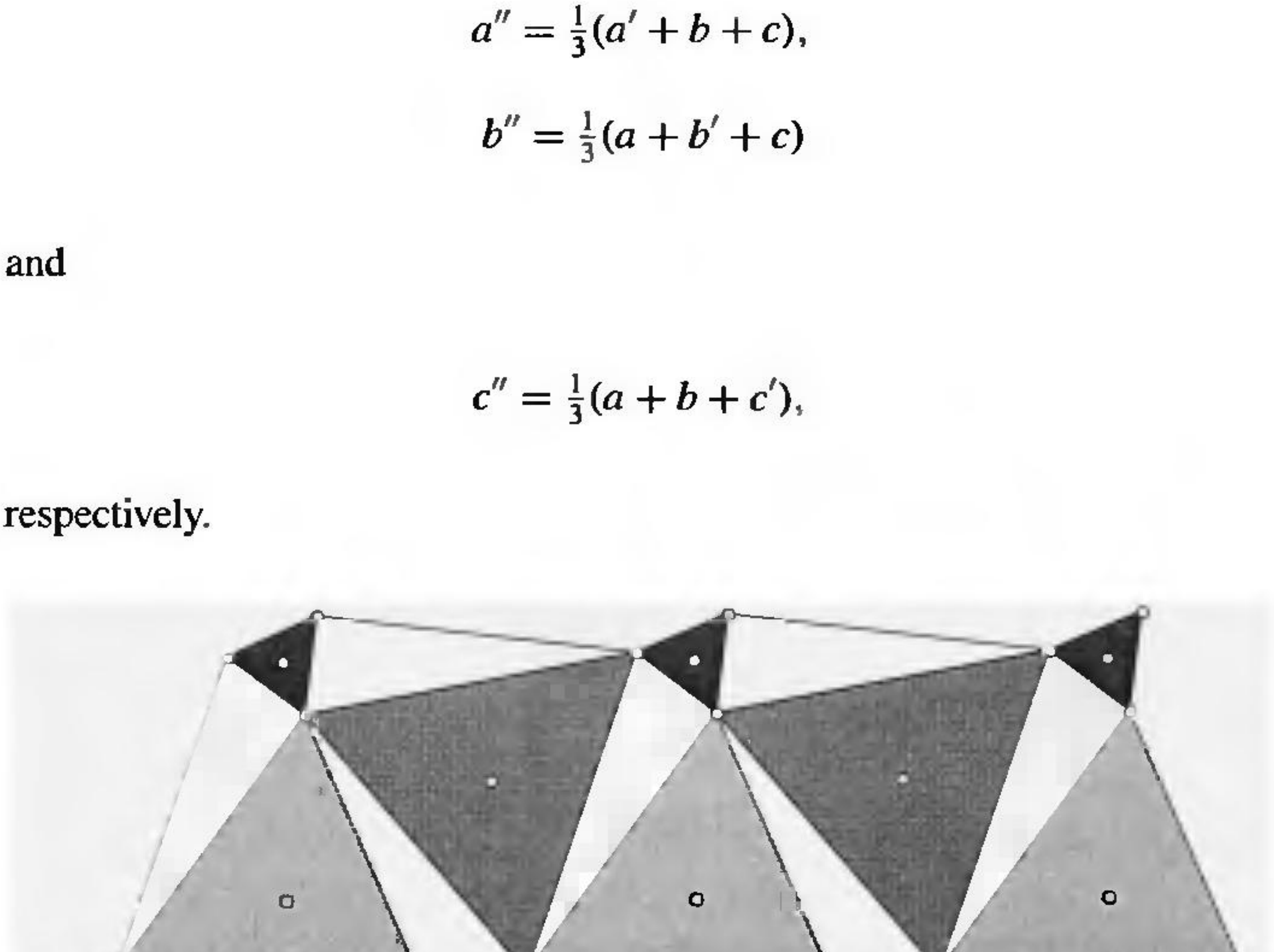

Figure 2.19

We have to check that $a'' + b''\varepsilon + c''\varepsilon^2 = 0$. Observe that

$$\begin{aligned} a'' + b''\varepsilon + c''\varepsilon^2 &= \tfrac{1}{3}(a' + b'\varepsilon + c'\varepsilon^2) \\ &\quad + \tfrac{1}{3}(b + c\varepsilon + a\varepsilon^2) + \tfrac{1}{3}(c + a\varepsilon + b\varepsilon^2) \\ &= \tfrac{1}{3}(a' + b'\varepsilon + c'\varepsilon^2) \\ &\quad + \frac{\varepsilon^2}{3}(a + b\varepsilon + c\varepsilon^2) + \frac{\varepsilon}{3}(a + b\varepsilon + c\varepsilon^2) \\ &= \tfrac{1}{3}(a' + b'\varepsilon + c'\varepsilon^2). \end{aligned}$$

If we multiply the last two of the equalities (**) with $\varepsilon$ and $\varepsilon^2$ and add all three results, we obtain

$$a' + b'\varepsilon + c'\varepsilon^2 + 2(a + b\varepsilon + c\varepsilon^2) = 0,$$

and

$$a'' + b''\varepsilon + c''\varepsilon^2 = \tfrac{1}{3}(a' + b'\varepsilon + c'\varepsilon^2) = 0,$$

as desired.

The reader who wants a "proof without words" for Napoleon's problem should carefully examine Figure 2.19 (which may also be of some interest to carpet manufacturers . . . )

**Problem 2.** In the exterior of the acute triangle $ABC$, three equilateral triangles $ABC'$, $BCA'$ and $CAB'$ are constructed. Prove that the segments $AA'$, $BB'$ and $CC'$ are concurrent. Also, prove that the circumcircles of the equilateral triangles pass through the same point. (Figure 2.20.)

**Solution.** We start with another problem: find the point $T$ in the interior of triangle $ABC$ for which the sum $TA + TB + TC$ is minimal. Rotate clockwise the points $A$ and $T$ around $C$ with 60°, to $B'$ and $T'$, respectively (see Figure 2.21). Then $B'T' = AT$ and since triangle $CTT'$ is equilateral, $TT' = TC$.

We deduce that the sum $TA + TB + TC$ is equal to $BT + TT' + T'B'$, and the latter is minimal when the points $B$, $T$, $T'$ and $B'$ are collinear. Thus, the point $T$ for which the sum $TA + TB + TC$ is minimal lies on $BB'$. Similarly, we can prove that $T$ lies on $AA'$ and $CC'$, so the three segments are concurrent. Also (see Figure 2.22) notice that we have

$$\angle ATC = \angle CT'B' = 180^\circ - \angle CT'T = 120^\circ$$

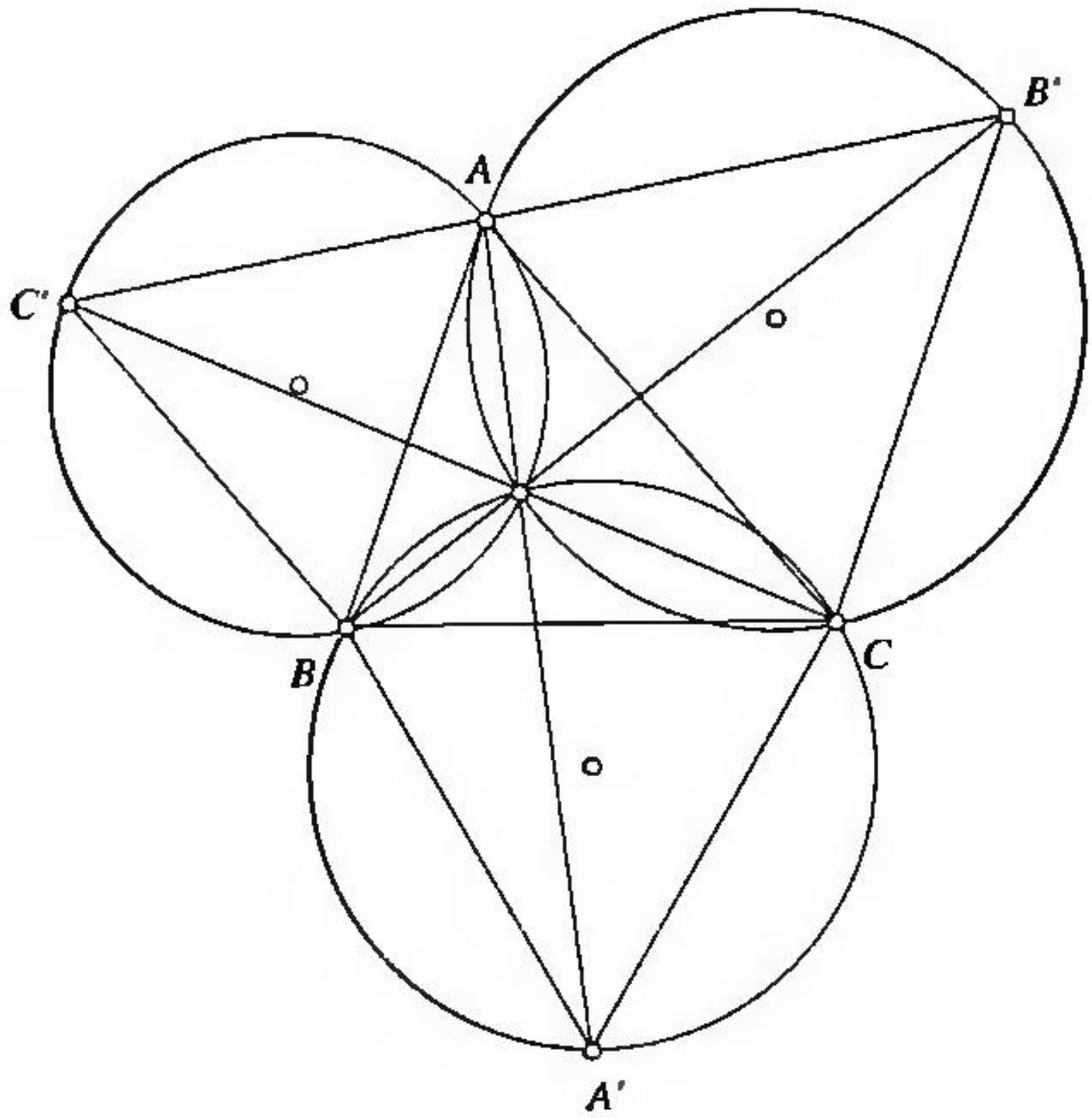


Figure 2.20

and

$$\angle BTC = 180^\circ - \angle CTT' = 120^\circ.$$

We deduce that the point $T$ is the point in the interior of the triangle with the property that

$$\angle ATB = \angle BTC = \angle CTA = 120^\circ$$

(this point is called Toricelli's point). In this case the quadrilaterals $ATCB'$, $ATBC'$ and $BCTA'$ are cyclic, so the circumcircles of the equilateral triangles pass through $T$.

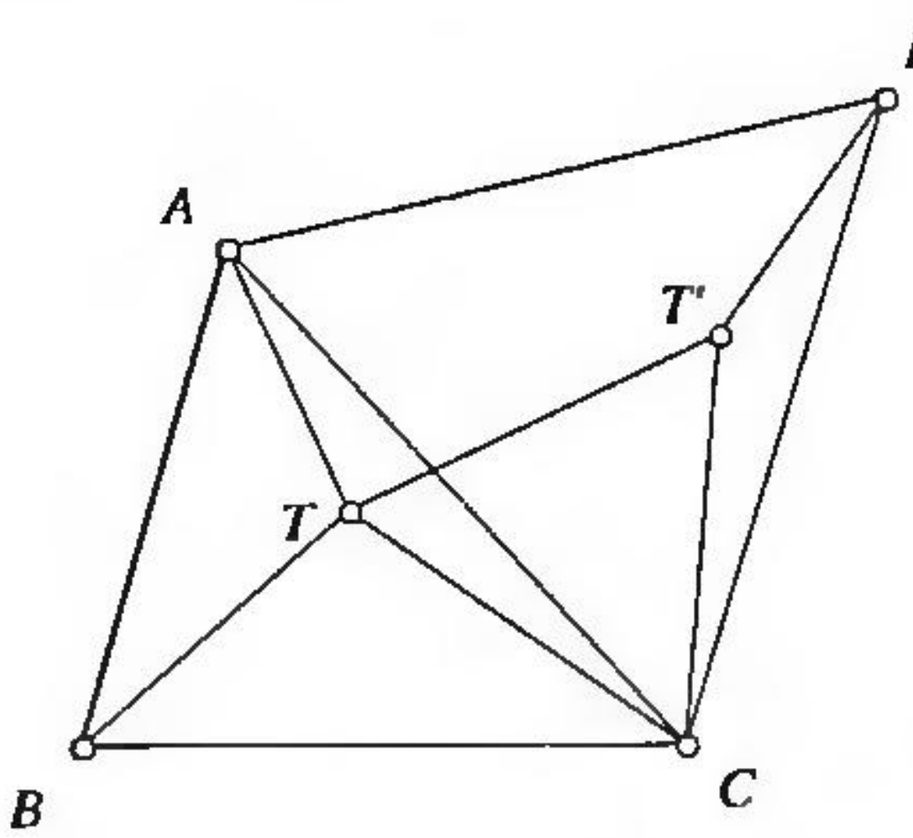


Figure 2.21

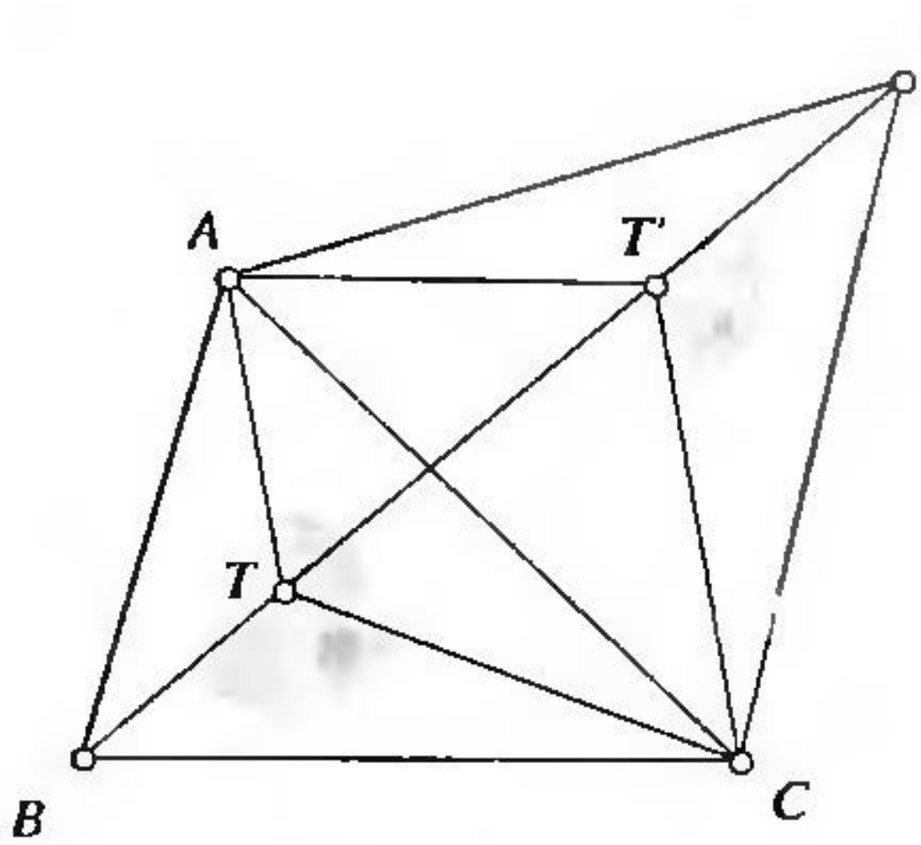


Figure 2.22

Here are some more problems:

**Problem 3.** Let $M$ be a point in the interior of the equilateral triangle $ABC$ and let $A'$, $B'$, and $C'$ be its projections onto the sides $BC$, $CA$, and $AB$, respectively.

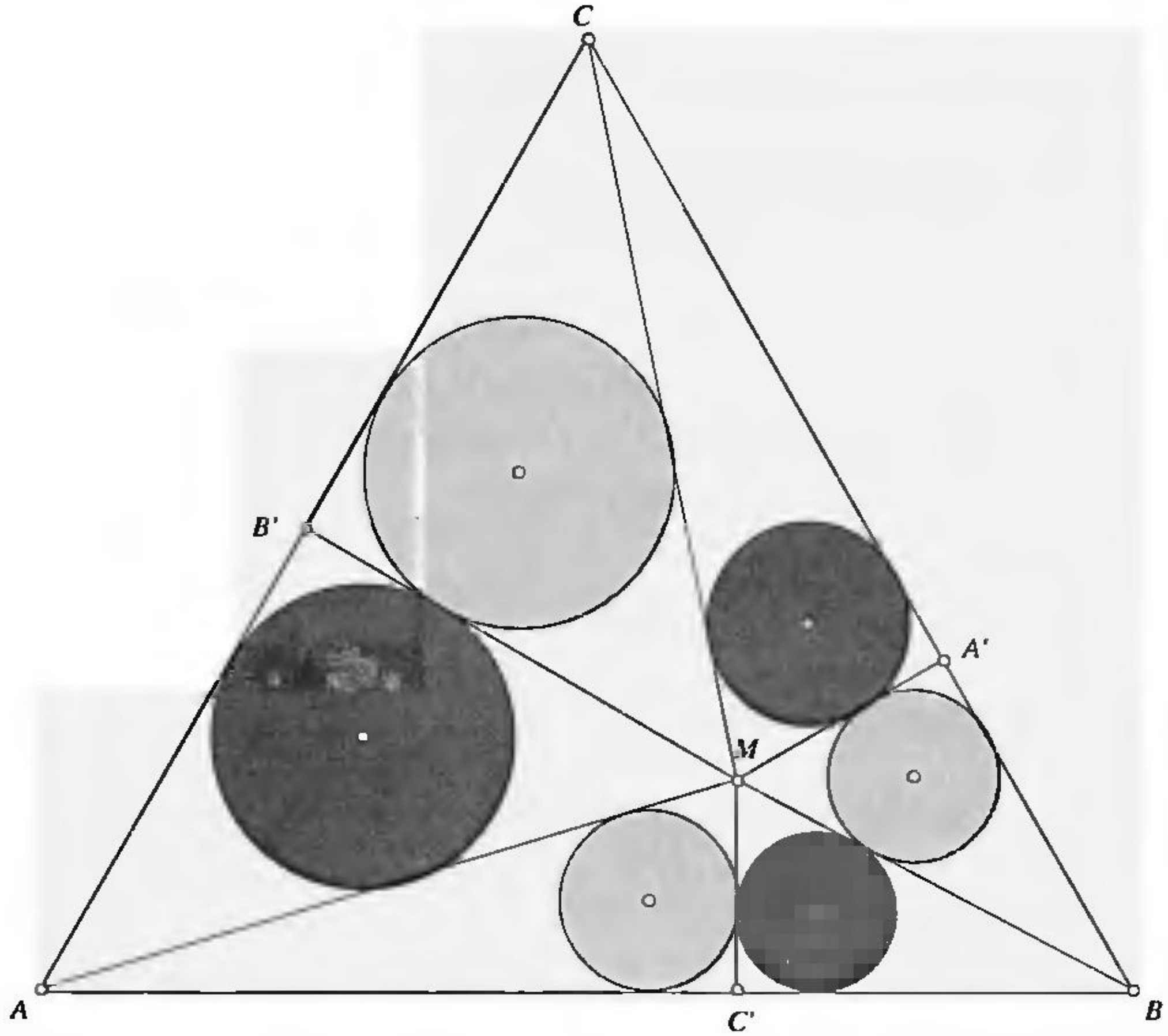


Figure 2.23

Prove that the sum of lengths of the inradii of triangles $MAC'$, $MBA'$, and $MCB'$ equals the sum of lengths of the inradii of triangles $MAB'$, $MBC'$, and $MCA'$. (Figure 2.23.)

**Problem 4.** Let $I$ be the incenter of the triangle $ABC$. It is known that for every point $M \in (AB)$, one can find the points $N \in (BC)$ and $P \in (AC)$ such that $I$ is the centroid of triangle $MNP$. Prove that $ABC$ is an equilateral triangle.

**Problem 5.** Let $ABC$ be an acute triangle. The interior bisectors of the angles $\angle B$ and $\angle C$ meet the opposite sides at the points $L$ and $M$, respectively.

Prove that there exists a point $K$ in the interior of the side $BC$ such that the triangle $KLM$ is equilateral if and only if $\angle A = 60°$.

**Problem 6.** Let $P_1P_2\ldots P_n$ be a convex polygon with the following property: for any two vertices $P_i$ and $P_j$, there exists a vertex $P_k$ such that the segment $P_iP_j$ is seen from $P_k$ under an angle of $60°$. Prove that the polygon is an equilateral triangle.

**Problem 7.** From a point on the circumcircle of an equilateral triangle $ABC$, parallels to the sides $BC, CA$, and $AB$ are drawn, intersecting the sides $CA$, $AB$, and $BC$ at the points $M$, $N$, $P$, respectively. Prove that the points $M$, $N$, and $P$ are collinear.

**Problem 8.** Let $P$ be a point on the circumcircle of the equilateral triangle $ABC$. Prove that the projections of any point $Q$ onto the lines $PA$. $PB$ and $PC$ are the vertices of an equilateral triangle.

## 2.6 The "carpets" theorem

We start with a textbook problem:

**Problem 1.** Let $M$ and $N$ be the midpoints of the sides $AB$ and $BC$ of the square $ABCD$. Let $P = AN \cap DM$, $Q = AN \cap CM$ and $R = CM \cap DN$.

Prove the equality

$$[AMP] + [BMQN] + [CNR] = [DPQR].$$

**Solution.** Let us try brute force. (Figure 2.24.) Assume with no loss of generality that the square's area equals 1. Since the figure is symmetric with respect to $BD$, it suffices to prove that

$$[AMP] + [BMQ] = [DPQ].$$

The triangles $AQD$ and $NQB$ are similar, so

$$\frac{AQ}{QN} = \frac{AD}{BN} = 2.$$

It follows that

$$[AQB] = \tfrac{2}{3}[ABN] = \tfrac{1}{5}$$

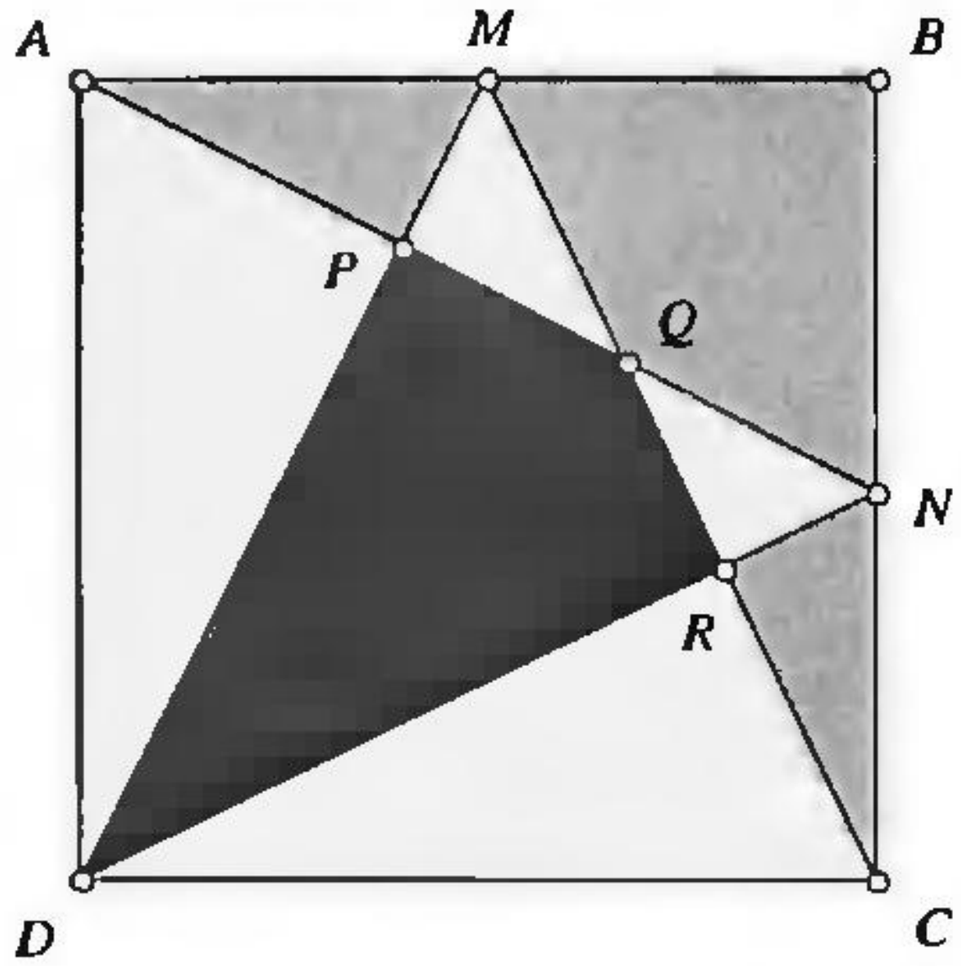


Figure 2.24

and then,

$$[BMQ] = \tfrac{1}{12}.$$

Let $N' = AN \cap CD$. Then triangles $AMP$ and $N'DP$ are also similar and

$$\frac{PM}{PD} = \frac{AM}{DN'} = \tfrac{1}{4}.$$

Consequently,

$$[AMP] = \tfrac{1}{5}[AMD] = \tfrac{1}{20}.$$

Finally, it is easy to see that

$$\frac{PQ}{AD} = \tfrac{4}{15},$$

and

$$[PQD] = \tfrac{4}{15}[AND] = \tfrac{2}{15}.$$

Now,

$$\tfrac{1}{12} + \tfrac{1}{20} = \tfrac{2}{15}$$

and we are done.

However, this is an inelegant solution. Fortunately, there exists a much simpler one, with no computations and that works for arbitrary points $M$ and $N$ on the respective sides!

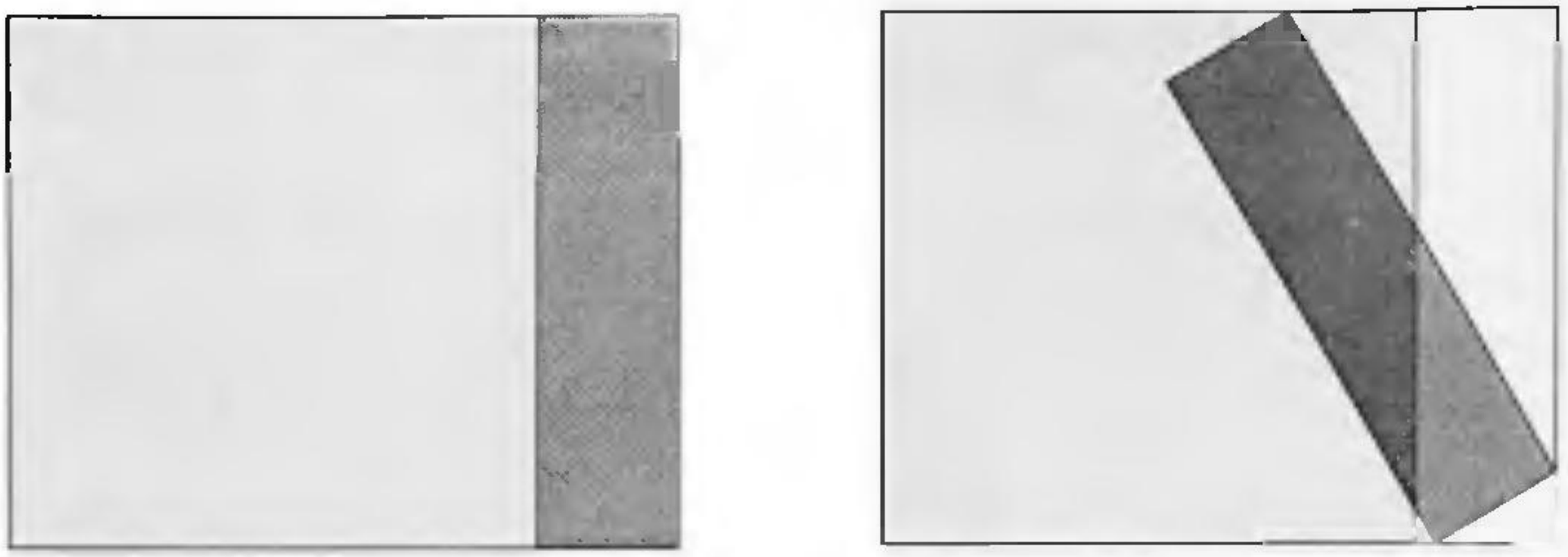

Figure 2.25

First, we look at the "carpets" theorem. Suppose that the floor of a rectangular room is completely covered by a collection of nonoverlapping carpets. If we move one of the carpets, then clearly the overlapping area is equal to the uncovered area of the floor (see Figure 2.25).

Of course, the shapes of the room or the carpets are irrelevant.

Now, let us return to Problem 1, taking $M$ and $N$ at arbitrary positions:

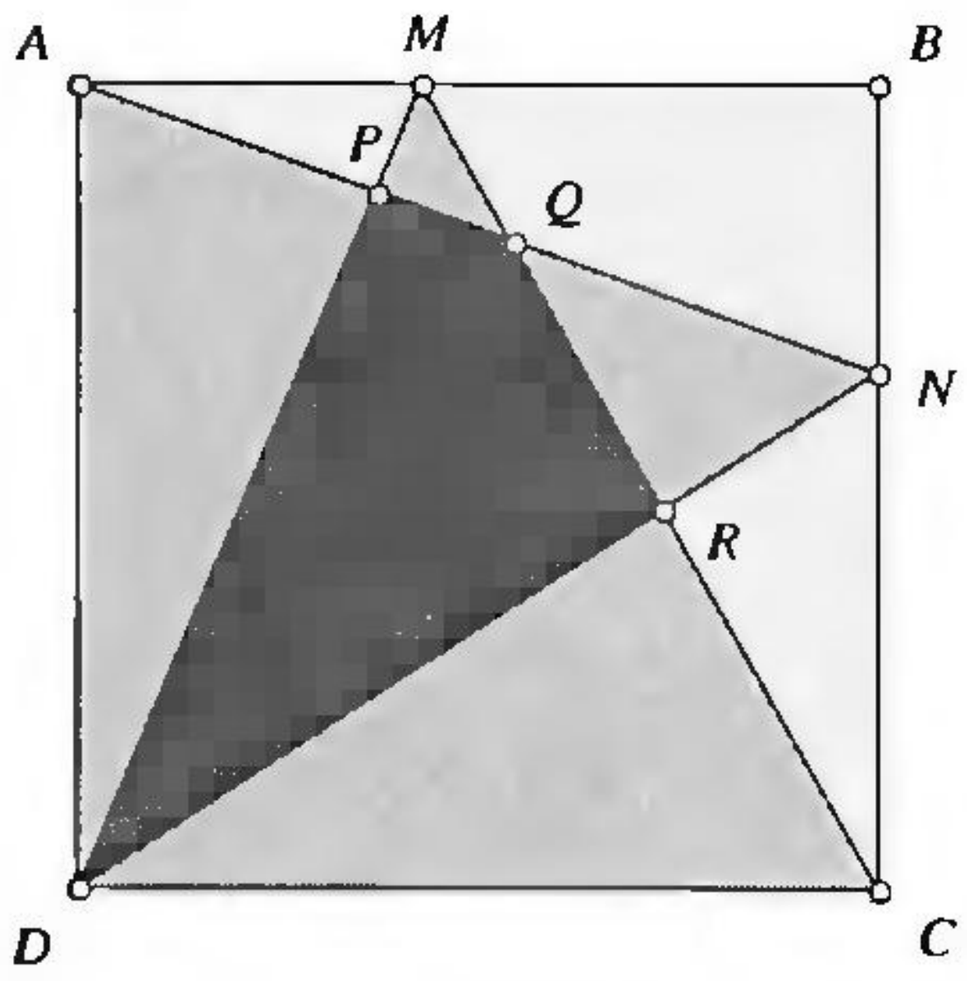


Figure 2.26

The "room" is $ABCD$ and the "carpets" are triangles $ADN$ and $CDM$. (Figure 2.26.) Since

$$[ADN] = [CDM] = \tfrac{1}{2}[ABCD],$$

the two carpets would completely cover the room if they didn't overlap. Hence the area of the overlapping surface, that is, $[DPQR]$, equals the area of the uncovered surface, i.e.,

$$[AMP] + [BMQN] + [CNR].$$

Arrange some carpets to solve the following problems.

**Problem 2.** Let $ABCD$ be a parallelogram. The points $M$, $N$, and $P$ are chosen on the segments $BD$, $BC$ and $CD$, respectively, such that $CNMP$ is a parallelogram. Let $E = AN \cap BD$ and $F = AP \cap BD$. Prove that

$$[AEF] = [DFP] + [BEN].$$

**Problem 3.** Consider the quadrilateral $ABCD$. The points $M$, $N$, $P$, and $Q$ are the midpoints of the sides $AB$, $BC$, $CD$, and $DA$. (Figure 2.27.)

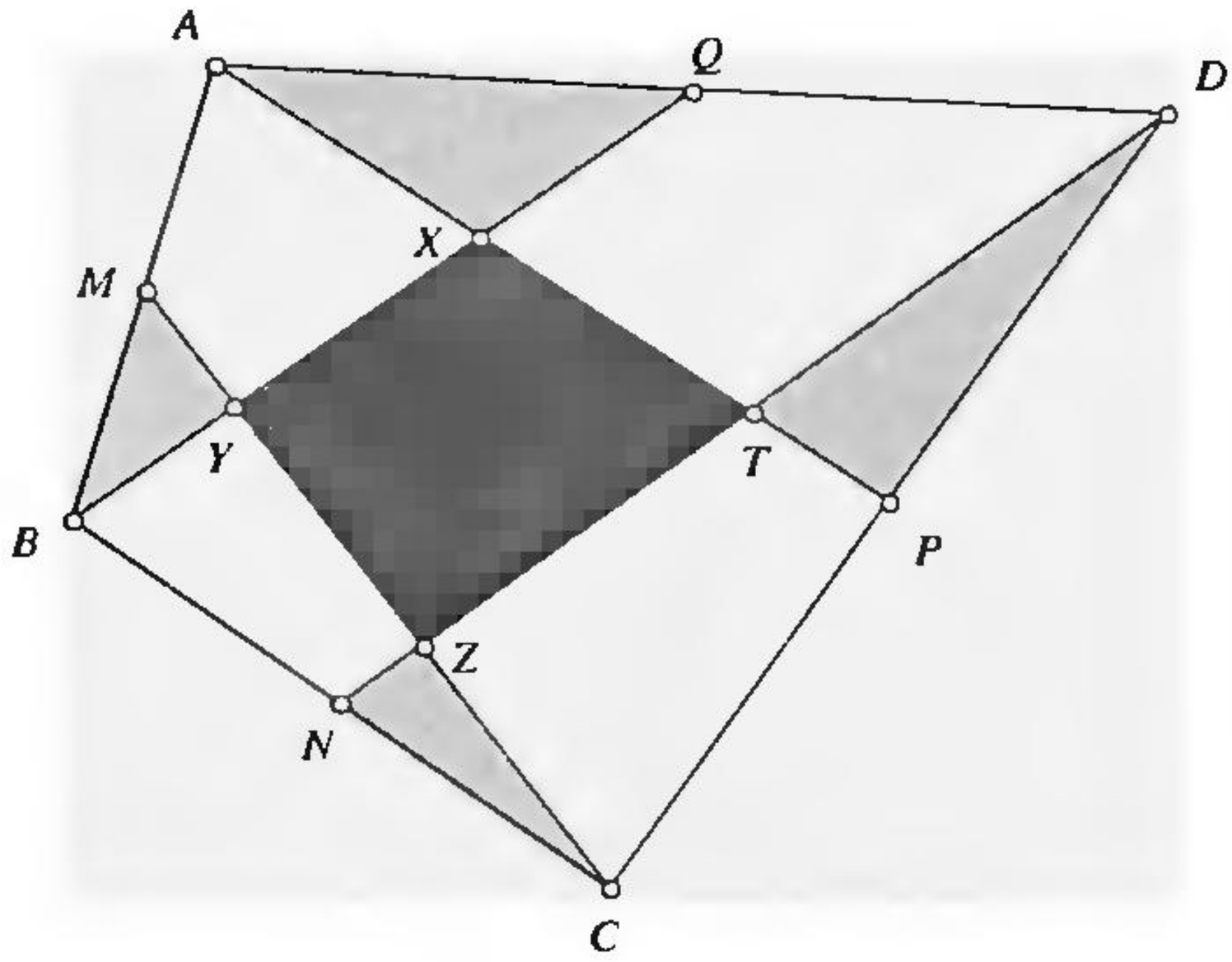


Figure 2.27

Let $X = AP \cap BQ$, $Y = BQ \cap CM$, $Z = CM \cap DN$, and $T = DN \cap AP$. Prove that

$$[XYZT] = [AQX] + [BMY] + [CNZ] + [DPT].$$

**Problem 4.** Through the vertices of the smaller base $AB$ of the trapezoid $ABCD$ two parallel lines are drawn, intersecting the segment $CD$. These lines and the trapezoid's diagonals divide it into seven triangles and a pentagon (see Figure 2.28).

Show that the area of the pentagon equals the sum of the areas of the three triangles that share a common side with the trapezoid.

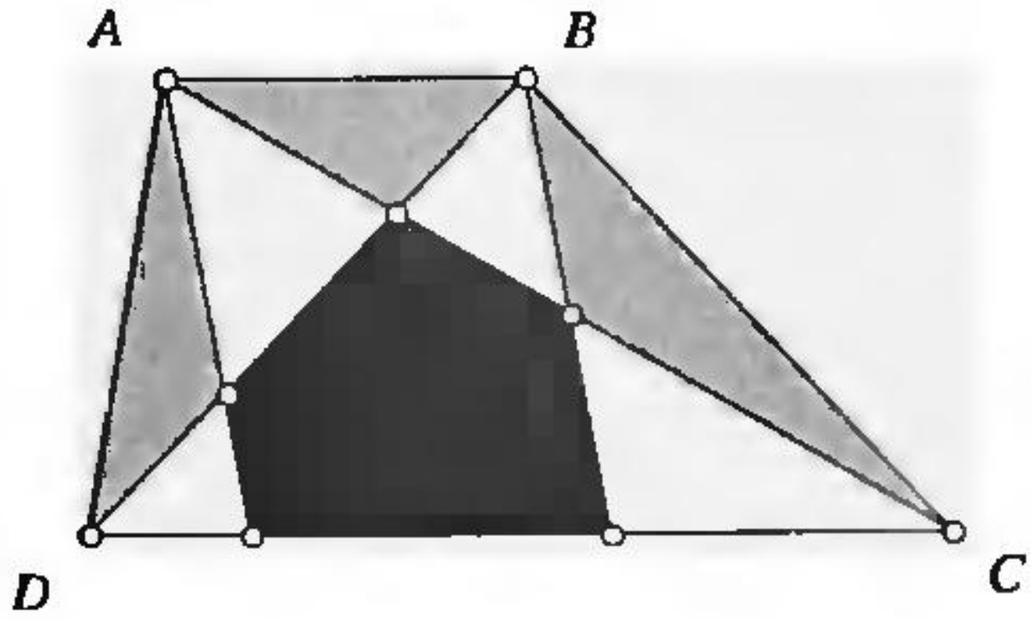


Figure 2.28

**Problem 5.** Let $M$ be a point in the interior of triangle $ABC$. Three lines are drawn through $M$, parallel to triangle's sides, thereby producing three trapezoids. Suppose a diagonal is drawn in each trapezoid in such a way that the diagonals have no common endpoints. These three diagonals divide $ABC$ into seven parts, four of them being triangles. Prove that the area of one of the four triangles equals the sum of the areas of the other three. (Figure 2.29.)

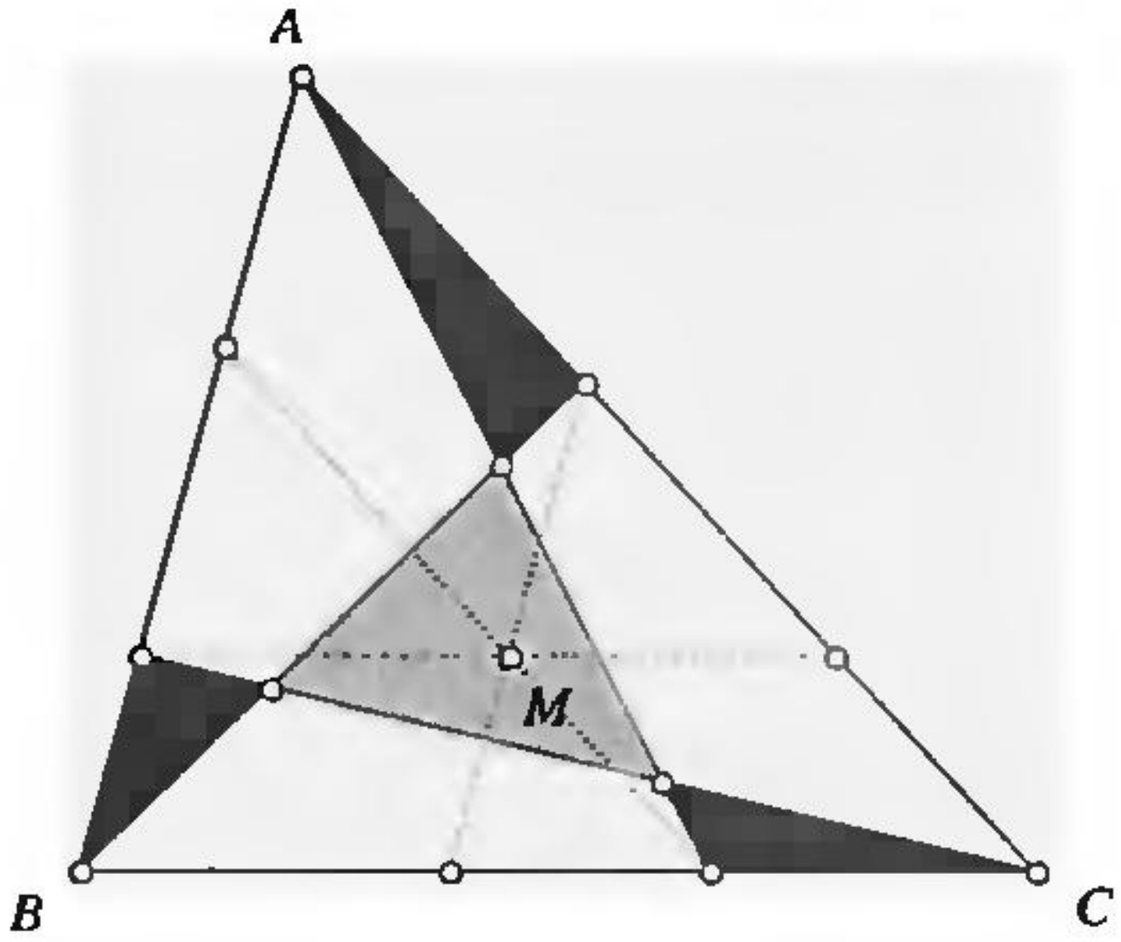


Figure 2.29

Prove that the area of one of the four triangles equals the sum of the areas of the other three.

## 2.7 Quadrilaterals with an inscribed circle

Everybody knows that in any triangle one can inscribe a circle whose center is at the intersection point of the angles' bisectors.

What can we say about a quadrilateral? If there exists a circle touching all the quadrilateral's sides, then its center is equidistant from them, hence it lies

on all four angle bisectors. We deduce that a necessary and sufficient condition for the existence of an inscribed circle is that the quadrilateral's angle bisectors are concurrent (in fact, this works for arbitrary convex polygons).

This does not happen in every quadrilateral. We can always draw a circle tangent to three of the four sides (its center being the point of intersection of two of the bisectors). (Figure 2.30.)

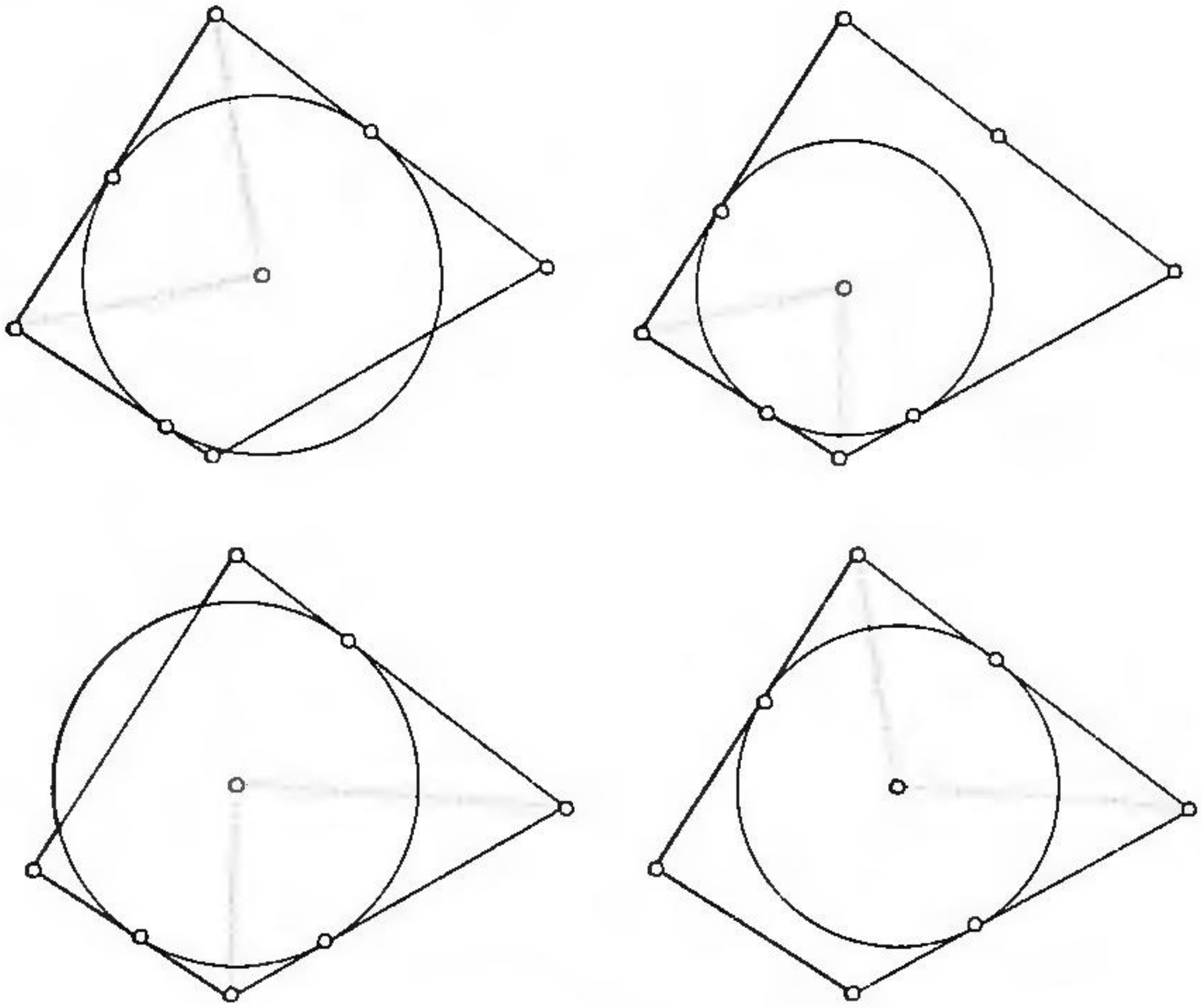

Figure 2.30

If three of the four angle bisectors meet at one point, it is easy to see that the fourth one will also pass through that point and that a circle can be inscribed in the quadrilateral.

Another necessary and sufficient condition for the existence of an inscribed circle in a quadrilateral is given by the following theorem, due to Pithot.

**Theorem.** Let $ABCD$ be a convex quadrilateral. There exists a circle inscribed in $ABCD$ if and only if:

$$AB + CD = AD + BC.$$

**Proof.** Suppose there exists a circle inscribed in the quadrilateral, touching the sides $AB$, $BC$, $CD$ and $DA$ at the points $K$, $L$, $M$, $N$, respectively.

Then, since the tangents from a point to a circle have equal lengths, we have: $AK = AN$, $BK = BL$, $CM = CL$ and $DM = DN$. (Figure 2.31.)

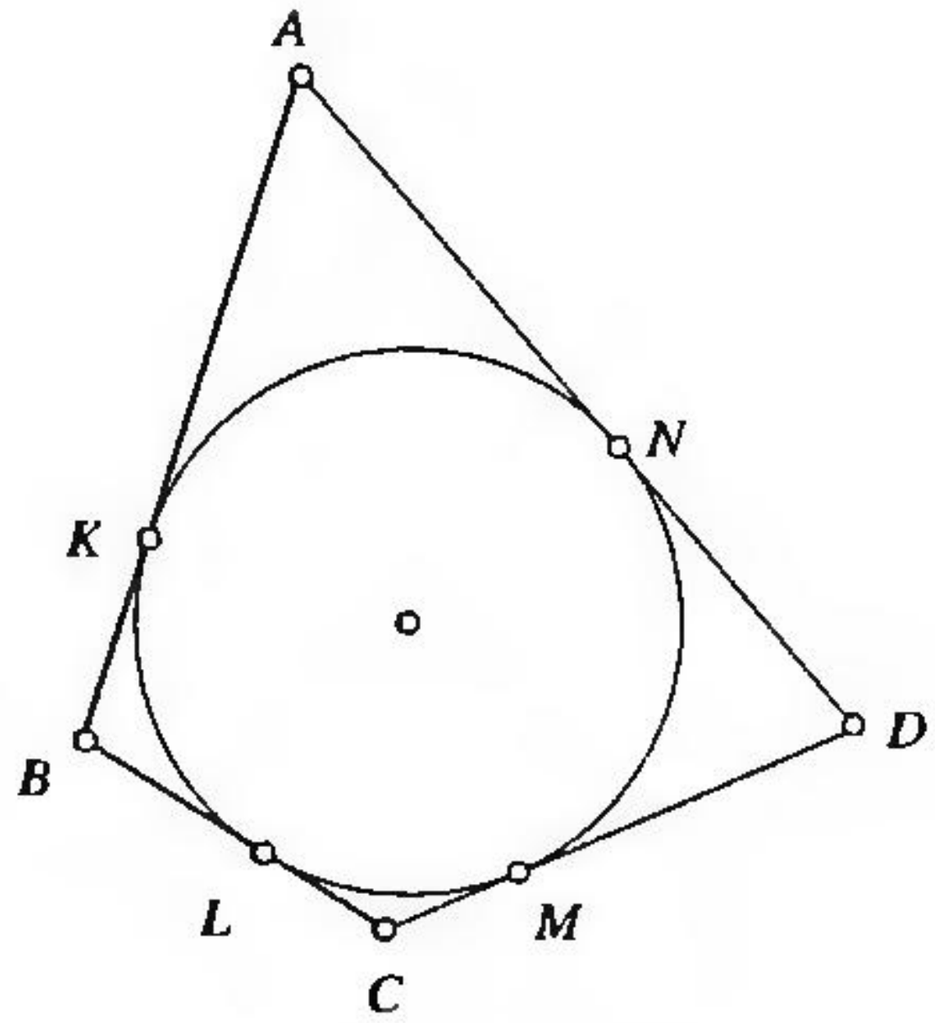


Figure 2.31

If we add up these equalities, we obtain the desired result.

Conversely, suppose $AB + CD = AD + BC$. Draw a circle tangent to $AB$, $BC$ and $CD$. If the circle is not tangent to $AD$, draw from $A$ the tangent to the circle and let $E$ be the point of intersection with $CD$. Suppose, for instance, that $E$ lies in the interior of $CD$. (Figure 2.32.)

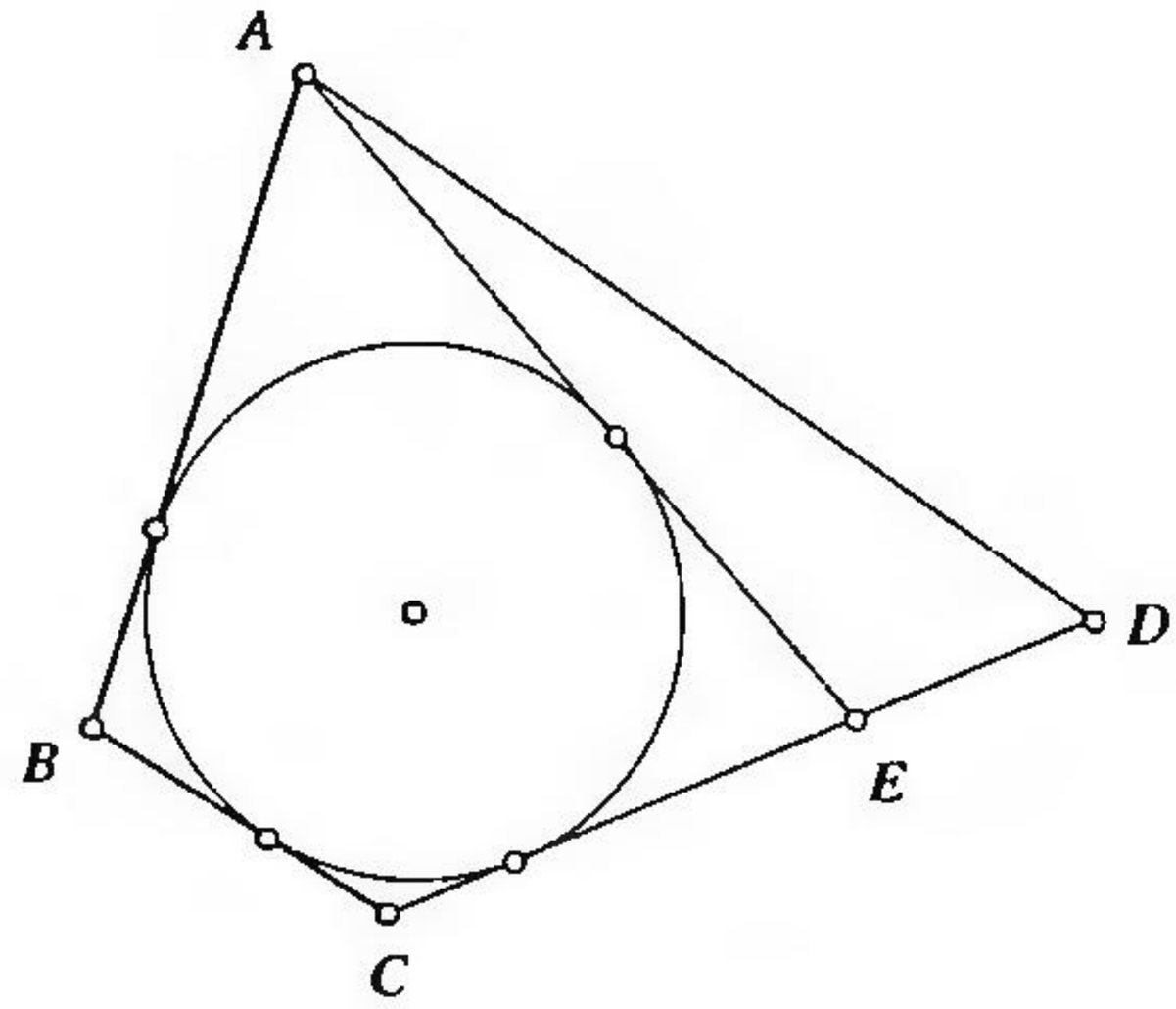


Figure 2.32

Since the circle is inscribed in the quadrilateral $ABCE$, we have that $AB + CE = AE + BC$. On the other hand, from the hypothesis, we have

that $AB + CD = AD + BC$, or $AB + CE + ED = AD + BC$. From these, we derive $ED + AE = AD$, which is impossible. It follows that the circle is also tangent to $AD$, hence it is inscribed in $ABCD$.

Another nice proof of the converse is the following: if $AB = AD$, then $BC = CD$ and the conclusion is immediate. Suppose, with no loss of generality, that $AB < AD$, and let $X \in AD$ be such that $AB = AX$. Let $Y \in CD$ be such that $DX = DY$. Since $AB + CD = AD + BC$, it follows that $CY = BC$. (Figure 2.33.)

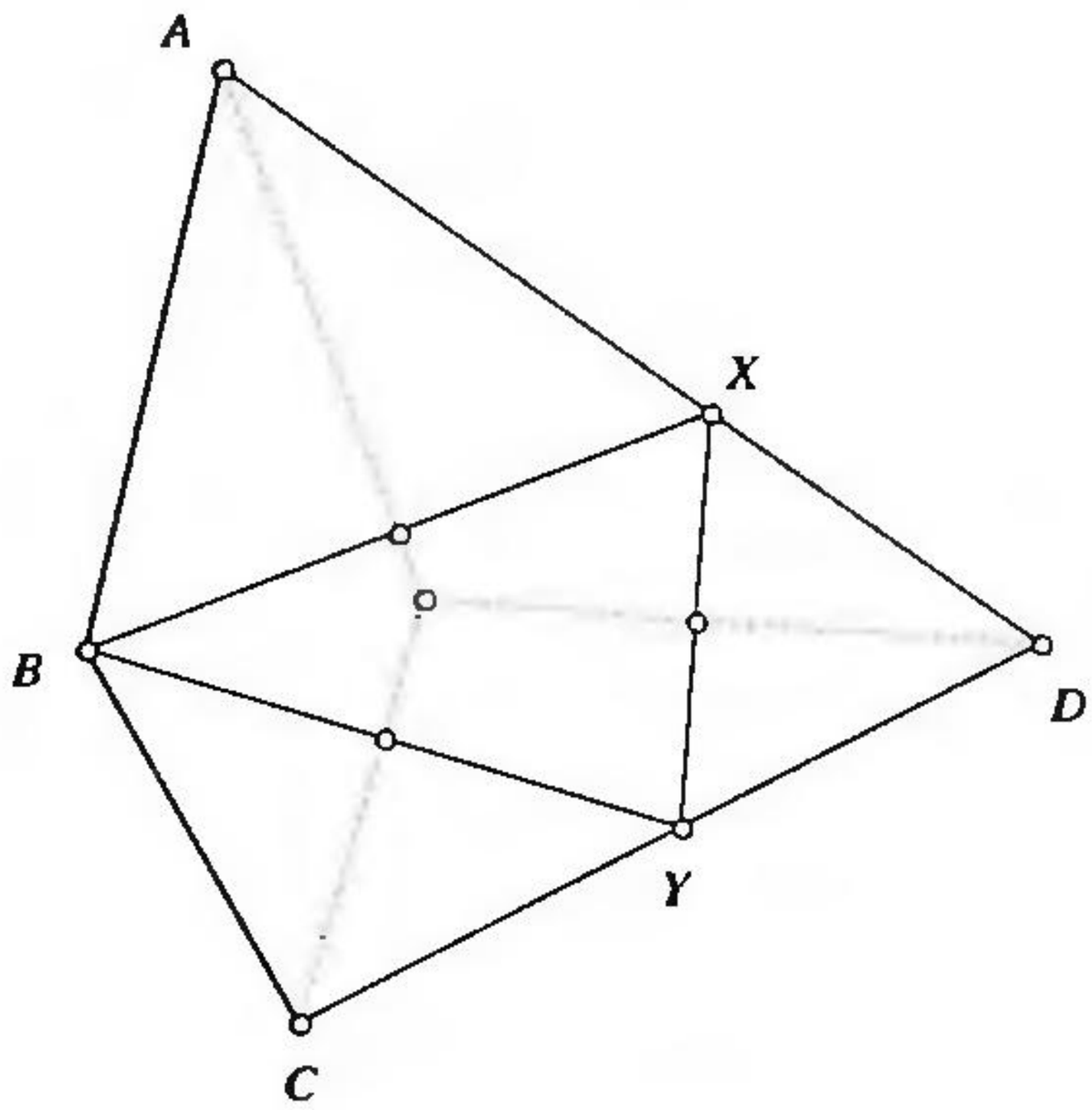


Figure 2.33

Thus, the triangles $ABX$, $DXY$ and $CYB$ are isosceles, so the perpendicular bisectors of the sides $BX$, $XY$ and $YB$ of triangle $BXY$ are also the angle bisectors of $\angle A$, $\angle D$, and $\angle C$ of the quadrilateral. Since the perpendicular bisectors of the sides of a triangle are concurrent, we obtain the desired conclusion.

**Problem 1.** Suppose that in the quadrilateral $ABCD$ we inscribe a circle with center $O$. Prove that the sum of the angles $\angle AOB$ and $\angle COD$ equals $180^\circ$.

**Problem 2.** Let $ABCD$ be a quadrilateral with an inscribed circle. Prove that the circles inscribed in triangles $ABC$ and $ADC$ are tangent to each other.

**Problem 3.** Let $ABCD$ be a convex quadrilateral. Suppose that the lines $AB$ and $CD$ intersect at $E$ and the lines $AD$ and $BC$ intersect at $F$. Prove that the following statements are equivalent:

(i) a circle is inscribed in $ABCD$;

(ii) $BE + BF = DE + DF$;

(iii) $AE - AF = CE - CF$.

**Problem 4.** Let $ABCD$ be a convex quadrilateral. Suppose that the lines $AB$ and $CD$ intersect at $E$, and the lines $AD$ and $BC$ intersect at $F$. Let $M$ and $N$ be two arbitrary points on the line segments $AB$ and $BC$, respectively. The line $EN$ intersects $AF$ and $MF$ at $P$ and $R$. The line $MF$ intersects $CE$ at $Q$. Prove that if the quadrilaterals $AMRP$ and $CNRQ$ have inscribed circles, then $ABCD$ has an inscribed circle. (Figure 2.34.)

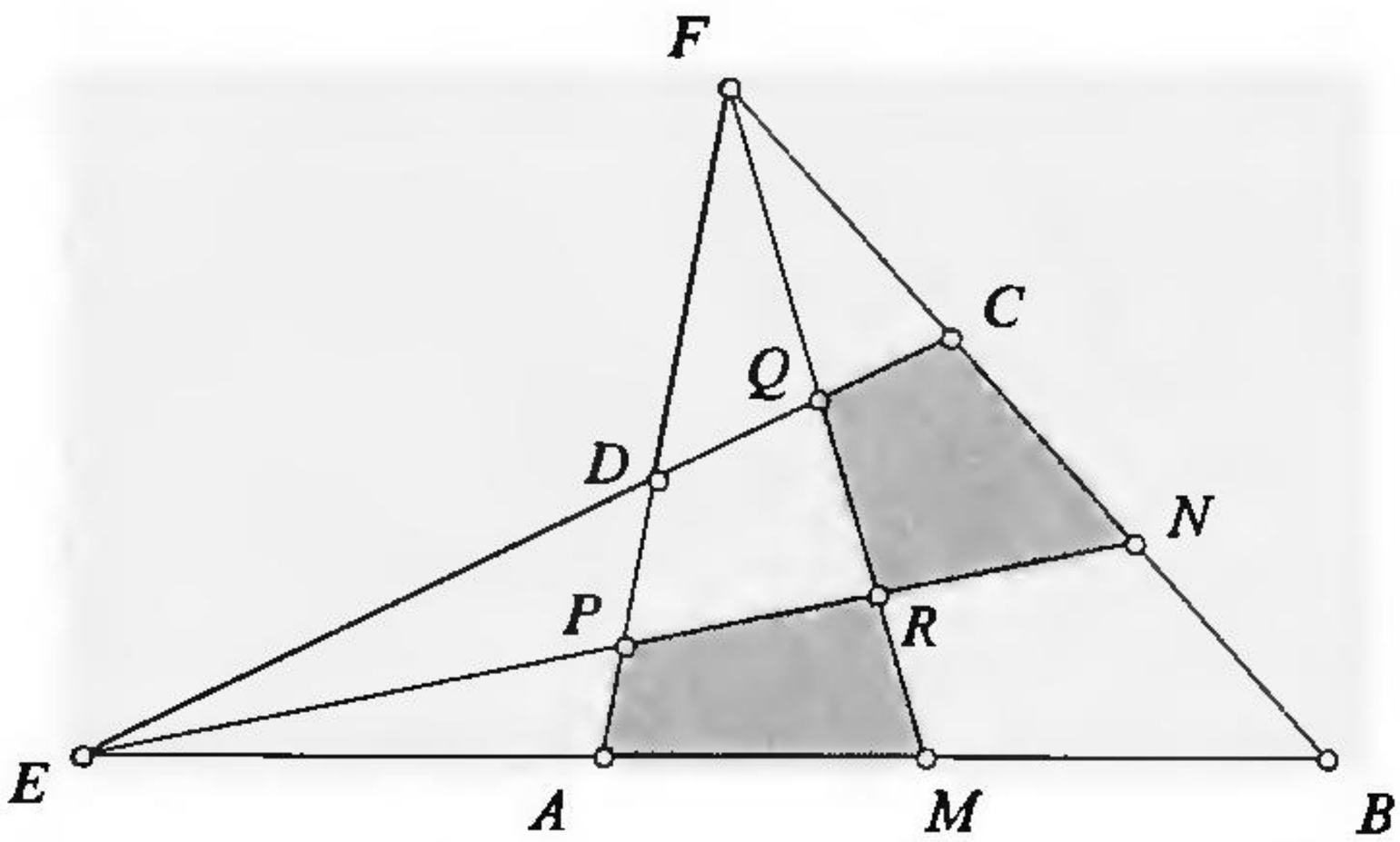


Figure 2.34

**Problem 5.** The points $A_1$, $A_2$, $C_1$, and $C_2$ are chosen in the interior of the sides $CD$, $BC$, $AB$, and $AD$ of the convex quadrilateral $ABCD$. Denote by $M$ the point of intersection of the lines $AA_2$ and $CC_1$, and denote by $N$ the point of intersection of the lines $AA_1$ and $CC_2$. Prove that if one can inscribe circles in three of the four quadrilaterals $ABCD$, $A_2BC_1M$, $AMCN$, and $A_1NC_2D$, then a circle can be also inscribed in the fourth one.

**Problem 6.** A line cuts a quadrilateral with an inscribed circle into two polygons with equal areas and equal perimeters. Prove that the line passes through the center of the inscribed circle.

## 2.8 Dr. Trig learns complex numbers

It is known that every complex number $z = a + bi$ can be written in the form

$$z = r(\cos\theta + i\sin\theta),$$

where $r = \sqrt{a^2 + b^2}$ is the absolute value of $z$ and $\theta$ is its argument. For instance, $i = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2}$, $1 + i = \frac{\sqrt{2}}{2}(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})$, etc. Also, if $z = r(\cos\theta + i\sin\theta)$, then a simple inductive argument proves that

$$z^n = r^n(\cos n\theta + i\sin n\theta)$$

(known as de Moivre's formula).

Now, if $z = \cos\theta + i\sin\theta$ (that is, its absolute value equals 1), then

$$\frac{1}{z} = \frac{1}{\cos\theta + i\sin\theta} = \frac{\cos\theta - i\sin\theta}{\cos^2\theta + \sin^2\theta} = \cos\theta - i\sin\theta.$$

We obtain the following useful formulae:

$$\cos\theta = \tfrac{1}{2}\left(z + \frac{1}{z}\right), \quad \sin\theta = \frac{1}{2i}\left(z - \frac{1}{z}\right).$$

Moreover, using de Moivre's formula, we have:

$$\cos n\theta = \tfrac{1}{2}\left(z^n + \frac{1}{z^n}\right), \quad \sin n\theta = \frac{1}{2i}\left(z^n - \frac{1}{z^n}\right).$$

These formulae may be very useful in solving a lot of Dr. Trig's problems. We might even forget some of his formulae. For instance, we have:

$$\begin{aligned}\cos 2\theta &= \tfrac{1}{2}\left(z^2 + \frac{1}{z^2}\right)\\ &= \tfrac{1}{2}\left(z + \frac{1}{z}\right)^2 - 1 = 2\left[\tfrac{1}{2}\left(z + \frac{1}{z}\right)\right]^2 - 1\\ &= 2\cos^2\theta - 1.\end{aligned}$$

Also:

$$\left[\tfrac{1}{2}\left(z + \frac{1}{z}\right)\right]^3 = \tfrac{1}{8}\left(z^3 + 3z + \frac{3}{z} + \frac{1}{z^3}\right) = \tfrac{1}{8}\left(z^3 + \frac{1}{z^3}\right) + \tfrac{3}{8}\left(z + \frac{1}{z}\right).$$

We deduce

$$\cos^3\theta = \frac{1}{4}\cos 3\theta + \frac{3}{4}\cos\theta,$$

a formula sometimes written as:

$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta.$$

Now, let us put some of this to work.

**Problem 1.** Let $a, b, c$ be real numbers such that

$$\cos a + \cos b + \cos c = \sin a + \sin b + \sin c = 0.$$

Prove that

$$\cos 2a + \cos 2b + \cos 2c = \sin 2a + \sin 2b + \sin 2c = 0.$$

**Solution.** Let $x = \cos a + i\sin a$, $y = \cos b + i\sin b$ and $z = \cos c + i\sin c$. From the hypothesis we have

$$x + y + z = 0,$$

and also

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = (\cos a - i\sin a) + (\cos b - i\sin b) + (\cos c - i\sin c) = 0.$$

It follows that $xy + xz + yz = 0$ so

$$x^2 + y^2 + z^2 = (x + y + z)^2 - 2(xy + xz + yz) = 0.$$

But then

$$\cos 2a + \cos 2b + \cos 2c + i\,(\sin 2a + \sin 2b + \sin 2c) = 0,$$

and we are done.

**Problem 2.** Prove the equality

$$\cos\frac{\pi}{=} + \cos\frac{3\pi}{=} + \cos\frac{5\pi}{=} = \tfrac{1}{2}.$$

**Solution.** Let

$$z = \cos\frac{\pi}{7} + i\sin\frac{\pi}{7}.$$

Then $z^7 = \cos\pi + i\sin\pi = -1$, hence $z^7 + 1 = 0$. On the other hand, we have:

$$\cos\frac{\pi}{7} + \cos\frac{3\pi}{7} + \cos\frac{5\pi}{7} = \tfrac{1}{2}\left(z + \frac{1}{z}\right) + \tfrac{1}{2}\left(z^3 + \frac{1}{z^3}\right) + \tfrac{1}{2}\left(z^5 + \frac{1}{z^5}\right)$$
$$= \frac{z^{10} + z^8 + z^6 + z^4 + z^2 + 1}{2z^5}.$$

Since $z^7 + 1 = 0$, we have $z^{10} = -z^3$ and $z^8 = -z$. It follows that:

$$z^{10} + z^8 + z^6 + z^4 + z^2 + 1 = z^6 + z^4 - z^3 + z^2 - z + 1$$
$$= z^6 - z^5 + z^4 - z^3 + z^2 - z + 1 + z^5 = \frac{z^7 + 1}{z + 1} + z^5 = z^5,$$

hence

$$\cos\frac{\pi}{7} + \cos\frac{3\pi}{7} + \cos\frac{5\pi}{7} = \frac{z^5}{2z^5} = \tfrac{1}{2}.$$

**Problem 3.** Find a closed form for the sum

$$S_n = \sin a + \sin 2a + \cdots + \sin na.$$

**Solution.** Let us ask for more: we will find a closed form for both $S_n$ and

$$C_n = \cos a + \cos 2a + \cdots + \cos na.$$

Let $z = \cos a + i\sin a$. Then

$$C_n + iS_n = z + z^2 + \cdots + z^n = z\frac{z^n - 1}{z - 1}.$$

Since $\cos x - 1 = -2\sin^2\frac{x}{2}$ and $\sin x = 2\sin\frac{x}{2}\cos\frac{x}{2}$, we obtain:

$$\frac{z^n - 1}{z - 1} = \frac{\cos na + i\sin na - 1}{\cos a + i\sin a - 1} = \frac{-2\sin^2\frac{na}{2} + 2i\sin\frac{na}{2}\cos\frac{na}{2}}{-2\sin^2\frac{a}{2} + 2i\sin\frac{a}{2}\cos\frac{a}{2}}$$
$$= \frac{\sin\frac{na}{2}}{\sin\frac{a}{2}}\left(\frac{\cos\frac{na}{2} + i\sin\frac{na}{2}}{\cos\frac{a}{2} + i\sin\frac{a}{2}}\right) = \frac{\sin\frac{na}{2}}{\sin\frac{a}{2}}\left(\cos\frac{(n-1)a}{2} + i\sin\frac{(n-1)a}{2}\right).$$

Thus

$$C_n + iS_n = (\cos a + i \sin a)\frac{\sin \frac{na}{2}}{\sin \frac{a}{2}}\left(\cos \frac{(n-1)a}{2} + i \sin \frac{(n-1)a}{2}\right)$$
$$= \frac{\sin \frac{na}{2}}{\sin \frac{a}{2}}\left(\cos \frac{(n+1)a}{2} + i \sin \frac{(n+1)a}{2}\right).$$

Finally

$$S_n = \frac{\sin \frac{na}{2} \sin \frac{(n+1)a}{2}}{\sin \frac{a}{2}},\ C_n = \frac{\sin \frac{na}{2} \cos \frac{(n+1)a}{2}}{\sin \frac{a}{2}}.$$

Now, help Dr. Trig to solve the following problems.

**Problem 4.** Let $a, b, c$ be real numbers such that

$$\cos a + \cos b + \cos c = \sin a + \sin b + \sin c = 0.$$

Prove that

$$\cos(a+b+c) = \tfrac{1}{3}(\cos 3a + \cos 3b + \cos 3c),$$
$$\sin(a+b+c) = \tfrac{1}{3}(\sin 3a + \sin 3b + \sin 3c).$$

**Problem 5.** Find the value of the product of $\cos 20^\circ$, $\cos 40^\circ$, and $\cos 80^\circ$.

**Problem 6.** Prove that

$$\frac{1}{\cos 6^\circ} + \frac{1}{\sin 24^\circ} + \frac{1}{\sin 48^\circ} = \frac{1}{\sin 12^\circ}.$$

**Problem 7.** Prove that

$$\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} + \tfrac{1}{2} = 0.$$

**Problem 8.** Prove the equality

$$\sin \frac{\pi}{n} \sin \frac{2\pi}{n} \cdots \sin \frac{(n-1)\pi}{n} = \frac{\sqrt{n}}{2^{n-1}}.$$

**Problem 9.** Solve the equation

$$\sin x + \sin 2x + \sin 3x = \cos x + \cos 2x + \cos 3x.$$

**Problem 10.** Prove that

$$\cos \frac{\pi}{5} = \frac{1+\sqrt{5}}{4}.$$

# Chapter 3

# Number Theory and Combinatorics

## 3.1 Arrays of numbers

Many Olympiad problems refer to arrays of numbers. Let us start with some examples.

**Problem 1.** The numbers $1, 2, \ldots, n^2$ are arranged in an $n \times n$ array in the following way:

| | | | | |
|---|---|---|---|---|
| 1 | 2 | 3 | $\cdots$ | $n$ |
| $n+1$ | $n+2$ | $n+3$ | $\cdots$ | $2n$ |
| $\vdots$ | | | | $\vdots$ |
| $n^2-n+1$ | $n^2-n+2$ | $n^2-n+3$ | $\cdots$ | $n^2$ |

Pick $n$ numbers from the array such that any two numbers are in different rows and different columns. Find the sum of these numbers.

**Solution.** If we denote by $a_{ij}$ the number in the $i$-th row and $j$-th column, then

$$a_{ij} = (i-1)n + j$$

for all $i, j = 1, 2, \ldots, n$. Because any two numbers are in different rows and different columns, it follows that from each row and each column exactly one number is chosen. Let $a_{1j_1}, a_{2j_2}, \ldots, a_{nj_n}$ be the chosen numbers, where $j_1, j_2, \ldots, j_n$ is a permutation of indices $1, 2, \ldots, n$. We have

$$\sum_{k=1}^{n} a_{kj_k} = \sum_{k=1}^{n} ((k-1)n + j_k) = n\sum_{k=1}^{n}(k-1) + \sum_{k=1}^{n} j_k.$$

But

$$\sum_{k=1}^{n} j_k = \frac{n(n+1)}{2}$$

since $j_1, j_2, \ldots, j_n$ is a permutation of indices $1, 2, \ldots, n$. It follows that

$$\sum_{k=1}^{n} a_{kj_k} = \frac{n(n^2+1)}{2}.$$

**Problem 2.** The entries of an $m \times n$ array of numbers are denoted by $x_{ij}$, with $1 \leq i, j \leq n$. For all $i, j, k$, $1 \leq i, j, k \leq n$ the following equality holds:

$$x_{ij} + x_{jk} + x_{ki} = 0.$$

Prove that there exist numbers $t_1, t_2, \ldots, t_n$ such that

$$x_{ij} = t_i - t_j$$

for all $i, j$, $1 \leq i, j \leq n$.

**Solution.** Setting $i = j = k$ in the given condition yields $3x_{ii} = 0$, hence $x_{ii} = 0$ for all $i$, $1 \leq i \leq n$. For $k = j \neq i$ we obtain

$$x_{ij} + x_{jj} + x_{ji} = 0,$$

hence $x_{ij} = -x_{ji}$, for all $i, j$. Now fix $i$ and $j$ and add up the equalities

$$x_{ij} + x_{jk} + x_{ki} = 0$$

for $k = 1, 2, \ldots, n$. It follows that

$$nx_{ij} + \sum_{k=1}^{n} x_{jk} + \sum_{k=1}^{n} x_{ki} = 0$$

or

$$nx_{ij} + \sum_{k=1}^{n} x_{jk} - \sum_{k=1}^{n} x_{ik} = 0.$$

If we define

$$t_i = \frac{1}{n} \sum_{k=1}^{n} x_{ik},$$

we obtain

$$x_{ij} = t_i - t_j,$$

for all $i, j$, as desired.

**Problem 3.** Prove that among any 10 entries of the table

$$\begin{array}{cccccc} 0 & 1 & 2 & 3 & \ldots & 9 \\ 9 & 0 & 1 & 2 & \ldots & 8 \\ 8 & 9 & 0 & 1 & \ldots & 7 \\ \vdots & & & & & \\ 1 & 2 & 3 & 4 & \ldots & 0 \end{array}$$

situated in different rows and different columns, at least two are equal.

**Solution.** Denote by $a_{ij}$ the entries of the table, $1 \leq i, j \leq 10$ and observe that $a_{ij} = a_{hk}$ if and only if $i - j \equiv h - k \pmod{10}$. Let $a_{ij_i}$, $i = 1, 2, \ldots, 10$ be 10 entries situated in different rows and different columns. If these entries are all different, then the differences $i - j_i$ give distinct residues mod 10, hence

$$\sum_{i=1}^{10}(i - j_i) \equiv 0 + 1 + \cdots + 9 \equiv 5 \pmod{10}.$$

On the other hand, since $j_1, j_2, \ldots, j_{10}$ is a permutation of the indices 1, 2, ..., 10, we have

$$\sum_{i=1}^{10}(i - j_i) = \sum_{i=1}^{10} i - \sum_{i=1}^{10} j_i = 0,$$

hence $0 \equiv 5 \pmod{10}$, a contradiction.

Here are some proposed problems.

**Problem 4.** Prove that the sum of any $n$ entries of the table

$$\begin{array}{cccccc} 1 & \frac{1}{2} & \frac{1}{3} & \cdots & \frac{1}{n} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \cdots & \frac{1}{n+1} \\ \vdots & & & & \\ \frac{1}{n} & \frac{1}{n+1} & \frac{1}{n+2} & \cdots & \frac{1}{2n-1} \end{array}$$

situated in different rows and different columns is not less than 1.

**Problem 5.** The entries of an $n \times n$ array of numbers are denoted by $a_{ij}$, $1 \leq i, j \leq n$. The sum of any $n$ entries situated on different rows and different columns is the same. Prove that there exist numbers $x_1, x_2, \ldots, x_n$ and $y_1, y_2, \ldots, y_n$, such that

$$a_{ij} = x_i + y_j,$$

for all $i, j$.

**Problem 6.** In an $n \times n$ array of numbers, all rows are different (two rows are different if they differ in at least one entry). Prove that there is a column which can be deleted in such a way that the remaining rows are still different.

**Problem 7.** The positive integers from 1 to $n^2$ are placed arbitrarily on squares of an $n \times n$ chessboard. Prove that there exist two adjacent squares (having a common vertex or a common side) such that the difference of the numbers placed on them is not less than $n + 1$.

**Problem 8.** A positive integer is written in each square of an $n^2 \times n^2$ chess board. The difference between the numbers in any two squares sharing an edge is less than or equal to $n$. Prove that at least $\left\lfloor \frac{n}{2} \right\rfloor + 1$ squares contain the same number.

**Problem 9.** The numbers $1, 2, \ldots, 100$ are arranged in the squares of a $10 \times 10$ table in the following way: the numbers $1, \ldots, 10$ are in the bottom row in increasing order, numbers $11, \ldots, 20$ are in the next row in increasing order, and so on. One can choose any number and two of its neighbors in two opposite directions (horizontal, vertical, or diagonal). Then either the number is increased by 2 and its neighbors are decreased by 1, or the number is decreased by 2 and its neighbors are increased by 1. After several such operations the table again contains all the numbers $1, 2, \ldots, 100$. Prove that they are in the original order.

## 3.2 Functions defined on sets of points

Several olympiad problems deal with functions defined on certain sets of points. These problems are interesting in that they combine both geometrical and algebraic ideas.

**Problem 1**. Let $n > 2$ be an integer and $f : P \to \mathbf{R}$ a function defined on the set of points in the plane, with the property that for any regular $n$-gon $A_1A_2 \ldots A_n$,

$$f(A_1) + f(A_2) + \cdots + f(A_n) = 0.$$

Prove that $f$ is the zero function.

**Solution.** Let $A$ be an arbitrary point. Consider a regular $n$-gon $AA_1A_2 \ldots A_{n-1}$. Let $k$ be an integer, $0 \leq k \leq n-1$. A rotation with center $A$ of angle $\frac{2k\pi}{n}$ sends the polygon $AA_1A_2 \ldots A_{n-1}$ to $A_{k0}A_{k1} \ldots A_{k,n-1}$, where $A_{k0} = A$ and $A_{ki}$ is the image of $A_i$, for all $i = 1, 2, \ldots, n-1$.

From the condition of the statement, we have

$$\sum_{k=0}^{n-1}\sum_{i=0}^{n-1} f(A_{ki}) = 0.$$

Observe that in the sum the number $f(A)$ appears $n$ times, therefore

$$nf(A) + \sum_{k=0}^{n-1}\sum_{i=1}^{n-1} f(A_{ki}) = 0.$$

On the other hand, we have

$$\sum_{k=0}^{n-1}\sum_{i=1}^{n-1} f(A_{ki}) = \sum_{i=1}^{n-1}\sum_{k=0}^{n-1} f(A_{ki}) = 0,$$

since the polygons $A_{0i}A_{1i} \ldots A_{n-1,i}$ are all regular $n$-gons. From the two equalities above we deduce $f(A) = 0$, hence $f$ is the zero function.

**Problem 2.** Let $n \geq 4$ be an integer and let $p \geq 2n-3$ be a prime number. Let $S$ be a set of $n$ points in the plane, no three of which are collinear, and let $f : S \to \{0, 1, \ldots, p-1\}$ be a function such that

1. there exists a unique point $A \in S$ such that $f(A) = 0$;
2. if $C(X, Y, Z)$ is the circle determined by the distinct points $X, Y, Z \in S$, then

$$\sum_{P \in S \cap C(X,Y,Z)} f(P) \equiv 0 \pmod{p}.$$

Prove that all points of $S$ lie on a circle.

**Solution.** Suppose there exist points $B$ and $C$ such that no other point of $S$ lies on $C(A, B, C)$. We have

$$f(B) + f(C) \equiv 0 \pmod{p},$$

so, if $f(B) = i \neq 0$, then $f(C) = p - i$. Let

$$\sigma = \sum_{X \in S} f(X).$$

If a number of $b$ circles pass through $A$, $B$ and other points of $S$ and a number of $c$ circles pass through $A$, $C$ and other points of $S$, applying the condition from the hypothesis to all these circles, we obtain

$$\sigma + (b-1)i \equiv 0 \pmod{p},$$
$$\sigma + (c-1)(p-i) \equiv 0 \pmod{p},$$

hence

$$a + b - 2 \equiv 0 \pmod{p}.$$

Since $1 \leq b, c \leq n-2$, we have $2 \leq b + c \leq 2n - 4 < p$, hence $a = b = 1$, which is a contradiction.

It follows that for any points $B, C \in S$, there exists at least one more point of $S$ lying on $C(A, B, C)$. This implies the fact that all the points of $S$ lie on a circle. Indeed, consider an inversion $I$ of pole $A$. The set of points $S - \{A\}$ is transformed into the set of points $I(S - \{A\}) = N$ and the circles $C(A, X, Y)$, with $X, Y \in S$ are transformed into lines $I_X I_Y$ through points of $N$. The above condition then reduces to the following: for any two points $I_X$, $I_Y$ of $N$, there exists at least one other point of $N$ lying on the line $I_X I_Y$. We can prove that all points of $N$ are collinear, whence the points of $S$ lie on a circle.

Indeed , suppose the points are not collinear and choose points $A$, $B$, $C$ such that the distance from $C$ to the line $AB$ is minimal. Let $C'$ be the projection of $C$ on the line $AB$. From the hypothesis it follows that there exists another point $D \in N$ on the line $AB$. The point $C'$ divides the line $AB$ into two half lines and at least two of the points $A$, $B$, $D$ lie in one of these half lines. Suppose these points are $B$ and $D$ located as in the Figure 3.1.

If $B''$ is a point on $CD$ such that $BB''$ is parallel to $CC'$ and $B'$ is the projection of $B$ on $CD$, it is not difficult to see that $BB' < BB'' < CC'$, contradicting the minimality of $CC'$.

Here are more examples

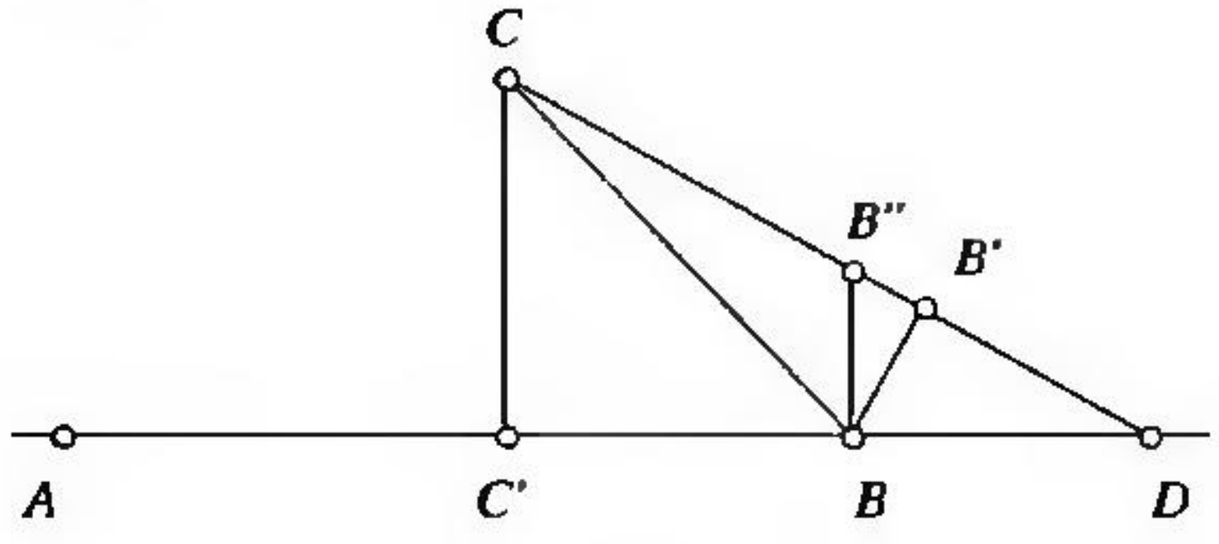


Figure 3.1

**Problem 3.** Let $D$ be a set of $n \geq 1$ concentric circles in the plane. Suppose that the function $f : D \to D$ satisfies

$$d(f(A), f(B)) \geq d(A, B)$$

for every $A, B \in D$, where $d(M, N)$ is the distance between the points $M$ and $N$.

Prove that

$$d(f(A), f(B)) = d(A, B)$$

for every $A, B \in D$.

**Problem 4.** Let $S$ be a set of $n \geq 4$ points in the plane, with the property that no three of them are collinear and not all of them lie on a circle. Find all functions $f : S \to \mathbf{R}$ with the property that for any circle $C$ containing at least three points of $S$,

$$\sum_{P \in C \cap S} f(P) = 0.$$

**Problem 5.** Let $P$ be the set of all points in the plane and $L$ be the set of all lines of the plane. Find, with proof, whether there exists a bijective function $f : P \to L$ such that for any three collinear points $A, B, C$, the lines $f(A)$, $f(B)$, and $f(C)$ are either parallel or concurrent.

**Problem 6.** Let $S$ be the set of interior points of a sphere and $C$ be the set of interior points of a circle. Find, with proof, whether there exists a function $f : S \to C$ such that for any points $A, B \in S$, $d(A, B) \leq d(f(A), f(B))$.

**Problem 7.** Let $S$ be the set of all polygons in the plane. Prove that there exists a function $f : S \to (0, +\infty)$ such that

1. $f(P) < 1$, for any $P \in S$;
2. If $P_1, P_2 \in S$ have disjoint interiors and $P_1 \cup P_2 \in S$, then $f(P_1 \cup P_2) = f(P_1) + f(P_2)$.

## 3.3 Count twice!

Many interesting results can be obtained by counting the elements of a set in two different ways. Moreover, counting twice can be a good problem solving strategy. Let us consider the following examples.

**Problem 1.** In how many ways can a committee of $k$ persons with a chairman be chosen from a set of $n$ persons?

**Solution.** We can choose the $k$ members of the committee in $\binom{n}{k}$ ways and then the chairman in $k$ ways, so that the answer is $k\binom{n}{k}$. On the other hand, we can first choose the chairman (this can be done in $n$ ways) and next the rest of the $k-1$ members from the remaining $n-1$ persons, leading to a total of $n\binom{n-1}{k-1}$ possibilities. Thus, we obtain the identity

$$k\binom{n}{k} = n\binom{n-1}{k-1}.$$

**Problem 2.** Consider a finite sequence of real numbers for which the sum of any seven consecutive terms is negative and the sum of any 11 consecutive terms is positive. Find the greatest number of terms of such a sequence.

**Solution.** It is not difficult to see that such a sequence cannot have 77 or more terms. For that, we add up in two ways the first 77 terms, obtaining a contradiction:

$$\begin{aligned}(x_1 + x_2 + \cdots + x_7) + (x_8 + x_9 + \cdots + x_{14}) \\ + \cdots + (x_{71} + x_{72} + \cdots + x_{77}) < 0,\end{aligned}$$

$$\begin{aligned}(x_1 + x_2 + \cdots + x_{11}) + (x_{12} + x_{13} + \cdots + x_{22}) \\ + \cdots + (x_{67} + x_{68} + \cdots + x_{77}) > 0.\end{aligned}$$

In fact, the number of terms is much less. The sequence cannot contain more than 16 terms. To prove that, we refine our double counting technique.

Suppose the sequence has at least 17 terms. We can arrange them in a $7 \times 11$ table as follows:

$$\begin{pmatrix} x_1 & x_2 & \cdots & x_{10} & x_{11} \\ x_2 & x_3 & \cdots & x_{11} & x_{12} \\ \vdots & \vdots & & \vdots & \vdots \\ x_6 & x_7 & \cdots & x_{15} & x_{16} \\ \cdots & \cdots & & \cdots & \cdots \end{pmatrix}$$

Now, add up all the entries in two ways: by rows and by columns. We again reach a contradiction.

Finally, there is an example of such a sequence with 16 terms:

$$5, 5, -13, 5, 5, 5, -13, 5, 5, -13, 5, 5, 5, -13, 5, 5.$$

Try counting twice in the following problems.

**Problem 3.** Find how many committees with a chairman can be chosen from a set of $n$ persons. Derive the identity

$$\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n} = n2^{n-1}.$$

**Problem 4.** In how many ways can one choose $k$ balls from a set containing $n-1$ red balls and a blue one? Derive the identity

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

**Problem 5.** Let $S$ be a set of $n$ persons such that:
(i) any person is acquainted with exactly $k$ other persons in $S$;
(ii) any two persons that are acquainted have exactly $l$ common acquaintances in $S$;
(iii) any two persons that are not acquainted have exactly $m$ common acquaintances in $S$.

Prove that

$$m(n-k) - k(k-l) + k - m = 0.$$

**Problem 6.** Let $n$ be an odd integer greater than 1 and let $c_1, c_2, \ldots, c_n$ be integers. For each permutation $a = (a_1, a_2, \ldots, a_n)$ of $\{1, 2, \ldots, n\}$, define

$$S(a) = \sum_{i=1}^{n} c_i a_i.$$

Prove that there exist permutations $a \neq b$ of $\{1, 2, \ldots, n\}$ such that $n!$ is a divisor of $S(a) - S(b)$.

**Problem 7.** Let $a_1 \leq a_2 \leq \cdots \leq a_n = m$ be positive integers. Denote by $b_k$ the number of those $a_i$ for which $a_i \geq k$. Prove that

$$a_1 + a_2 + \cdots + a_n = b_1 + b_2 + \cdots + b_m.$$

**Problem 8.** In how many ways can one fill an $m \times n$ table with $\pm 1$ such that the product of the entries in each row and each column equals $-1$?

## 3.4 Sequences of integers

Sequences of integers are a favorite of olympiad problem writers, since such sequences involve several different mathematical concepts, including, for example, algebraic techniques, recursive relations, divisibility and primality.

**Problem 1.** Consider the sequence $(a_n)_{n\geq 1}$ defined by $a_1 = a_2 = 1$, $a_3 = 199$ and

$$a_{n+1} = \frac{1989 + a_n a_{n-1}}{a_{n-2}}$$

for all $n \geq 3$. Prove that all the terms of the sequence are positive integers.

**Solution.** We have

$$a_{n+1}a_{n-2} = 1989 + a_n a_{n-1}.$$

Replacing $n$ by $n-1$ yields

$$a_n a_{n-3} = 1989 + a_{n-1}a_{n-2}$$

and we obtain

$$a_{n+1}a_{n-2} - a_n a_{n-1} = a_n a_{n-3} - a_{n-1}a_{n-2}.$$

This is equivalent to

$$a_{n-2}(a_{n+1} + a_{n-1}) = a_n(a_{n-1} + a_{n-3})$$

or

$$\frac{a_{n+1} + a_{n-1}}{a_n} = \frac{a_{n-1} + a_{n-3}}{a_{n-2}}$$

for all $n \geq 4$. If $n$ is even, we obtain

$$\frac{a_{n+1} + a_{n-1}}{a_n} = \frac{a_{n-1} + a_{n-3}}{a_{n-2}} = \cdots = \frac{a_3 + a_1}{a_2} = 200,$$

while if $n$ is odd,

$$\frac{a_{n+1} + a_{n-1}}{a_n} = \frac{a_{n-1} + a_{n-3}}{a_{n-2}} = \cdots = \frac{a_4 + a_2}{a_3} = 11.$$

It follows that

$$a_{n+1} = \begin{cases} 200a_n - a_{n-1}, & \text{if } n \text{ is even;} \\ 11a_n - a_{n-1}, & \text{if } n \text{ is odd.} \end{cases}$$

An inductive argument shows that all $a_n$ are positive integers.

**Problem 2.** Let $a, b, c$ be positive real numbers. The sequence $(a_n)_{n\geq 1}$ is defined by $a_1 = a, a_2 = b$ and

$$a_{n+1} = \frac{a_n^2 + c}{a_{n-1}}$$

for all $n \geq 2$. Prove that the terms of the sequence are all positive integers if and only if $a, b$ and $\frac{a^2+b^2+c}{ab}$ are positive integers.

**Solution.** Clearly, all the terms of the sequence are positive numbers. Write the recursive relation as

$$a_{n+1}a_{n-1} = a_n^2 + c.$$

Replacing $n$ by $n-1$ yields

$$a_n a_{n-2} = a_{n-1}^2 + c,$$

and by subtracting the two equalities we deduce

$$a_{n-1}(a_{n+1} + a_{n-1}) = a_n(a_n + a_{n-2}).$$

Therefore

$$\frac{a_{n+1} + a_{n-1}}{a_n} = \frac{a_n + a_{n-2}}{a_{n-1}}$$

for all $n \geq 3$. It follows that the sequence $b_n = \frac{a_{n+1}+a_{n-1}}{a_n}$ is constant, say $b_n = k$, for all $n \geq 2$. Then the sequence $(a_n)$ satisfies the recursive relation

$$a_{n+1} = ka_n - a_{n-1}$$

for all $n \geq 2$ and since $a_3 = \frac{b^2+c}{a} = kb - a$, we derive that

$$k = \frac{a^2 + b^2 + c}{ab}.$$

Now, if $a, b$ and $k$ are positive integers, it follows inductively that $a_n$ is a positive integer for all $n \geq 1$. Conversely, suppose that $a_n$ is a positive integer for all $n \geq 1$. Then $a, b$ are positive integers and $k = \frac{a_3+a}{b}$ is a rational number. Let $k = \frac{p}{q}$, where $p$ and $q$ are relatively prime positive integers. We want to prove that $q = 1$. Suppose that $q > 1$. From the recursive relation we obtain

$$q(a_{n+1} + a_{n-1}) = pa_n,$$

and hence $q$ divides $a_n$ for all $n \geq 2$. We prove by induction on $s$ that $q^s$ divides $a_n$ for all $n \geq s+1$. We have seen that this is true for $s = 1$. Suppose $q^{s-1}$ divides $a_n$ for all $n \geq s$. We have

$$a_{n+2} = \frac{p}{q} a_{n+1} - a_n,$$

which is equivalent to

$$\frac{a_{n+2}}{q^{s-1}} = p\frac{a_{n+1}}{q^s} - \frac{a_n}{q^{s-1}}.$$

If $n \geq s$, then $q^{s-1}$ divides $a_n$ and $a_{n+2}$, hence $q^s$ divides $a_{n+1}$. It follows that $q^s$ divides $a_n$ for all $n \geq s+1$. Finally, we have

$$a_{s+2} = \frac{a_{s+1}^2 + c}{a_s}$$

or

$$c = a_s a_{s+2} - a_s^2,$$

which implies that $c$ is divisible by $q^{2(s-1)}$ for all $s \geq 1$. Because $c > 0$, this is a contradiction.

**Problem 3.** Consider the sequence $(a_n)_{n\geq 0}$ given by the following relation: $a_0 = 4$, $a_1 = 22$, and for all $n \geq 2$,

$$a_n = 6a_{n-1} - a_{n-2}.$$

Prove that there exist sequences of positive integers $(x_n)_{n\geq 0}$, $(y_n)_{n\geq 0}$ such that

$$a_n = \frac{y_n^2 + 7}{x_n - y_n},$$

for all $n \geq 0$.

**Solution.** Observe that $a_n$ is an even positive integer for all $n$ and that the sequence $(a_n)_{n\geq 0}$ is increasing. The last assertion follows inductively if we write the recursive relation under the form

$$a_n - a_{n-1} = 5(a_{n-1} - a_{n-2}) + 4a_{n-2}.$$

Define

$$x_n = \frac{a_n + a_{n-1}}{2}, \quad y_n = \frac{a_n - a_{n-1}}{2}$$

with $x_0 = 3$, $y_0 = 1$. Observe that

$$x_n = 3x_{n-1} + 4y_{n-1}$$

and

$$y_n = 2x_{n-1} + 3y_{n-1}.$$

We have $a_n = x_n + y_n$, hence it is sufficient to prove that

$$x_n + y_n = \frac{y_n^2 + 7}{x_n - y_n}$$

or

$$x_n^2 = 2y_n^2 + 7$$

for all $n$. We use induction. The equality is true for $n = 0$. Suppose that $x_{n-1}^2 = 2y_{n-1}^2 + 7$. Then

$$x_n^2 - 2y_n^2 - 7 = (3x_{n-1} + 4y_{n-1})^2 - 2(2x_{n-1} + 3y_{n-1})^2 - 7$$

$$= x_{n-1}^2 - 2y_{n-1}^2 - 7 = 0,$$

which proves our claim.

Try your hand at the following problems.

**Problem 4.** Prove that there exist sequences of odd positive integers $(x_n)_{n\geq 3}$, $(y_n)_{n\geq 3}$ such that

$$7x_n^2 + y_n^2 = 2^n$$

for all $n \geq 3$.

**Problem 5.** Let $x_1 = x_2 = 1$, $x_3 = 4$, and

$$x_{n+3} = 2x_{n+2} + 2x_{n+1} - x_n$$

for all $n \geq 1$. Prove that $x_n$ is a square for all $n \geq 1$.

**Problem 6.** The sequence $(a_n)_{n\geq 0}$ is defined by $a_0 = a_1 = 1$ and

$$a_{n+1} = 14a_n - a_{n-1}$$

for all $n \geq 1$. Prove that the number $2a_n - 1$ is a square for all $n \geq 0$.

**Problem 7.** The sequence $(x_n)_{n\geq 1}$ is defined by $x_1 = 0$ and

$$x_{n+1} = 5x_n + \sqrt{24x_n^2 + 1}$$

for all $n \geq 1$. Prove that all $x_n$ are positive integers.

**Problem 8.** Let $(a_n)_{n\geq 1}$ be an increasing sequence of positive integers such that

1. $a_{2n} = a_n + n$ for all $n \geq 1$;
2. if $a_n$ is a prime, then $n$ is a prime.

Prove that $a_n = n$, for all $n \geq 1$.

## 3.5 Equations with infinitely many solutions

In this section we will examine several Diophantine equations that have an infinite number of solutions.

**Problem 1.** Find integer solutions for the equation

$$4x^2 + 9y^2 = 72z^2.$$

**Solution.** We first notice that $x$ must be divisible by 3 and that $y$ is an even integer. Setting $x = 3u$ and $y = 2v$ yields

$$36u^2 + 36v^2 = 72z^2$$

or

$$2z^2 = u^2 + v^2.$$

We deduce that $u, v$ have the same parity, so that $u + v$ and $u - v$ are even integers. It follows that

$$z^2 = \left(\frac{u+v}{2}\right)^2 + \left(\frac{u-v}{2}\right)^2.$$

This is the well-known Pythagorean equation, whose solutions are

$$z = k(m^2 + n^2),\ \frac{u+v}{2} = 2kmn,\ \frac{u-v}{2} = k(m^2 - n^2),$$

yielding

$$x = 3k(2mn + m^2 - n^2),\ y = 2k(2mn - m^2 + n^2),\ z = k(m^2 + n^2).$$

**Problem 2.** Prove that the equation

$$x^2 + (x+1)^2 = y^2$$

has infinitely many solutions in positive integers.

**Solution.** Observe that $x = 3$, $y = 5$ is a solution. We define the sequences $(x_n)_{n\geq 1}$, $(y_n)_{n\geq 1}$ by

$$x_{n+1} = 3x_n + 2y_n + 1,$$

$$y_{n+1} = 4x_n + 3y_n + 2,$$

with $x_1 = 3$ and $y_1 = 5$. Suppose that $(x_n, y_n)$ is a solution of the equation. We claim that $(x_{n+1}, y_{n+1})$ is also a solution and since the sequences are clearly increasing, the equation has infinitely many solutions.

Indeed, we have

$$x_{n+1}^2 + (x_{n+1} + 1)^2 = (3x_n + 2y_n + 1)^2 + (3x_n + 2y_n + 2)^2.$$

Using the equality

$$x_n^2 + (x_n + 1)^2 = y_n^2$$

yields

$$(3x_n + 2y_n + 1)^2 + (3x_n + 2y_n + 2)^2 = (4x_n + 3y_n + 2)^2$$

and hence

$$x_{n+1}^2 + (x_{n+1} + 1)^2 = y_{n+1}^2,$$

thus proving the claim.

Here are some proposed problems.

**Problem 3.** Find all triples of integers $(x, y, z)$ such that

$$x^2 + xy = y^2 + xz.$$

**Problem 4.** Let $n$ be an integer. Prove that the equation

$$x^2 + y^2 = n + z^2$$

has infinitely many integer solutions.

**Problem 5.** Let $m$ be a positive integer. Find all pairs of integers $(x, y)$ such that

$$x^2(x^2 + y) = y^{m+1}.$$

**Problem 6.** Let $m$ be a positive integer. Find all pairs of integers $(x, y)$ such that

$$x^2(x^2 + y^2) = y^{m+1}.$$

**Problem 7.** Find all nonnegative integers $a, b, c, d, n$ that satisfy

$$a^2 + b^2 + c^2 + d^2 = 7 \cdot 4^n.$$

**Problem 8.** Show that there are infinitely many positive integers $x, y, z, t$, with no common divisor greater than 1, satisfying the equality

$$x^3 + y^3 + z^2 = t^4.$$

**Problem 9.** Let $k \geq 6$ be an integer number. Prove that the system of equations

$$\begin{cases} x_1 + x_2 + \cdots + x_{k-1} = x_k, \\ x_1^3 + x_2^3 + \cdots + x_{k-1}^3 = x_k, \end{cases}$$

has infinitely many integral solutions.

## 3.6 Equations with no solutions

One of the most famous problems in number theory is Fermat's last theorem. The theorem states that for $n \geq 3$, the equation

$$x^n + y^n = z^n$$

has no nonzero integer solutions. Hundreds of articles and books were written about this theorem, which was finally proved by Andrew Wiles in 1994.

For our purposes, however, we will consider much simpler Diophantine equations with no solutions.

**Problem 1.** Prove that the equation

$$x^3 + y^3 + z^3 = 2002$$

has no solutions in integers.

**Solution.** We first notice that a cube of an integer is congruent to either $0, 1$ or $-1 \pmod 9$. Then the sum of two cubes can be congruent to $0, 1, -1, 2$ or $-2$ and the sum of three cubes to $0, 1, -1, 2, -2, 3$ or $-3$. Since 2002 is congruent to $4 \pmod 9$ the equation has no solutions.

**Problem 2.** Prove that the equation

$$4x^3 - 7y^3 = 2003$$

has no solutions in integers.

**Solution.** Because $2003 = 1 + 7 \cdot 286$, the equation is equivalent to

$$4x^3 - 1 = 7\left(y^3 + 286\right).$$

Notice that if $x$ is an integer, then $x^3 \equiv 0, 1$ or $6 \pmod 7$, so that $4x^3 \equiv 0, 4$ or $3 \pmod 7$. We conclude that $4x^3 - 1$ is not divisible by 7, hence the equation has no integer solutions.

**Problem 3.** Prove that the equation

$$x^3 + y^4 = 7$$

has no solution in integers.

**Solution.** The residues of a cube modulo 13 are $0, 1, 5, 8$ or 12 while the residues of a fourth power are $0, 1, 3$ or 9 modulo 13. This leads us to conclude that the equation $x^3 + y^4 \equiv 7 \pmod{13}$ cannot hold.

Here are some proposed problems.

**Problem 4.** Prove that the equation

$$4xy - x - y = z^2$$

has no positive integer solutions.

**Problem 5.** Prove that the equation

$$6(6a^2 + 3b^2 + c^2) = 5d^2$$

has no solution in nonzero integers.

**Problem 6.** Prove that the system of equations

$$\begin{cases} x^2 + 6y^2 = z^2 \\ 6x^2 + y^2 = t^2 \end{cases}$$

has no positive integer solutions.

**Problem 7.** Let $k$ and $n$ be positive integers, with $n > 2$. Prove that the equation

$$x^n - y^n = 2^k$$

has no positive integer solutions.

**Problem 8.** Prove that the equation

$$\frac{x^{2000} - 1}{x - 1} = y^2$$

has no positive integer solutions.

**Problem 9.** Prove that the equation

$$4\left(x_1^4 + x_2^4 + \cdots + x_{14}^4\right) = 7\left(x_1^3 + x_2^3 + \cdots + x_{14}^3\right)$$

has no solution in positive integers.

## 3.7 Powers of 2

Many problems deal in one way or another with powers of 2 and related ideas like binary representation and divisibility.

**Problem 1.** Prove that the numbers $F_n = 2^{2^n} + 1$, $n = 0, 1, 2, \ldots$, are pairwise coprime.

**Solution.** Let $m > n$ and $d = \gcd(F_m, F_n)$. Now, $F_m - 2 = 2^{2^m} - 1 = \left(2^{2^{n+1}}\right)^{2^{m-n-1}} - 1$ is divisible by $2^{2^{n+1}} - 1$, since $x^k - 1$ is always divisible by $x - 1$. But $2^{2^{n+1}} - 1 = \left(2^{2^n} - 1\right)\left(2^{2^n} + 1\right) = \left(2^{2^n} - 1\right) F_n$, hence $F_n$ divides $F_m - 2$. We deduce that $d$ is a divisor of 2 and, since all $F_n$ are odd numbers, $d$ must be equal to 1, thus proving the claim. Observe that the above property incidentally also proves that there exist infinitely many prime numbers.

**Problem 2.** The sequence $(a_n)_{n\geq 1}$ satisfies $a_1 = 2$, $a_2 = 3$ and, for each $n \geq 2$, either $a_{n+1} = 2a_{n-1}$ or $a_{n+1} = 3a_n - 2a_{n-1}$. Prove that no integer between 1600 and 2000 can be a term of the sequence.

**Solution.** Clearly, the sequence is not uniquely determined by the above conditions. A short analysis shows that the first six terms could be

(a) 2, 3, 4, 6, 8, 12;

(b) 2, 3, 4, 6, 10, 12;

(c) 2, 3, 4, 6, 10, 18;

(d) 2, 3, 5, 6, 10, 12;

(e) 2, 3, 5, 6, 10, 18;

(f) 2, 3, 5, 6, 8, 12;

(g) 2, 3, 5, 9, 10, 18;

(h) 2, 3, 5, 9, 10, 12;

(i) 2, 3,5, 9, 17, 18;

(j) 2, 3, 5, 9, 17, 33.

The key observation is that each of the numbers 2, 3, 4, 5, 6, 8, 9, 10, 12, 17, 18 and 33 can be written as a sum of two powers of 2 (or, equivalently, that their binary representation contains exactly two 1's). Thus, we may conjecture that $a_n = 2^{x_n} + 2^{y_n}$, for some integers $x_n$, $y_n$ and try induction. Unfortunately, the attempt fails unless we observe something more. Let's take, for instance, the case h) and write the terms as sums of powers of 2: $2 = 2^0 + 2^0$, $3 = 2^0 + 2^1$, $5 = 2^0 + 2^2$, $9 = 2^0 + 2^3$, $10 = 2^1 + 2^3$, $12 = 2^2 + 2^3$. If you don't see the pattern, let's take case (c): $2 = 2^0 + 2^0$, $3 = 2^0 + 2^1$, $4 = 2^1 + 2^1$, $6 = 2^1 + 2^2$, $10 = 2^1 + 2^3$, $18 = 2^1 + 2^4$.

We can see that at each step, exactly one of the numbers $x_n$ and $y_n$ increases with one unit. Now, we can state a stronger induction hypothesis which is easier to prove (this is not uncommon!): for each $n$, $a_n = 2^{x_n} + 2^{y_n}$, with $x_n \leq x_{n+1}$, $y_n \leq y_{n+1}$ and $x_n + y_n + 1 = x_{n+1} + y_{n+1}$. This obviously holds for $n = 1, 2$. Suppose it holds for all $k \leq n$ and look at $a_{n+1}$.

If $a_{n+1} = 2a_{n-1}$, then

$$a_{n+1} = 2\left(2^{x_n - 1} + 2^{y_n - 1}\right) = 2^{x_n - 1 + 1} + 2^{y_n - 1 + 1}.$$

If $x_n = x_{n-1}+1$, $y_n = y_{n-1}$, then let $x_{n+1} = x_n$ and $y_{n+1} = y_n+1$. We obtain $a_{n+1} = 2^{x_{n+1}} + 2^{y_{n+1}}$. If $x_n = x_{n-1}$, $y_n = y_{n-1} + 1$, then let $x_{n+1} = x_n + 1$ and $y_{n+1} = y_n$. Also, we get $a_{n+1} = 2^{x_{n+1}} + 2^{y_{n+1}}$.

If $a_{n+1} = 3a_n - 2a_{n-1}$, then

$$a_{n+1} = 3\left(2^{x_n} + 2^{y_n}\right) - 2\left(2^{x_{n-1}} + 2^{y_{n-1}}\right).$$

If $x_n = x_{n-1} + 1$, $y_n = y_{n-1}$, then

$$a_{n+1} = 3\left(2^{x_{n-1}+1} + 2^{y_{n-1}}\right) - 2\left(2^{x_{n-1}} + 2^{y_{n-1}}\right) = 2^{x_{n-1}+2} + 2^{y_{n-1}},$$

so that if we take $x_{n+1} = x_n+1$ and $y_{n+1} = y_n$, we obtain $a_{n+1} = 2^{x_{n+1}}+2^{y_{n+1}}$. The case $x_n = x_{n-1}$, $y_n = y_{n-1} + 1$ is similar; thus our assertion is proved.

Finally, we notice that the binary representations of the numbers 1600 and 2000 are 11001000000 and 11111010000, respectively. Clearly, no number between them has a binary representation with only two 1's. In fact, we can replace 1600 and 2000 with 1537 and 2047.

Here are some proposed problems:

**Problem 3.** Let $n$ be a positive integer such that $2^n + 1$ is a prime number. Prove that $n = 2^k$, for some integer $k$.

**Problem 4.** Let $n$ be a positive integer such that $2^n - 1$ is a prime number. Prove that $n$ is a prime number.

**Problem 5.** Prove that the number $A = 2^{1992} - 1$ can be written as a product of six integers greater than $2^{248}$.

**Problem 6.** Determine the remainder when $3^{2^n} - 1$ is divided by $2^{n+3}$.

**Problem 7.** Prove that for each $n$, there exists a number $A_n$, divisible by $2^n$, whose decimal representation contains $n$ digits, each of them equal to 1 or 2.

**Problem 8.** Using only the digits 1 and 2, suppose we write down numbers with $2^n$ digits such that the digits of every two of them differ in at least $2^{n-1}$ places. Prove that no more than $2^{n+1}$ such numbers exist.

**Problem 9.** Does there exist a natural number $N$ which is a power of 2 whose digits (in the decimal representation) can be permuted to form a different power of 2?

**Problem 10.** For a positive integer $N$, let $s(N)$ be the sum of its digits, in the decimal representation. Prove that there are infinitely many $n$ for which $s(2^n) > s(2^{n+1})$.

**Problem 11.** Find all integers of the form $2^n$ (where $n$ is a positive integer) with the property that, after deleting the first digit of its decimal representation, we again get a power of 2.

**Problem 12.** Let $a_0 = 0$, $a_1 = 1$ and, for $n \geq 2$, $a_n = 2a_{n-1} + a_{n-2}$. Prove that $a_n$ is divisible by $2^k$ if and only if $n$ is divisible by $2^k$.

**Problem 13.** If $A = \{a_1, a_2, \ldots, a_p\}$ is a set of real numbers such that $a_1 > a_2 > \cdots > a_p$, we define

$$s(A) = \sum_{k=1}^{p} (-1)^{k-1} a_k.$$

Let $M$ be a set of $n$ positive integers. Prove that $\sum_{A \subseteq M} s(A)$ is divisible by $2^{n-1}$.

## 3.8 Progressions

We call a sequence $(a_n)_{n \geq 1}$ of real numbers an arithmetical progression (sequence) if there exists a real number $r$ such that

$$a_{n+1} = a_n + r,$$

for every $n \geq 1$. Then $r$ is called the common difference of the sequence and the terms of the progression are uniquely determined by $a_1$ and $r$. It is easy to see that

$$a_n = a_1 + (n-1)r,$$

for every $n \geq 1$ and that

$$S_n = a_1 + a_2 + \cdots + a_n = \frac{a_1 + a_n}{2} \cdot n.$$

Also, notice that for every $n \geq 2$,

$$a_n = \frac{a_{n-1} + a_{n+1}}{2}.$$

A sequence $(b_n)_{n\geq 1}$ of nonzero real numbers is called a geometrical sequence (progression) if there exists $q \neq 0$ such that for every $n \geq 1$,

$$b_{n+1} = b_n \cdot q.$$

Here, $q$ is called the common ratio of the progression.

We have $b_n = b_1 \cdot q^{n-1}$ and

$$b_1 + b_2 + \cdots + b_n = b_1 \frac{q^n - 1}{q - 1},$$

for every $n \geq 1$, if $q \neq 1$.

We will focus on integer progressions.

**Problem 1.** Prove that if an arithmetical progression of positive integers contains a square, then it contains infinitely many squares.

**Solution.** Assume with no loss of generality that the first term is a square, say $a^2$, and let $r$ be the common difference. Hence the progression is:

$$a^2, a^2 + r, a^2 + 2r, \ldots.$$

Eventually we get to

$$a^2 + (2a + r)r = (a + r)^2.$$

By repeating the argument we can find infinitely many squares among the progression's terms.

**Problem 2.** Prove that one can eliminate some terms of an arithmetical progression of positive integers in such a way that the remaining terms form a geometric progression.

**Solution.** Let $a$ be the first term and $r$ the common difference of the arithmetical progression.Thus the progression is given by

$$a, a + r, a + 2r, \ldots.$$

One of the terms is

$$a + ar = a(1 + r).$$

Another one is

$$a + (2a + ar)r = a(1 + r)^2.$$

It is not difficult to see that all numbers of the form $a(1 + r)^n$ , $n \geq 1$ are terms of a geometrical progression.

**Problem 3.** The set of positive integers is partitioned into several arithmetical progressions. Prove that in at least one of them the first term is divisible by the common difference.

**Solution.** Let $a_i, r_i$, $i = 1, 2, \ldots, n$ be the first terms and the common differences of the progressions. Consider the number

$$r = r_1 r_2 \cdots r_n.$$

This number must belong to one of the progressions, thus, for some $i$ and $k$ we have

$$r_1 r_2 \cdots r_n = a_i + k r_i.$$

It follows that $a_i$ is divisible by $r_i$.

Try to solve the following problems.

**Problem 4.** Partition the set of positive integers into two subsets such that none of them contains a nonconstant arithmetical progression.

**Problem 5.** Prove that among the terms of the progression 3,7,11, ... there are infinitely many prime numbers.

**Problem 6.** Does there exist an (infinite) nonconstant arithmetical progression whose terms are all prime numbers?

**Problem 7.** Consider an arithmetical progression of positive integers. Prove that one can find infinitely many terms whose decimal digits add up to the same sum.

**Problem 8.** The set of positive integers is partitioned into $n$ arithmetical progressions, with common differences $r_1, r_2, \ldots, r_n$. Prove that

$$\frac{1}{r_1} + \frac{1}{r_2} + \cdots + \frac{1}{r_n} = 1.$$

**Problem 9.** Prove that for every positive integer $n$, one can find $n$ integers in arithmetical progression, all of them powers of integers, but one cannot find an infinite sequence with this property.

**Problem 10.** Prove that for any integer $n$, $n \geq 3$, there exist $n$ positive integers in arithmetical progression $a_1, a_2, \ldots, a_n$ and $n$ positive integers in geometric progression $b_1, b_2, \ldots, b_n$, such that

$$b_1 < a_1 < b_2 < a_2 < \cdots < b_n < a_n.$$

# Part II

# Solutions

# Chapter 4

# Algebra

## 4.1 An algebraic identity

**Problem 4.** Factor $(a+2b-3c)^3+(b+2c-3a)^3+(c+2a-3b)^3$.

**Solution.** Observe that $(a+2b-3c)+(b+2c-3a)+(c+2a-3b)=0$. Because $x+y+z=0$ implies $x^3+y^3+z^3=3xyz$, we obtain

$$\begin{aligned}&(a+2b-3c)^3+(b+2c-3a)^3+(c+2a-3b)^3\\&=3(a+2b-3c)(b+2c-3a)(c+2a-3b).\end{aligned}$$

**Problem 5.** Let $x, y, z$ be integers such that

$$(x-y)^2+(y-z)^2+(z-x)^2=xyz.$$

Prove that $x^3+y^3+z^3$ is divisible by $x+y+z+6$.

**Solution.** We first notice that at least one of $x, y, z$ is even. Indeed, if $x, y, z$ are all odd, the left-hand side of the given equality is even while the right-hand side is odd, a contradiction. Because

$$x^3+y^3+z^3-3xyz=(x+y+z)(x^2+y^2+z^2-xy-yz-zx),$$

and

$$x^2+y^2+z^2-xy-yz-zx$$

$$=\tfrac{1}{2}\left((x-y)^2+(y-z)^2+(z-x)^2\right)=\frac{xyz}{2},$$

we obtain

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)\frac{xyz}{2},$$

hence

$$x^3 + y^3 + z^3 = \frac{xyz}{2}(x + y + z + 6),$$

and since $\frac{xyz}{2}$ is an integer, the conclusion follows.

**Problem 6.** Let $a, b, c$ be distinct real numbers. Prove that the following equality cannot hold:

$$\sqrt[3]{a-b} + \sqrt[3]{b-c} + \sqrt[3]{c-a} = 0.$$

**Solution.** Suppose $\sqrt[3]{a-b} + \sqrt[3]{b-c} + \sqrt[3]{c-a} = 0$. Then

$$(a-b) + (b-c) + (c-a) = 3\sqrt[3]{(a-b)(b-c)(c-a)}.$$

This implies

$$(a-b)(b-c)(c-a) = 0,$$

which contradicts the hypothesis that $a, b, c$ are distinct.

**Problem 7.** Prove that the number

$$\sqrt[3]{45 + 29\sqrt{2}} + \sqrt[3]{45 - 29\sqrt{2}}$$

is a rational number.

**Solution.** Let $a = \sqrt[3]{45 + 29\sqrt{2}} + \sqrt[3]{45 - 29\sqrt{2}}$. Because

$$a - \sqrt[3]{45 + 29\sqrt{2}} - \sqrt[3]{45 - 29\sqrt{2}} = 0,$$

we have

$$a^3 - (45 + 29\sqrt{2}) - (45 - 29\sqrt{2}) = 3a\sqrt[3]{\left(45 + 29\sqrt{2}\right)\left(45 - 29\sqrt{2}\right)},$$

which is equivalent to

$$a^3 - 21a - 90 = 0$$

or

$$(a-6)(a^2 + 6a + 15) = 0.$$

The equation $a^2 + 6a + 15 = 0$ has no real roots; hence

$$\sqrt[3]{45 + 29\sqrt{2}} + \sqrt[3]{45 - 29\sqrt{2}} = 6,$$

a rational number.

**Problem 8.** Let $a, b, c$ be rational numbers such that

$$a + b\sqrt[3]{2} + c\sqrt[3]{4} = 0.$$

Prove that $a = b = c = 0$.

**Solution.** Notice first that we can assume that $a, b, c$ are integers, otherwise we can multiply the equality by the least common multiple of the denominators. Next, if $a, b, c$ are not equal to 0, we can assume that they are not all even; if they were, we could simplify the equation by dividing both sides by the greatest common power of two.

We have seen that $x + y + z = 0$ implies

$$x^3 + y^3 + z^3 = 3xyz.$$

Taking $x = a$, $y = b\sqrt[3]{2}$ and $z = c\sqrt[3]{4}$ yields

$$a^3 + 2b^3 + 4c^3 = 6abc,$$

thus $a$ must be even, say $a = 2a_1$. Plugging this in the above equality yields

$$8a_1^3 + 2b^3 + 4c^3 = 12a_1bc,$$

or

$$4a_1^3 + b^3 + 2c^3 = 6a_1bc,$$

hence $b$ is also even.

By similar arguments, $c$ is also even; which is a contradiction.

**Problem 9.** Let $r$ be a real number such that

$$\sqrt[3]{r} + \frac{1}{\sqrt[3]{r}} = 3.$$

Determine the value of

$$r^3 + \frac{1}{r^3}.$$

**Solution.** From $\sqrt[3]{r} + \frac{1}{\sqrt[3]{r}} - 3 = 0$, we obtain

$$r + \frac{1}{r} - 27 = 3\sqrt[3]{r}\frac{1}{\sqrt[3]{r}}(-3) = -9.$$

Therefore,

$$r + \frac{1}{r} - 18 = 0.$$

It follows that

$$r^3 + \frac{1}{r^3} - 18^3 = 3r\frac{1}{r}(-18) = -54.$$

Finally,

$$r^3 + \frac{1}{r^3} = 18^3 - 54 = 5778.$$

**Problem 10.** Find the locus of points $(x, y)$ for which

$$x^3 + y^3 + 3xy = 1.$$

**Solution.** Observe that the given equality is equivalent to

$$x^3 + y^3 + (-1)^3 - 3xy(-1) = 0,$$

which can be written as

$$(x + y - 1)(x^2 + y^2 - xy + x + y + 1) = 0.$$

Note that we can write the second factor more conveniently:

$$x^2 + y^2 - xy + x + y + 1 = \tfrac{1}{2}\left[(x - y)^2 + (x + 1)^2 + (y + 1)^2\right];$$

it follows that the locus consists of the line $x + y - 1 = 0$ and the point $(-1, -1)$.

**Problem 11.** Let $n$ be a positive integer. Factor

$$3^{3^n}\left(3^{3^n} + 1\right) + 3^{3^n+1} - 1.$$

**Solution.** Observe that

$$3^{3^n}\left(3^{3^n} + 1\right) + 3^{3^n+1} - 1 = a^3 + b^3 + c^3 - 3abc,$$

where $a = 3^{3^{n-1}}$, $b = 9^{3^{n-1}}$ and $c = -1$. Thus, the number factors as

$$\left(3^{3^{n-1}} + 9^{3^{n-1}} - 1\right)\left(9^{3^{n-1}} + 81^{3^{n-1}} + 1 - 27^{3^{n-1}} + 3^{3^{n-1}} + 9^{3^{n-1}}\right).$$

**Problem 12.** Let $S$ be the set of integers $x$ such that $x = a^3 + b^3 + c^3 - 3abc$, for some integers $a, b, c$. Prove that if $x, y \in S$, then $xy \in S$.

**Solution.** We have seen that

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a + b\omega + c\omega^2)(a + b\omega^2 + c\omega),$$

where $\omega = \frac{-1+i\sqrt{3}}{2}$ is one of the primitive cubic roots of unity. Let the polynomial $P(X) = a + bX + cX^2$. We obtain

$$a^3 + b^3 + c^3 - 3abc = P(1)P(\omega)P(\omega^2).$$

We observe that $x \in S$ if and only if there exists a polynomial $P$ with integer coefficients and degree at most 2 such that

$$x = P(1)P(\omega)P(\omega^2).$$

Now, let $x, y \in S$, and let $P$ and $Q$ be polynomials with integer coefficients, of degree at most 2 such that $x = P(1)P(\omega)P(\omega^2)$ and $y = Q(1)Q(\omega)Q(\omega^2)$. Then $xy = R(1)R(\omega)R(\omega^2)$, where $R(X) = P(X)Q(X)$ is a polynomial of degree at most 4. If we divide $R$ by $X^3 - 1$, we obtain a remainder $R_1(X)$, with integer coefficients and degree at most 2 (this follows from the division algorithm). Thus

$$R(X) = (X^3 - 1)C(X) + R_1(X)$$

for some polynomial $C(X)$.

But then $R(1) = R_1(1)$, $R(\omega) = R_1(\omega)$, and $R(\omega^2) = R_1(\omega^2)$, since $1, \omega$ and $\omega^2$ are the roots of $X^3 - 1$.

It follows that $xy = R_1(1)R_1(\omega)R_1(\omega^2)$, where $R_1$ is a polynomial with integer coefficients, of degree at most 2, hence $xy \in S$.

*Observation.* Another proof can be given if we recall that

$$a^3 + b^3 + c^3 - 3abc = \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}.$$

Let

$$x = \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix} \text{ and } y = \begin{vmatrix} a' & b' & c' \\ c' & a' & b' \\ b' & c' & a' \end{vmatrix},$$

where $a, b, c, a', b', c'$ are integers. Then

$$xy = \begin{vmatrix} aa'+bc'+cb' & ab'+ba'+cc' & ac'+bb'+ca' \\ ac'+bb'+ca' & aa'+bc'+cb' & ab'+ba'+cc' \\ ab'+ba'+cc' & ac'+bb'+ca' & aa'+bc'+cb' \end{vmatrix} \in S.$$

**Problem 13.** Let $a, b, c$ be distinct positive integers and let $k$ be a positive integer such that

$$ab+bc+ca \geq 3k^2-1.$$

Prove that

$$\frac{a^3+b^3+c^3}{3} - abc \geq 3k.$$

**Solution.** The desired inequality is equivalent to

$$a^3+b^3+c^3-3abc \geq 9k.$$

Assume, with no loss of generality, that $a > b > c$. Then $a-b \geq 1, b-c \geq 1$ and $a-c \geq 2$. It follows that

$$\begin{aligned} a^2+b^2+c^2-ab-bc-ac &= \tfrac{1}{2}\left((a-b)^2+(b-c)^2+(a-c)^2\right) \geq \tfrac{1}{2}(1+1+4) = 3. \end{aligned}$$

We obtain

$$\begin{aligned} a^3+b^3+c^3-3abc &= (a+b+c)\left(a^2+b^2+c^2-ab-bc-ac\right) \\ &\geq 3(a+b+c), \end{aligned}$$

so it suffices to prove that

$$3(a+b+c) \geq 9k,$$

or

$$a+b+c \geq 3k.$$

But

$$\begin{aligned} (a+b+c)^2 &= a^2+b^2+c^2+2ab+2bc+2ac \\ &= a^2+b^2+c^2-ab-bc-ac+3(ab+bc+ac) \\ &\geq 3+3(3k^2-1) = 9k^2, \end{aligned}$$

and the conclusion follows.

**Problem 14.** Let $a$, $b$,and $c$ be the side lengths of a triangle. Prove that

$$\sqrt[3]{\frac{a^3+b^3+c^3+3abc}{2}} \geq \max(a,b,c).$$

**Solution.** Assume, with no loss of generality, that $a \geq b \geq c$. We have to prove that

$$\sqrt[3]{\frac{a^3+b^3+c^3+3abc}{2}} \geq a,$$

which is equivalent to

$$-a^3+b^3+c^3+3abc \geq 0.$$

Since

$$-a^3+b^3+c^3+3abc = (-a)^3+b^3+c^3-3(-a)bc,$$

the latter expression factors into

$$\tfrac{1}{2}(-a+b+c)\left((a+b)^2+(a+c)^2+(b-c)^2\right).$$

The conclusion now follows from the triangle inequality: namely, $b+c>a$ in any triangle.

## 4.2 Cauchy–Schwartz revisited

**Problem 3.** Let $x, y, z > 0$. Prove that

$$\frac{2}{x+y}+\frac{2}{y+z}+\frac{2}{z+x} \geq \frac{9}{x+y+z}.$$

**Solution.** Writing the left-hand side as

$$\frac{(\sqrt{2})^2}{x+y}+\frac{(\sqrt{2})^2}{y+z}+\frac{(\sqrt{2})^2}{z+x},$$

and using the lemma at the bottom of page 8, we deduce that

$$\frac{(\sqrt{2})^2}{x+y}+\frac{(\sqrt{2})^2}{y+z}+\frac{(\sqrt{2})^2}{z+x} \geq \frac{(3\sqrt{2})^2}{2(x+y+z)} = \frac{9}{x+y+z}.$$

Equality occurs when $x=y=z$.

**Problem 4.** Let $a, b, x, y, z$ be positive real numbers. Prove that

$$\frac{x}{ay+bz}+\frac{y}{az+bx}+\frac{z}{ax+by}\geq\frac{3}{a+b}.$$

**Solution.** We have

$$\begin{aligned}\frac{x}{ay+bz}+\frac{y}{az+bx}+\frac{z}{ax+by}&=\frac{x^2}{axy+bxz}+\frac{y^2}{ayz+bxy}+\frac{z^2}{axz+byz}\\&\geq\frac{(x+y+z)^2}{(a+b)(xy+xz+yz)}\geq\frac{3}{a+b}.\end{aligned}$$

**Problem 5.** Let $a, b, c > 0$. Prove that

$$\frac{a^2+b^2}{a+b}+\frac{b^2+c^2}{b+c}+\frac{a^2+c^2}{a+c}\geq a+b+c.$$

**Solution.** We write the left-hand side as

$$\frac{a^2}{a+b}+\frac{b^2}{b+c}+\frac{c^2}{a+c}+\frac{b^2}{a+b}+\frac{c^2}{b+c}+\frac{a^2}{a+c}.$$

Applying the same lemma as in Problem 3, we find that this expression is greater than or equal to

$$\frac{(2a+2b+2c)^2}{4(a+b+c)}=a+b+c.$$

**Problem 6.** Let $a, b, c$ be positive numbers such that $abc = 1$. Prove that

$$\frac{1}{a^3(b+c)}+\frac{1}{b^3(a+c)}+\frac{1}{c^3(a+b)}\geq\frac{3}{2}.$$

**Solution.** We see that

$$\begin{aligned}\frac{1}{a^3(b+c)}+\frac{1}{b^3(a+c)}+\frac{1}{c^3(a+b)}&=\frac{\frac{1}{a^2}}{ab+ac}+\frac{\frac{1}{b^2}}{ab+bc}+\frac{\frac{1}{c^2}}{ac+bc}\\&\geq\frac{\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)^2}{2(ab+bc+ac)}.\end{aligned}$$

The latter inequality comes from the lemma used in Problem 3.

Because $abc = 1$, it follows that

$$\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)^2 = \frac{(ab+bc+ca)^2}{(abc)^2} = (ab+bc+ca)^2.$$

Hence

$$\frac{1}{a^3(b+c)}+\frac{1}{b^3(a+c)}+\frac{1}{c^3(a+b)} \geq \frac{(ab+bc+ca)}{2} \geq \frac{3\sqrt[3]{(abc)^2}}{2} = \tfrac{3}{2}.$$

**Problem 7.** Let $x, y, z > 0$. Prove that

$$\frac{x}{x+2y+3z}+\frac{y}{y+2z+3x}+\frac{z}{z+2x+3y} \geq \tfrac{1}{2}.$$

**Solution.** We write the left-hand side as

$$\frac{x^2}{x^2+2xy+3xz}+\frac{y^2}{y^2+2yz+3xy}+\frac{z^2}{z^2+2xz+3yz}$$

and apply the lemma. This yields

$$\frac{x}{x+2y+3z}+\frac{y}{y+2z+3x}+\frac{z}{z+2x+3y} \geq \frac{(x+y+z)^2}{x^2+y^2+z^2+5(xy+xz+yz)}.$$

Now it suffices to prove that

$$\frac{(x+y+z)^2}{x^2+y^2+z^2+5(xy+xz+yz)} \geq \tfrac{1}{2},$$

but this is equivalent to

$$x^2+y^2+z^2 \geq xy+xz+yz,$$

proved in Problem 2.

**Problem 8.** Let $x, y, z > 0$. Prove that

$$\frac{x^2}{(x+y)(x+z)}+\frac{y^2}{(y+z)(y+x)}+\frac{z^2}{(z+x)(z+y)} \geq \tfrac{3}{4}.$$

**Solution.** We apply the lemma again to obtain

$$\frac{x^2}{(x+y)(x+z)} + \frac{y^2}{(y+z)(y+x)} + \frac{z^2}{(z+x)(z+y)} \geq \frac{(x+y+z)^2}{x^2+y^2+z^2+3(xy+xz+yz)}.$$

The inequality

$$\frac{(x+y+z)^2}{x^2+y^2+z^2+3(xy+xz+yz)} \geq \frac{3}{4}$$

is equivalent to

$$x^2+y^2+z^2 \geq xy+xz+yz.$$

**Problem 9.** Let $a, b, c, d, e$ be positive real numbers. Prove that

$$\frac{a}{b+c} + \frac{b}{c+d} + \frac{c}{d+e} + \frac{d}{e+a} + \frac{e}{a+b} \geq \frac{5}{2}.$$

**Solution.** We have

$$\frac{a}{b+c} + \frac{b}{c+d} + \frac{c}{d+e} + \frac{d}{e+a} + \frac{e}{a+b} = \frac{a^2}{ab+ac} + \frac{b^2}{bc+bd} + \frac{c^2}{cd+ce} + \frac{d^2}{de+ad} + \frac{e^2}{ae+be} \geq \frac{(a+b+c+d+e)^2}{\sum ab}.$$

Because

$$(a+b+c+d+e)^2 = \sum a^2 + 2\sum ab,$$

we have to prove that

$$2\sum a^2 + 4\sum ab \geq 5\sum ab,$$

which is equivalent to

$$2\sum a^2 \geq \sum ab.$$

The last inequality follows from $\sum (a-b)^2 \geq 0$.

**Problem 10.** Let $a, b, c$ be positive real numbers such that

$$ab + bc + ca = \tfrac{1}{3}.$$

Prove that

$$\frac{a}{a^2 - bc + 1} + \frac{b}{b^2 - ca + 1} + \frac{c}{c^2 - ab + 1} \geq \frac{1}{a + b + c}.$$

**Solution.** We apply our lemma:

$$\sum \frac{a}{a^2 - bc + 1} = \sum \frac{a^2}{a^3 - abc + a} \geq \frac{(a + b + c)^2}{\sum a^3 + \sum a - 3abc}.$$

Because

$$\sum a^3 - 3abc = (a + b + c)\left(\sum a^2 - \sum ab\right) = (a + b + c)\left(\sum a^2 - \tfrac{1}{3}\right)$$

it follows that

$$\frac{(a + b + c)^2}{\sum a^3 + \sum a - 3abc} = \frac{\sum a^2 + 2\sum ab}{(a + b + c)\left(\sum a^2 - \frac{1}{3} + 1\right)}$$

$$= \frac{\sum a^2 + \frac{2}{3}}{(a + b + c)\left(\sum a^2 + \frac{2}{3}\right)} = \frac{1}{a + b + c},$$

as desired.

## 4.3 Easy ways through absolute values

**Problem 6.** Solve the equation $|x - 3| + |x + 1| = 4$.

**Solution.** Using the inequality $|a| + |b| \geq |a - b|$, we obtain

$$|x - 3| + |x + 1| \geq |x - 3 - (x + 1)| = 4.$$

Equality occurs when $(x - 3)(x + 1) \leq 0$; that is, when $x \in [-1, 3]$.

**Problem 7.** Show that the equation $|2x-3|+|x+1|+|5-x| = 0.99$ has no solutions.

**Solution.** We use the inequality

$$|\pm a_1 \pm a_2 \pm \cdots \pm a_n| \le |a_1| + |a_2| + \cdots + |a_n|,$$

which is valid for all numbers $a_1, a_2, \ldots, a_n$. This yields

$$0.99 = |2x-3|+|x+1|+|5-x| \ge |(2x-3)-(x+1)+(5-x)| = 1,$$

which is impossible. It follows that the given equation has no solutions.

**Problem 8.** Let $a, b > 0$. Find the values of $m$ for which the equation

$$|x-a|+|x-b|+|x+a|+|x+b| = m(a+b)$$

has at least one real solution.

**Solution.** Suppose the equation has at least one real solution $x$. Then

$$m(a+b) = |x-a|+|x+b|+|x-b|+|x+a|$$

$$\ge |(x-a)-(x+b)|+|(x-b)-(x+a)| = 2(a+b),$$

and since $a+b > 0$, it follows that $m \ge 2$. Conversely, if we assume that $m \ge 2$, then the equation has at least one real solution. Indeed, define

$$f(x) = |x-a|+|x-b|+|x+a|+|x+b|.$$

We observe that $f(0) = 2(a+b) \le m(a+b)$, and $f(ma+mb) = 4m(a+b) > m(a+b)$. The existence of a value $x$ such that $f(x) = m(a+b)$ follows from the fact that $f$ is continuous.

**Problem 9.** Find all possible values of the expression

$$E(x,y,z) = \frac{|x+y|}{|x|+|y|} + \frac{|y+z|}{|y|+|z|} + \frac{|z+x|}{|z|+|x|},$$

where $x$, $y$, $z$ are nonzero real numbers.

**Solution.** Two of the three numbers, say $x$ and $y$, must have the same sign; therefore $|x+y| = |x|+|y|$. It follows that

$$E(x,y,z) = 1 + + \frac{|y+z|}{|y|+|z|} + \frac{|z+x|}{|z|+|x|} \ge 1.$$

On the other hand,

$$E(x, y, z) = \frac{|x+y|}{|x|+|y|} + \frac{|y+z|}{|y|+|z|} + \frac{|z+x|}{|z|+|x|} \leq 1+1+1 = 3.$$

We show that the set of values of $E(x, y, z)$ is the interval $[1, 3]$. Indeed, $E(1, 1, -1) = 1$ and $E(1, 1, 1) = 3$, and for each $m \in (1, 3)$, we take $x = 1$, $y = z = \frac{m-3}{m+1}$. A short computation shows that in this case, $E(x, y, z) = m$.

**Problem 10.** Find all positive real numbers $x, x_1, x_2, \ldots, x_n$ such that

$$|\log(xx_1)| + |\log(xx_2)| + \cdots + |\log(xx_n)| + \left|\log\left(\frac{x}{x_1}\right)\right| + \left|\log\left(\frac{x}{x_2}\right)\right| + \cdots + \left|\log\left(\frac{x}{x_n}\right)\right| = |\log x_1 + \log x_2 + \cdots + \log x_n|.$$

**Solution.** Observe that

$$|\log(xx_1)| + \left|\log\left(\frac{x}{x_1}\right)\right| \geq \left|\log(xx_1) - \log\left(\frac{x}{x_1}\right)\right| = \left|\log x_1^2\right| = 2\,|\log x_1|.$$

It follows that the left-hand side of the equality is greater than or equal to

$$2\,(|\log x_1| + |\log x_2| + \cdots + |\log x_n|).$$

However, the right-hand side is less than or equal to

$$|\log x_1| + |\log x_2| + \cdots + |\log x_n|.$$

Thus, equality cannot hold unless

$$|\log x_1| = |\log x_2| = \cdots = |\log x_n| = 0,$$

hence $x_1 = x_2 = \cdots = x_n = 1$. It is not difficult to see that $x = 1$ as well.

**Problem 11.** Prove that for all real numbers $a$, $b$, we have:

$$\frac{|a+b|}{1+|a+b|} \leq \frac{|a|}{1+|a|} + \frac{|b|}{1+|b|}.$$

**Solution.** If $a$ and $b$ have the same sign, then $|a+b| = |a| + |b|$, and we obtain

$$\frac{|a+b|}{1+|a+b|} = \frac{|a|+|b|}{1+|a|+|b|}$$
$$= \frac{|a|}{1+|a|+|b|} + \frac{|b|}{1+|a|+|b|} \leq \frac{|a|}{1+|a|} + \frac{|b|}{1+|b|}.$$

Suppose $a$ and $b$ have opposite signs; assume $|a| \geq |b|$. Then $|a+b| \leq |a|$, and

$$\frac{|a+b|}{1+|a+b|} \leq \frac{|a|}{1+|a|} \leq \frac{|a|}{1+|a|} + \frac{|b|}{1+|b|}.$$

**Problem 12.** Let $n$ be an odd positive integer and let $x_1, x_2, \ldots, x_n$ be distinct real numbers. Find all one-to-one functions

$$f : \{x_1, x_2, \ldots, x_n\} \to \{x_1, x_2, \ldots, x_n\}$$

such that

$$|f(x_1) - x_1| = |f(x_2) - x_2| = \cdots = |f(x_n) - x_n|.$$

**Solution.** Let $f$ be a function with the stated property and let

$$a = |f(x_1) - x_1| = |f(x_2) - x_2| = \cdots = |f(x_n) - x_n|.$$

Then for each $k$, $1 \leq k \leq n$, we have

$$f(x_k) = x_k + \varepsilon_k a,$$

with either $\varepsilon_k = 1$ or $\varepsilon_k = -1$. Adding up these equalities yields

$$\sum_{k=1}^{n} f(x_k) = \sum_{k=1}^{n} x_k + a\sum_{k=1}^{n} \varepsilon_k.$$

Because $f$ is a one-to-one function, $\sum_{k=1}^{n} f(x_k) = \sum_{k=1}^{n} x_k$, and, consequently, $a\sum_{k=1}^{n} \varepsilon_k = 0$. But the sum of an odd number of odd integers cannot be equal to zero, so $\sum_{k=1}^{n} \varepsilon_k \neq 0$. It follows that $a = 0$ and $f(x_k) = x_k$ for all $k$.

**Problem 13.** Suppose the sequence $a_1, a_2, \ldots, a_n$ satisfies the following conditions:

$$a_1 = 0.\ |a_2| = |a_1 + 1|, \ldots, |a_n| = |a_{n-1} + 1|.$$

Prove that

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \geq -\frac{1}{2}.$$

**Solution.** Define $a_{n+1}$ as follows: $a_{n+1} = |a_n + 1|$. The statements $|x| = |y|$ and $x^2 = y^2$ are equivalent; hence

$$\begin{aligned} a_2^2 &= a_1^2 + 2a_1 + 1, \\ a_3^2 &= a_2^2 + 2a_2 + 1, \\ &\vdots \\ a_n^2 &= a_{n-1}^2 + 2a_{n-1} + 1, \\ a_{n+1}^2 &= a_n^2 + 2a_n + 1. \end{aligned}$$

Adding all these equalities gives

$$a_{n+1}^2 = 2(a_1 + a_2 + \cdots + a_n) + n,$$

so that

$$2(a_1 + a_2 + \cdots + a_n) + n \geq 0,$$

and the result follows.

**Problem 14.** Find real numbers $a, b, c$ such that

$$|ax + by + cz| + |bx + cy + az| + |cx + ay + bz| = |x| + |y| + |z|,$$

for all real numbers $x, y, z$.

**Solution.** Plugging $x = y = z = 1$ in the given equality yields

$$|a + b + c| = 1.$$

For $x = 1$ and $y = z = 0$, we obtain

$$|a| + |b| + |c| = 1.$$

Since $|a+b+c| = |a| + |b| + |c|$, we deduce that $a$, $b$ and $c$ have the same sign.

Now, let $x = 1$, $y = -1$ and $c = 0$. It follows that

$$|a-b| + |b-c| + |c-a| = 2.$$

But $|a-b| \leq |a| + |b|$, with equality if and only if $a$ and $b$ have opposite signs or if one of them is zero. Writing the analogous inequalities, we obtain

$$2 = |a-b| + |b-c| + |c-a| \leq 2(|a| + |b| + |c|) = 2.$$

Thus in each pair $(a, b)$, $(b, c)$, $(c, a)$, one of the numbers must be zero, and we conclude that the required triples $(a, b, c)$ are $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$, $(-1, 0, 0)$, $(0, -1, 0)$, $(0, 0, -1)$.

## 4.4 Parameters

**Problem 3.** Solve the equation

$$x = \sqrt{a - \sqrt{a+x}},$$

where $a > 0$ is a parameter.

**Solution.** We must have $x \geq 0$, $a + x \geq 0$ and $a - \sqrt{a+x} \geq 0$. Squaring both sides and rearranging yields

$$x^2 = a - \sqrt{a+x},$$

or

$$\sqrt{a+x} = a - x^2.$$

It follows that $a - x^2 \geq 0$, and thus

$$a + x = (a - x^2)^2,$$

which is equivalent to the fourth-degree equation

$$P(x) = 0,$$

where

$$P(x) = x^4 - 2ax^2 - x + a^2 - a.$$

Let us consider $a$ as the unknown and $x$ as the parameter. Then we obtain a quadratic equation in $a$ as follows:

$$a^2 - (2x^2 + 1)a + x^4 - x = 0.$$

Its discriminant is $\Delta = (2x^2 + 1)^2 - 4(x^4 - x) = (2x + 1)^2$; hence the roots are $a_1 = x^2 - x$ and $a_2 = x^2 + x + 1$. This leads to the factorization of $P$ as

$$P(x) = (x^2 - x - a)(x^2 + x + 1 - a).$$

The positive root of the equation $x^2 - x - a = 0$ does not satisfy the condition $a - x^2 \geq 0$, since $a - x^2 = -x$. The roots of the equation $x^2 + x + 1 - a = 0$ are $x_1 = (-1 + \sqrt{4a - 3})/2$ and $x_2 = (-1 - \sqrt{4a - 3})/2$. Only $x_1$ can be nonnegative, and this happens if and only if $a \geq 1$.

**Problem 4.** Let $a$ be a nonzero real number. Solve the equation

$$a^3x^4 + 2a^2x^2 + x + a + 1 = 0.$$

**Solution.** Multiplying the equation by $a$ and setting $t = ax$ yields

$$t^4 + 2at^2 + t + a^2 + a = 0.$$

Considering $a$ as the variable, we obtain the quadratic

$$a^2 + (2t^2 + 1)a + t^4 + t = 0.$$

This is very similar to the equation in the previous problem. We obtain the equations $t^2 + t + a = 0$ and $t^2 - t + 1 + a = 0$. If $a \leq \frac{1}{4}$, then the first of these two equations has solutions $x = \frac{t}{a} = \frac{1}{2a}\left(-1 \pm \sqrt{(1 - 4a)}\right)$, and if $a \leq -\frac{3}{4}$, then the second equation has solutions $x = \frac{1}{a}\left(\frac{1}{2} \pm \frac{1}{2}\sqrt{(-3 - 4a)}\right)$.

**Problem 5.** Let $a \in \left(0, \frac{1}{4}\right)$. Solve the equation

$$x^2 + 2ax + \tfrac{1}{16} = -a + \sqrt{a^2 + x - \tfrac{1}{16}}.$$

**Solution.** We write the equation as

$$x^2 + 2ax + a + \tfrac{1}{16} = \sqrt{a^2 + x - \tfrac{1}{16}}$$

and square both sides. Rearranging the terms yields

$$x^4 + 4ax^3 + \left(4a^2 + 2a + \frac{1}{8}\right)x^2 + \left(4a^2 + \frac{a}{4} - 1\right)x + \frac{a}{8} + \frac{17}{256} = 0.$$

As before, considering $x$ as a parameter and $a$ as an unknown, we obtain

$$4a^2x(x+1)+\tfrac{1}{8}a(2x+1)\left(16x^2+1\right)+x^4+\tfrac{1}{8}x^2-x+\tfrac{17}{256}=0.$$

The discriminant of the equation equals

$$\left(\frac{16x^2+32x-1}{8}\right)^2,$$

and the procedure is identical to that of the previous problem.

*Observation.* Another approach is possible. Suppose we view $a$ as the fixed parameter and $x$ as the variable. Let us look at the function

$$f:[-a,+\infty)\to[-a^2+\tfrac{1}{16},+\infty),\quad f(x)=x^2+2ax+\tfrac{1}{16}.$$

It is not difficult to see that the inverse of $f$ is given by $f^{-1}(x)=-a+\sqrt{a^2+x-\frac{1}{16}}$; hence the equation

$$f(x)=f^{-1}(x),$$

is equivalent to

$$f(f(x))=x.$$

We claim that if $x_0$ is a solution of this equation, then $f(x_0)=x_0$. Indeed, since $f$ is increasing, if $f(x_0)<x_0$, then $x_0=f(f(x_0))<f(x_0)<x_0$, and if $f(x_0)>x_0$, then $x_0=f(f(x_0))>f(x_0)>x_0$. In both cases we reach a contradiction. The conclusion is that the initial equation is equivalent to

$$x^2+2ax+\tfrac{1}{16}=x,$$

which is a straightforward quadratic equation.

**Problem 6.** Find the positive solutions of the following system of equations:

$$\begin{cases}\frac{a^2}{x^2}-\frac{b^2}{y^2}=8(y^4-x^4)\\ ax-by=x^4-y^4\end{cases},$$

where $a,b>0$ are parameters.

**Solution.** We solve the system for $a, b$ instead of $x, y$. Multiplying the first equation by $x^4$ yields

$$a^2x^2 = \frac{b^2x^4}{y^2} + 8x^4(y^4 - x^4),$$

and from the second equation we deduce

$$a^2x^2 = (by + x^4 - y^4)^2 = b^2y^2 + 2by(x^4 - y^4) + (x^4 - y^4)^2.$$

It follows that

$$\frac{b^2(x^4 - y^4)}{y^2} - 2by(x^4 - y^4) - (x^4 - y^4)(9x^4 - y^4) = 0.$$

If $x \neq y$, we can divide by $x^4 - y^4$ to obtain a quadratic equation in $b$ :

$$b^2 - 2by^3 - y^2(9x^4 - y^4) = 0.$$

The solutions are $y^3 + 3x^2y$ and $y^3 - 3x^2y$. If $b = y^3 - 3x^2y$, the second equation gives $a = x^3 - 3xy^2$. Because $a, b > 0$, it follows that $x^2 > 3y^2$ and $y^2 > 3x^2 > 9y^2$; this is a contradiction. Thus $b = y^3 + 3x^2y$ and $a = x^3 + 3xy^2$. We observe that $a + b = (x + y)^3$ and $a - b = (x - y)^3$; hence $x + y = \sqrt[3]{a + b}$ and $x - y = \sqrt[3]{a - b}$, yielding

$$(x, y) = \left( \frac{\sqrt[3]{a+b} + \sqrt[3]{a-b}}{2}, \frac{\sqrt[3]{a+b} - \sqrt[3]{a-b}}{2} \right).$$

**Problem 7.** Let $a, b, c > 0$. Solve the system of equations

$$\begin{cases} ax - by + \frac{1}{xy} = c \\ bz - cx + \frac{1}{zx} = a \\ cy - az + \frac{1}{yz} = b \end{cases}.$$

**Solution.** Considering $a, b, c$ as unknowns, we have a linear system of equations. Multiply the first equation successively by $x$ and $y$ and substitute $cx$ in the second equation and $cy$ in the last one. We obtain

$$\begin{cases} a(x^2 + 1) - b(xy + z) = \frac{1}{xz} - \frac{1}{y} & (i) \\ a(xy - z) - b(y^2 + 1) = -\frac{1}{x} - \frac{1}{yz} & (ii) \end{cases}.$$

Now multiply equation $(i)$ by $y^2+1$, and equation $(ii)$ by $-(xy+z)$, and add them up. It follows that

$$a(x^2+y^2+z^2+1)=\frac{x^2+y^2+z^2+1}{xz},$$

yielding $a=\frac{1}{xz}$. Analogously, we obtain $b=\frac{1}{yz}$ and $c=\frac{1}{xy}$. Hence $abc=\frac{1}{(xyz)^2}$, so $xyz=\pm\frac{1}{\sqrt{abc}}$. The solutions of the system are

$$\left(\frac{b}{\sqrt{abc}},\frac{a}{\sqrt{abc}},\frac{c}{\sqrt{abc}}\right),\left(\frac{-b}{\sqrt{abc}},\frac{-a}{\sqrt{abc}},\frac{-c}{\sqrt{abc}}\right).$$

**Problem 8.** Solve the equation

$$x+a^3=\sqrt[3]{a-x},$$

where $a$ is a real parameter.

**Solution.** Write the equation in the equivalent form

$$\sqrt[3]{\sqrt[3]{a-x}-x}=a.$$

If we consider $a$ as a variable and define the function $f(a)=\sqrt[3]{a-x}$, we see that the equation can be written as

$$f(f(a))=a,$$

and since $f$ is an increasing function, this is equivalent to

$$f(a)=a,$$

(see the observation in the solution of Problem 5). It is not difficult now to obtain the solution $x=a-a^3$.

## 4.5 Take the conjugate!

**Problem 5.** Let $a$ and $b$ be distinct positive numbers and let $A=\frac{a+b}{2}$, $B=\sqrt{ab}$. Prove the inequality

$$B<\frac{(a-b)^2}{8(A-B)}<A.$$

**Solution.** Observe that

$$\frac{(a-b)^2}{8\,(A-B)} = \frac{(a-b)^2\,(A+B)}{8\left(A^2-B^2\right)}$$

and

$$A^2 - B^2 = \frac{a^2+b^2+2ab}{4} - ab = \frac{a^2+b^2-2ab}{4} = \frac{(a-b)^2}{4}.$$

It follows that

$$\frac{(a-b)^2}{8\,(A-B)} = \frac{A+B}{2},$$

and since $A > B$, we have

$$B < \frac{A+B}{2} < A,$$

as desired.

**Problem 6.** Let $m, n$ be positive integers with $m < n$. Find a closed form for the sum

$$\frac{1}{\sqrt{m}+\sqrt{m+1}} + \frac{1}{\sqrt{m+1}+\sqrt{m+2}} + \cdots + \frac{1}{\sqrt{n-1}+\sqrt{n}}.$$

**Solution.** By taking the conjugate for each term of the sum, we obtain

$$\frac{\sqrt{m+1}-\sqrt{m}}{m+1-m} + \frac{\sqrt{m+2}-\sqrt{m+1}}{m+2-m-1} + \cdots + \frac{\sqrt{n}-\sqrt{n-1}}{n-n+1} =$$

$$\sqrt{m+1}-\sqrt{m}+\sqrt{m+2}-\sqrt{m+1}+\cdots+\sqrt{n}-\sqrt{n-1} = \sqrt{n}-\sqrt{m}.$$

**Problem 7.** For any positive integer $n$, let

$$f(n) = \frac{4n+\sqrt{4n^2-1}}{\sqrt{2n+1}+\sqrt{2n-1}}.$$

Evaluate the sum $f(1) + f(2) + \cdots + f(40)$.

**Solution.** Let $x = \sqrt{2n+1}$ and $y = \sqrt{2n-1}$. Then $x^2 + y^2 = 4n$, $xy = \sqrt{4n^2-1}$ and $x^2 - y^2 = 2$. Hence

$$f(n) = \frac{x^2+y^2+xy}{x+y} = \frac{x^3-y^3}{x^2-y^2} = \frac{1}{2}\left(\sqrt{(2n+1)^3} - \sqrt{(2n-1)^3}\right).$$

Adding up the equalities for $n = 1, 2, \ldots, 40$ yields

$$f(1) + f(2) + \cdots + f(40) = \tfrac{1}{2}\left(\sqrt{81^3} - \sqrt{1^3}\right) = 364.$$

**Problem 8.** Let $a$ and $b$ be distinct real numbers. Solve the equation

$$\sqrt{x - b^2} - \sqrt{x - a^2} = a - b.$$

**Solution.** We must have $x \geq a^2$ and $x \geq b^2$. Squaring both sides leads to rather complicated computations. We take the conjugate instead, which gives

$$\frac{a^2 - b^2}{\sqrt{x - b^2} + \sqrt{x - a^2}} = a - b$$

and we obtain the equivalent form

$$\sqrt{x - b^2} + \sqrt{x - a^2} = a + b.$$

Adding this to the original equation yields

$$\sqrt{x - b^2} = a,$$

and the solution is

$$x = \sqrt{a^2 + b^2}.$$

The conditions $x \geq a^2$ and $x \geq b^2$ are clearly satisfied.

**Problem 9.** Solve the following equation, where $m$ is a real parameter:

$$\sqrt{x + \sqrt{x}} - \sqrt{x - \sqrt{x}} = m\sqrt{\frac{x}{x + \sqrt{x}}}.$$

**Solution.** We first notice that $x$ must be a positive number. By taking the conjugate on the left-hand side, we obtain

$$\frac{2\sqrt{x}}{\sqrt{x + \sqrt{x}} + \sqrt{x - \sqrt{x}}} = \frac{m\sqrt{x}}{\sqrt{x + \sqrt{x}}},$$

so $m$ must be a positive number, as well. Simplifying by $\sqrt{x}$ yields the equivalent equation

$$(2 - m)\sqrt{x + \sqrt{x}} = m\sqrt{x - \sqrt{x}}.$$

It follows that $2 - m \geq 0$, and then

$$(2-m)^2(x+\sqrt{x}) = m^2(x-\sqrt{x})$$

or

$$(2-m)^2(\sqrt{x}+1) = m^2(\sqrt{x}-1).$$

We obtain

$$\sqrt{x} = \frac{m^2-2m+2}{2(m-1)};$$

therefore $m > 1$ and the solution of the equation is

$$x = \frac{(m^2-2m+2)^2}{4(m-1)^2},$$

for $1 < m \leq 2$.

**Problem 10.** Prove that for every positive integer $k$, there exists a positive integer $n_k$ such that $\left(\sqrt{3}-\sqrt{2}\right)^k = \sqrt{n_k} - \sqrt{n_k - 1}$.

**Solution.** If we take the conjugate in the above equality, we obtain

$$\frac{1}{(\sqrt{3}+\sqrt{2})^k} = \frac{1}{\sqrt{n_k}+\sqrt{n_k-1}},$$

and hence

$$(\sqrt{3}+\sqrt{2})^k = \sqrt{n_k}+\sqrt{n_k-1}.$$

We deduce that

$$\sqrt{n_k} = \tfrac{1}{2}\left(\left(\sqrt{3}-\sqrt{2}\right)^k + \left(\sqrt{3}+\sqrt{2}\right)^k\right)$$

and thus

$$n_k = \frac{1}{4}\left(\left(5-2\sqrt{6}\right)^k + \left(5+2\sqrt{6}\right)^k + 2\right).$$

All we have to do now is to prove that the number in the right-hand side is an integer.

Setting $x_k = \left(5-2\sqrt{6}\right)^k + \left(5+2\sqrt{6}\right)^k$, we observe that the sequence $(x_k)_{k\geq 1}$ satisfies the recursive relation:

$$x_{k+2} = 10x_{k+1} - x_k$$

for all $k \geq 0$ (more generally, if $x_k = a^k + b^k$, then $x_{k+2} = (a+b)x_{k+1} - abx_k$, for all $k$). Since $x_0 = 2$ and $x_1 = 10$, it follows that $x_k$ is an integer for all $k \geq 0$. Moreover, using the recursive relation, we can prove inductively that $x_k \equiv 2 \pmod 4$ for all $k$. Indeed, this holds for $k = 0$ and $k = 1$. Assuming that $x_k \equiv 2 \pmod 4$ and $x_{k+1} \equiv 2 \pmod 4$, it follows that $x_{k+2} \equiv 10 \cdot 2 - 2 \equiv 2 \pmod 4$.

We conclude that $n_k = \frac{1}{4}(x_k + 2)$ is an integer as claimed.

**Problem 11.** Let $a$ and $b$ be nonzero integers with $|a| \leq 100$, $|b| \leq 100$. Prove that

$$\left|a\sqrt{2} + b\sqrt{3}\right| \geq \frac{1}{350}.$$

**Solution.** We take the conjugate:

$$|a\sqrt{2} + b\sqrt{3}| = \frac{|2a^2 - 3b^2|}{|a\sqrt{2} - b\sqrt{3}|},$$

and observe that $2a^2 - 3b^2 \neq 0$; if it were, $\sqrt{\frac{3}{2}}$ would be a rational number, a contradiction. Because $2a^2 - 3b^2$ is an integer, it follows that $2a^2 - 3b^2 \geq 1$. On the other hand, we have

$$|a\sqrt{2} - b\sqrt{3}| \leq |a|\sqrt{2} + |b|\sqrt{3} \leq 100(\sqrt{2} + \sqrt{3}) < 350.$$

It follows that

$$|a\sqrt{2} + b\sqrt{3}| = \frac{|2a^2 - 3b^2|}{|a\sqrt{2} - b\sqrt{3}|} \geq \frac{1}{350}.$$

**Problem 12.** Let $n$ be a positive integer. Prove that

$$\left\lfloor \left(\frac{1+\sqrt{5}}{2}\right)^{4n-2} \right\rfloor - 1$$

is a perfect square.

**Solution.** Let

$$x_n = \left(\frac{1+\sqrt{5}}{2}\right)^{4n-2} + \left(\frac{1-\sqrt{5}}{2}\right)^{4n-2}.$$

Observe that

$$x_n = \frac{4}{6+2\sqrt{5}}\left(\frac{7+3\sqrt{5}}{2}\right)^n + \frac{4}{6-2\sqrt{5}}\left(\frac{7-3\sqrt{5}}{2}\right)^n,$$

and hence $x_n$ satisfies the recursive relation

$$x_{n+1} = 7x_n - x_{n-1},$$

with $x_0 = x_1 = 3$. It follows that $x_n$ is a positive integer for all $n \geq 0$. Because

$$0 < \left(\frac{1-\sqrt{5}}{2}\right)^{4n-2} < 1,$$

we deduce that

$$\left\lfloor \left(\frac{1+\sqrt{5}}{2}\right)^{4n-2} \right\rfloor - 1 = x_n - 2.$$

But

$$x_n - 2 = \left(\left(\frac{1+\sqrt{5}}{2}\right)^{2n-1} + \left(\frac{1-\sqrt{5}}{2}\right)^{2n-1}\right)^2$$

and a similar argument shows that

$$\left(\frac{1+\sqrt{5}}{2}\right)^{2n-1} + \left(\frac{1-\sqrt{5}}{2}\right)^{2n-1}$$

is a positive integer for all $n \geq 0$. The claim is proved.

## 4.6 Inequalities with convex functions

**Problem 4.** Let $a, b > 0$ and let $n$ be a positive integer. Prove the inequality:

$$\frac{a^n + b^n}{2} \geq \left(\frac{a+b}{2}\right)^n.$$

**Solution.** We can try induction on $n$. The assertion holds for $n = 1$ and, assuming it holds for $n$, multiply both sides by $\frac{a+b}{2}$. It follows that

$$\frac{(a^n + b^n)(a+b)}{4} \geq \left(\frac{a+b}{2}\right)^{n+1},$$

and therefore it is sufficient to prove that

$$\frac{a^{n+1} + b^{n+1}}{2} \geq \frac{(a^n + b^n)(a+b)}{4}.$$

A short computation shows that this is equivalent to

$$(a-b)\left(a^n - b^n\right) \geq 0,$$

which is true since $a - b$ and $a^n - b^n$ have the same sign.

A shorter solution uses convex functions. Consider $n \geq 2$ and let $f : (0, +\infty) \to (0, +\infty)$; $f(x) = x^n$. Because $f''(x) = n(n-1)x^{n-2} > 0$, $f$ is convex; thus for all $a, b > 0$ we have

$$\frac{f(a) + f(b)}{2} \geq f\left(\frac{a+b}{2}\right).$$

This is exactly the inequality we had to prove.

**Problem 5.** Prove that

$$\sqrt[3]{3 - \sqrt[3]{3}} + \sqrt[3]{3 + \sqrt[3]{3}} < 2\sqrt[3]{3}.$$

**Solution.** Consider the function $f(x) = \sqrt[3]{x}$. It is not difficult to see that $f$ is concave on the interval $(0, +\infty)$. Then, for any two distinct numbers $a, b > 0$ we have

$$\frac{f(a) + f(b)}{2} < f\left(\frac{a+b}{2}\right).$$

If we take $a = 3 - \sqrt[3]{3}$ and $b = 3 + \sqrt[3]{3}$, we obtain the desired result.

**Problem 6.** Prove the $AM - GM$ inequality

$$\frac{x_1 + x_2 + \cdots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \ldots x_n},$$

for all $x_1, x_2, \ldots, x_n > 0$.

**Solution.** There are many proofs of the $AM - GM$ inequality. One of them uses convexity. Consider the function $f : (0, +\infty) \to (0, +\infty)$, $f(x) = \log x$. This is a concave function, so we have the inequality

$$\frac{f(x_1) + f(x_2) + \cdots + f(x_n)}{n} \leq f\left(\frac{x_1 + x_2 + \cdots + x_n}{n}\right).$$

for all $x_1, x_2, \ldots, x_n > 0$. Using the properties of the logarithmic function, we obtain

$$\frac{\log x_1 + \log x_2 + \cdots + \log x_n}{n} = \frac{\log (x_1 x_2 \ldots x_n)}{n}$$
$$= \log (x_1 x_2 \ldots x_n)^{\frac{1}{n}} = \log \sqrt[n]{x_1 x_2 \ldots x_n},$$

and hence

$$\log \sqrt[n]{x_1 x_2 \ldots x_n} \leq \log \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

Because $f$ is increasing, this is equivalent to the $AM - GM$ inequality.

**Problem 7.** Let $a_1 < a_2 < \cdots < a_{2n+1}$ be positive real numbers. Prove the inequality

$$\sqrt[n]{a_1 - a_2 + a_3 - \cdots - a_{2n} + a_{2n+1}} \geq \sqrt[n]{a_1} - \sqrt[n]{a_2} + \cdots + \sqrt[n]{a_{2n+1}}.$$

**Solution.** We will prove inductively a stronger statement. Let $m \geq 2$ and $n \geq 1$ be integers. Then the following inequality holds for any positive real numbers $a_1 < a_2 < \cdots < a_{2n+1}$:

$$\sqrt[m]{a_1 - a_2 + a_3 - \cdots - a_{2n} + a_{2n+1}} \geq \sqrt[m]{a_1} - \sqrt[m]{a_2} + \cdots + \sqrt[m]{a_{2n+1}}.$$

Now, we fix $m$ and prove the assertion by induction on $n$. For $n = 1$ we have to prove that

$$\sqrt[m]{a_1 - a_2 + a_3} \geq \sqrt[m]{a_1} - \sqrt[m]{a_2} + \sqrt[m]{a_3},$$

for all $0 < a_1 < a_2 < a_3$. This follows from Problem 1, for the concave function $f : (0, +\infty) \to (0, +\infty)$, $f(x) = \sqrt[m]{x}$. Suppose the inequality

holds for $n$ and let $0 < a_1 < a_2 < \cdots < a_{2n+1} < a_{2n+2} < a_{2n+3}$. Then, from the induction hypothesis, we have

$$\sqrt[m]{a_1 - a_2 + \cdots + a_{2n+1} - a_{2n+2} + a_{2n+3}}$$

$$\geq \sqrt[m]{a_1} - \sqrt[m]{a_2} + \cdots + \sqrt[m]{a_{2n+1} - a_{2n+2} + a_{2n+3}},$$

since $a_1 < a_2 < \cdots < a_{2n} < a_{2n+1} - a_{2n+2} + a_{2n+3}$. But

$$\sqrt[m]{a_{2n+1} - a_{2n+2} + a_{2n+3}} \geq \sqrt[m]{a_{2n+1}} - \sqrt[m]{a_{2n+2}} + \sqrt[m]{a_{2n+3}},$$

and the result follows.

**Problem 8.** Let $x, y, z > 0$. Prove that

$$\frac{x}{2x+y+z} + \frac{y}{x+2y+z} + \frac{z}{x+y+2z} \leq \frac{3}{4}.$$

**Solution.** Let $s = x + y + z$. The inequality becomes

$$\frac{x}{s+x} + \frac{y}{s+y} + \frac{z}{s+z} \leq \frac{3}{4}.$$

Consider the function $f : (0, +\infty) \to (0, +\infty)$ defined by

$$f(t) = \frac{t}{s+t}.$$

We see that

$$f''(x) = -\frac{2s}{(s+t)^3} < 0,$$

for all $t > 0$, so $f$ is concave. It follows that

$$\frac{f(x) + f(y) + f(z)}{3} \leq f\left(\frac{x+y+z}{3}\right).$$

But

$$f\left(\frac{x+y+z}{3}\right) = f\left(\frac{s}{3}\right) = \frac{\frac{s}{3}}{s+\frac{s}{3}} = \frac{1}{4},$$

and thus

$$f(x) + f(y) + f(z) \leq \frac{3}{4},$$

as desired.

**Problem 9.** Prove that if $a, b, c, d > 0$ and $a \leq 1, a+b \leq 5, a+b+c \leq 14, a+b+c+d \leq 30$, then

$$\sqrt{a}+\sqrt{b}+\sqrt{c}+\sqrt{d} \leq 10.$$

**Solution.** The function $f : (0, +\infty) \to (0, +\infty)$, defined by $f(x) = \sqrt{x}$, is concave, and therefore, for any positive real numbers $\lambda_1, \lambda_2, \ldots, \lambda_n$ such that $\lambda_1 + \lambda_2 + \cdots + \lambda_n = 1$, we have

$$\lambda_1 f(x_1) + \lambda_2 f(x_2) + \cdots + \lambda_n f(x_n) \leq f(\lambda_1 x_1 + \lambda_2 x_2 + \cdots + \lambda_n x_n).$$

Now, take $n = 4$ and $\lambda_1 = \frac{1}{10}, \lambda_2 = \frac{2}{10}, \lambda_3 = \frac{3}{10}, \lambda_4 = \frac{4}{10}$. It follows that

$$\frac{1}{10}\sqrt{a} + \frac{2}{10}\sqrt{\frac{b}{4}} + \frac{3}{10}\sqrt{\frac{c}{9}} + \frac{4}{10}\sqrt{\frac{d}{16}} \leq \sqrt{\frac{a}{10} + \frac{b}{20} + \frac{c}{30} + \frac{d}{40}},$$

so

$$\sqrt{a}+\sqrt{b}+\sqrt{c}+\sqrt{d} \leq 10\sqrt{\frac{12a+6b+4c+3d}{120}}.$$

But

$$\begin{aligned} 12a+6b+4c+3d &= 3(a+b+c+d)+(a+b+c)+2(a+b)+6a \\ &\leq 3 \cdot 30 + 14 + 2 \cdot 5 + 6 \cdot 1 = 120, \end{aligned}$$

and the claim is proved.

## 4.7 Induction at work

**Problem 4.** Let $n$ be a positive integer. Prove the inequality

$$\left(1+\frac{1}{2}\right)\left(1+\frac{1}{2^2}\right)\left(1+\frac{1}{2^3}\right)\ldots\left(1+\frac{1}{2^n}\right) < \frac{5}{2}.$$

**Solution.** As in Problem 2, we will prove a stronger statement:

$$\left(1+\frac{1}{2}\right)\left(1+\frac{1}{2^2}\right)\left(1+\frac{1}{2^3}\right)\ldots\left(1+\frac{1}{2^n}\right) \leq \frac{5}{2}\left(1-\frac{1}{2^n}\right)$$

for all $n \geq 2$. For $n = 2$, both sides of the above inequality equal $\frac{15}{8}$. Suppose the inequality holds for $n > 2$ and multiply both sides by $1 + \frac{1}{2^{n+1}}$. We have to prove that

$$\tfrac{5}{2}\left(1 - \frac{1}{2^n}\right)\left(1 + \frac{1}{2^{n+1}}\right) \leq \tfrac{5}{2}\left(1 - \frac{1}{2^{n+1}}\right),$$

and a short computation shows that the latter is equivalent to the obvious $\frac{1}{2^{2n+1}} > 0$.

**Problem 5.** Let $n$ be a positive integer. Prove that the number

$$a_n = 2^{2^n} - 1$$

has at least $n$ distinct prime divisors.

**Solution.** Note that $a_1 = 3$ has one prime divisor and suppose that for some $n > 1$, $a_n$ has at least $n$ distinct prime divisors $p_1, p_2, \ldots, p_n$. Now,

$$a_{n+1} = 2^{2^{n+1}} - 1 = \left(2^{2^n}\right)^2 - 1 = a_n\left(2^{2^n} + 1\right),$$

and hence $p_1, p_2, \ldots, p_n$ are distinct prime divisors of $a_{n+1}$ as well. The numbers $2^{2^n} + 1$ and $2^{2^n} - 1$ are coprime since any of their common divisors greater than 1 must be odd and must divide $2^{2^n} + 1 - \left(2^{2^n} - 1\right) = 2$, which is impossible. Therefore any prime divisor of $2^{2^n} + 1$ is different from $p_1, p_2, \ldots, p_n$. It follows that $a_{n+1}$ has at least $n + 1$ distinct prime divisors.

**Problem 6.** Let $a$ and $n$ be positive integers such that $a$ is less than or equal to $n!$. Prove that $a$ can be represented as a sum of at most $n$ distinct divisors of $n!$.

**Solution.** The assertion is trivial for $n = 1$, so suppose it holds true for some $n > 1$. We have $a \leq (n + 1)!$. By the division algorithm, there exist $b, c$ such that $a = b(n + 1) + c$, with $0 \leq c \leq n$. Clearly, $b \leq n!$; and thus, using the induction hypothesis, $b$ can be represented as a sum of at most $n$ distinct divisors of $n!$, say

$$b = d_1 + d_2 + \cdots + d_k,$$

with $k \leq n$. But then

$$a = d_1(n + 1) + d_2(n + 1) + \cdots + d_k(n + 1) + c,$$

and the numbers $d_1(n+1), d_2(n+1), \ldots, d_k(n+1)$ and $c$ are distinct divisors of $(n + 1)!$.

**Problem 7.** Let $x_1, x_2, \ldots, x_m, y_1, y_2, \ldots, y_n$ be positive integers such that the sums $x_1 + x_2 + \cdots + x_m$ and $y_1 + y_2 + \cdots + y_n$ are equal and less than $mn$. Prove that in the equality

$$x_1 + x_2 + \cdots + x_m = y_1 + y_2 + \cdots + y_n$$

one can cancel some terms and obtain another equality.

**Solution.** Let $k = m + n$. We proceed by induction on $k$. Let $s = x_1 + x_2 + \cdots + x_m = y_1 + y_2 + \cdots + y_n$. Since $s \geq m$, $s \geq n$ and $s < mn$, it follows that $m, n \geq 2$, and hence $k \geq 4$. The case $k = 4$ is easy to discard, so let us suppose that $k > 4$. With no loss of generality, we may assume that $x_1 \geq x_2 \geq \cdots \geq x_m$ and $y_1 \geq y_2 \geq \cdots \geq y_n$. If $x_1 = y_1$, these terms cancel and we are done. Suppose, for instance, that $x_1 > y_1$. We want to apply the induction hypothesis to the equality

$$(x_1 - y_1) + x_2 + \cdots + x_m = y_2 + y_3 + \cdots + y_n$$

in which there are $m + (n - 1) = k - 1$ terms. For this, we have to check that the sum $s' = y_2 + y_3 + \cdots + y_n$ verifies the condition $s' < m(n - 1)$. Because $y_1 \geq y_2 \geq \cdots \geq y_n$, it follows that $y_1 \geq \frac{s}{n}$,so that

$$s' = s - y_1 \leq s - \frac{s}{n} = s\frac{(n-1)}{n} < mn\frac{(n-1)}{n} = m(n-1),$$

as needed.

**Problem 8.** The sequence $(x_n)_{n\geq 1}$ is defined by $x_1 = 1$, $x_{2n} = 1 + x_n$ and $x_{2n+1} = \frac{1}{x_{2n}}$ for all $n \geq 1$. Prove that for any positive rational number $r$ there exists a unique $n$ such that $r = x_n$.

**Solution.** Note that all the terms of the sequence are positive numbers and that $x_{2n} > 1$, $x_{2n+1} < 1$ for all $n \geq 1$. We prove by induction on $k \geq 2$ the following statement: for all positive integers $a, b$ such that $\gcd(a, b) = 1$ and $a + b \leq k$, there exists a term of the sequence equal to $\frac{a}{b}$. If $k = 2$, then $a = b = 1$ and $\frac{a}{b} = 1 = x_1$. Suppose that the statement is true for some $k > 2$, and let $a, b$ be coprime positive integers such that $a + b = k + 1$. If $a > b$, then we apply the induction hypothesis to the numbers $a - b$ and $b$. Clearly, $\gcd(a - b, b) = 1$ and $(a - b) + b = a \leq k$; therefore, there exists $n$ such that $x_n = \frac{a-b}{b}$. But then

$$x_{2n} = 1 + x_n = 1 + \frac{a-b}{b} = \frac{a}{b}.$$

If $a < b$, then we apply the induction hypothesis to the numbers $b - a$ and $a$. There exists $n$ such that $x_n = \frac{b-a}{a}$, and we obtain

$$x_{2n+1} = \frac{1}{x_{2n}} = \frac{1}{1 + x_n} = \frac{1}{1 + \frac{b-a}{a}} = \frac{a}{b}.$$

We proved that for any positive rational number $r$ there exists $n$ such that $r = x_n$, but we still have to prove that $n$ is unique. This follows from the fact that the terms of the sequence $(x_n)_{n \geq 1}$ are pairwise distinct. Indeed, suppose not, and choose $n \neq m$ such that $x_n = x_m$, with minimal $n$. The observation at the beginning of the solution shows that $n$ and $m$ have the same parity. If $n = 2n'$ and $m = 2m'$, then we obtain $x_{n'} = x_{m'}$, contradicting the minimality of $n$. The case $n = 2n' + 1$ and $m = 2m' + 1$ yields $x_{2n'} = x_{2m'}$, and we again reach a contradiction.

## 4.8 Roots and coefficients

**Problem 4.** Let $a, b, c$ be nonzero real numbers such that $a + b + c \neq 0$ and

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{a + b + c}.$$

Prove that for all odd integers $n$

$$\frac{1}{a^n} + \frac{1}{b^n} + \frac{1}{c^n} = \frac{1}{a^n + b^n + c^n}.$$

**Solution.** Let $f(X) = X^3 + mX^2 + nX + p$ be the monic polynomial with roots $a, b, c$. The given equality can be written as

$$(a + b + c)(ab + bc + ca) = abc,$$

so, by Viète's relations, we obtain $p = mn$. Thus

$$\begin{aligned} f(X) &= X^3 + mX^2 + nX + mn \\ &= X^2(X + m) + n(X + m) \\ &= (X + m)(X^2 + n). \end{aligned}$$

It follows that one of the roots, say $a$, equals $-m = a+b+c$; hence $b+c = 0$. Substituting $c = -b$, the desired equality becomes

$$\frac{1}{a^n} + \frac{1}{b^n} + \frac{1}{(-b)^n} = \frac{1}{a^n + b^n + (-b)^n},$$

which is clearly true if $n$ is an odd integer.

*Observation.* We could also ask the following question: if $m$ and $n$ are odd integers, prove that

$$\frac{1}{a^m}+\frac{1}{b^m}+\frac{1}{c^m}=\frac{1}{a^m+b^m+c^m}$$

if and only if

$$\frac{1}{a^n}+\frac{1}{b^n}+\frac{1}{c^n}=\frac{1}{a^n+b^n+c^n}.$$

**Problem 5.** Let $a \le b \le c$ be real numbers such that

$$a+b+c=2$$

and

$$ab+bc+ca=1.$$

Prove that

$$0 \le a \le \tfrac{1}{3} \le b \le 1 \le c \le \frac{4}{3}.$$

**Solution.** Let $f(X)=X^3+mX^2+nX+p$ be the monic polynomial with roots $a, b, c$. Viète's relations yield $m=-2$ and $n=1$, hence

$$f(X)=X^3-2X^2+X+p.$$

The derivative of $f$ is

$$f'(X)=3X^2-4X+1,$$

with roots $\frac{1}{3}$ and 1. Using the sign of $f'$, we deduce that $f$ increases on the interval $(-\infty, \frac{1}{3}]$ from $-\infty$ to $f(\frac{1}{3})=p+\frac{4}{27}$, and then decreases on the interval $[\frac{1}{3}, 1]$ from $p+\frac{4}{27}$ to $f(1)=p$. Finally, $f$ increases again on the interval $[1, +\infty)$ from $p$ to $+\infty$. It follows that $f$ has three real roots if and only if $p+\frac{4}{27} \ge 0$ and $p \le 0$.

Observe that $f(0)=p$ and $f(\frac{4}{3})=p+\frac{4}{27}$, and hence $f$ changes its sign on each of the intervals $[0, \frac{1}{3}]$, $[\frac{1}{3}, 1]$ and $[1, \frac{4}{3}]$. We deduce that each of these intervals contains a root of $f$. The claim follows.

**Problem 6.** Prove that two of the four roots of the polynomial $X^4+12X-5$ add up to 2.

**Solution.** Let $x_1, x_2, x_3$ and $x_4$ be the roots of the polynomial. Writing Viète's relations in terms of $s = x_1 + x_2$, $s' = x_3 + x_4$, $p = x_1x_2$ and $p' = x_3x_4$, we obtain

$$\begin{aligned} s + s' &= 0, \\ p + p' + ss' &= 0, \\ ps' + p's &= -12, \\ pp' &= -5. \end{aligned}$$

Substituting $s' = -s$ into the second and third equalities yields

$$\begin{aligned} p + p' &= s^2, \\ -p + p' &= -\frac{12}{s}, \end{aligned}$$

and we obtain

$$p = \tfrac{1}{2}\left(s^2 + \frac{12}{s}\right), \; p' = \tfrac{1}{2}\left(s^2 - \frac{12}{s}\right).$$

From $pp' = -5$ it follows that

$$\tfrac{1}{4}\left(s^2 + \frac{12}{s}\right)\left(s^2 - \frac{12}{s}\right) = -5,$$

or

$$s^6 + 20s^2 - 144 = 0.$$

Hence $s = x_1 + x_2$ is a root of the polynomial

$$P(X) = X^6 + 20X^2 - 144.$$

Using similar arguments, we deduce that its other five roots are $x_1 + x_3$, $x_1 + x_4$, $x_2 + x_3$, $x_2 + x_4$ and $x_3 + x_4$. It suffices now to check that 2 is a root of $P$. Indeed,

$$P(2) = 64 + 20 \cdot 4 - 144 = 0.$$

**Problem 7.** Find $m$ and solve the following equation, knowing that its roots form a geometrical sequence

$$X^4 - 15X^3 + 70X^2 - 120X + m = 0.$$

**Solution.** Let $x_1, x_2, x_3, x_4$ be the roots of the polynomial and suppose they form a geometrical sequence in the order $x_1, x_3, x_4, x_2$. We again write Viète's relations in terms of $s = x_1 + x_2$, $s' = x_3 + x_4$, $p = x_1x_2$ and $p' = x_3x_4$. This gives

$$s + s' = 15,$$
$$p + p' + ss' = 70,$$
$$ps' + p's = 120,$$
$$pp' = m.$$

Then $x_1x_2 = x_3x_4$, and hence $p = p'$. It follows that $120 = ps' + p's = p(s + s') = 15p$ and thus $p = 8$. Then $m = 64$. and $ss' = 70 - 16 = 54$. The numbers $s$ and $s'$ are the roots of the equation $X^2 - 15X + 54 = 0$, hence $s = 6$ and $s' = 9$ or vice versa. It follows that the roots of the polynomial are 1, 2, 4 and 8.

**Problem 8.** Let $x_1, x_2, \ldots, x_n$ be the roots of the polynomial $X^n + X^{n-1} + \cdots + X + 1$. Prove that

$$\frac{1}{1 - x_1} + \frac{1}{1 - x_2} + \cdots + \frac{1}{1 - x_n} = \frac{n}{2}.$$

**Solution.** We look for the polynomial with roots

$$y_k = \frac{1}{1 - x_k}, k = 1, 2, \ldots, n.$$

From the above equality, it follows that

$$x_k = \frac{y_k - 1}{y_k},$$

and since $x_k$ is a root of $X^n + X^{n-1} + \cdots + X + 1$, we obtain

$$\left(\frac{y_k - 1}{y_k}\right)^n + \left(\frac{y_k - 1}{y_k}\right)^{n-1} + \cdots + \frac{y_k - 1}{y_k} + 1 = 0.$$

This is equivalent to

$$(y_k - 1)^n + y_k (y_k - 1)^{n-1} + \cdots + y_k^{n-1} (y_k - 1) + y_k^n = 0.$$

It follows that $y_k$ is a root of the polynomial

$$P(X) = (X - 1)^n + X (X - 1)^{n-1} + \cdots + X^{n-1} (X - 1) + X^n.$$

The desired sum equals

$$y_1 + y_2 + \cdots + y_n,$$

and its value can be determined by using Viète's relations. Observe that

$$P(X) = (n+1)X^n - X^{n-1}\left(\binom{n}{1} + \binom{n-1}{1} + \cdots + \binom{1}{1}\right) + \cdots.$$

and hence

$$y_1 + y_2 + \cdots + y_n = \frac{\binom{n}{1} + \binom{n-1}{1} + \cdots + \binom{1}{1}}{n+1} = \frac{n(n+1)}{2(n+1)} = \frac{n}{2},$$

as claimed.

**Problem 9.** Let $a, b, c$ be rational numbers and let $x_1, x_2, x_3$ be the roots of the polynomial $P(X) = X^3 + aX^2 + bX + c$. Prove that if $\frac{x_1}{x_2}$ is a rational number, different from 0 and $-1$, then $x_1, x_2, x_3$ are rational numbers.

**Solution.** Let $\frac{x_1}{x_2} = m \in \mathbf{Q}, m \neq 0, -1$. We have

$$x_1 + x_2 + x_3 = -a.$$

We claim that if one of the roots is a rational number, then so are all three of them. Indeed, if $x_1 \in \mathbf{Q}$, then $x_2 = \frac{x_1}{m} \in \mathbf{Q}$ and $x_3 = -a - x_1 - x_2 \in \mathbf{Q}$. If $x_2 \in \mathbf{Q}$, then $x_1 = mx_2 \in \mathbf{Q}$ and $x_3 \in \mathbf{Q}$. If $x_3 \in \mathbf{Q}$, then $x_2 = \frac{-a-x_3}{1+m} \in \mathbf{Q}$ and $x_1 = mx_2 \in \mathbf{Q}$. It suffices now to prove that $P$ has a rational root.

Substituting $x_1 = mx_2$ and $x_3 = -a - mx_2 - x_2$ in the equality

$$x_1x_2 + x_1x_3 + x_2x_3 = b$$

yields

$$\left(m^2 + m + 1\right)x_2^2 + a\,(m+1)\,x_2 + b = 0,$$

and consequently $x_2$ is a root of a second-degree polynomial $f$ with rational coefficients. Since $P$ and $f$ share a common root, their greatest common divisor is a nonconstant polynomial with rational coefficients. It follows that $P$ can be decomposed into factors with rational coefficients; since one of the factors must be of degree 1, it has at least one rational root.

**Problem 10.** Solve in real numbers the system

$$\begin{cases} x+y+z=0 \\ x^3+y^3+z^3=18 \\ x^7+y^7+z^7=2058 \end{cases}.$$

**Solution.** Consider the polynomial

$$P(t)=t^3+at^2+bt+c,$$

with roots $x, y, z$.

Since $x+y+z=0$, it follows that $a=0$, hence

$$P(t)=t^3+bt+c.$$

Because $x, y, z$ are the roots of $P$, we have

$$x^3+bx+c=0,$$
$$y^3+by+c=0,$$
$$z^3+bz+c=0.$$

Adding these equalities and using the fact that $x^3+y^3+z^3=18$, we obtain $c=-6$. Therefore

$$P(t)=t^3+bt-6.$$

Now, use the last equation of the system to find $b$. Multiply the previous equalities by $x^n$, $y^n$ , $z^n$ respectively and add them up to obtain

$$x^{n+3}+y^{n+3}+z^{n+3}+b\left(x^{n+1}+y^{n+1}+z^{n+1}\right)-6\left(x^n+y^n+z^n\right)=0.$$

Denoting $S_n=x^n+y^n+z^n$ for all $n\geq 1$, this equality becomes

$$S_{n+3}+bS_{n+1}-6S_n=0 \qquad (*)$$

for all positive integers $n$. We have $S_7=2058$. On the other hand, using (*), we obtain

$$S_7=-bS_5+6S_4=-b(-bS_3+6S_2)+6(-bS_2+6S_1)$$
$$=b^2S_3-12bS_2+36S_1.$$

Since $S_3 = 18$, $S_2 = (x+y+z)^2 - 2(xy+xz+yz) = -2b$ and $S_1 = 0$, it follows that

$$S_7 = 42b^2,$$

so $b = \pm 7$.

The equation $t^3 + 7t - 6 = 0$ has only one real root (observe that the function $f(t) = t^3+7t-6$ is strictly increasing). The equation $t^3-7t-6 = 0$ has roots $t_1 = -1$, $t_2 = -2$ and $t_3 = 3$, and thus the solutions of the system are $(-1, -2, 3)$ and all of its permutations.

**Problem 11.** Solve in real numbers the system of equations

$$\begin{cases} a+b=8 \\ ab+c+d=23 \\ ad+bc=28 \\ cd=12 \end{cases}$$

**Solution.** The expressions on the left-hand side remind us of the coefficients obtained when two polynomials are multiplied. Indeed, observe that

$$(x^2+ax+c)(x^2+bx+d) = x^4+(a+b)x^3+(ab+c+d)x^2+(ad+bc)x+cd.$$

We obtain the polynomial

$$P(x) = x^4+8x^3+23x^2+28x+12,$$

which, fortunately, has integer roots. We can find them by checking the divisors of 12. We obtain

$$P(x) = (x+1)(x+2)^2(x+3).$$

The polynomial factors in two ways as a product of quadratic polynomials:

$$P(x) = (x^2+4x+3)(x^2+4x+4)$$

and

$$P(x) = (x^2+3x+2)(x^2+5x+6).$$

Hence the solutions of the system are (4, 4, 3, 4), (4, 4, 4, 3), (3, 5, 2, 6), (5, 3, 6, 2).

# Chapter 5

# Geometry and Trigonometry

## 5.1 Geometric inequalities

**Problem 5.** Let $ABCD$ be a convex quadrilateral. Prove that

$$max(AB + CD, AD + BC) < AC + BD < AB + BC + CD + DA.$$

**Solution.** Let $O$ be the point of intersection of the diagonals $AC$ and $BD$. We have $AO + OB > AB$ and $CO + OD > CD$; thus $AC + BD > AB + CD$. Similarly, $AO + OD > AD$ and $BO + OC > BC$; thus $AC + BD > AD + BC$. It follows that

$$max(AB + CD, AD + BC) < AC + BD.$$

For the second inequality, note that $AC < AB + BC$ and $AC < AD + DC$; hence

$$AC < \tfrac{1}{2}(AB + BC + CD + DA).$$

Analogously,

$$BD < \tfrac{1}{2}(AB + BC + CD + DA),$$

and the result follows by adding these inequalities.

**Problem 6.** Let $M$ be the midpoint of the segment $AB$. Prove that if $O$ is an arbitrary point, then

$$|OA - OB| \leq 2OM.$$

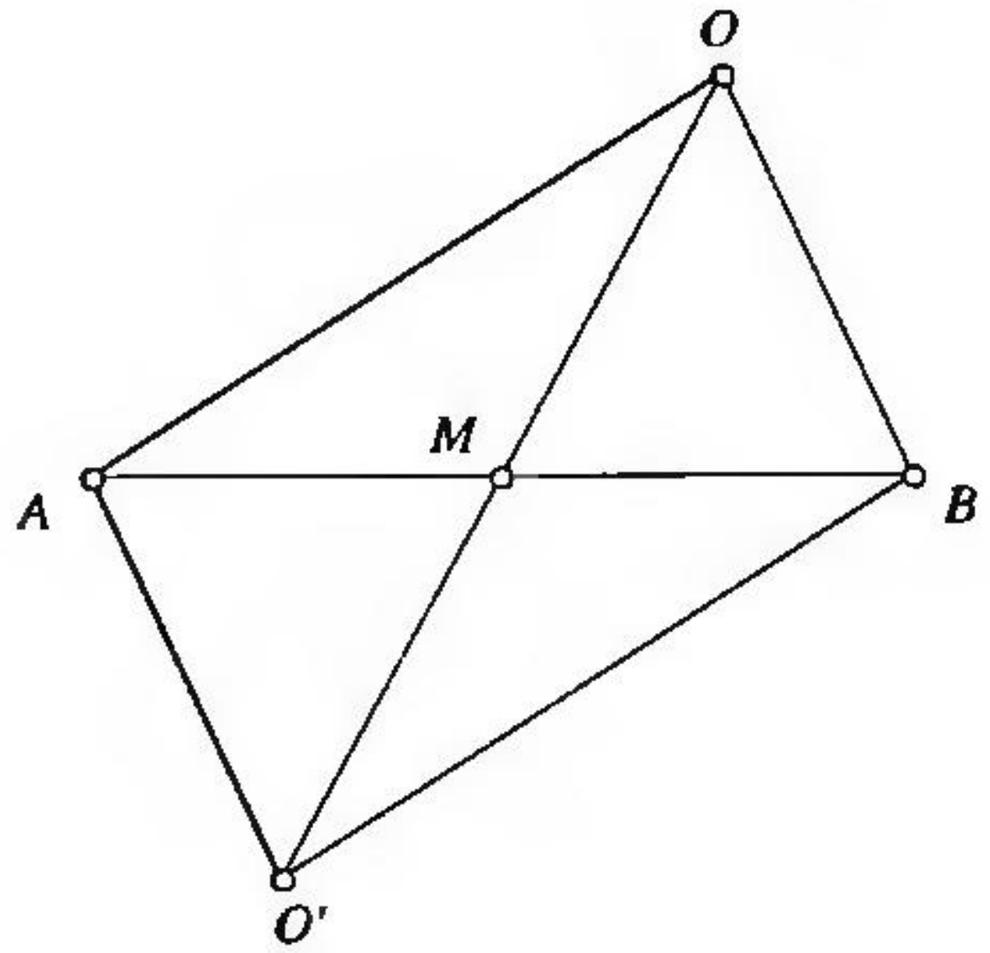


Figure 5.1

**Solution.** Suppose that $O$ does not lie on the line $AB$ and let the point be $O'$ the reflection of $O$ across $M$. (Figure 5.1.)

Because the segments $AB$ and $OO'$ have the same midpoint, the quadrilateral $AOBO'$ is a parallelogram, so $OA = BO'$. In the triangle $OBO'$, we have $OO' + OB > BO'$ and $OO' + BO' > OB$; hence $2OM > OA - OB$ and $2OM > OB - OA$.

We conclude that

$$2OM > |OA - OB|.$$

If $O$ lies on the line $AB$, we introduce coordinates on the line $AB$ such that $M$ is the origin and the points $A$ and $B$ have coordinates $-1$ and 1, respectively. If $x$ is the coordinate of the point $O$, we have to prove that

$$2|x| \geq ||x + 1| - |x - 1||.$$

Squaring both sides and rearranging terms, we find that the inequality is equivalent to $|x^2 - 1| \geq 1 - x^2$, which is obvious. We also deduce that the equality holds if and only if $1 - x^2 \geq 0$; that is, if and only if $O$ lies on the segment $AB$.

**Problem 7.** Prove that in an arbitrary triangle, the sum of the lengths of the altitudes is less than the triangle's perimeter.

**Solution.** It is not difficult to see that in a right triangle, the length of a leg is less than the length of the hypotenuse; this is an immediate consequence of the Pythagorean theorem, but it can also be proven by using simple inequalities.

Suppose $ABC$ is a right triangle, with $\angle A = 90°$. We prove, for instance, that $BC > AB$. Let $B'$ be the reflection of $B$ across $A$. Then $\angle B'AC = 90°$ and triangles $ABC$ and $AB'C$ are congruent. We have $BC + B'C > BB'$, hence $2BC > 2AB$. Returning to our problem, if $AA'$, $BB'$ and $CC'$ are the altitudes of triangle $ABC$, then $AA' < AB$, $BB' < BC$, and $CC' < CA$. We add these inequalities to obtain the desired conclusion.

**Problem 8.** Denote by $P$ the perimeter of triangle $ABC$. If $M$ is a point in the interior of the triangle, prove that

$$\tfrac{1}{2}P < MA + MB + MC < P.$$

**Solution.** In the triangle $MAB$, we have $MA + MB > AB$. Analogously, $MA + MC > AC$ and $MB + MC > BC$. The first inequality is obtained by adding these three inequalities. For the second one, we have seen in Problem 3 that $MB + MC < AB + AC$. Writing the similar inequalities and adding them up gives the desired result.

**Problem 9.** Prove that if $A'$, $B'$ and $C'$ are the midpoints of the sides $BC$, $CA$ and $AB$, respectively, then (Figure 5.2)

$$\tfrac{3}{4}P < AA' + BB' + CC' < P.$$

**Solution.** It is known that the medians of a triangle are concurrent at a point called the centroid of the triangle. Denote this point by $G$. Then

$$AA' + BB' + CC' > GA + GB + GC > \tfrac{1}{2}P,$$

which follows from the preceding problem by taking $M = G$, but this is not enough.

In order to obtain the required inequality, we have to use another property of the centroid: the point $G$ divides each median in the ratio 1:2, that is, $GA = \frac{2}{3}AA'$, etc. Then, in triangle $GBC$, we have $GB + GC > BC$, or $\frac{2}{3}(BB' + CC') > BC$. If we add this with the other two similar inequalities, we get

$$\tfrac{4}{3}(AA' + BB' + CC') > P,$$

and hence the desired result. For the second inequality, just use Problem 2.

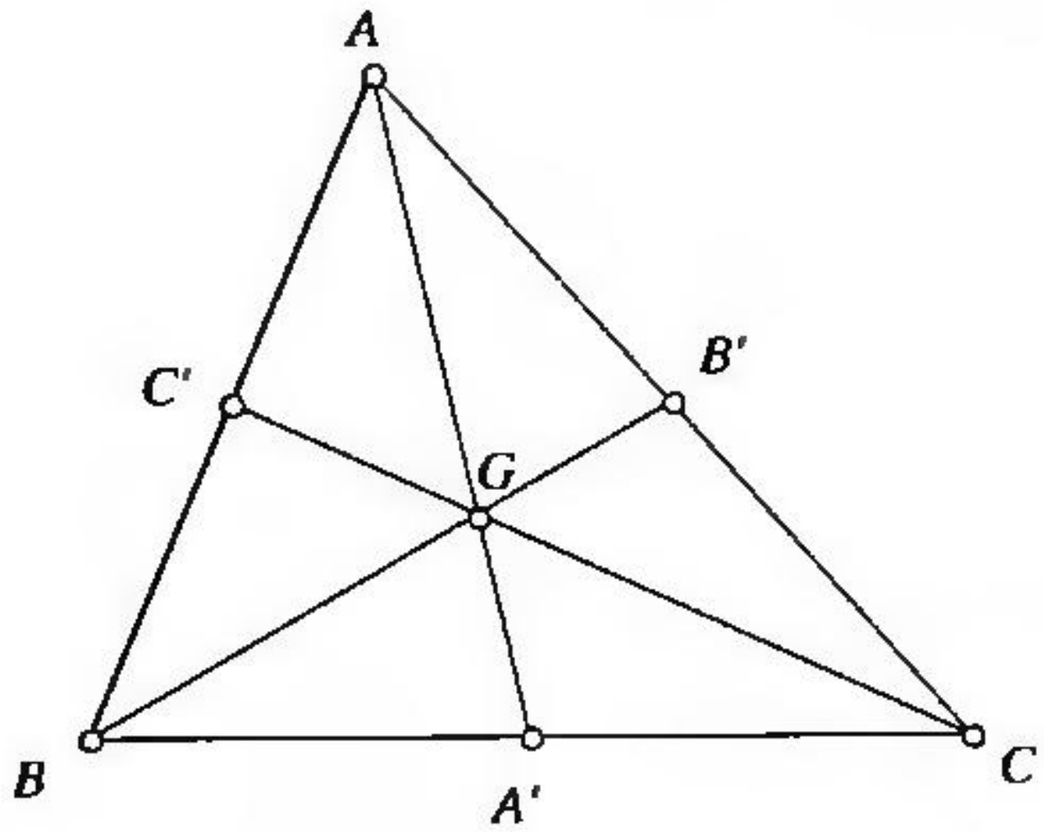


Figure 5.2

**Problem 10.** In the convex quadrilateral $ABCD$, we have

$$AB + BD \leq AC + CD.$$

Prove that $AB < AC$.

**Solution.** Let $O$ be the point of intersection of the diagonals $AC$ and $BD$. In triangles $AOB$ and $COD$, we have $AB < AO + OB$, $CD < CO + OD$, and hence $AB + CD < AC + BD$. Adding this inequality with $AB + BD \leq AC + CD$, we get $AB < AC$.

**Problem 11.** Consider $n$ red and $n$ blue points in the plane, no three of them being collinear. Prove that one can connect each red point to a blue one with a segment such that no two segments intersect.

**Solution.** There is a finite number of ways in which the $n$ red points can be connected with the $n$ blue ones. We choose the connection with the property that the sum of lengths of all connecting segments is minimal. We claim that in this case every two segments are disjoint. Indeed, suppose the points $A$ and $D$ are red, $B$ and $C$ are blue and the segments $AB$ and $CD$ intersect at $O$. Then we can replace the segments $AB$ and $CD$ by $AC$ and $BD$ (see Figure 5.3). Clearly, $AC + BD < AB + CD$, and this contradicts the assumption that the sum of lengths of all connecting segments is minimal.

**Problem 12.** Let $n$ be an odd positive integer. On some field, $n$ gunmen are placed such that all pairwise distances between them are different. At a signal, every gunman takes out his gun and shoots the closest gunman. Prove that:

(a) at least one gunman survives;

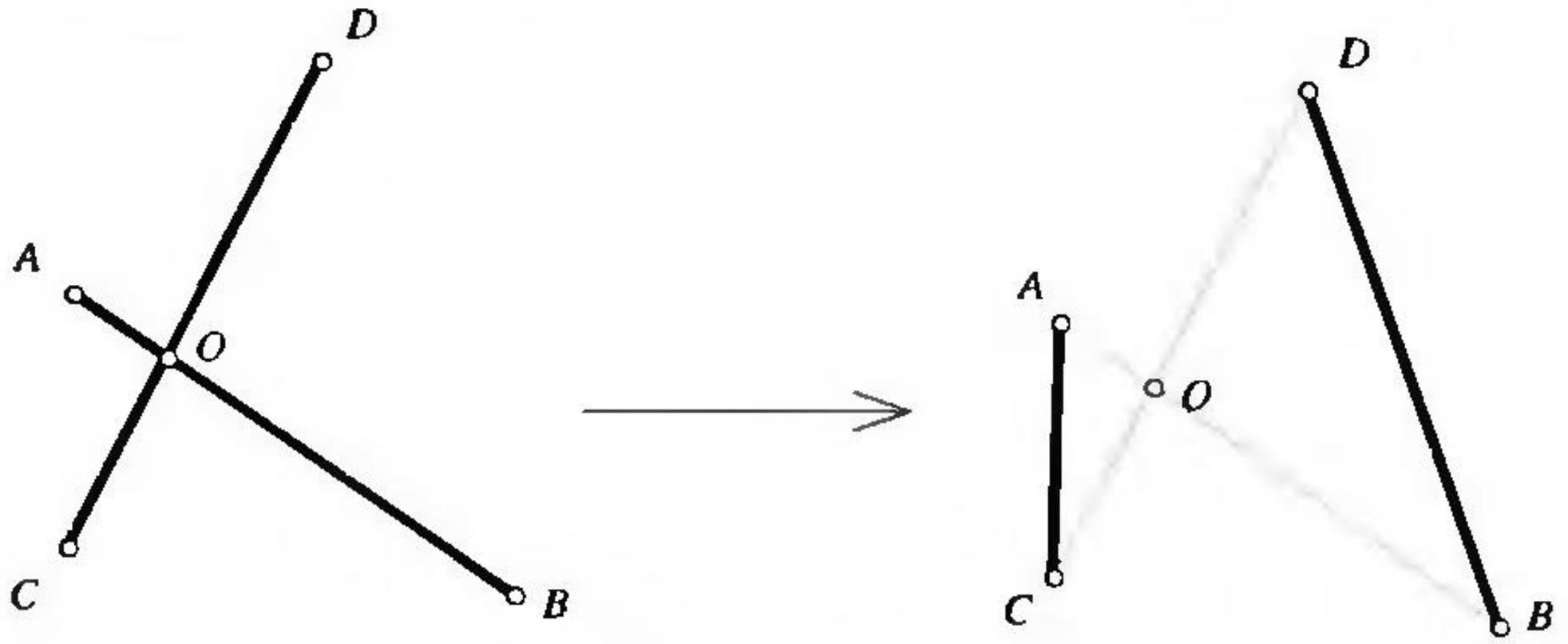


Figure 5.3

(b) no gunman is shot more than five times;

(c) the trajectories of the bullets do not intersect.

**Solution.** (a) Consider the gunmen closest to each other. They will shoot each other. If anybody else shoots at either of these two, a gunman will certainly survive (there are $n$ bullets shot, so if someone ends up with more than one bullet in his body, someone else survives). If not, discard these two and repeat the reasoning for the $n-2$ remaining gunmen. Because $n$ is odd, we eventually reach the case $n=3$, where the conclusion is obvious.

(b) Suppose a gunman $A$ is shot by at least six other gunmen, denoted $A_1$, $\ldots$, $A_6$. One of the six angles centered at $A$ is at most $60^\circ$. Suppose $\angle A_1AA_2$ is such an angle (see Figure 5.4). In triangle $A_1AA_2$, one of the other two angles, say $A_2$, is greater than $60^\circ$. Then $AA_1 > A_1A_2$, but since $A_1$ shot $A$ and not $A_2$, $AA_1 < A_1A_2$, which is a contradiction.

(c) Suppose $A$ shoots $B$, $C$ shoots $D$ and $AB \cap CD = O$. Then $AB < AD$ and $CD < BC$. Thus, in the convex quadrilateral $ACBD$, $AB + CD < AD + BC$, a contradiction.

**Problem 13.** Prove that the medians of a given triangle can form a triangle.

**Solution.** Let $A'$, $B'$ and $C'$ be the midpoints of the sides $BC$, $CA$, and $AB$ of the triangle $ABC$. We have $\overrightarrow{AA'} = \frac{1}{2}(\overrightarrow{AB} + \overrightarrow{AC})$. Adding this with the other two similar equalities, we obtain $\overrightarrow{AA'} + \overrightarrow{BB'} + \overrightarrow{CC'} = 0$. Since the three vectors are noncollinear, this shows that the segments $AA'$, $BB'$ and $CC'$ can form a triangle.

*Observation.* We can actually construct this triangle. Consider the point $M$ such that $B'$ is the midpoint of the segment $A'M$ (see Figure 5.5). Then

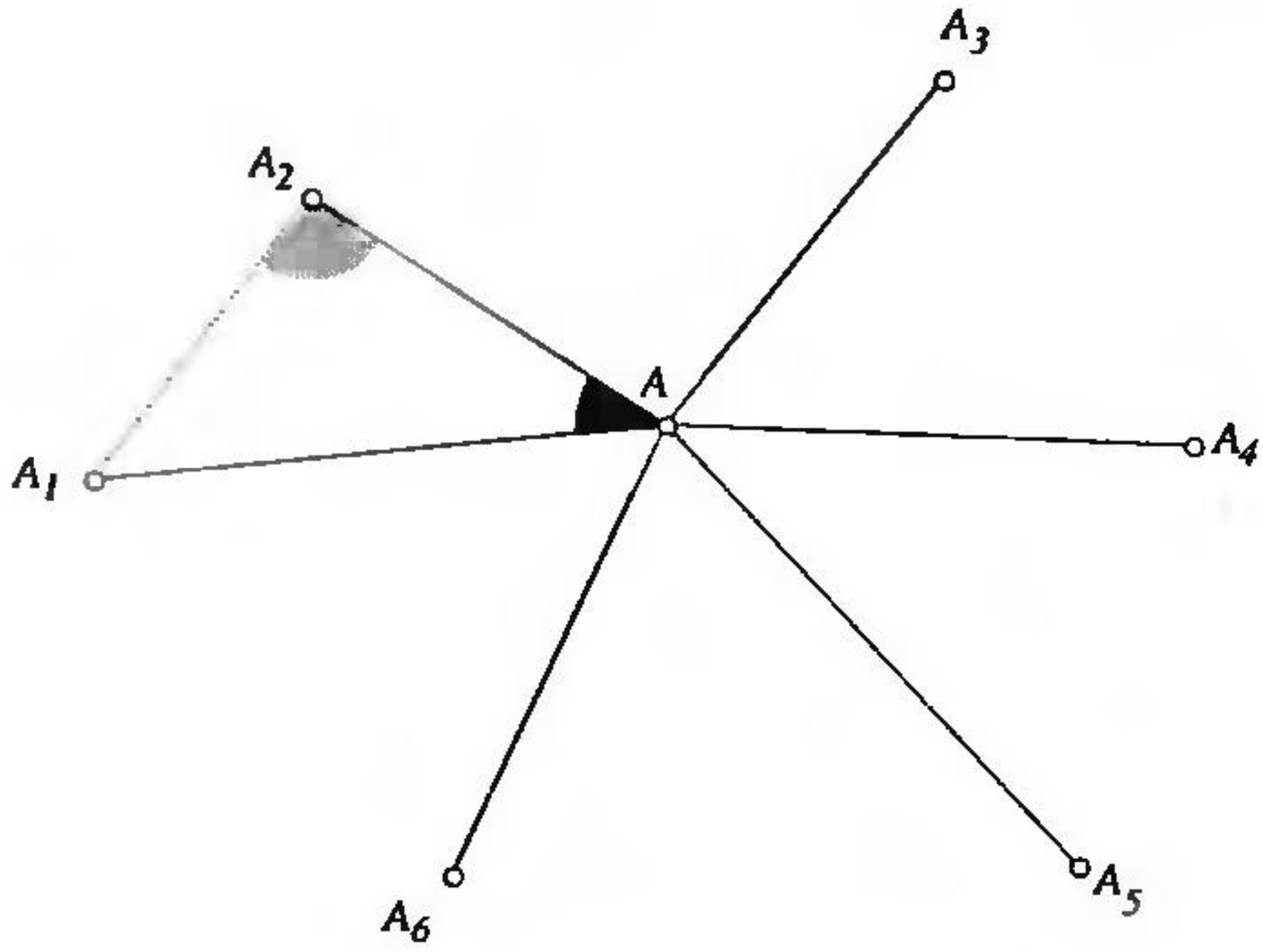


Figure 5.4

the quadrilateral $AA'CM$ is a parallelogram.

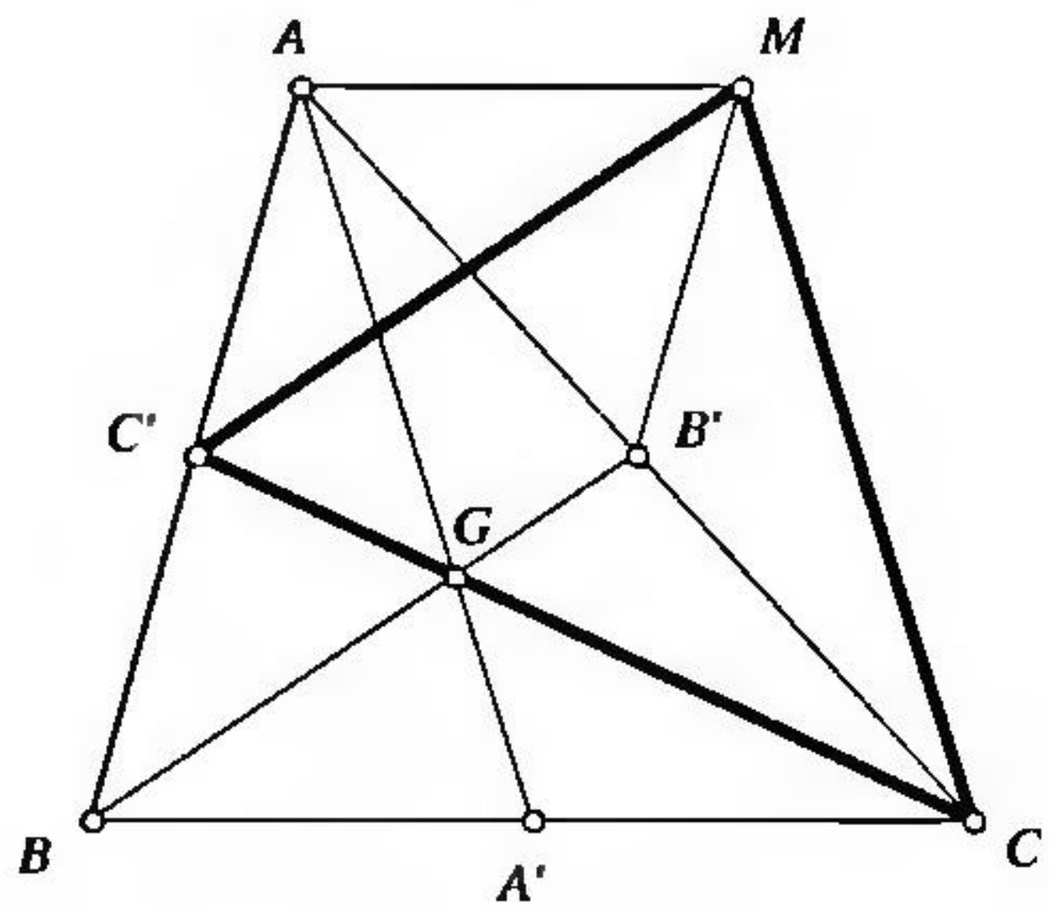


Figure 5.5

It is not difficult to see that in this case $BC'MB'$ is also a parallelogram, and the side lengths of triangle $CC'M$ are equal to the lengths of the medians $AA'$, $BB'$, and $CC'$.

**Problem 14.** Let $A$ and $B$ be two points situated on the same side of a line $XY$. Find the position of a point $M$ on the line such that the sum $AM + MB$ is minimal.

**Solution.** Let $B'$ be the reflection of $B$ across the line $XY$. We claim that $M$ is the point of intersection of $AB'$ and $XY$.

Indeed, let $M'$ be another point on $XY$ (see Figure 5.6). By symmetry

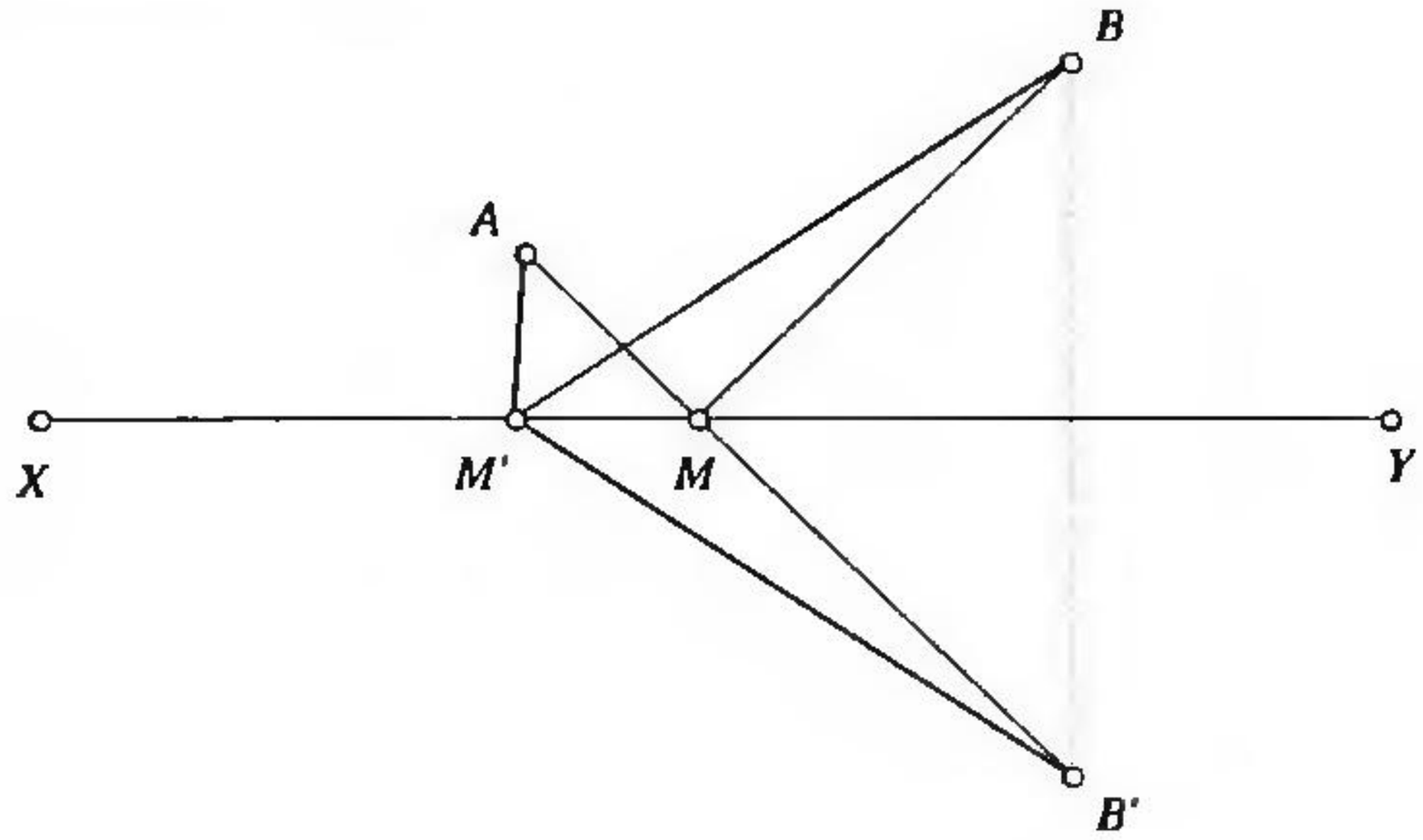


Figure 5.6

across $XY$, $MB' = MB$ and $M'B' = M'B$. Then

$$AM' + M'B = AM' + M'B' > AB' = AM + MB' = AM + MB.$$

**Problem 15.** Let $ABC$ be an acute triangle. Find the positions of the points $M$, $N$, $P$ on the sides $BC$, $CA$, $AB$, respectively, such that the perimeter of the triangle $MNP$ is minimal.

**Solution.** Let us fix $M$ on the side $BC$ and look for the positions of $P$ and $Q$ such that the perimeter of triangle $MNP$ is minimal. Reflect $M$ across $AB$ and $AC$ to obtain $M'$ and $M''$, respectively.

If $P'$ and $Q'$ are points on the sides $AB$ and $AC$, we have

$$MP' + P'Q' + Q'M = M'P' + P'Q' + Q'M''.$$

This sum is minimal when the points $M'$, $P'$, $Q'$ and $M''$ are collinear, so $P$ and $Q$ are the points of intersection between $M'M''$ and the sides $AB$ and $AC$ (see Figure 5.7). In this case, the perimeter of $MPQ$ equals $M'M''$.

Now, the problem can be rephrased in the following way: find the point $M$ on the side $BC$ such that the length of $M'M''$ is minimal.

Observe that

$$\angle M'AB = \angle BAM, \text{ and } \angle M''AC = \angle CAM,$$

therefore $\angle M'AM'' = 2\angle BAC$. Moreover, $M'A = M''A = MA$. It follows that regardless of the position of $M$ on $BC$, triangle $M'AM''$ is isosceles, with fixed angles.

All such triangles are similar to each other, so if $M'M''$ is minimal, the sides $M'A$ and $M''A$ are also minimal. Clearly, this happens when $MA$ is

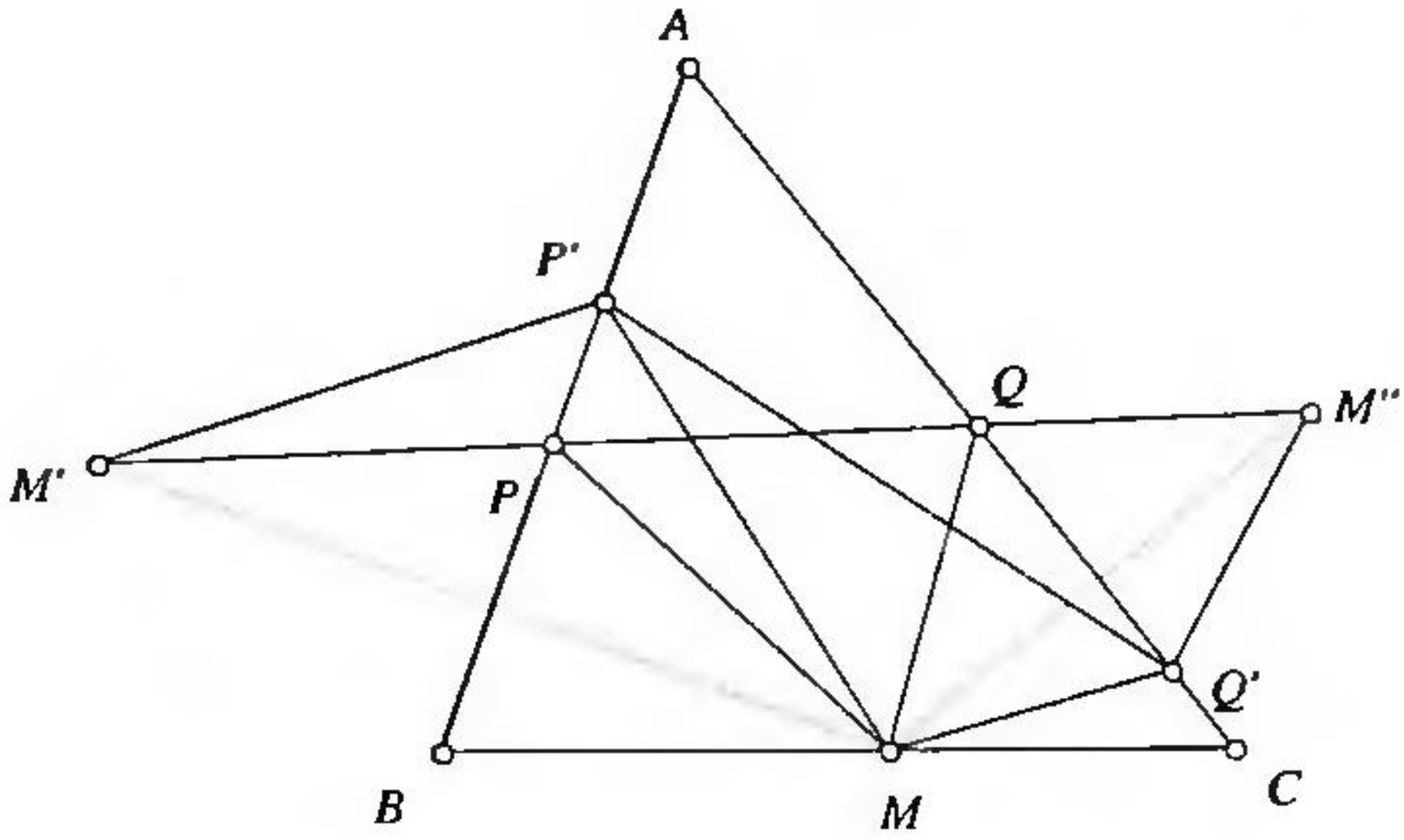


Figure 5.7

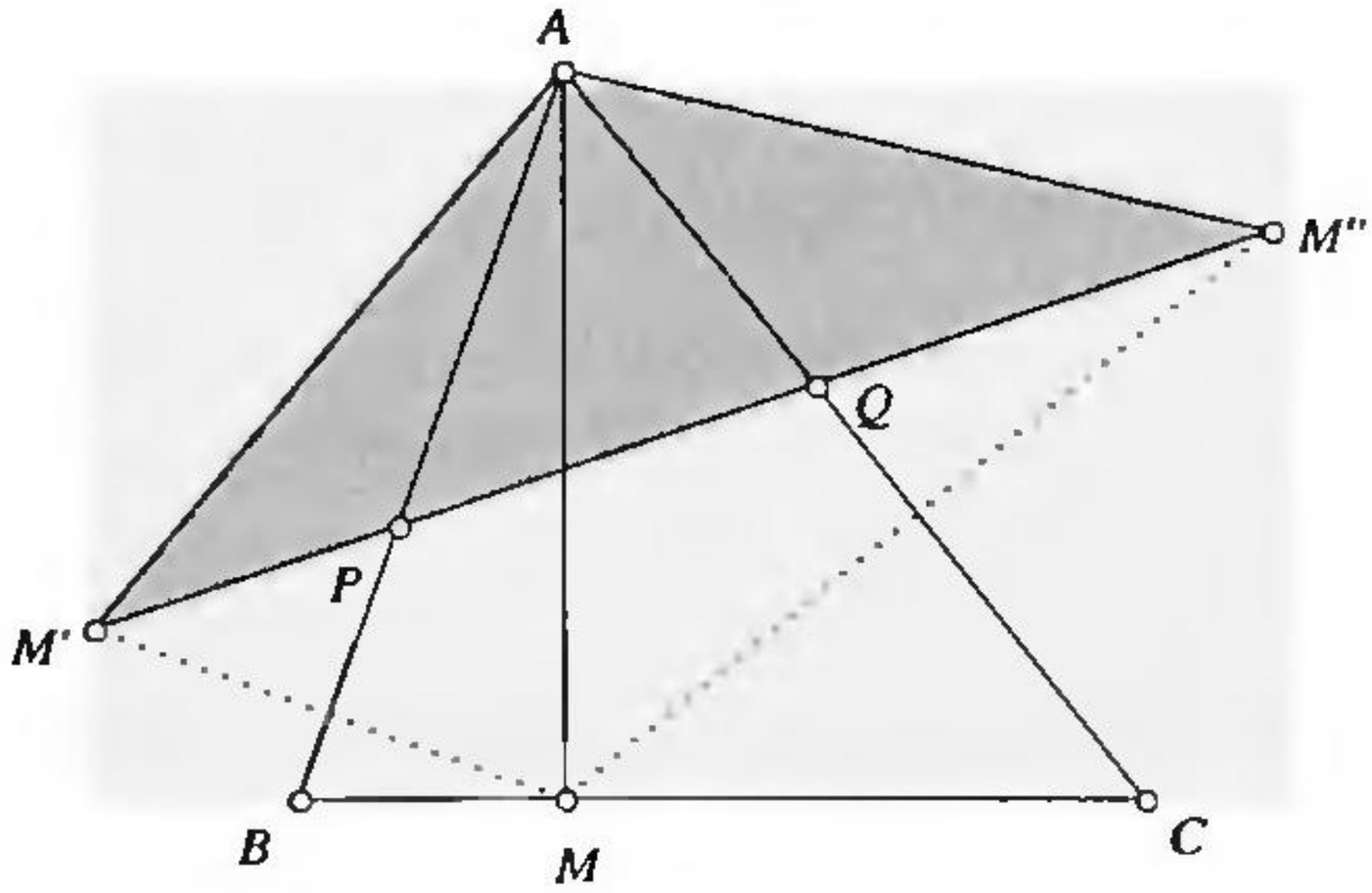


Figure 5.8

minimal, i.e., when $MA$ is an altitude of $ABC$ (see Figure 5.8). We conclude that the perimeter of $MPQ$ is minimal when $M$, $P$, and $Q$ are the feet of the altitudes of the triangle $ABC$ (the so-called orthic triangle).

**Problem 16.** Seven real numbers are given in the interval (1,13). Prove that at least three of them are the lengths of a triangle's sides.

**Solution.** Let the numbers be $a_1, a_2, \ldots, a_7$. We can assume that

$$a_1 \leq a_2 \leq \cdots \leq a_7.$$

Suppose by way of contradiction that no three of them are the lengths of a triangle's sides. We have $a_1 + a_3 > a_2$ and $a_2 + a_3 > a_1$. Then $a_1 + a_2 \leq a_3$, because if not, then there would exist a triangle with side lengths $a_1, a_2, a_3$.

We deduce $a_3 \geq 2$. Similarly, we have $a_4 \geq a_3 + a_2 \geq 3$, $a_5 \geq a_4 + a_3 \geq 5$, $a_6 \geq a_5 + a_4 \geq 8$ and, finally, $a_7 \geq a_6 + a_5 \geq 13$, which is a contradiction.

## 5.2 An interesting locus

**Problem 5.** Find the locus of points $M$ in the plane of triangle $ABC$ such that $[ABM] = 2[ACM]$.

**Solution.** Let $C'$ be the reflection of $A$ across $C$. Then $[AMC'] = 2[AMC]$, so we have to determine the locus of points $M$ for which $[ABM] = [AC'M]$. (Figure 5.9.)

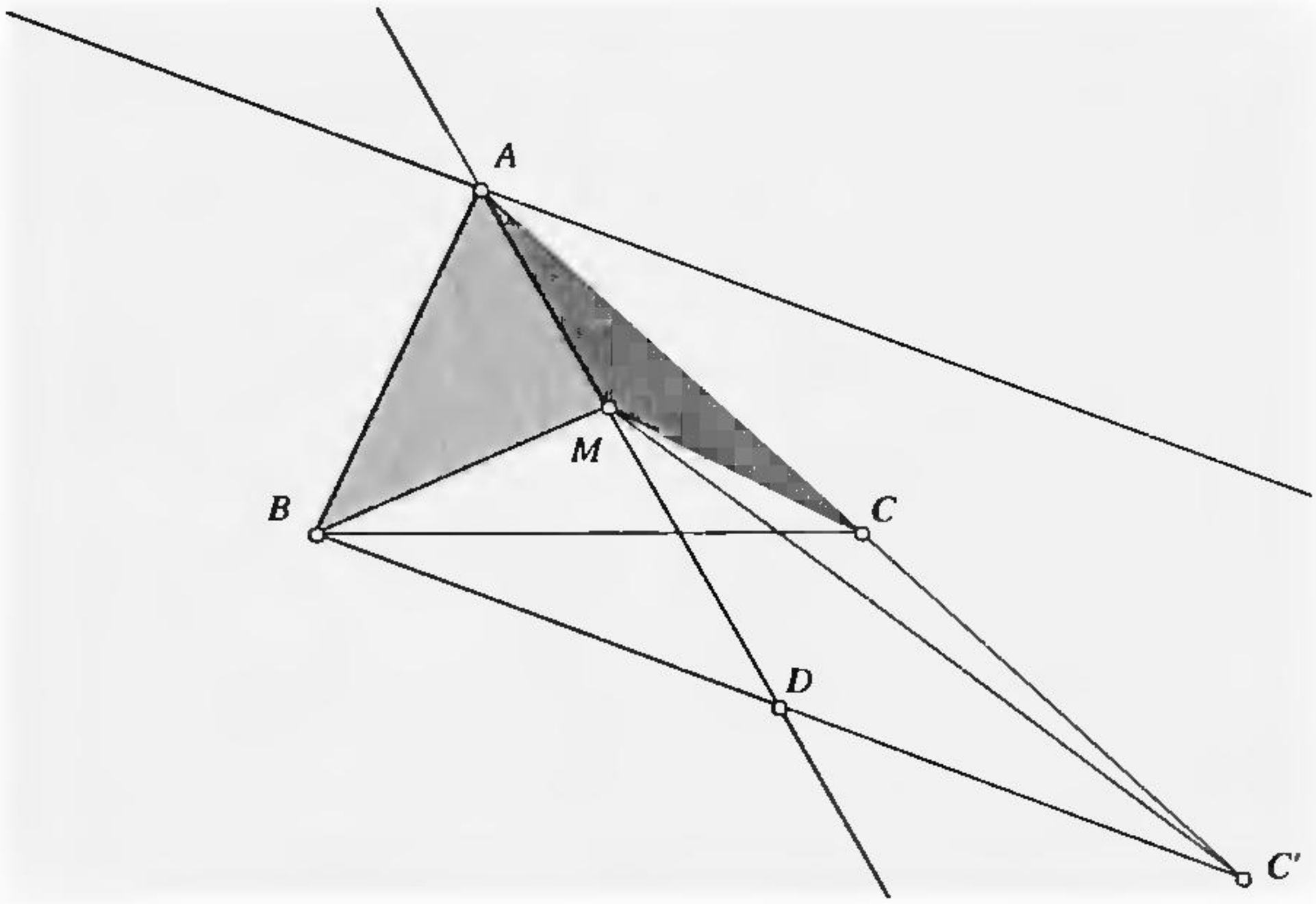


Figure 5.9

It follows that the required locus is the union of the line $AD$ ($D$ being the midpoint of $BC'$) and a parallel to $BC'$ containing the point $A$.

**Problem 6.** Let $D$ be a point on the side $BC$ of triangle $ABC$ and $M$ a point on $AD$. Prove that

$$\frac{[ABM]}{[ACM]} = \frac{BD}{CD}.$$

Deduce Ceva's theorem: If the segments $AD$, $BE$ and $CF$ are concurrent, then

$$\frac{BD}{CD} \cdot \frac{CE}{AE} \cdot \frac{AF}{BF} = 1.$$

**Solution.** Let $A'$ be the projection of $A$ onto $BC$. (See Figure 5.10.) Then $[ABD] = \frac{1}{2}AA' \cdot BD$, $[ACD] = \frac{1}{2}AA' \cdot CD$, hence

$$\frac{[ABD]}{[ACD]} = \frac{BD}{CD}.$$

Similarly, we obtain

$$\frac{[MBD]}{[MCD]} = \frac{BD}{CD}.$$

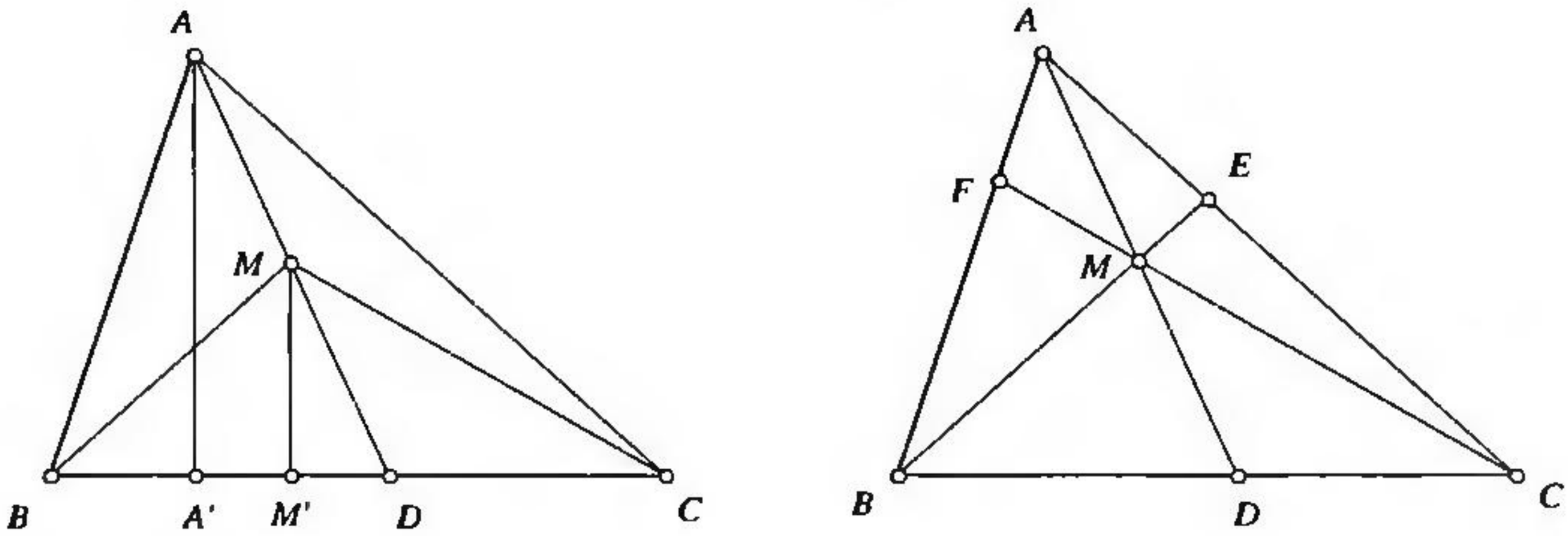


Figure 5.10

We deduce

$$\frac{BD}{CD} = \frac{[ABD] - [MBD]}{[ACD] - [MCD]} = \frac{[ABM]}{[ACM]},$$

as desired. For the second part, observe that

$$\frac{BD}{CD} \cdot \frac{CE}{AE} \cdot \frac{AF}{BF} = \frac{[ABM]}{[ACM]} \cdot \frac{[BCM]}{[ABM]} \cdot \frac{[ACM]}{[BCM]} = 1.$$

**Problem 7.** Let $ABCD$ be a convex quadrilateral and $M$ a point in its interior such that

$$[MAB] = [MBC] = [MCD] = [MDA].$$

Prove that one of the diagonals of $ABCD$ passes through the midpoint of the other diagonal.

**Solution.** Because $[MAB] = [MBC]$, the point $M$ lies on the median of the triangle $ABC$. (Figure 5.11.) Similarly, we deduce that $M$ lies on the median of the triangle $ACD$. If the two medians coincide, then they also coincide with $BD$, and then $BD$ passes through the midpoint of $AC$.

If not, then $M$ must lie on $AC$. Since $[MAB] = [MAD]$, $AC$ bisects $BD$, as desired.

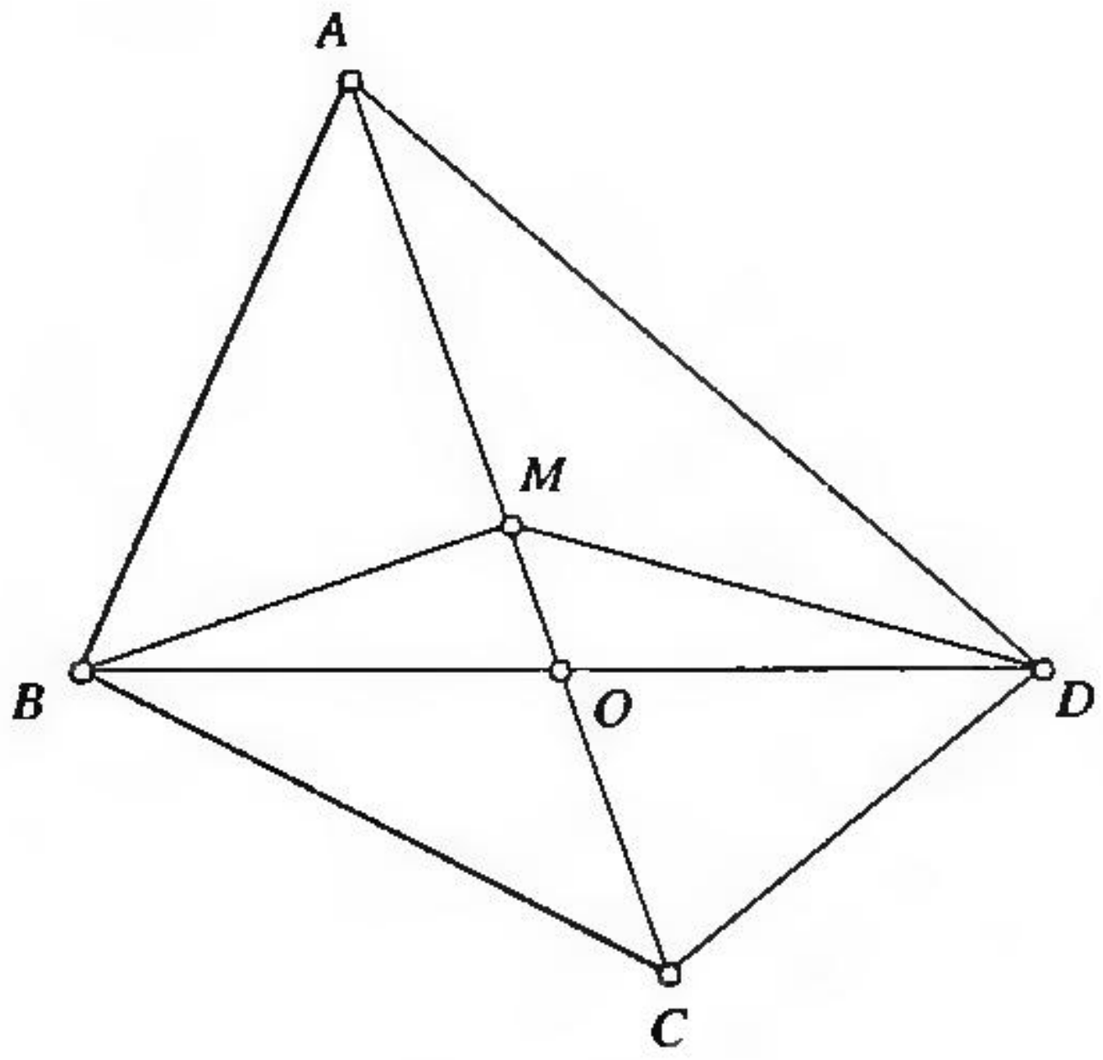


Figure 5.11

**Problem 8.** Let $ABCD$ be a convex quadrilateral. Find the locus of points $M$ in its interior such that

$$[MAB] = 2[MCD].$$

**Solution.** We apply the same method as in Problem 1 (see Figure 5.12).

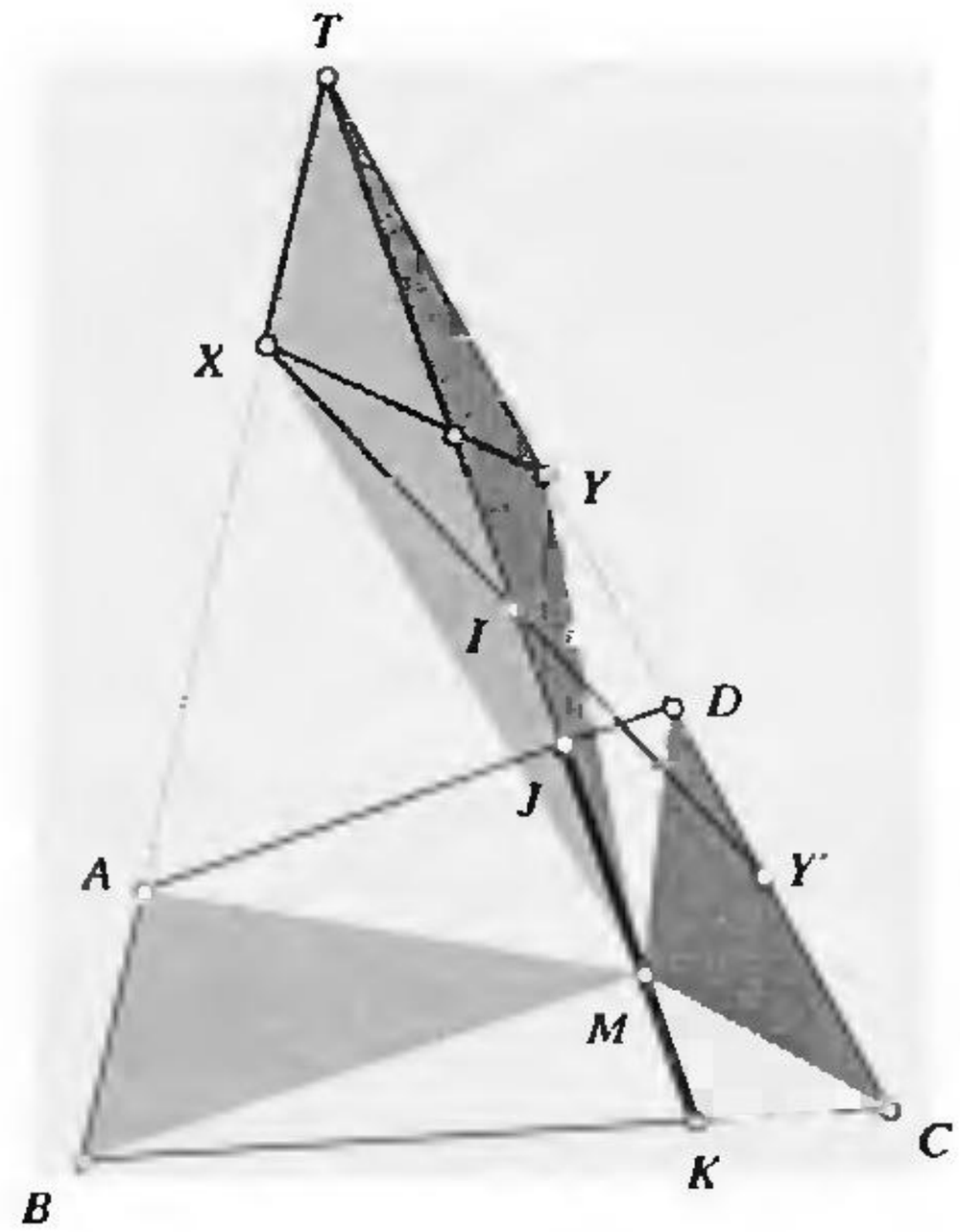


Figure 5.12

We have to find the locus of points $M$ such that $[MXT] = 2[MYT]$. As we have seen in Problem 5, if $Y'$ is a point on $YC$ such that $TY = YY'$,

then the locus of $M$ is the set of points in the interior of the quadrilateral $ABCD$ located on the line $TI$, where $I$ is the midpoint of $XY'$. Therefore, the required locus is the segment $JK$.

**Problem 9.** Let $ABCD$ be a convex quadrilateral and let $k > 0$ be a real number. Find the locus of points $M$ in its interior such that

$$[MAB] + 2[MCD] = k.$$

**Solution.** Combining Problems 3 and 5 and using the construction above, we deduce that the area of $XY'M$ is constant. (Figure 5.13.)

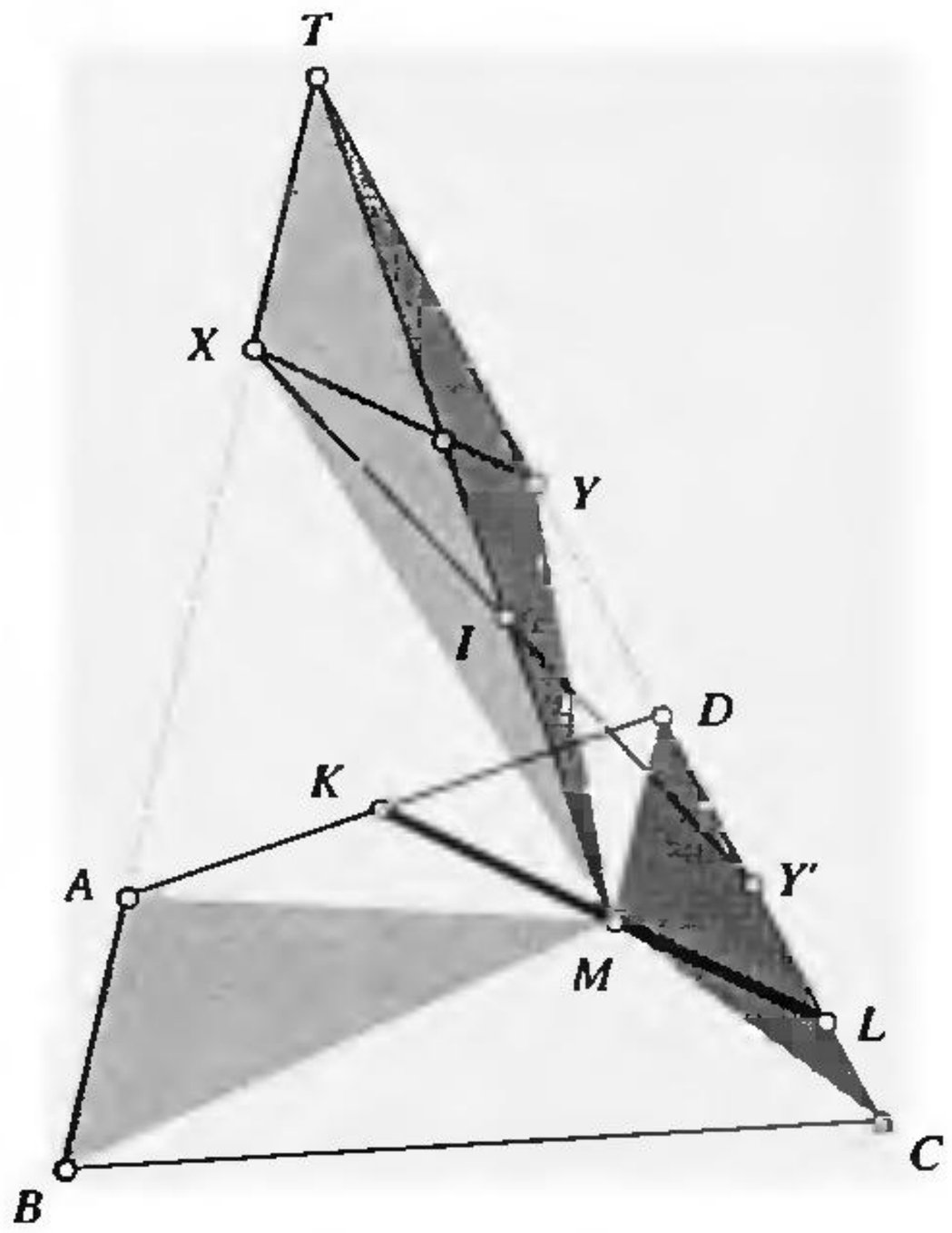


Figure 5.13

It results that the locus is (according to the value of $k$ ) either a segment $KL$ parallel to $XY'$, or a point, or the empty set.

**Problem 10.** Let $d, d'$ be two nonparallel lines in the plane and let $k > 0$. Find the locus of points, the sum of whose distances to $d$ and $d'$ is equal to $k$.

**Solution.** Let $O$ be the point of intersection between $d$ and $d'$. We consider the points $A$, $B$, $C$, $D$, as in Figure 5.14, such that $OA = OB = OC = OD = a > 0$. Suppose $M$ lies in the interior of the angle $AOB$.

If $M_A$ and $M_B$ are the projections of $M$ onto $OA$ and $OB$, we have

$$MM_A \perp MM_B = k$$

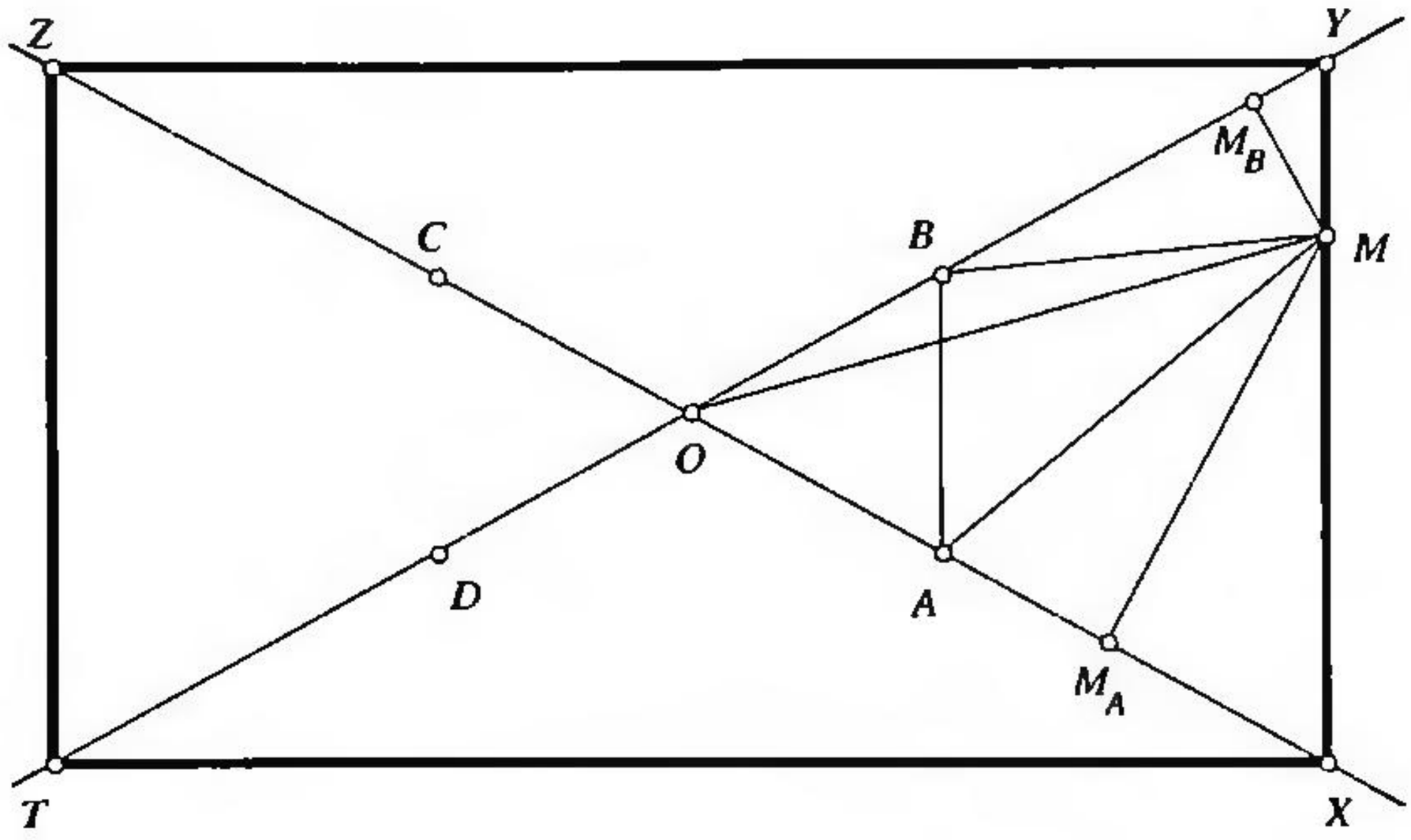


Figure 5.14

Multiplying by $a/2$, we obtain

$$[MOA] + [MOB] = \frac{ka}{2}.$$

But

$$[MOA] + [MOB] = [MAOB] = [OAB] + [MAB].$$

It follows that the area of $MAB$ is constant, and hence the locus of $M$ in the interior of $\angle AOB$ is a segment $XY$ parallel to $AB$.

Considering the other three possible locations of $M$, we deduce that the locus is a rectangle $XYZT$ whose diagonals are $d$ and $d'$.

*Observation.* It is worth mentioning that if $M$ lies, for instance, on the line $XY$ but not in the interior of the segment $XY$, another equality occurs.

If $M$ is on the half-line $XY$ as in Figure 5.15, we have

$$[MOA] - [MOB] = [OAB] + [MAB],$$

so we deduce that for those positions of $M$, the difference of the distances to $d$ and $d'$ equals $k$.

**Problem 11.** Let $ABCD$ be a convex quadrilateral and let $E$ and $F$ be the points of intersections of the lines $AB$, $CD$ and $AD$, $BC$, respectively. Prove that the midpoints of the segments $AC$, $BD$, and $EF$ are collinear.

**Solution.** Let $P$, $Q$, and $R$ be the midpoints of $AC$, $BD$, and $EF$. (Figure 5.16.) Denote by $S$ the area of $ABCD$. As we have seen, the locus of the

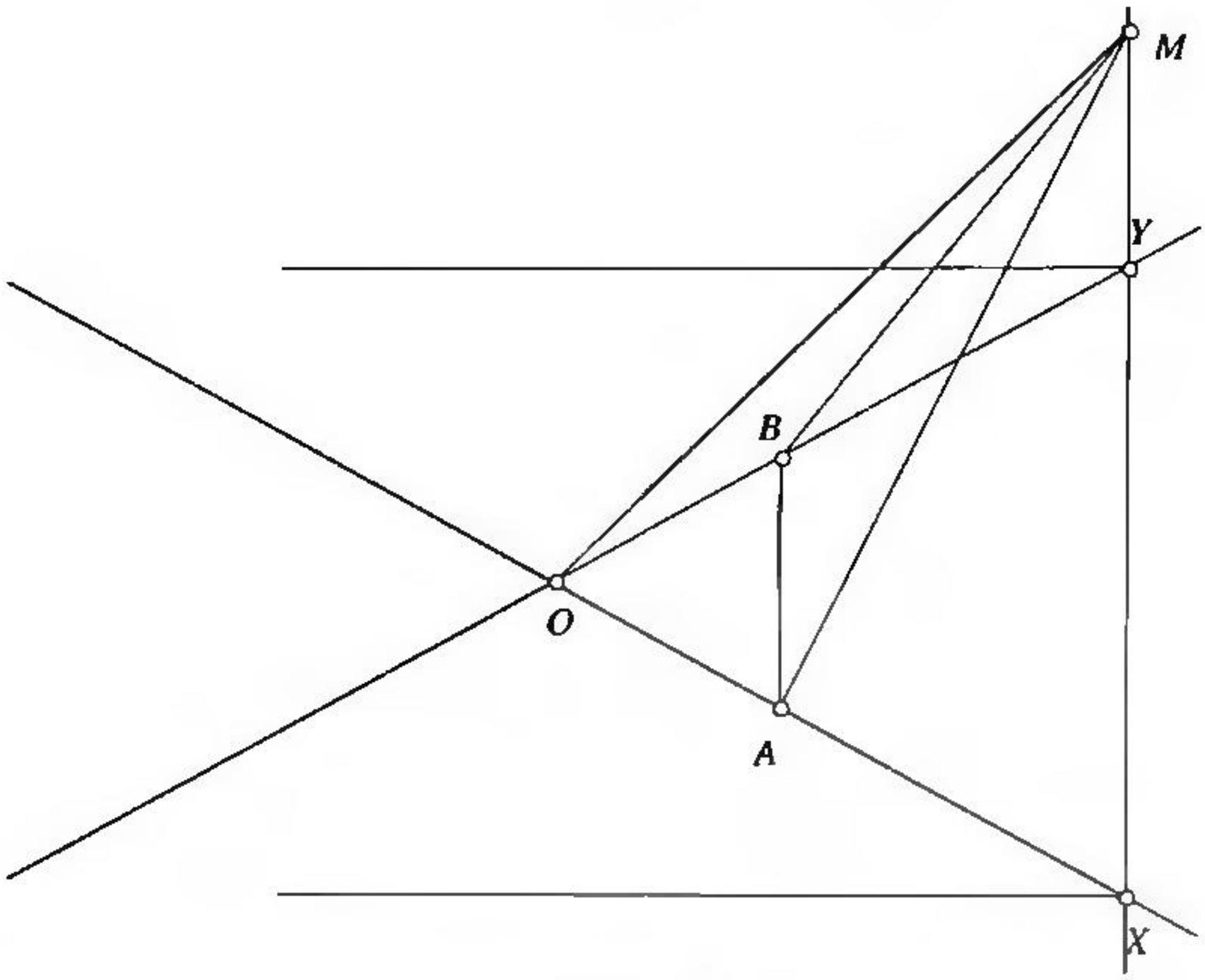


Figure 5.15

points $M$ in the interior of $ABCD$ for which

$$[MAB] + [MCD] = \tfrac{1}{2}S$$

is a segment. We see that $P$ and $Q$ belong to this segment. Indeed,

$$[PAB] + [PCD] = \tfrac{1}{2}[ABC] + \tfrac{1}{2}[ACD] = \tfrac{1}{2}S,$$

and

$$[QAD] + [QCD] = \tfrac{1}{2}[ABD] + \tfrac{1}{2}[BCD] = \tfrac{1}{2}S.$$

Now, we have $[RAB] = \frac{1}{2}[FAB]$, since the distance from $F$ to $AB$ is twice the distance from $R$ to $AB$. Similarly, $[RCD] = \frac{1}{2}[FCD]$. We obtain

$$[RAB] - [RCD] = \tfrac{1}{2}([FAB] - [FCD]) = \tfrac{1}{2}S.$$

Taking into account the observation in the solution to Problem 3, it follows that $P$, $Q$ and $R$ are collinear.

*Observation.* Another proof of the assertion in the problem can be obtained by using a simple (yet useful) lemma. We leave the proof as an exercise to the reader.

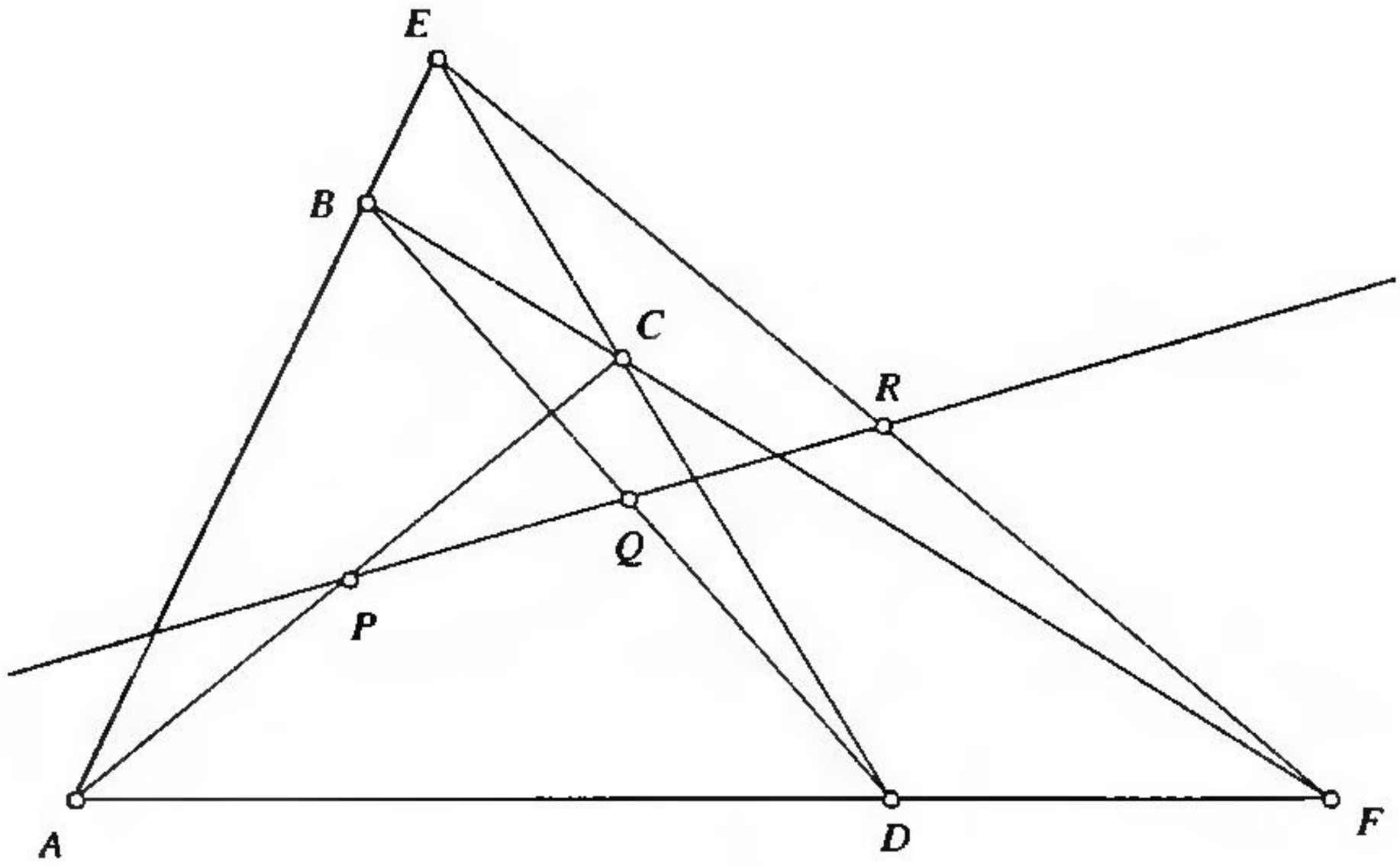


Figure 5.16

*Lemma.* Suppose that through the point $M$ lying in the interior of the parallelogram $ABCD$, two parallels to $AB$ and $AD$ are drawn, intersecting the sides of $ABCD$ at the points $P$, $Q$, $R$, $S$ (see Figure 5.17).

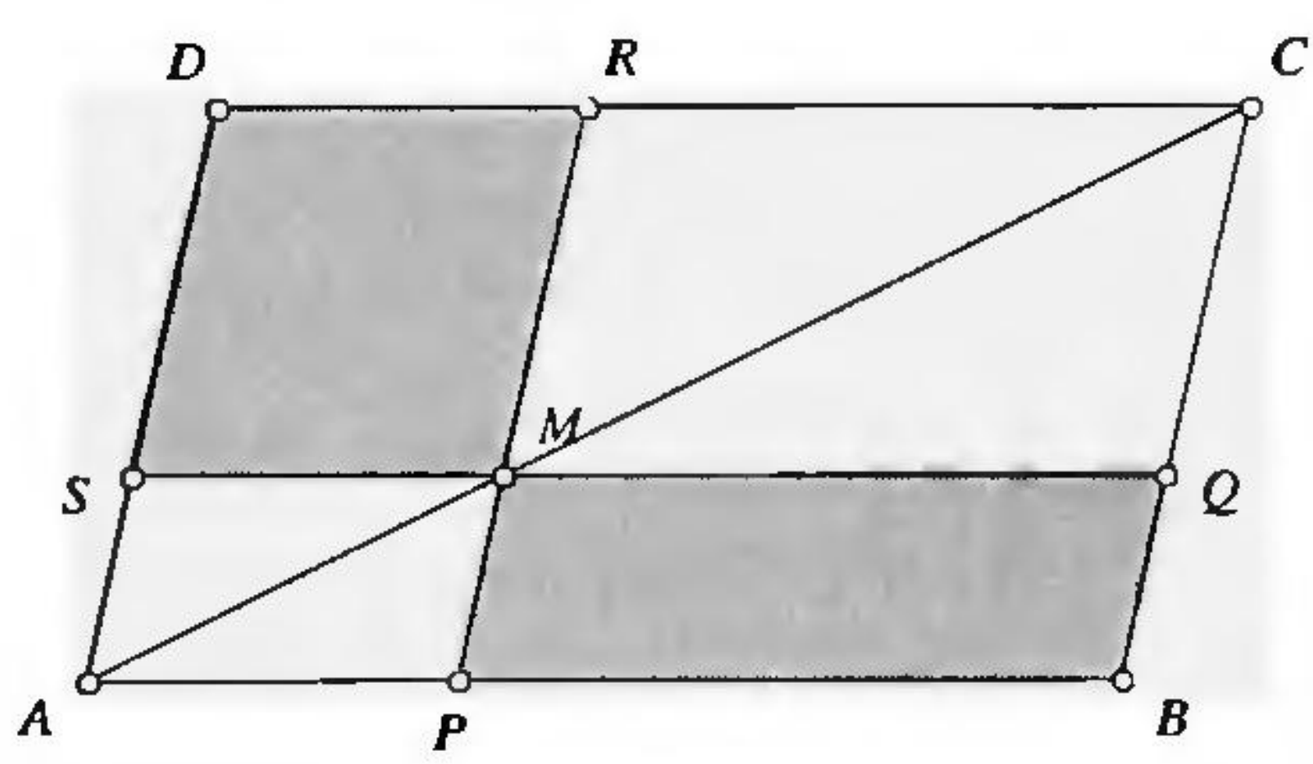


Figure 5.17

Then $M$ lies on the diagonal $AC$ if and only if $[MRDS] = [MPBQ]$.

Returning to our problem, let us draw through the points $B$, $C$, $D$, $E$, and $F$ parallels to $AB$ and $AD$ (see Figure 5.18). It is not difficult to see that the points $P$, $Q$, and $R$ are the images of $C$, $S$, and $T$ through a homothety of center $A$ and ratio 1/2, therefore it suffices to prove that $C$, $S$ and $T$ are collinear.

Now, we apply the lemma, using the fact that $C$ lies on both $ED$ and $BF$ (see Figure 5.19).

We obtain that $S$ lies on $CT$ (see Figure 5.20).

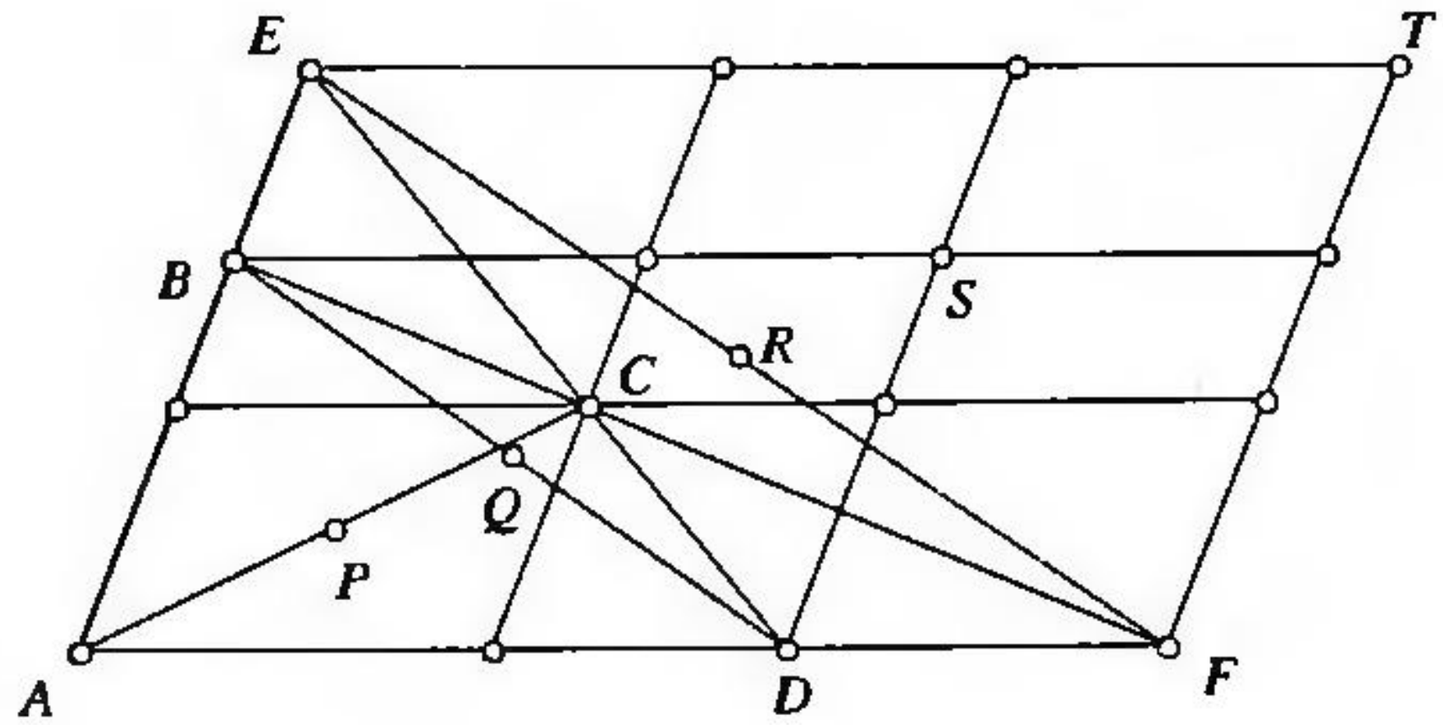


Figure 5.18

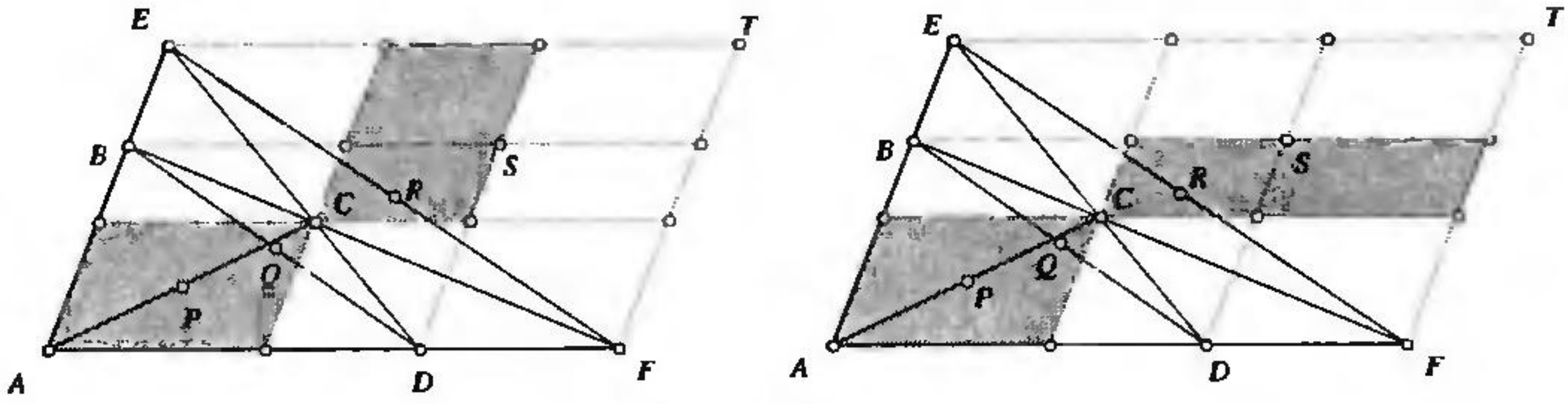


Figure 5.19

**Problem 12.** In the interior of a quadrilateral $ABCD$, consider a variable point $P$. Prove that if the sum of distances from $P$ to the sides is constant, then $ABCD$ is a parallelogram.

**Solution.** Consider the points $X$, $Y$, $Z$, and $T$ on the sides $AB$, $BC$, $CD$, and $DA$, respectively, such that

$$AX = AT = CY = CZ = a.$$

Let $x$, $y$, $z$, and $t$ be the distances from $M$ to the quadrilateral's sides. If $x + y + z + t$ is constant, then the expression

$$\begin{aligned}\tfrac{1}{2}a(x+y+z+t) &= [MAX] + [MCY] + [MCZ] + [MAT] \\ &= [ATX] + [CYZ] + [MXT] + [MYZ]\end{aligned}$$

is constant, as well. Because $[ATX]$ and $[CYZ]$ do not depend on the position of $M$, it follows that the sum $[MXT] + [MYZ]$ is constant. (See Figure 5.21.)

As we have seen, if $XT$ and $YZ$ are not parallel, the locus of points $M$ satisfying the above condition is a segment, and not all the interior points of the quadrilateral $XYZT$. It follows that $XT$ is parallel to$YZ$, hence the angle bisectors of $XAT$ and $YCZ$ are parallel. From here we deduce that $\angle B = \angle D$. A similar argument vields $\angle A = \angle C$. hence $ABCD$ is a parallelogram.

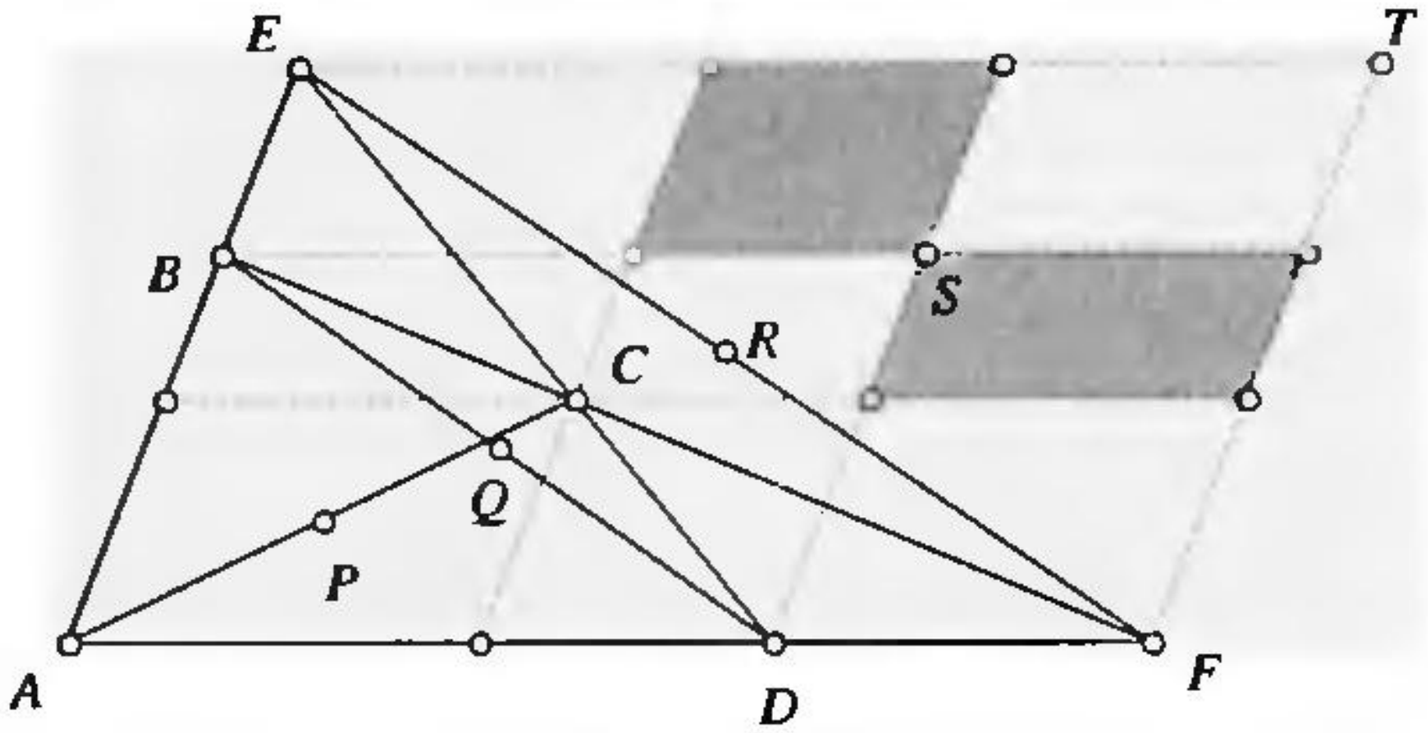


Figure 5.20

*Observation.* More generally, if $A_1A_2\cdots A_n$ is a convex polygon, then the sum of the distances from a variable interior point to its sides is constant if and only if the following equality holds:

$$\frac{1}{A_1A_2}\cdot\overline{A_1A_2}+\frac{1}{A_2A_3}\cdot\overline{A_2A_3}+\cdots+\frac{1}{A_nA_1}\cdot\overline{A_nA_1}=0.$$

## 5.3 Cyclic quads

**Problem 4.** Let $D$, $E$, and $F$ be the feet of the altitudes of the triangle $ABC$. Prove that the altitudes of $ABC$ are the angle bisectors of the triangle $DEF$. (Figure 5.22.)

**Solution.** We can see that the quadrilateral $AFHE$ is cyclic. Indeed, the angles $\angle AFH$ and $\angle AEH$ are right angles; thus $AFHE$ is inscribed in the circle of diameter $AH$. It follows that $\angle FAH=\angle FEH$.

Similarly, in the quadrilateral $CDHE$ we have $\angle DCH=\angle DEH$. But in the right triangles $ABD$ and $CBF$, the angles $\angle FAH$ and $\angle DCH$ equal $90^\circ-\angle B$, hence $\angle FAH=\angle DCH$. It follows that $\angle FEH=\angle DEH$; that is, $BE$ is the angle bisector of $\angle DEF$.

**Problem 5.** Let $ABCD$ be a convex quadrilateral. Prove that

$$AB\cdot CD+AD\cdot BC=AC\cdot BD$$

if and only if $ABCD$ is cyclic (Ptolemy's theorem).

**Solution.** Consider the point $A'$ with the property that the triangles $ABD$ and $A'BC$ are similar. (Figure 5.23.) Then

$$\frac{AD}{BD}=\frac{A'C}{BC},$$

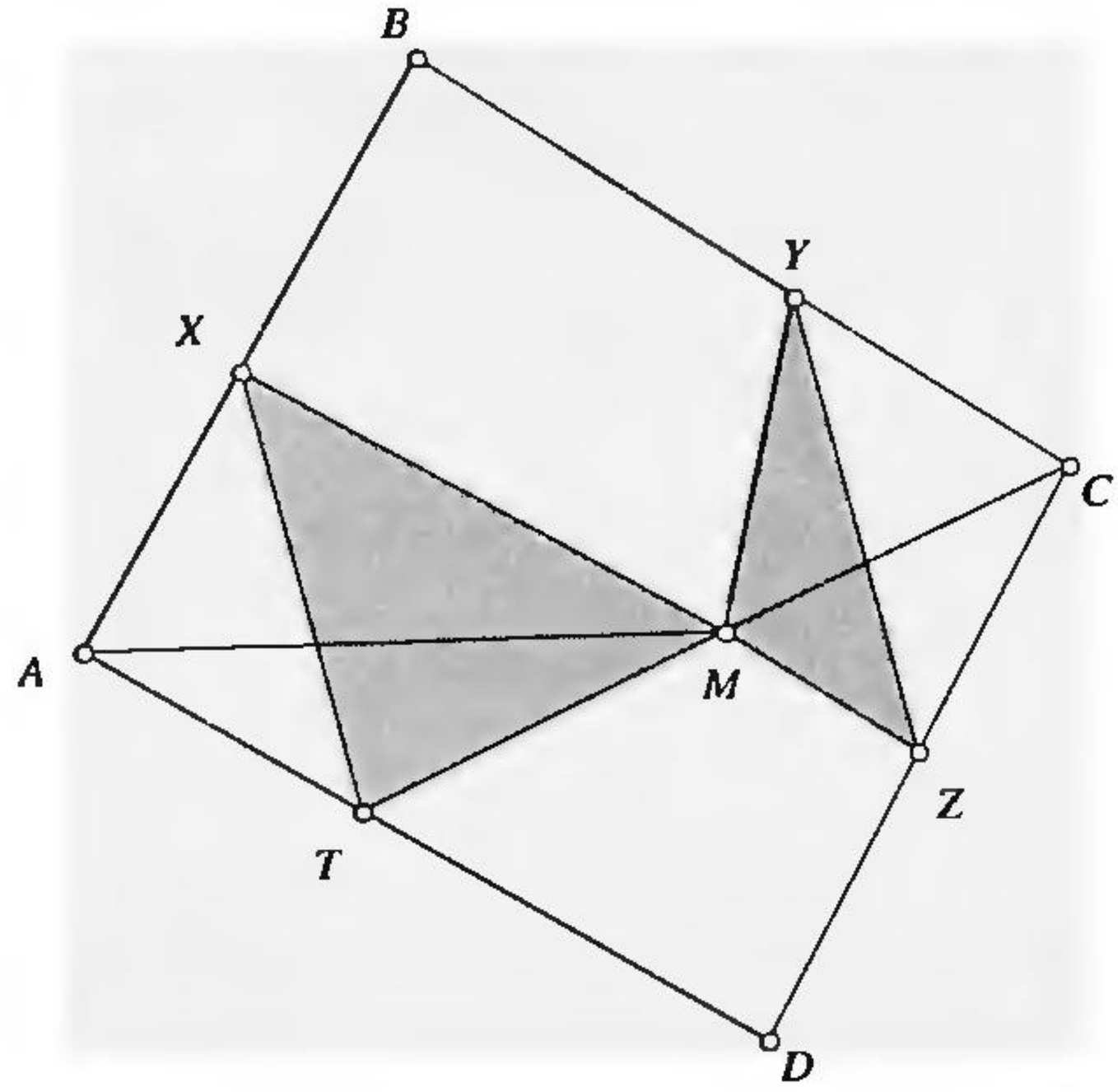


Figure 5.21

and hence

$$AD \cdot BC = A'C \cdot BD. \tag{1}$$

We also have

$$\frac{AB}{A'B} = \frac{BD}{BC}$$

and since $\angle ABA' = \angle DBC$, it follows that the triangles $ABA'$ and $DBC$ are similar, too.

We deduce that

$$\frac{AB}{AA'} = \frac{BD}{DC},$$

so

$$AB \cdot DC = AA' \cdot BD. \tag{2}$$

Adding up (1) and (2) we obtain:

$$AD \cdot BC + AB \cdot CD = (A'C + AA') \cdot BD.$$

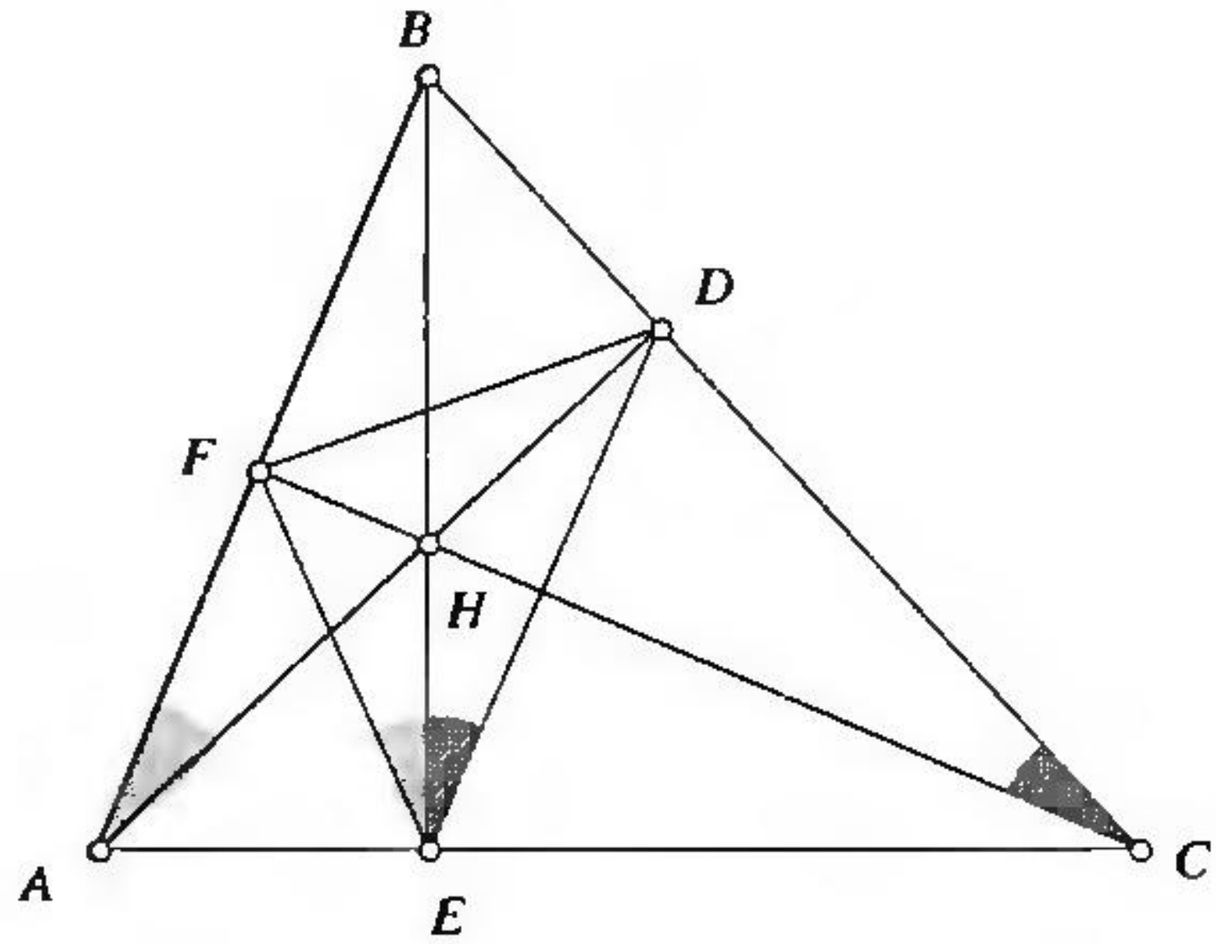


Figure 5.22

But in triangle $AA'C$, $A'C + AA' \geq AC$, and therefore

$$AD \cdot BC + AB \cdot CD \geq AC \cdot BD.$$

This inequality holds in every convex quadrilateral, with equality if and only if the triangle $AA'C$ is degenerate; that is, the point $A'$ lies on $AC$. This happens if and only if $\angle A'CB = \angle ACB$. But since $\angle A'CB = \angle ADB$, the equality holds if and only if $\angle ADB = \angle ACB$; this is the condition that $ABCD$ is cyclic.

**Problem 6.** Let $A'$, $B'$ and $C'$ be points on the sides $BC$, $CA$, and $AB$ of the triangle $ABC$. Prove that the circumcircles of the triangles $AB'C'$, $BA'C'$, and $CA'B'$ have a common point.

**Solution.** Let $M$ be the point of intersection of the circumcircles of triangles $AB'C'$ and $BC'A'$. Because the quadrilateral $AB'MC'$ is cyclic, $\angle MC'A = \angle MB'C$. Similarly, since $BC'MA'$ is cyclic, $\angle MA'B = \angle MC'A$. It follows that $\angle MB'C = \angle MA'B$, and thus $MA'CB'$ is also cyclic. This means that the circumcircle of the triangle $CA'B'$ passes through $M$. (See Figure 5.24.)

*Observation.* The property holds even if the points $A'$, $B'$ and $C'$ are collinear (clearly, in this case, the points do not all lie on the triangle's sides). Suppose that $A'$ and $B'$ lie on the sides $BC$ and $AC$, and $C'$ lies on the line $AB$ in such a way that the three points are collinear. Let $M$ be the point of intersection between the circumcircles of the triangles $AB'C'$ and $BC'A'$. Then $\angle AC'M = \angle MB'C$ ( $AB'MC'$ is cyclic) and $\angle AC'M = \angle MA'C$ ($BA'MC'$ is cyclic). It follows that $\angle MB'C = \angle MA'C$, and thus $MB'A'C$ is also cyclic.

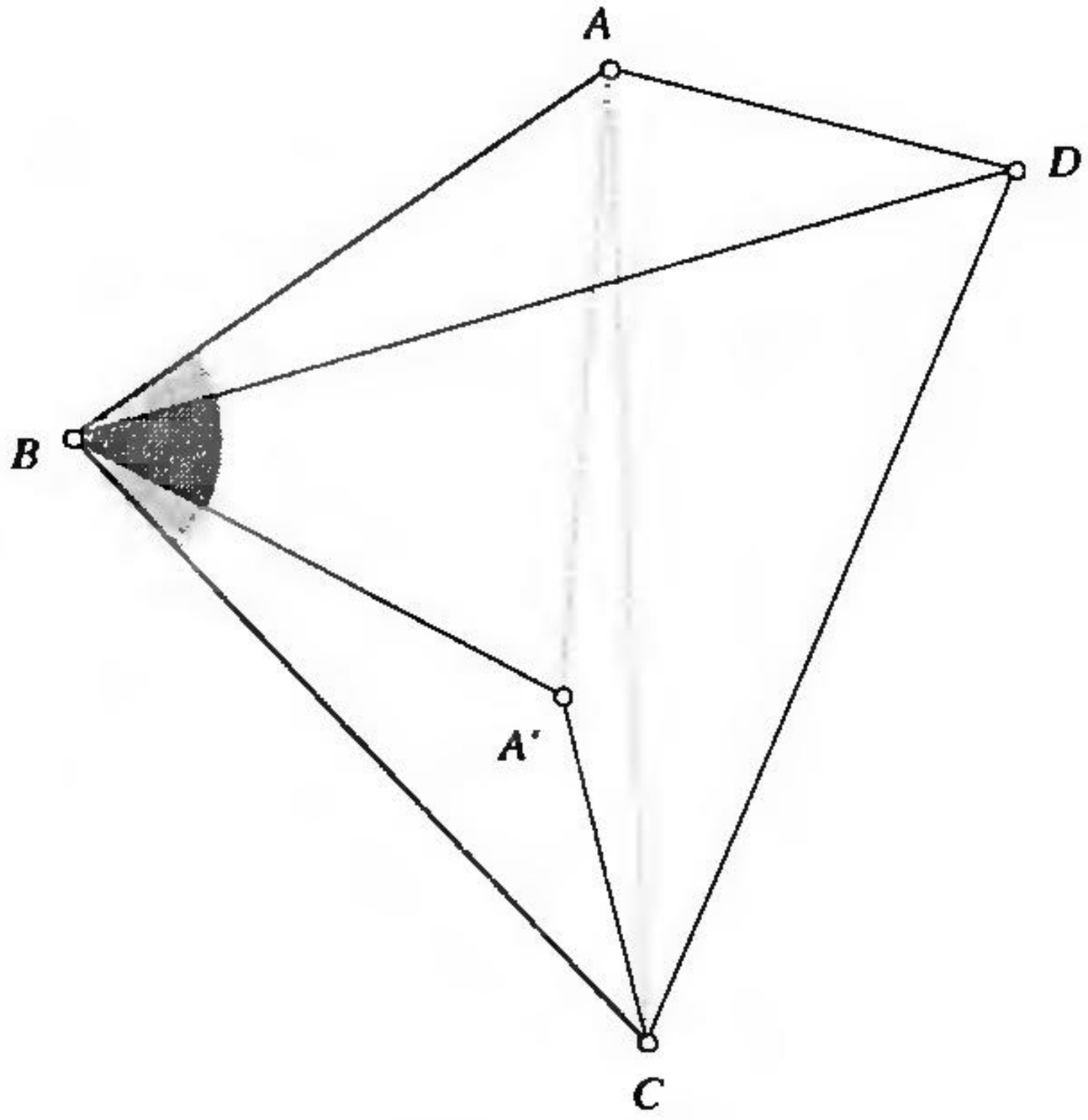


Figure 5.23

Moreover, in the cyclic quadrilateral $AB'MC'$, we have $\angle AMC' = \angle AB'C'$. Similarly, $\angle A'MC = \angle A'B'C$. Because $\angle AB'C' = \angle A'B'C$, it follows that $\angle AMC' = \angle A'MC$, hence $\angle A'MC' = \angle AMC$. But $\angle A'MC' + \angle ABC = 180^\circ$ ( $A'BC'M$ is cyclic) so $\angle AMC + \angle ABC = 180^\circ$. This means that $ABCM$ is cyclic, hence the circumcircle of triangle $ABC$ also passes through $M$. (See Figure 5.25.)

**Problem 7.** Let $ABCD$ be a cyclic quadrilateral. Prove that the orthocenters of the triangles $ABC$, $BCD$, $CDA$, and $DAB$ are the vertices of a quadrilateral congruent to $ABCD$ and prove that the centroids of the same triangles are the vertices of a cyclic quadrilateral.

**Solution.** We first prove a very useful result.

*Lemma.* Let $H$ be the orthocenter of the triangle $ABC$ and $O$ its circumcenter. Then the vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$, $\overrightarrow{OC}$ and $\overrightarrow{OH}$ satisfy the following equality:

$$\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC} = \overrightarrow{OH}.$$

*Proof.* Let $A'$ be a point on the circumcircle such that $AA'$ is a diameter. Then the quadrilateral $A'BHC$ is a parallelogram. Indeed, since $AA'$ is a diameter, $\angle A'CA = 90^\circ$, so that $A'C$ is perpendicular to $AC$. But we also have $BH$ pependicular to $AC$, and thus $A'C$ and $BH$ are parallel. Similarly, it follows that $A'B$ and $CH$ are parallel, so $A'BHC$ is a parallelogram.

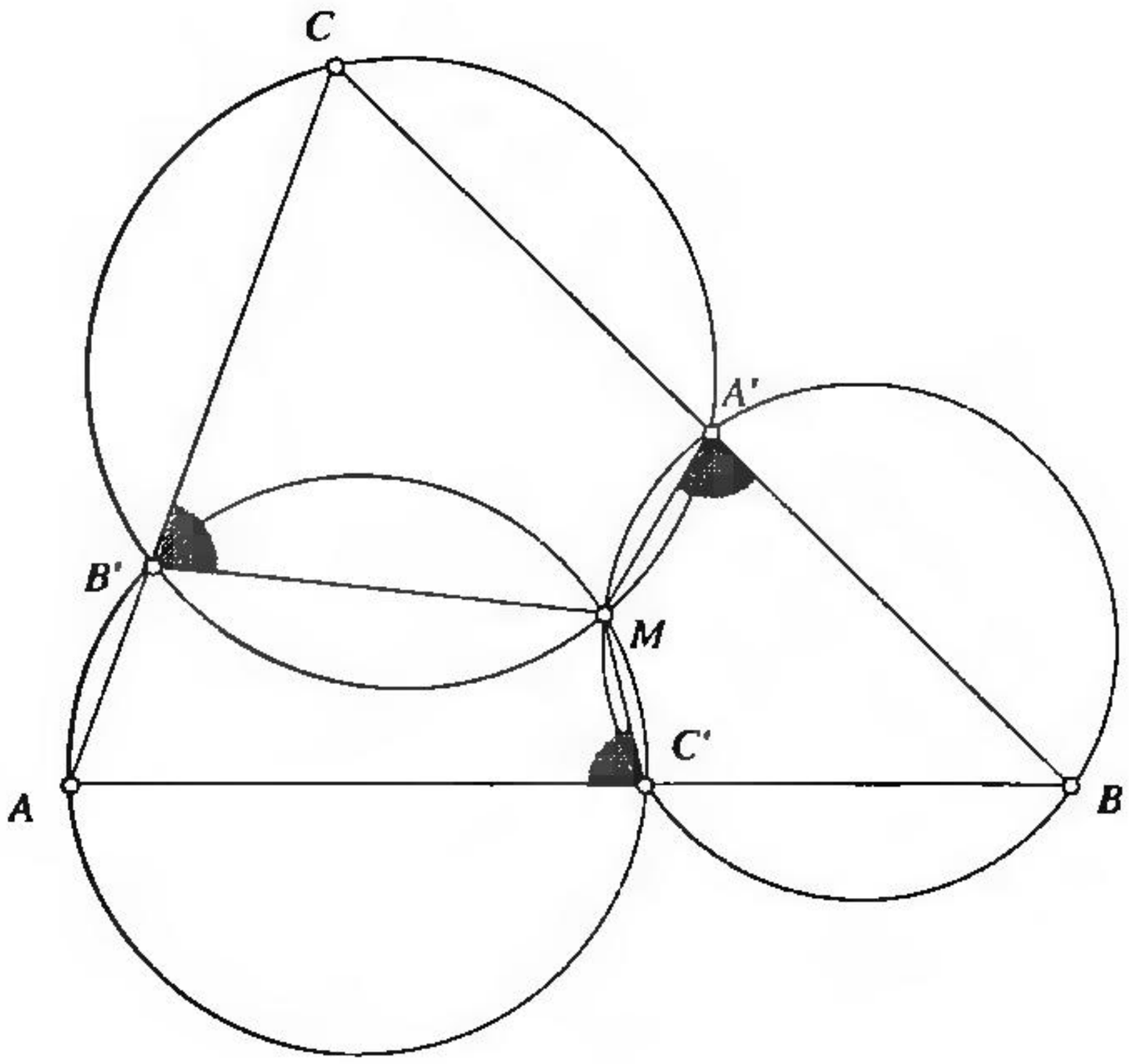


Figure 5.24

We have

$$\overline{HC} + \overline{HB} = \overline{HA'}$$

and

$$\overline{HA'} + \overline{HA} = 2\overline{HO}$$

($O$ is the midpoint of $AA'$), so that

$$\overline{HA} + \overline{HB} + \overline{HC} = 2\overline{HO}.$$

But then

$$\overline{OA} + \overline{OB} + \overline{OC} = \overline{OH} + \overline{HA} + \overline{OH} + \overline{HB} + \overline{OH} + \overline{HC}$$
$$= 3\overline{OH} + 2\overline{HO} = \overline{OH},$$

as desired.

It is worth mentioning that if $G$ is the centroid of triangle $ABC$, then

$$\overline{OA} + \overline{OB} + \overline{OC} = 3\overline{OG},$$

and hence $\overline{OH} = 3\overline{OG}$. This means that the points $O$, $G$, and $H$ are collinear and $OH = 3OG$.

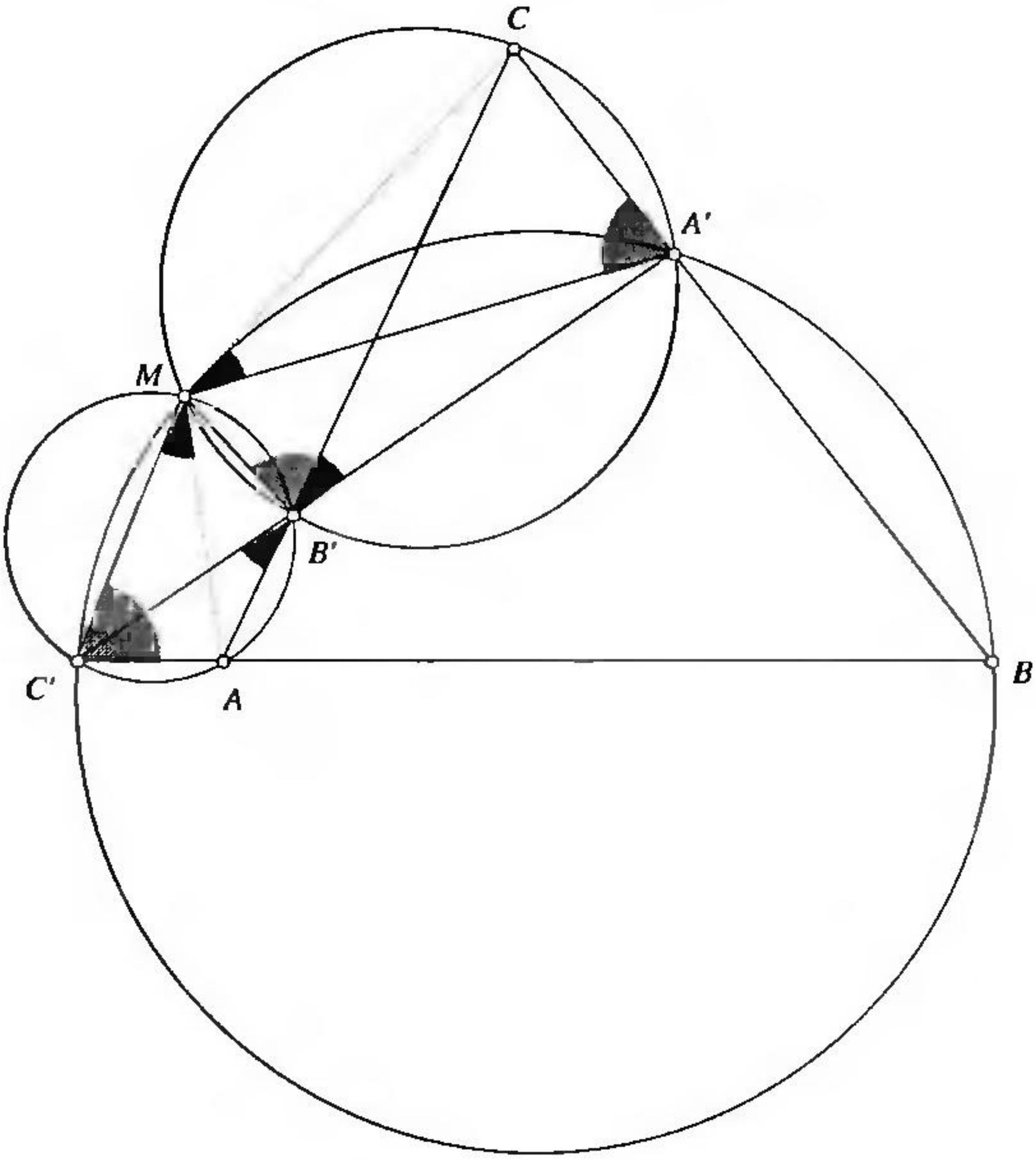


Figure 5.25

Returning to the problem, let $O$ be circumcenter of the quadrilateral $ABCD$ and let $H_A$, $H_B$, $H_C$, $H_D$ be the orthocenters of the triangles $BCD$, $CDA$, $DAB$, and $ABC$, respectively. Using the lemma, we have:

$$\begin{aligned}\overline{H_AH_B} = \overline{OH_B} - \overline{OH_A} &= \left(\overline{OC} + \overline{OD} + \overline{OA}\right) - \left(\overline{OB} + \overline{OC} + \overline{OD}\right) \\ &= \overline{OA} - \overline{OB} = \overline{BA},\end{aligned}$$

and consequently the segments $H_AH_B$ and $AB$ are parallel and equal in length. We conclude that the quadrilaterals $ABCD$ and $H_AH_BH_CH_D$ are congruent.

For the second claim, let $G_A$, $G_B$, $G_C$, and $G_D$ denote the centroids. It follows from the observation above that $G_AG_BG_CG_D$ is obtained from $H_AH_BH_CH_D$ by a homothety of center $O$ and ratio 1/3. Because $H_AH_BH_CH_D$ is cyclic, the same is true of $G_AG_BG_CG_D$. (See Figure 5.27.)

**Problem 8.** Let $K$, $L$ $M$ and $N$ be the midpoints of the sides $AB$, $BC$,

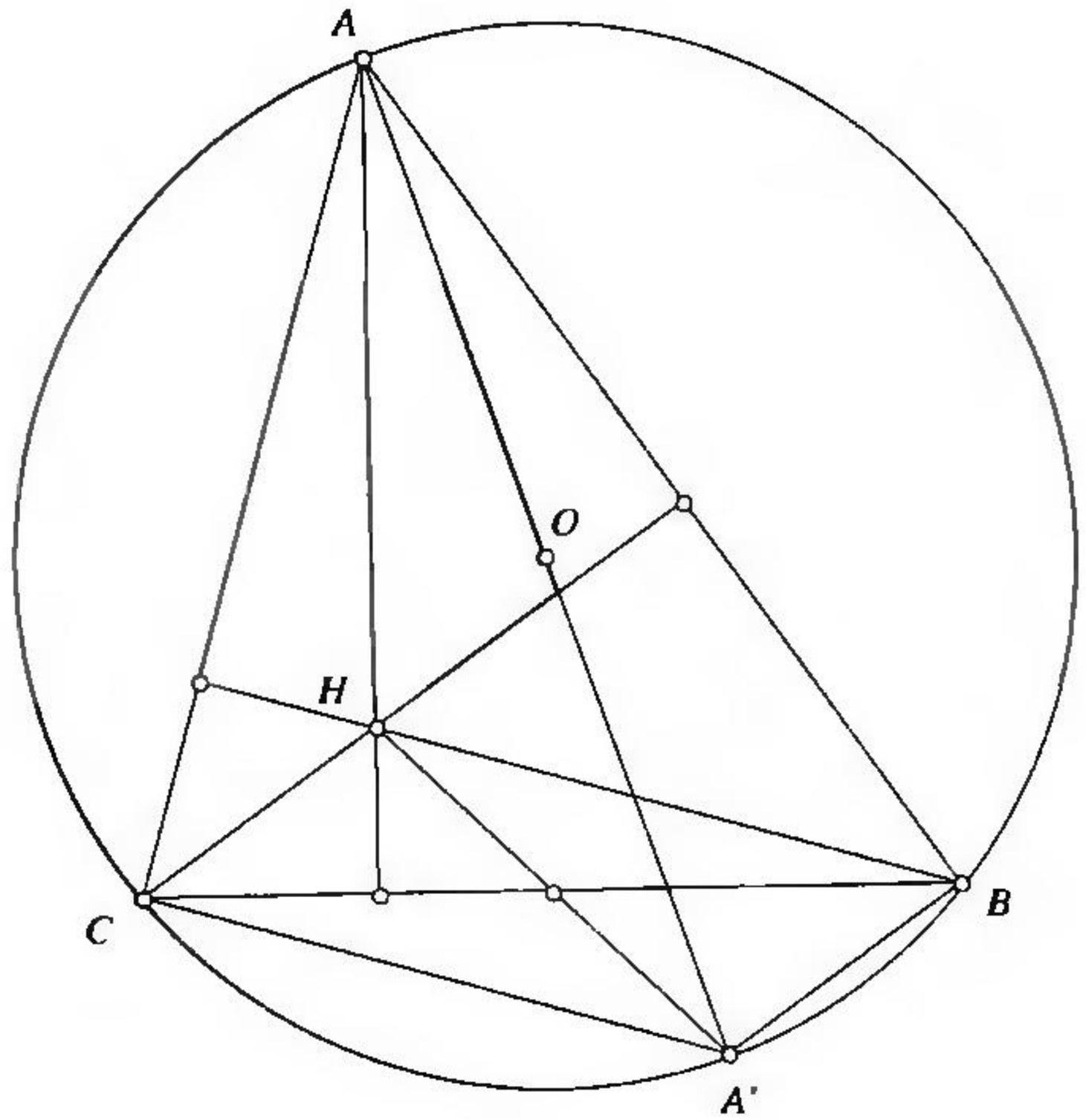


Figure 5.26

$CD$, and $DA$, respectively, of a cyclic quadrilateral $ABCD$. Prove that the orthocenters of the triangles $AKN$, $BKL$, $CLM$, and $DMN$ are the vertices of a parallelogram.

**Solution.** Denote by $H_A$, $H_B$, $H_C$, and $H_D$ the orthocenters of the triangles $AKN$, $BKL$, $CLM$, and $DMN$. If $O$ is the circumcenter of $ABCD$ and $H$ is the orthocenter of $ABD$, we have from the previous lemma that:

$$\overline{OH} = \overline{OA} + \overline{OB} + \overline{OD}.$$

Because $K$ and $N$ are the midpoints of $AB$ and $AD$, the triangle $AKN$ is the image of $ABD$ through a homothety of center $A$ and ratio 1/2. Thus, $\overline{AH_A} = \frac{1}{2}\overline{AH}$. Then we have

$$\begin{aligned}\overline{OH_A} &= \overline{OA} + \overline{AH_A} \\ &= \overline{OA} + \tfrac{1}{2}\overline{AH} = \overline{OA} + \tfrac{1}{2}\left(\overline{OH} - \overline{OA}\right) \\ &= \overline{OA} + \tfrac{1}{2}\left(\overline{OB} + \overline{OD}\right).\end{aligned}$$

Similarly,

$$\overline{OH_C} = \overline{OC} + \tfrac{1}{2}\left(\overline{OB} + \overline{OD}\right),$$

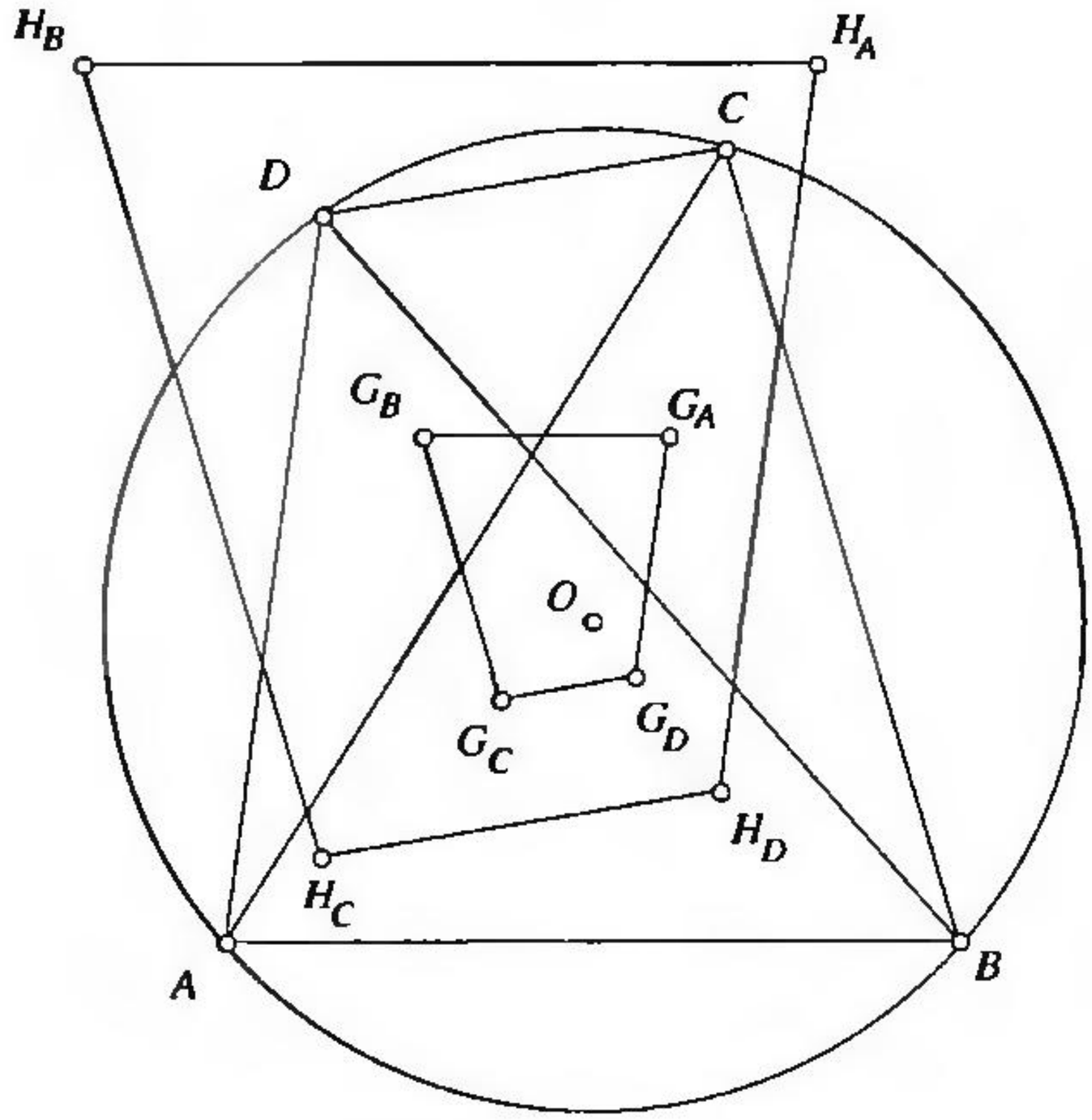


Figure 5.27

and by adding these equalities, we obtain

$$\overline{OH_A} + \overline{OH_C} = \overline{OA} + \overline{OB} + \overline{OC} + \overline{OD}.$$

From the symmetry of the right-hand side, it follows that

$$\overline{OH_A} + \overline{OH_C} = \overline{OH_B} + \overline{OH_D},$$

and this equality implies that the segments $H_AH_C$ and $H_BH_D$ have the same midpoint, hence $H_AH_BH_CH_D$ is a parallelogram. (See Figure 5.28.)

**Problem 9.** Prove that the perpendiculars dropped from the midpoints of the sides of a cyclic quadrilateral to the opposite sides are concurrent.

**Solution.** Let $K$ and $M$ be the midpoints of $AB$ and $CD$ and let $X$ be the intersection point of the perpendiculars dropped from these points to the opposite sides. Let $O$ be the circumcenter of $ABCD$. Because $OK$ is perpendicular to $AB$ and $OM$ is perpendicular to $CD$, the quadrilateral $OKXM$ is a parallelogram, hence the midpoint of $OX$ coincides with the midpoint of $KM$.

Let $L$ and $N$ the midpoints of $BC$ and $AD$ and $Y$ the intersection point of the perpendiculars dropped from these points to the opposite sides. It results that the midpoint of $OY$ coincides with the midpoint of $LN$. It is not difficult

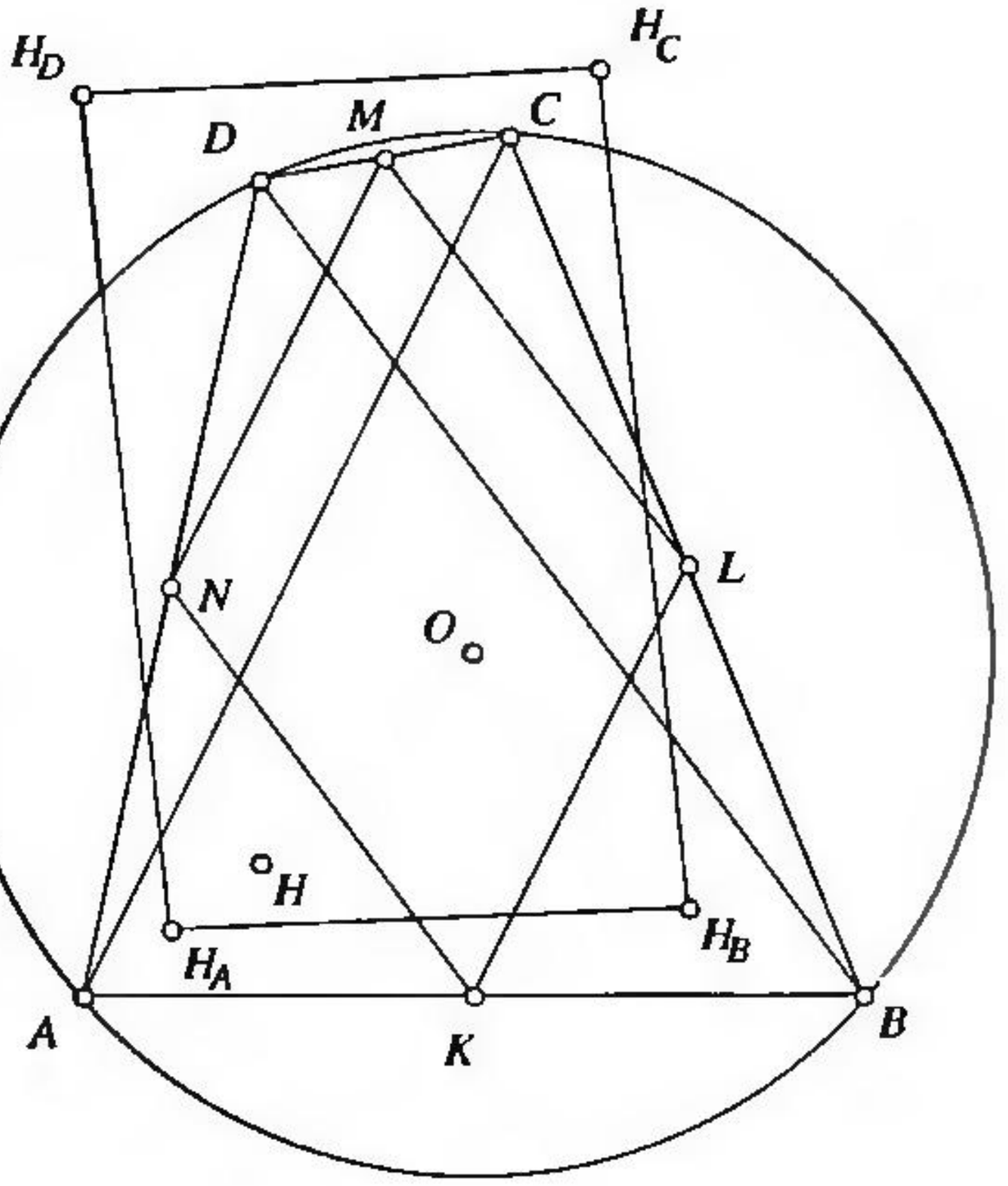


Figure 5.28

to see that $KLMN$ is a parallelogram, hence $KM$ and $LN$ have the same midpoint. It follows that $X = Y$ and the claim is proved. (See Figure 5.29.)

**Problem 10.** In the convex quadrilateral $ABCD$ the diagonals $AC$ and $BD$ intersect at $O$ and are perpendicular. Prove that projections of $O$ on the quadrilateral's sides are the vertices of a cyclic quadrilateral.

**Solution.** Let $K, L, M$, and $N$ be the projections of $O$ on the sides $AB, BC, BD$, and $DA$, respectively. The quadrilateral $AKON$ is cyclic, hence $\angle OAN = \angle OKN$. (See Figure 5.30.)

In the same way we obtain $\angle OBL = \angle OKL$, $\angle ODN = \angle OMN$, $\angle OCL = \angle OML$. If we add these equalities, it follows that

$$\begin{aligned}\angle LKN + \angle LMN &= \angle OAD + \angle ODA + \angle OBC + OCB \\ &= 90^\circ + 90^\circ = 180^\circ,\end{aligned}$$

and hence $KLMN$ is cyclic.

## 5.4 Equiangular polygons

**Problem 4.** Let $ABCDE$ be an equiangular pentagon whose side lengths are rational numbers. Prove that the pentagon is regular.

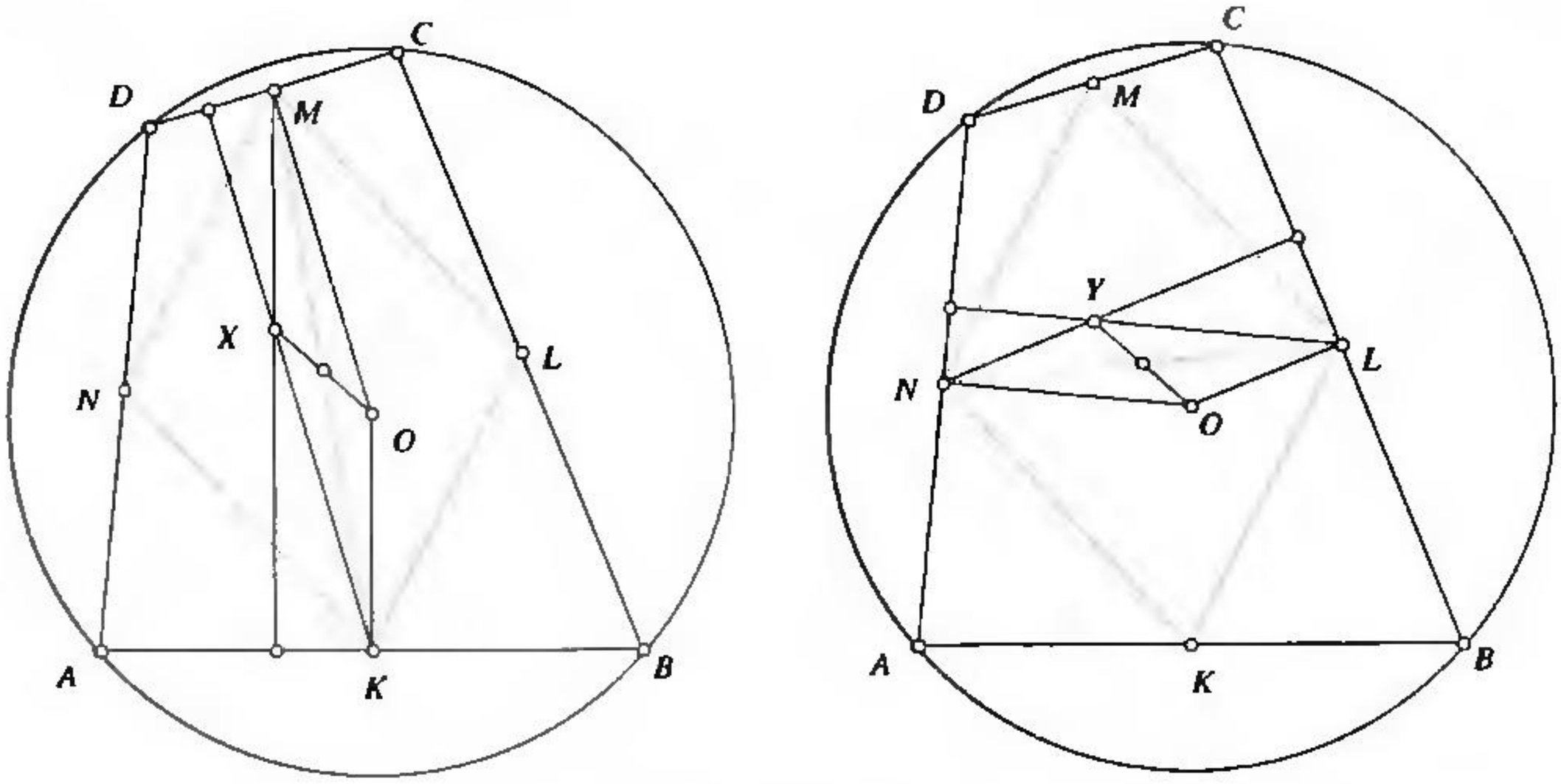


Figure 5.29

**Solution.** Let $M$ and $N$ be the intersection points of the line $AE$ with $BC$ and $CD$, respectively.

Because $ABCDE$ is equiangular, the triangles $AMB$ and $DNE$ are isosceles, with $\angle M = \angle N = 36^\circ$. Therefore, the triangle $CMN$ is also isosceles and $CM = CN$. It follows that $BC + BM = CD + DN$. But $BM = \frac{AB}{2\cos 72^\circ}$ and $DN = \frac{DE}{2\cos 72^\circ}$, hence $BC - CD = \frac{DE-AB}{2\cos 72^\circ}$. If $DE \neq AB$, it follows that $\cos 72^\circ$ is a rational number, which is a contradiction, since $\cos 72^\circ = \frac{\sqrt{5}-1}{4}$. The conclusion is that $DE = AB$. By similar arguments, all the pentagon's sides have equal lengths. (See Figure 5.31.)

**Problem 5.** Prove that $p$ is a prime number if and only if every equiangular polygon with $p$ sides of rational lengths is regular.

**Solution.** The proof uses some advanced knowledge in the algebra of polynomials. Suppose $p$ is a prime number and let the rational numbers $a_1, a_2, \ldots, a_p$ be the side lengths of an equiangular polygon. We have seen that

$$\zeta = \cos\frac{2\pi}{p} + i\sin\frac{2\pi}{p}$$

is a root of the polynomial

$$P(X) = a_1 + a_2X + \cdots + a_pX^{p-1}.$$

On the other hand, $\zeta$ is also a root of the polynomial

$$Q(X) = 1 + X + X^2 + \cdots + X^{p-1}.$$

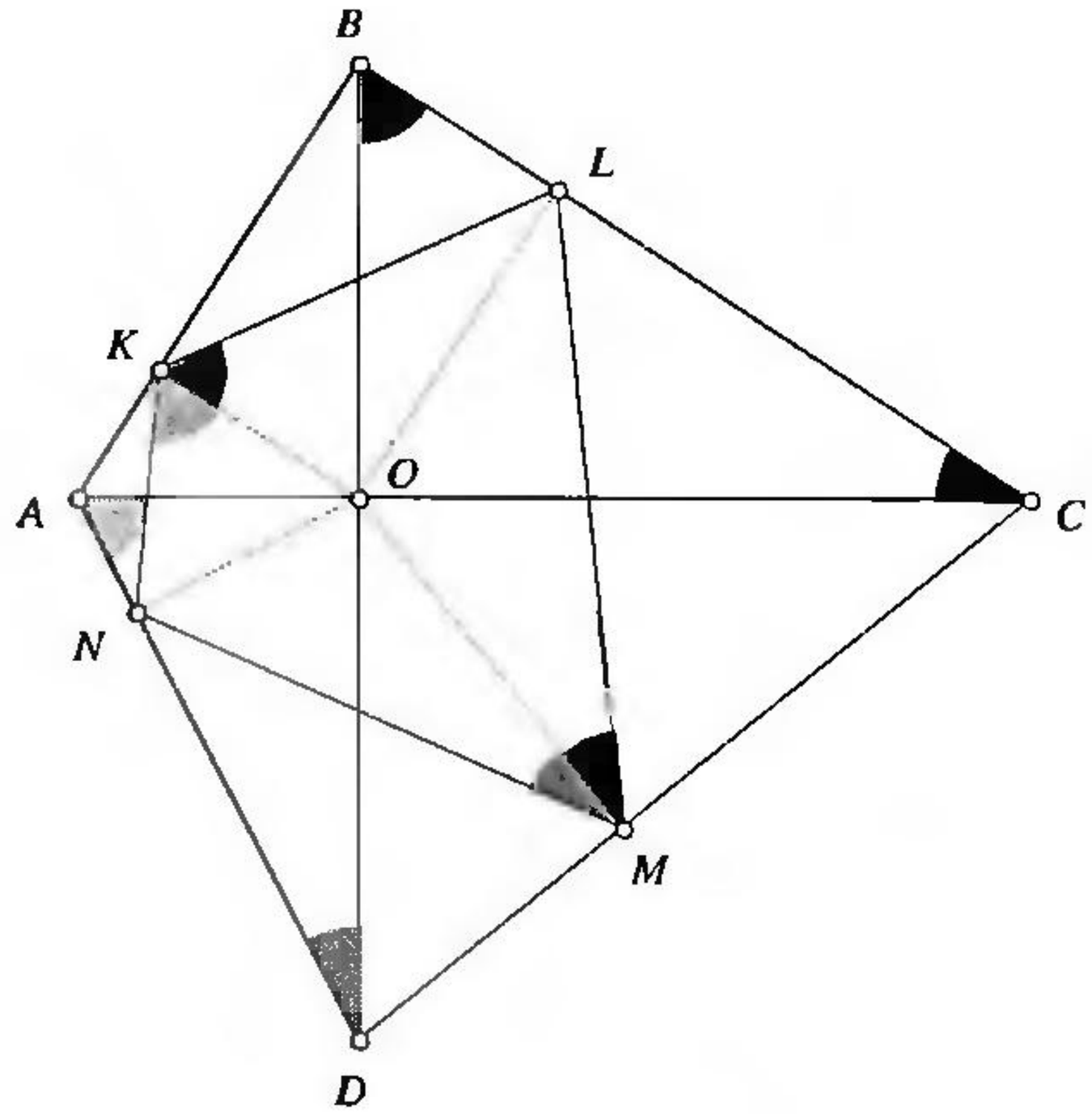


Figure 5.30

Because the two polynomials share a common root, their greatest common divisor must be a nonconstant polynomial with rational coefficients.

This implies that $Q$ can be factorized as a product of two nonconstant polynomials with rational coefficients, which is impossible (to prove that, one applies Eisenstein's criterion to the polynomial $Q(X+1)$ ).

Conversely, suppose $p$ is not a prime number and let $p = mn$ for some positive integers $m, n > 1$. It results that $\zeta^n$ is a root of order $m$ of unity, and hence $1 + \zeta^n + \zeta^{2n} + \cdots + \zeta^{(m-1)n} = 0$. If we add this equality to $1+\zeta+\zeta^2+\zeta^3+\cdots+\zeta^{p-1} = 0$, we deduce that $\zeta$ is the root of a polynomial

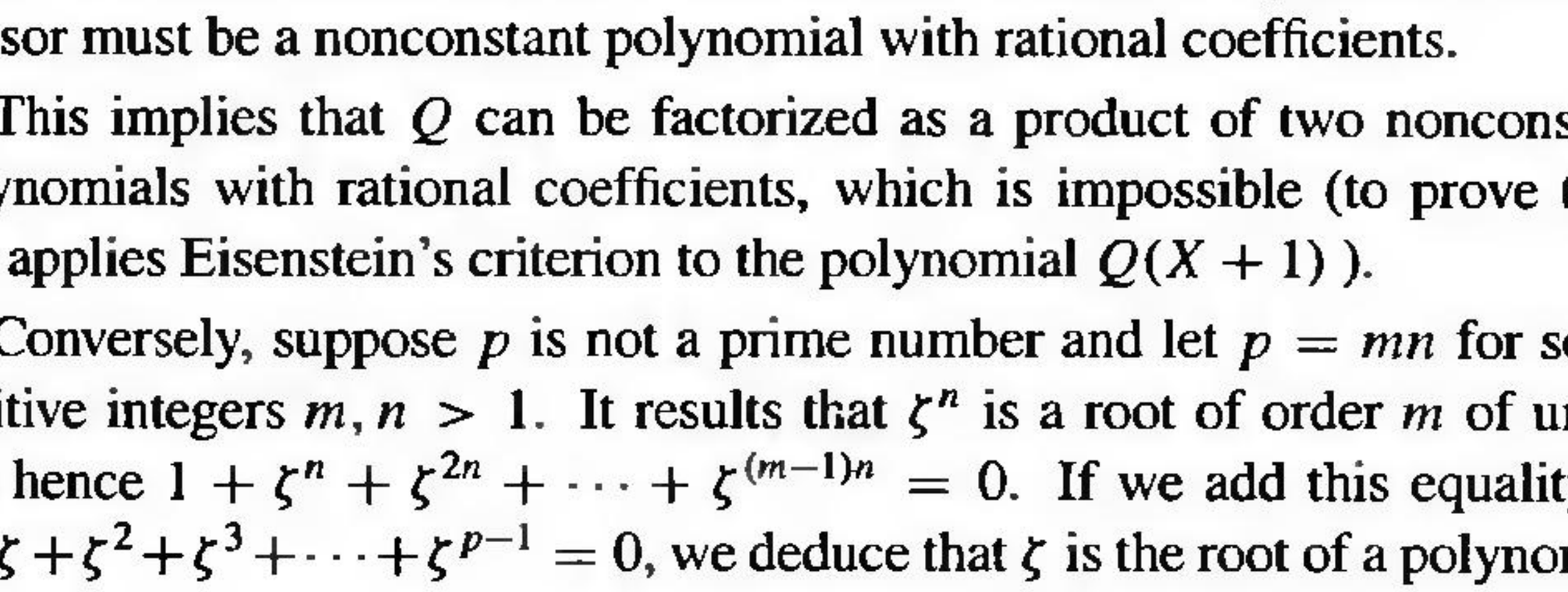


Figure 5.31

of degree $p-1$ in which some coefficients are equal to 1 and others are equal to 2. This means that there exists an equiangular polygon with $p$ sides, some of length 1 and the rest of length 2. Because such a polygon is not regular, our claim is proved.

*Observation.* We can examine, for instance, the case $p = 6 = 2 \cdot 3$. If

$$\zeta = \cos\frac{\pi}{3} + i \sin\frac{\pi}{3},$$

then

$$1 + \zeta^2 + \zeta^4 = 0$$

and

$$1 + \zeta + \zeta^2 + \zeta^3 + \zeta^4 + \zeta^5 = 0,$$

so that $\zeta$ is a root of the polynomial

$$2 + X + 2X^2 + X^3 + 2X^4 + X^5.$$

This means that there exists an equiangular hexagon with side lengths 2, 1, 2, 1, 2, 1, as can be seen in Figure 5.32.

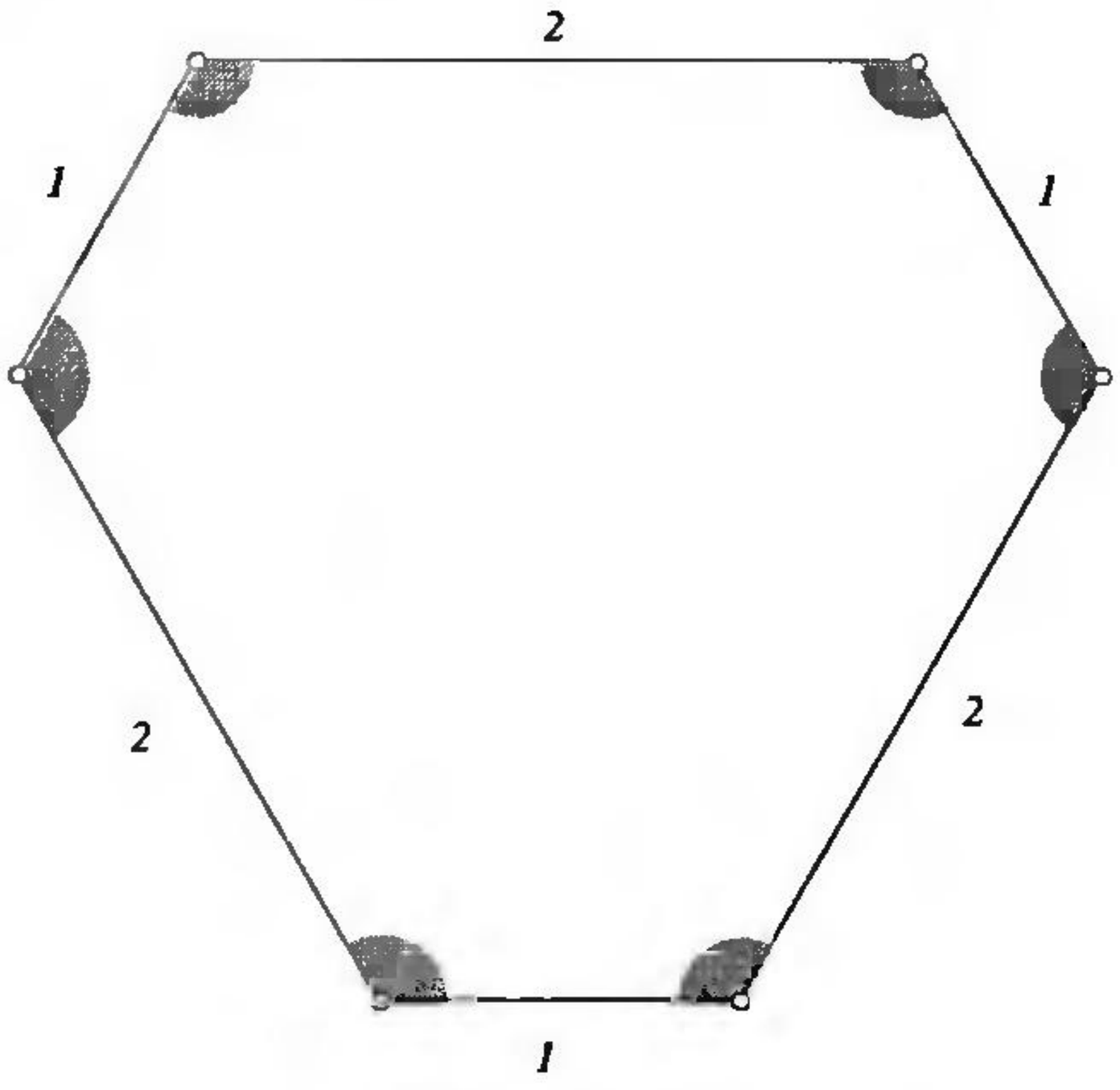


Figure 5.32

**Problem 6.** An equiangular polygon with an odd number of sides is inscribed in a circle. Prove that the polygon is regular.

**Solution.** Let $A_0A_1 \ldots A_{2n}$ be the polygon. (Figure 5.33.) Then the triangles $A_0A_1A_2$ and $A_1A_2A_3$ are congruent. Indeed, $A_1A_2$ is a common side; the angles $\angle A_0A_1A_2$ and $\angle A_1A_2A_3$ are congruent since the polygon is equiangular; and the angles $\angle A_1A_0A_2$ and $\angle A_1A_3A_2$ are also congruent (their measure is $\frac{1}{2}\overset{\frown}{A_1A_2}$). It follows that $A_0A_1 = A_2A_3$.

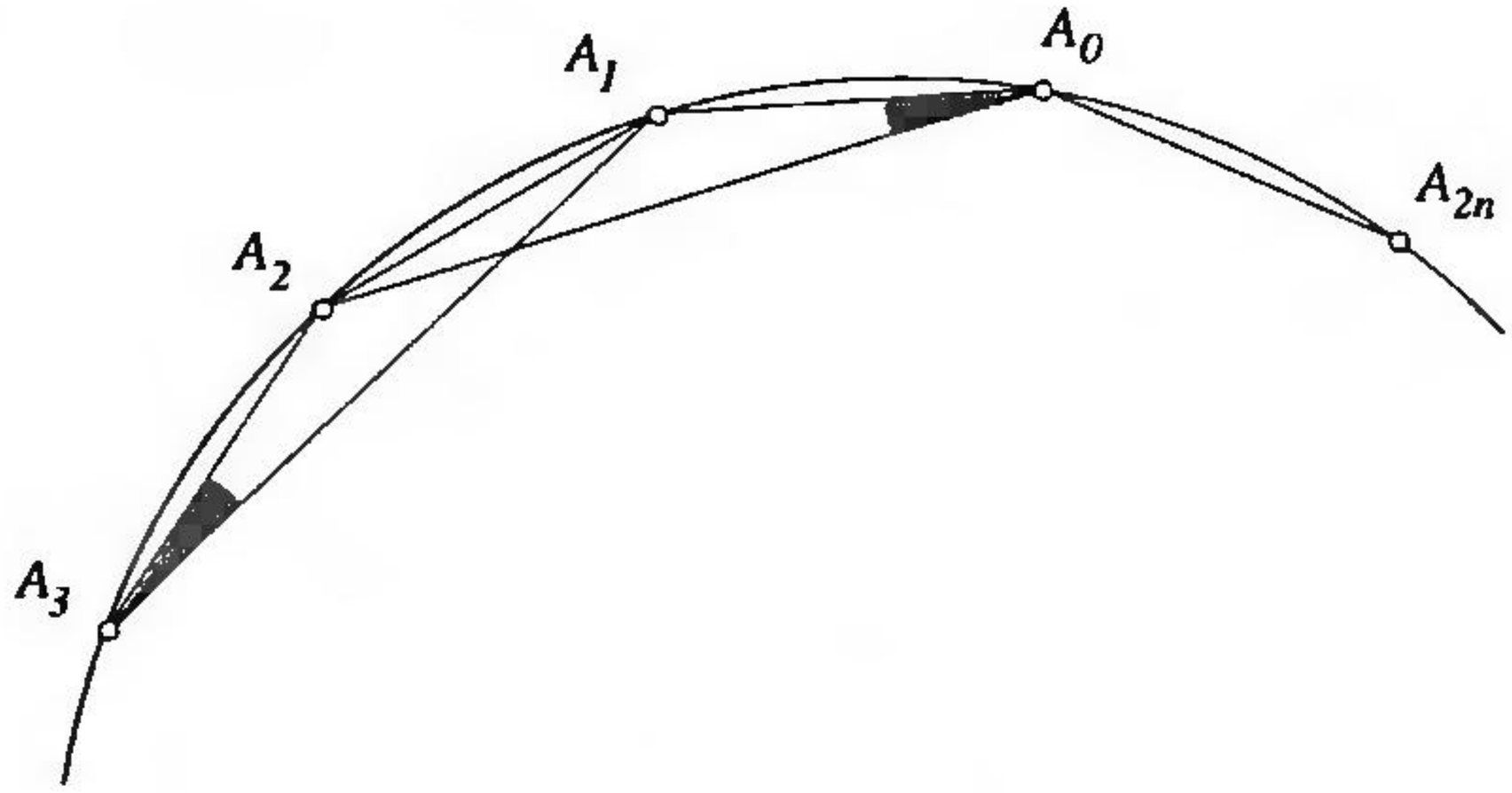


Figure 5.33

In the same way we obtain

$$A_2A_3 = A_4A_5 = \cdots = A_{2n-2}A_{2n-1} = A_{2n}A_0 = A_1A_2 = \cdots = A_{2n-1}A_{2n},$$

and hence the polygon is regular.

**Problem 7.** Let $a_1, a_2, \ldots, a_n$ be the side lengths of an equiangular polygon. Prove that if $a_1 \geq a_2 \geq \cdots \geq a_n$, then the polygon is regular.

**Solution.** The first approach is geometrical. We examine two cases: $n$ odd and $n$ even. If $n$ is odd, say $n = 2k + 1$, consider the angle bisector of $\angle A_{2k+1}A_1A_2$. It is not difficult to see that it is perpendicular to the side $A_{k+1}A_{k+2}$. Project all the sides of the polygon onto this line. If we denote by $x_i$ the length of the projection of the side $A_iA_{i+1}$ (with the usual convention $A_{2k+2} = A_1$), then

$$x_1 + x_2 + \cdots + x_k = x_{k+2} + x_{k+3} + \cdots + x_{2k+1} = A_1B$$

(see Figure 5.34). On the other hand, the angle between $A_iA_{i+1}$ and $A_1B$ is equal to the angle between $A_{2k+2-i}A_{2k+3-i}$ and $A_1B$, thus $x_i \geq x_{2k+2-i}$ for

all $1 \leq i \leq k$. It follows that the above equality can be reached only if the sides of the polygon are equal.

If $n$ is even, there is a similar argument, but instead of the angle bisector of $\angle A_{2k+1}A_1A_2$, consider the perpendicular to the sides $A_1A_2$ and $A_{k+1}A_{k+2}$ (it is easy to see that these two sides are in this case parallel).

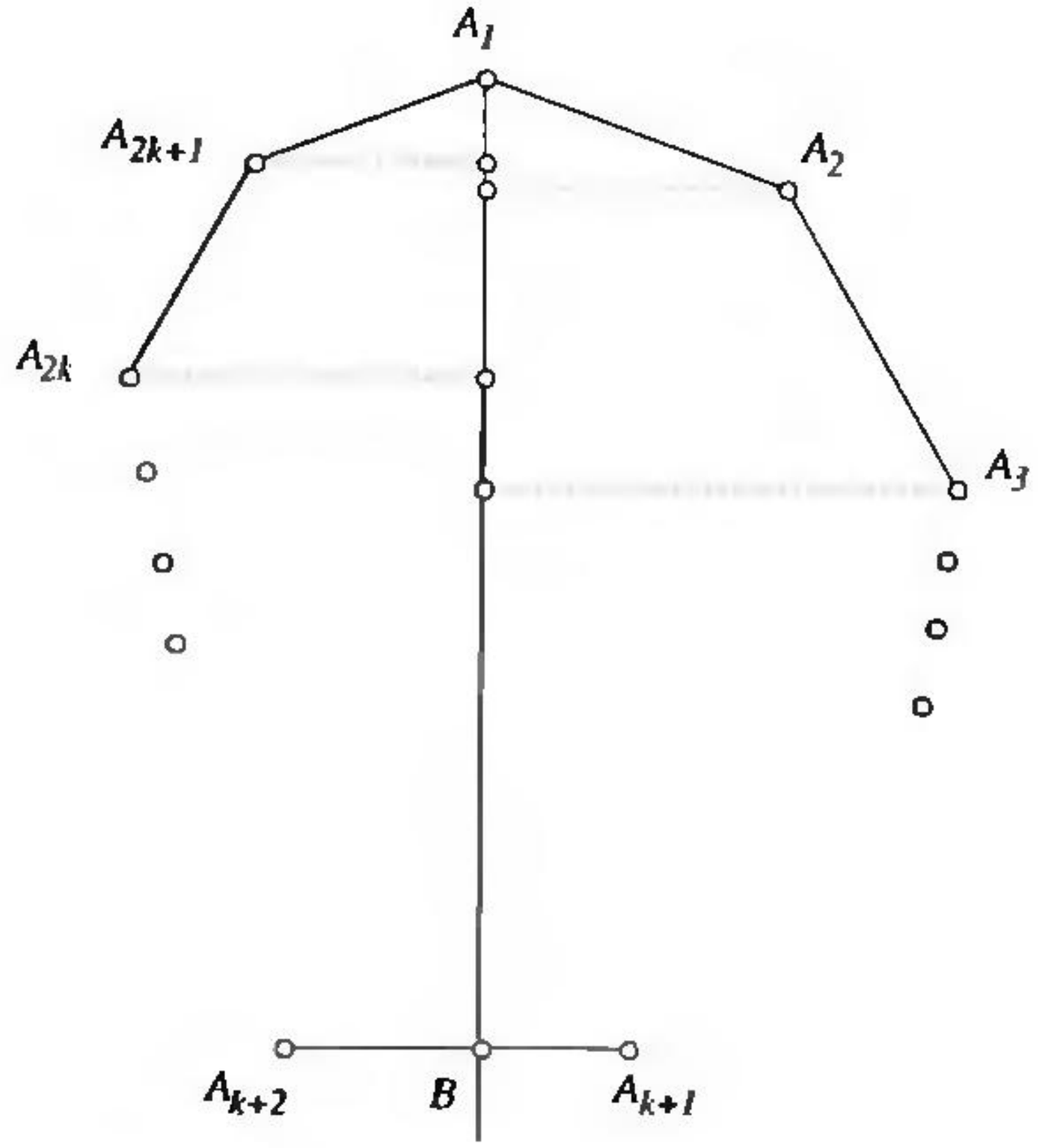


Figure 5.34

The second approach is algebraic. Let

$$\varepsilon = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$$

be a primitive root of unity. Then $\varepsilon$ is a root of the polynomial

$$P(X) = a_1 + a_2X + \cdots + a_nX^{n-1}.$$

The conclusion is obtained from the following:

*Lemma.* Let $P(X) = a_1 + a_2X + \cdots + a_nX^{n-1}$, where $a_1 \geq a_2 \geq \cdots \geq a_n > 0$. If $\alpha$ is a root of $P$, then $|\alpha| \geq 1$, and $|\alpha| = 1$ only if $a_1 = a_2 = \cdots = a_n$.

*Proof.* We have

$$a_1 + a_2\alpha + \cdots + a_n\alpha^{n-1} = 0.$$

If we multiply this equality with $\alpha - 1$, we obtain

$$-a_1 + \alpha(a_1 - a_2) + \alpha^2(a_2 - a_3) + \cdots + \alpha^{n-1}(a_{n-1} - a_n) + a_n\alpha^n = 0,$$

or, equivalently,

$$a_1 = \alpha(a_1 - a_2) + \alpha^2(a_2 - a_3) + \cdots + \alpha^{n-1}(a_{n-1} - a_n) + a_n\alpha^n.$$

Now, suppose that $|\alpha| \leq 1$. We obtain

$$\begin{aligned} a_1 &= |\alpha(a_1 - a_2) + \alpha^2(a_2 - a_3) + \cdots + \alpha^{n-1}(a_{n-1} - a_n) + a_n\alpha^n| \\ &\leq |\alpha|(a_1 - a_2) + |\alpha|^2(a_2 - a_3) + \cdots + |\alpha|^{n-1}(a_{n-1} - a_n) + a_n|\alpha|^n \\ &\leq (a_1 - a_2) + (a_2 - a_3) + \cdots + (a_{n-1} - a_n) + a_n = a_1. \end{aligned}$$

Consequently, all inequalities must be equalities. Because $\alpha \notin \mathbf{R}$, this is possible only if $a_1 = a_2 = \cdots = a_n$; that is, if the polygon is regular.

**Problem 8.** The side lengths of an equiangular octagon are rational numbers. Prove that the octagon has a symmetry center.

**Solution.** The angles of an equiangular octagon are equal to 135°, thus, the lines containing the segments $AB$, $CD$, $EF$ and $GH$ determine a rectangle.

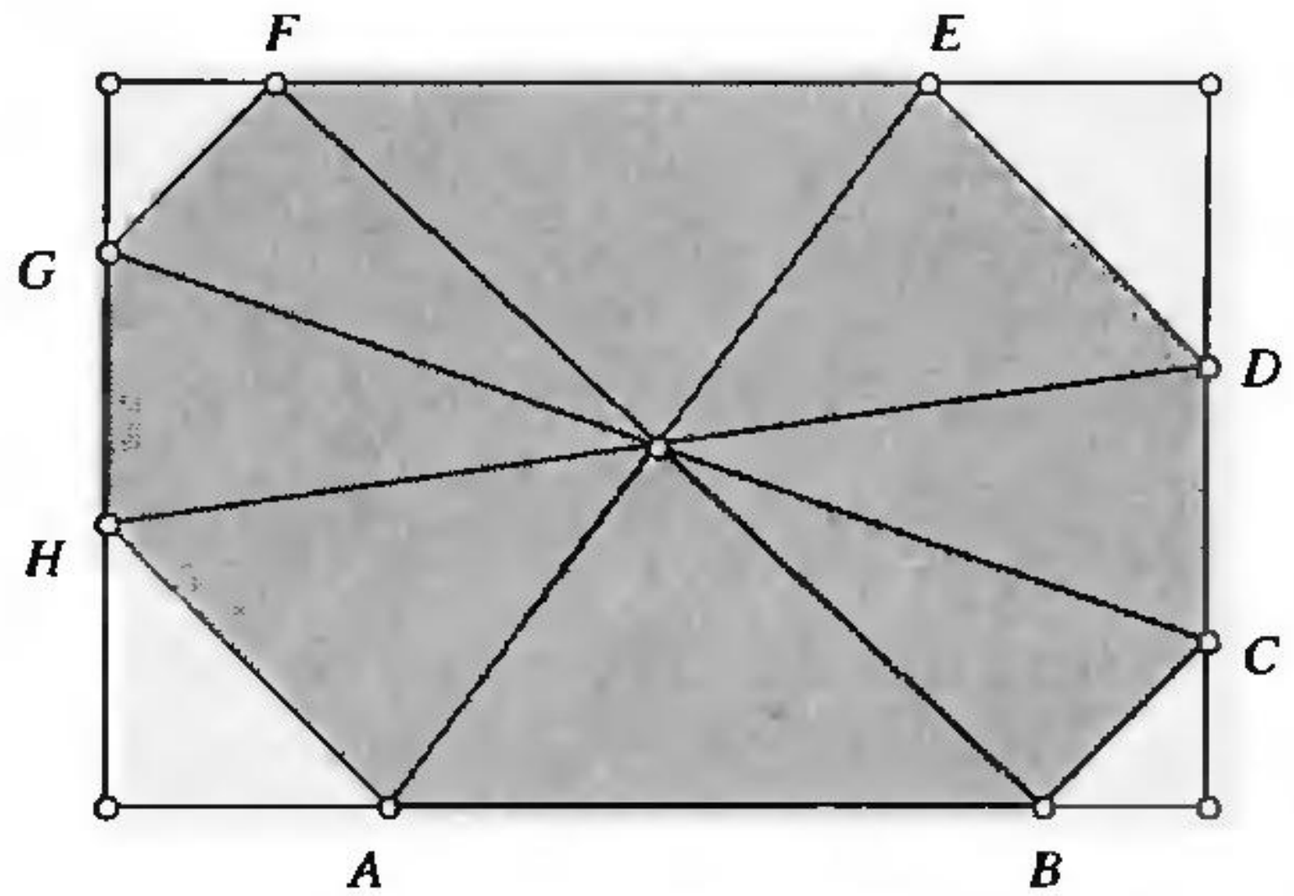


Figure 5.35

Because the opposite sides of this rectangle are equal, we obtain

$$AB + \frac{\sqrt{2}}{2}(AH + BC) = EF + \frac{\sqrt{2}}{2}(DE + FG),$$

or, equivalently,

$$AB - EF = \frac{\sqrt{2}}{2}(DE + FG - AH - BC).$$

Because the side lengths of the octagon are rational numbers, the above equality can hold if and only if

$$AB - EF = DE + FG - AH - BC = 0.$$

In a similar way, we obtain

$$CD - GH = FG + AH - DE - BC = 0.$$

From these equalities, it follows that $AB = EF$, $CD = GH$, $BC = FG$, and $DE = AH$, so the opposite sides of the octagon are equal and parallel. It follows that the quadrilaterals $ABEF$, $BCFG$, $CDGH$, and $DEHA$ are parallelograms, and therefore the midpoints of the segments $AE, BF, CG, DH$ and $DH$ coincide. Obviously, this common point is a symmetry center of the octagon.

**Problem 9.** A convex hexagon is given in which any two opposite sides have the following property: the distance between their midpoints is $\sqrt{3}/2$ times the sum of their lengths. Prove that the hexagon is equiangular.

**Solution.** Let $ABCDEF$ be the hexagon and let $M$ and $N$ be the midpoints of $AB$ and $DE$. Consider the points $M'$ and $M''$ such that the quadrilaterals $MBEM'$ and $MADM''$ are parallelograms (see Figure 5.36).

It follows that the segments $DE$ and $M'M''$ have the same midpoint. Moreover,

$$M'M'' \leq M'E + ED + DM'' = AB + DE,$$

and the equality holds only if $AB$ is parallel to $DE$.

Thus, in the triangle $MM'M''$, the median $MN$ satisfies the inequality:

$$MN \geq \frac{\sqrt{3}}{2}M'M''.$$

It is not difficult to derive that $\angle M'MM'' \leq 60°$; thus, the angle between the diagonals $AD$ and $BE$ is less than or equal to 60°. In a similar way we obtain that the other two angles between the long diagonals of the hexagon are less than or equal to 60°. This is possible only if all these angles equal

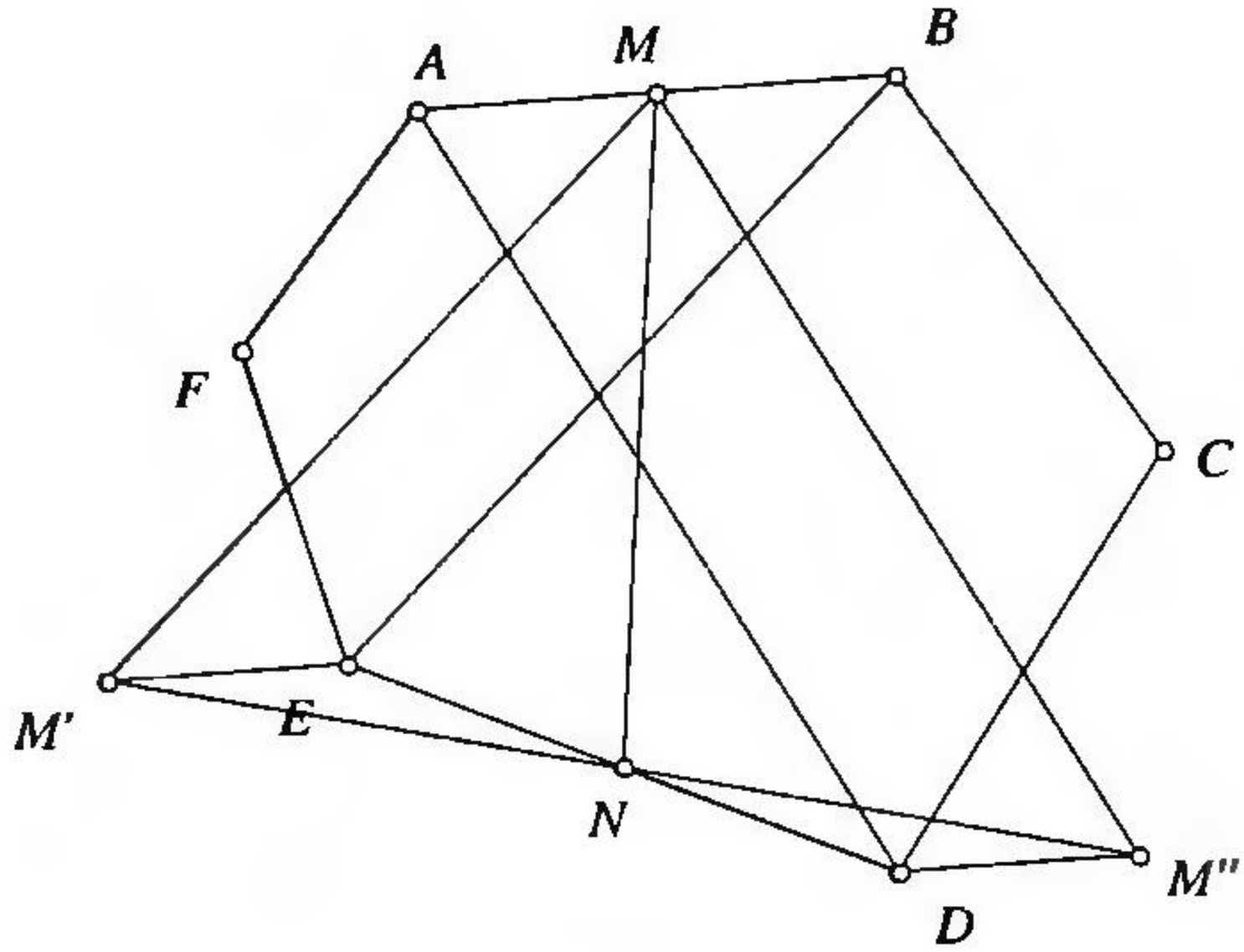


Figure 5.36

60°, and hence the previous inequalities become equalities. Consequently, the opposite sides of the hexagon are parallel. It follows that the segments $AD$, $BE$, and $MN$ are concurrent. Let $X$ be their common point.

In the triangle $AXB$, we have $\angle AXB = 60°$. This implies that $MX \leq \frac{\sqrt{3}}{2}AB$, with equality if and only if the triangle is equilateral. Similarly, $NX \leq \frac{\sqrt{3}}{2}BD$; but since $MN = MX + NX = \frac{\sqrt{3}}{2}(AB + DE)$ we conclude that the triangles $ABX$ and $BDX$ are both equilateral triangles. Applying the same argument for the other two pairs of long diagonals of the hexagon we obtain that all the hexagon's angles equal 120°.

*Observation.* There are several other ways to prove that the angles between the hexagon's long diagonals equal 60°. For instance, suppose that the angle between $AD$ and $BE$ is greater than 60°. Then the point $X = AD \cap BE$ must lie in the interior of the circumcircles of the equilateral triangles $ABR$ and $DES$ (see Figure 5.38); and hence, the distance between their centers is less than the sum of their radii.

It follows that $MN \leq MO + OO' + O'N < \frac{\sqrt{3}}{2}(AB + DE)$, a contradiction.

**Problem 10.** Prove that the segments joining the midpoints of the opposite sides of an equiangular hexagon are concurrent.

**Solution.** We will prove a more general statement: In a hexagon with parallel opposite sides, the segments joining the midpoints of the opposite sides are concurrent. Clearly, the opposite sides of an equiangular hexagon are parallel.

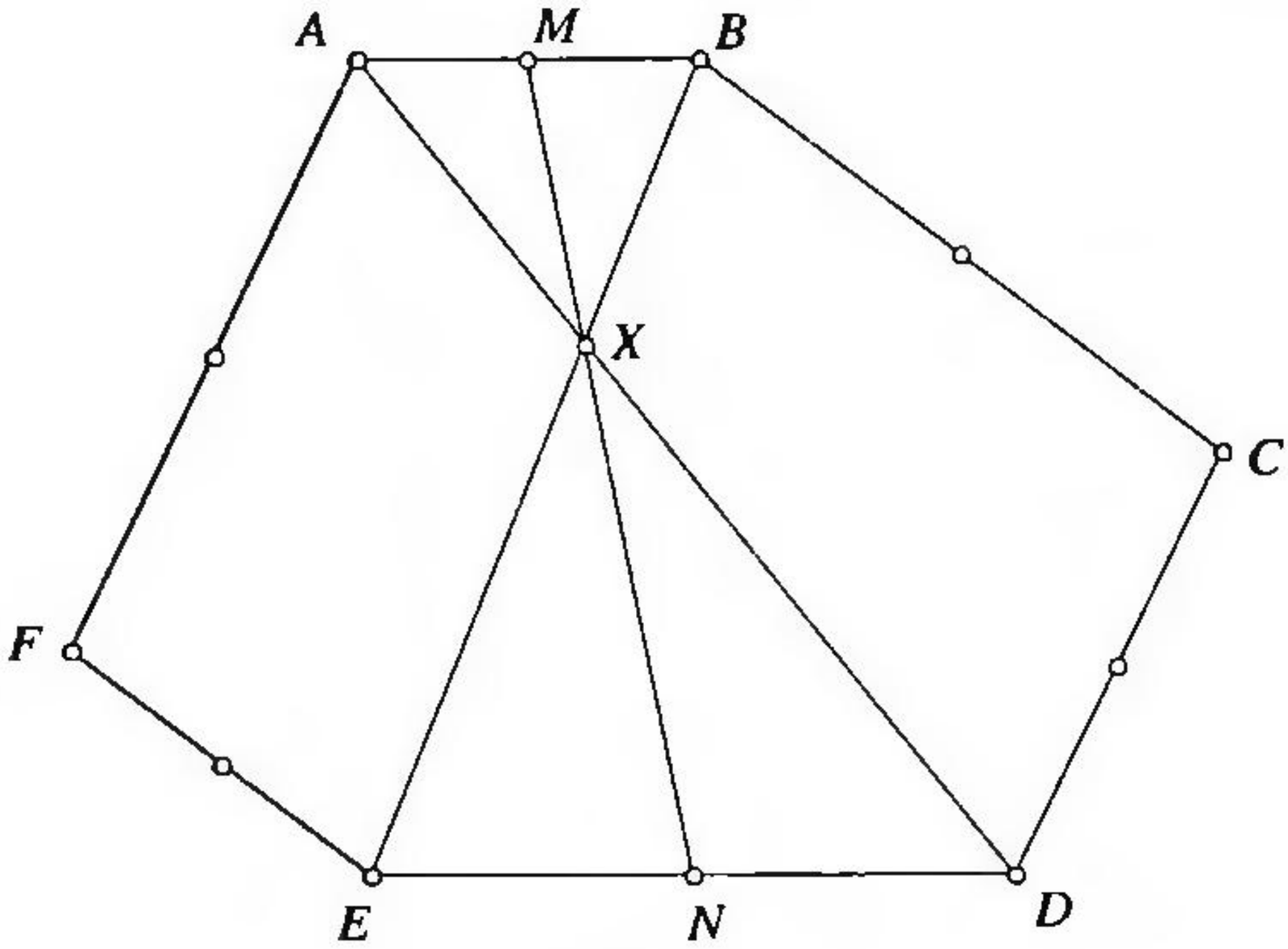


Figure 5.37

Let $ABCDEF$ be a hexagon with parallel opposite sides. If two opposite sides have equal lengths, say $AB = DE$, it is not difficult to see that the point $O$ of intersection between $AD$ and $BE$ is a symmetry center of the hexagon and all three segments joining the midpoints of the opposite sides pass through $O$.

Now, suppose that $AB \neq DE$, $BC \neq EF$, and $CD \neq FA$. Let $M$ and $N$ be the midpoints of $AB$ and $DE$, $T = AE \cap BD$ and $Z = AD \cap BE$ (see Figure 5.39). From Problem 2 of Section 2.2, it follows that the points $M$, $N$, $T$, and $Z$ are collinear. We will obtain the desired conclusion applying Ceva's theorem in the triangle $XYZ$.

We have $[AMZ] = [BMZ]$, hence

$$\tfrac{1}{2}AZ \cdot MZ \cdot \sin \angle AZM = \tfrac{1}{2}BZ \cdot MZ \cdot \sin \angle BZM.$$

We obtain

$$\frac{\sin \angle AZM}{\sin \angle BZM} = \frac{BZ}{AZ}.$$

Similarly,

$$\frac{\sin \angle DZN}{\sin \angle EZN} = \frac{EZ}{DZ}.$$

Since $\angle AZM = \angle DZN$ and $\angle BZM = \angle EZN$, it follows that

$$\frac{\sin \angle DZN}{\sin \angle EZN} = \frac{BZ}{AZ} = \frac{EZ}{DZ} = \frac{BZ + EZ}{AZ + DZ} = \frac{BE}{AD}.$$

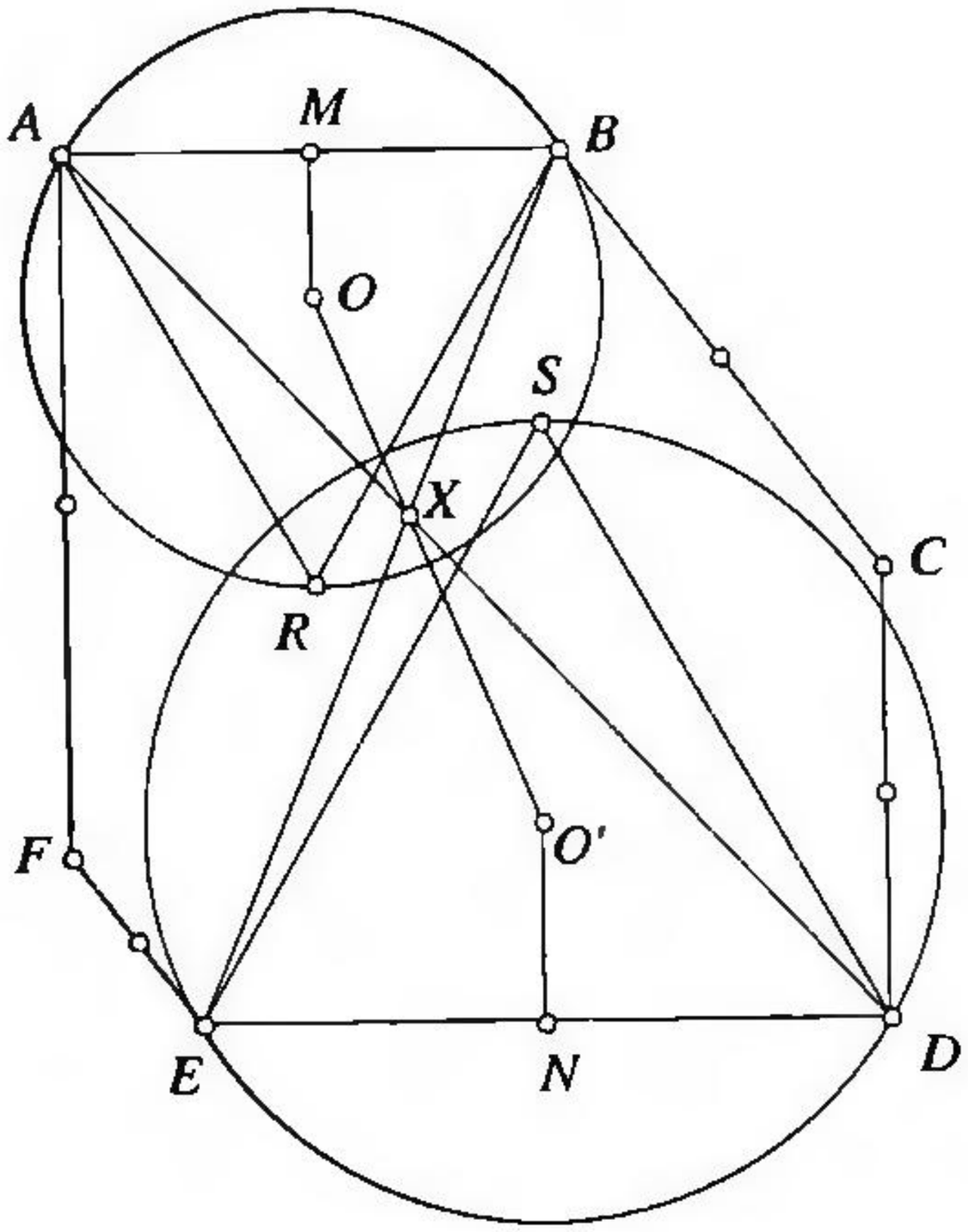


Figure 5.38

Thus, in the triangle $XYZ$ we have

$$\frac{\sin \angle YZZ'}{\sin \angle XZZ'} = \frac{BE}{AD}.$$

In a similar way we obtain

$$\frac{\sin \angle ZXX'}{\sin \angle YXX'} = \frac{CF}{BE}$$

and

$$\frac{\sin \angle XYY'}{\sin \angle ZYY'} = \frac{AD}{CF}.$$

The conclusion follows from the trigonometric form of Ceva's theorem, by multiplying these equalities.

## 5.5 More on equilateral triangles

**Problem 3.** Let $M$ be a point in the interior of the equilateral triangle $ABC$ and let $A'$, $B'$, and $C'$ be its projections onto the sides $BC$, $CA$, and $AB$, respectively. Prove that the sum of lengths of the inradii of triangles $MAC'$,

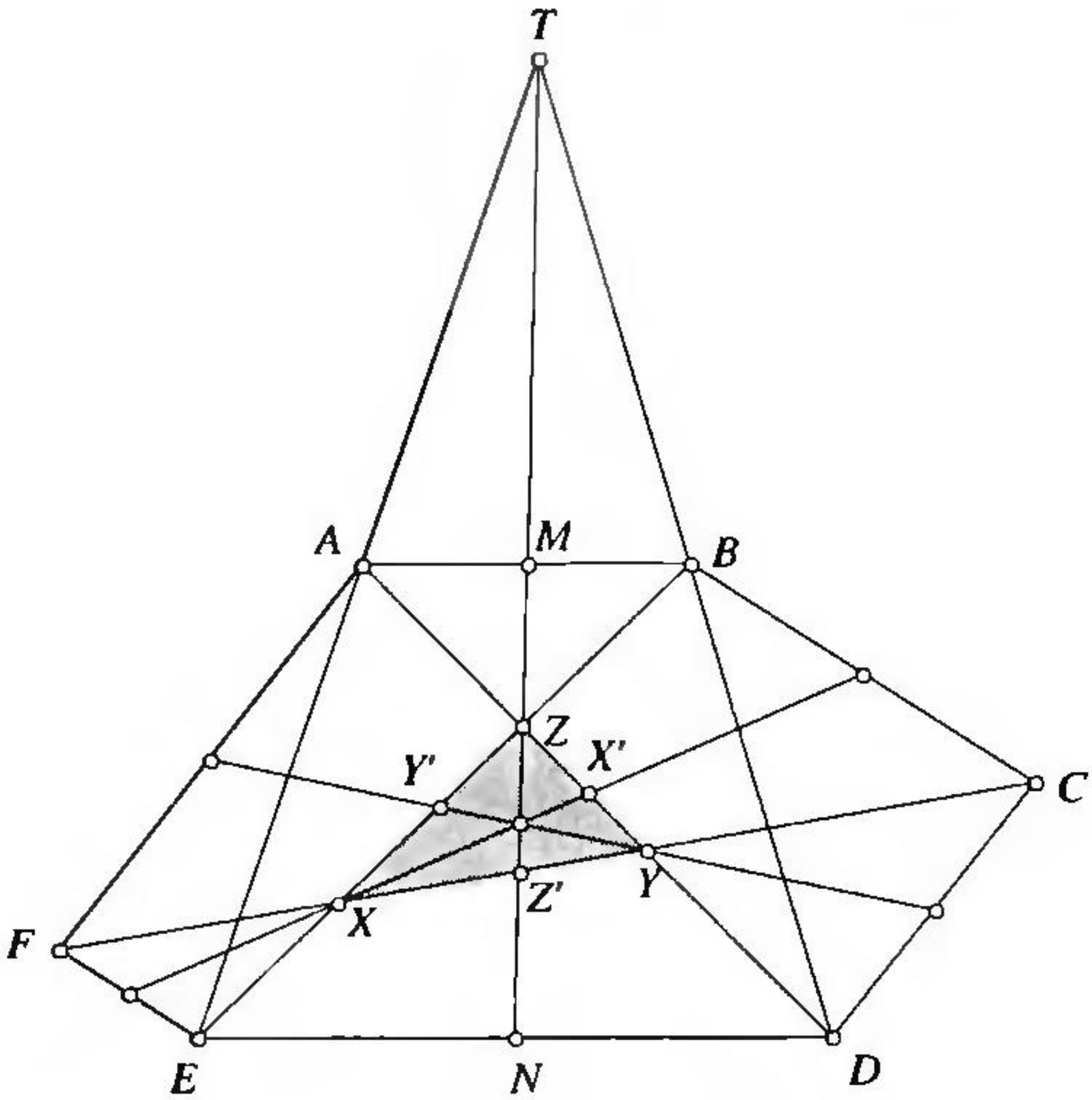


Figure 5.39

$MBA'$ and $MCB'$ equals the sum of lengths of the inradii of triangles $MAB'$, $MBC'$ and $MCA'$.

**Solution.** We start by proving an additional result.

*Lemma.* Let $XYZ$ be a right triangle, with $\angle X = 90^\circ$ and side lengths $x, y, z$, respectively. If $r$ is the length of the inradius of $XYZ$, then $r = \frac{1}{2}(y + z - x)$.

*Proof.* Let $X', Y', Z'$ be the tangency points of the incircle and the sides of the triangle. (Figure 5.40.) Then $r = XY' = XZ'$. Denote by $s = ZY' = ZX'$ and $t = YX' = YZ'$.

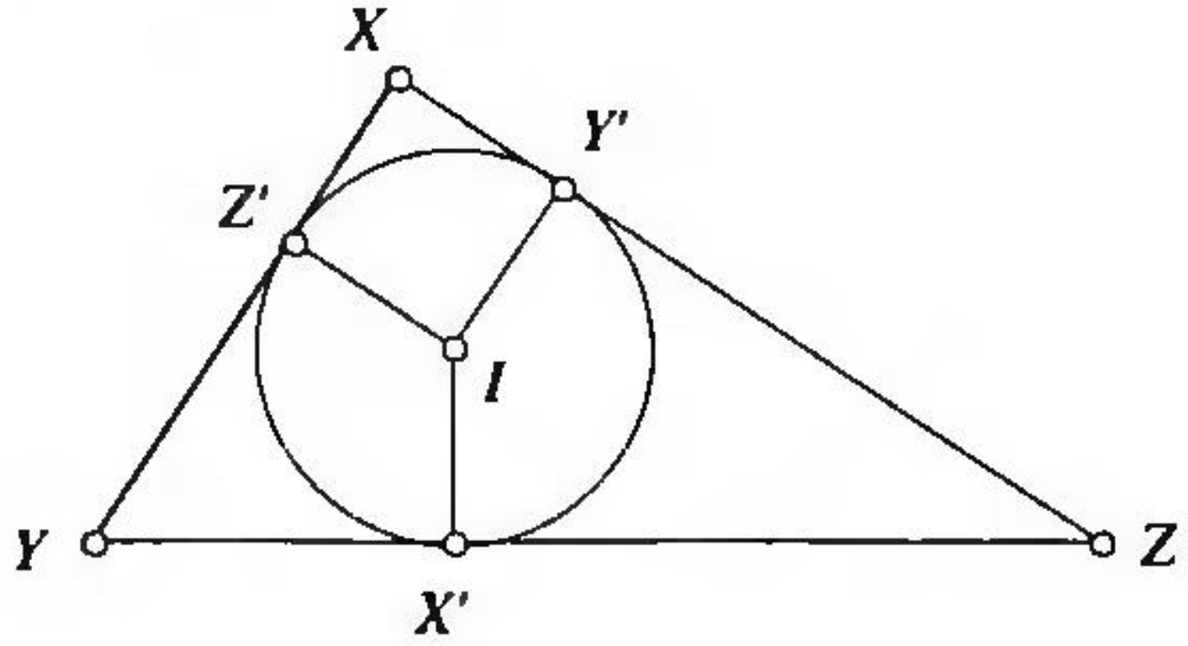


Figure 5.40

Then $r + s = y$, $r + t = z$ and $s + t = x$. Solving for $r$, we obtain $r = \frac{1}{2}(y + z - x)$.

Applying the lemma in our problem, we have to prove that

$$(AC' + MC' - MA) + (BA' + MA' - MB) + (CB' + MB' - MC)$$
$$= (AB' + MB' - MA) + (BC' + MC' - MB) + (CA' + MA' - MC),$$

which is equivalent to

$$AB' + BC' + CA' = AC' + BA' + CB'.$$

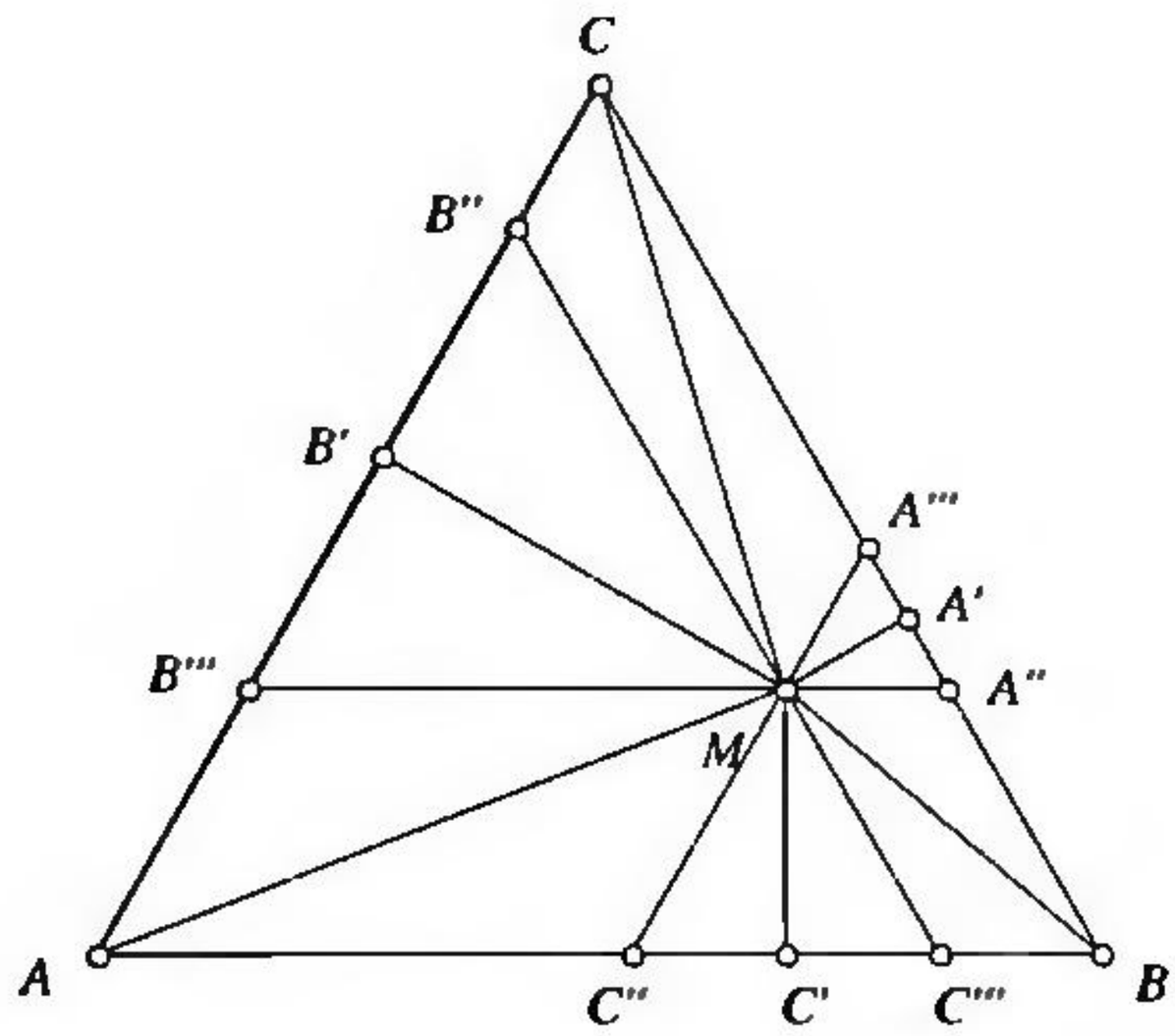


Figure 5.41

To prove this equality, we draw through $M$ parallels to the triangle's sides, as in Figure 5.41. Then $AB' = AB''' + BB'$, but $AB''' = BA''$ and $B'''B' = B'B''$. Writing the similar equalities, the result follows by adding them.

*Observation.* Another way to deal with the last part of the problem is to use the Pythagorean theorem. We have $A'M^2 = BM^2 - A'B^2 = CM^2 - A'C^2$, and hence $BM^2 - CM^2 = A'B^2 - A'C^2 = (A'B - A'C) \cdot a$, where $a$ is the side length of the triangle. We get the desired result by writing the other two similar equalities and adding them.

**Problem 4.** Let $I$ be the incenter of the triangle $ABC$. It is known that for every point $M \in (AB)$, one can find the points $N \in (BC)$ and $P \in (AC)$ such that $I$ is the centroid of triangle $MNP$. Prove that $ABC$ is an equilateral triangle.

**Solution.** Let $a, b, c$ be the lengths of the triangle's sides and let $m, n$, and $p$ be real numbers such that $\overline{AM} = m\overline{AB}$, $\overline{BN} = n\overline{BC}$, and $\overline{CP} = p\overline{CA}$. If $G$ and $G'$ are the centroids of the triangles $ABC$ and $MNP$, then

$$\begin{aligned}\overline{GG'} &= \tfrac{1}{3}\left(\overline{GM} + \overline{GN} + \overline{GP}\right)\\ &= \tfrac{1}{3}\left(\overline{GA} + \overline{AM} + \overline{GB} + \overline{BN} + \overline{GC} + \overline{CP}\right)\\ &= \tfrac{1}{3}\left(\overline{AM} + \overline{BN} + \overline{CP}\right) = \tfrac{1}{3}\left(m\overline{AB} + n\overline{BC} + p\overline{CA}\right)\\ &= \frac{m-p}{3}\overline{AB} + \frac{n-p}{3}\overline{BC}.\end{aligned}$$

Because $I$ is the incenter of $ABC$, we have

$$\overline{GI} = \frac{a\overline{GA} + b\overline{GB} + c\overline{GC}}{a+b+c} = \frac{b+c-2a}{3(a+b+c)}\overline{AB} + \frac{2c-a-b}{3(a+b+c)}\overline{BC}.$$

It results that $G' = I$ if and only if

$$\begin{cases} m - p = \dfrac{b+c-2a}{a+b+c} \\ n - p = \dfrac{2c-a-b}{a+b+c} \end{cases}$$

or, equivalently,

$$\begin{cases} n = m + 1 - \dfrac{3b}{a+b+c} \\ p = m - 1 + \dfrac{3a}{a+b+c} \end{cases}. \qquad (*)$$

Therefore, for any point $M \in AB$, one can find points $N \in BC$ and $P \in CA$ such that the centroid of $MNP$ is the point $I$. The problem is that $N$ and $P$ must lie in the interior of the segments $BC$ and $CA$, respectively. This is equivalent to the following: for every $m \in (0, 1)$, the numbers $n$ and $p$ given by (*) also belong to the interval (0,1). It is not difficult to see that this happens if and only if

$$1 - \frac{3b}{a+b+c} = -1 + \frac{3a}{a+b+c} = 0,$$

and we deduce that $a = b = c$.

**Problem 5.** Let $ABC$ be an acute triangle. The interior bisectors of the angles $\angle B$ and $\angle C$ meet the opposite sides at the points $L$ and $M$, respectively. Prove that there exists a point $K$ in the interior of the side $BC$ such that the triangle $KLM$ is equilateral if and only if $\angle A = 60°$.

**Solution.** Let us first notice that if $M$ is a point on the bisector of the angle $XOY$ and if $P \in OX$ and $Q \in OY$ with $MP = MQ$, then either $\angle OPM = \angle OQM$, or $\angle OPM + \angle OQM = 180°$ (see Figure 5.42).

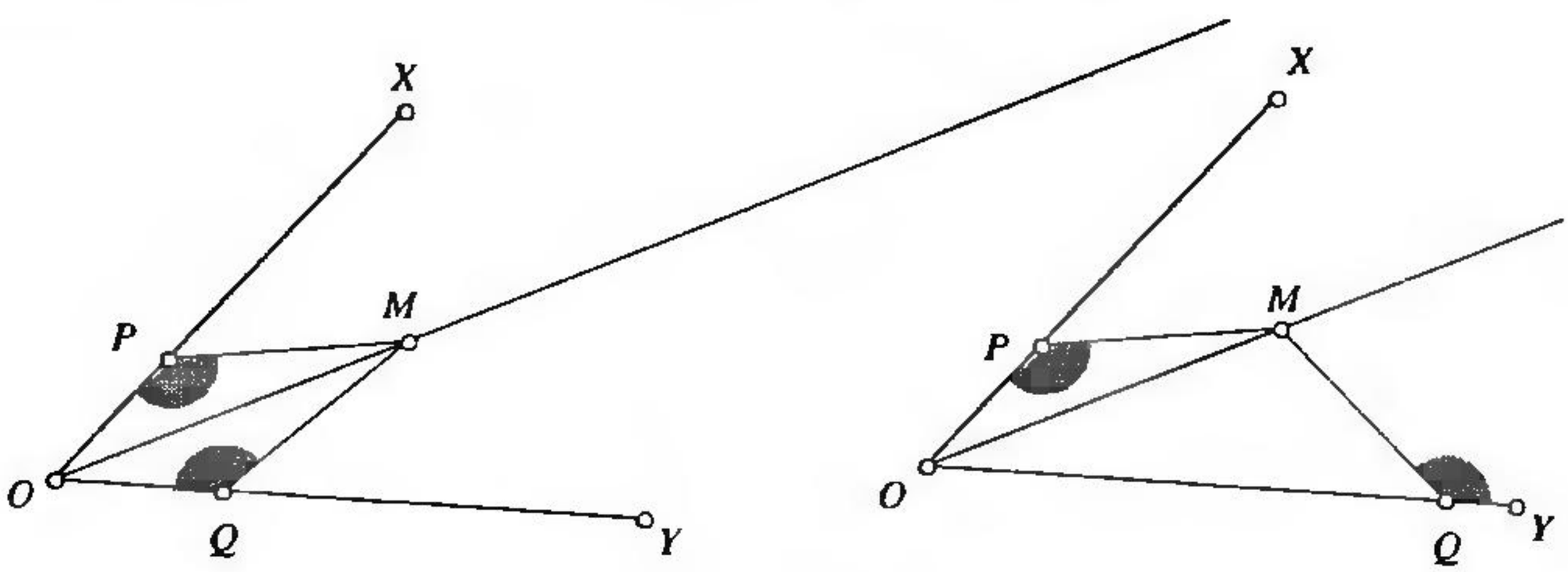


Figure 5.42

Returning to the problem, let us suppose that such a point $K$ exists. (Figure 5.43.) Then $M$ lies on the bisector of $\angle ACB$ and $ML = MK$, so either $\angle ALM = \angle BKM$ or $\angle ALM + \angle BKM = 180°$. But in the latter case, it follows that the quadrilateral $MKCL$ is cyclic, so $\angle C = 180° - \angle KML = 120°$, which is a contradiction since triangle $ABC$ is acute.

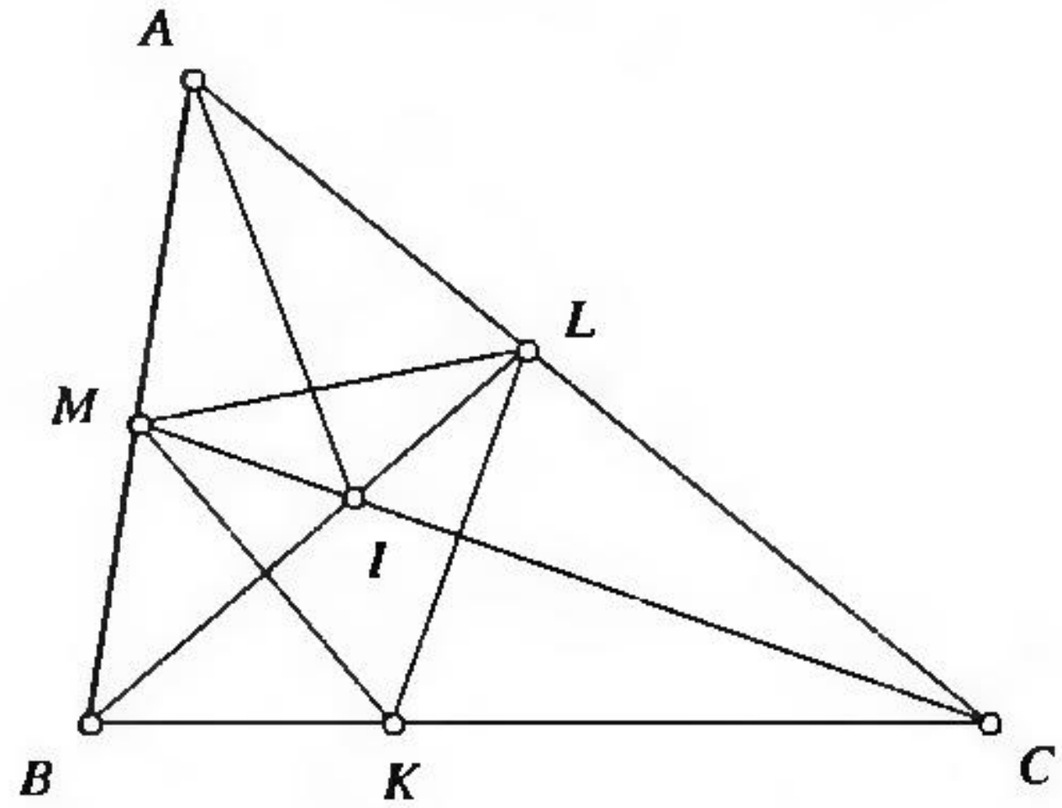


Figure 5.43

It follows that $\angle ALM = \angle BKM$, and, similarly, $\angle LKC = \angle LMA$. In the triangle $AML$, we have

$$\angle AML + \angle ALM + \angle A = 180°.$$

On the other hand,

$$\angle BKM + \angle LKC + \angle MKL = 180^\circ.$$

We derive that $\angle A = \angle MKL = 60^\circ$.

Conversely, suppose $\angle A = 60^\circ$. Let $K \in BC$ such that $MK \perp BL$. Because $BL$ is the bisector of $\angle B$, it follows that $BL$ is the perpendicular bisector of $MK$, so $LM = LK$.

Denote by $I$ the intersection point of $BL$ and $CM$. Clearly, $AI$ bisects $\angle A$ and a short computation shows that $\angle MIL = 120^\circ$, and therefore the quadrilateral $AMIL$ is cyclic. Then $\angle MLI = \angle MAI = 30^\circ$, so we have $\angle KLM = 60^\circ$. It follows that $KLM$ is an isosceles triangle with a $60^\circ$ angle; hence it is equilateral.

**Problem 6.** Let $P_1P_2\ldots P_n$ be a convex polygon with the following property: for any two vertices $P_i$ and $P_j$, there exists a vertex $P_k$ such that the segment $P_iP_j$ is seen from $P_k$ under an angle of $60^\circ$. Prove that the polygon is an equilateral triangle.

**Solution.** Consider the vertices $P_i$ and $P_j$ such that the segment $P_iP_j$ has minimal length and $P_k$ such that $\angle P_iP_kP_j = 60^\circ$. We have $P_iP_k \geq P_iP_j$ and $P_jP_k \geq P_iP_j$, so $\angle P_iP_jP_k \geq 60^\circ$ and $\angle P_jP_iP_k \geq 60^\circ$. It follows that triangle $P_iP_kP_j$ is equilateral. Denote it by $ABC$. Similarly, let $A'B'C'$ be the equilateral triangle with sides of maximal length. Because the polygon is convex, each of the points $A'$, $B'$, and $C'$ must lie in the interior of the set $D_A \cup D_B \cup D_C$ (see Figure 5.44) or coincide with either $A$, $B$ or $C$.

Suppose that the triangles $ABC$ and $A'B'C'$ have no common vertices. If at least two of the points $A'$, $B'$ or $C'$ lie in the same set $D_A$, $D_B$ or $D_C$ we get a contradiction. Indeed, assume that $A'$ and $B'$ lie in $D_C$. Then $\angle A'CB' < 60^\circ$ and hence $A'C > A'B'$ or $B'C > A'B'$, which is impossible, since $A'B'$ has maximal length. If the points $A'$, $B'$ and $C'$ lie in distinct sets we again reach a contradiction: since the polygon is convex, the points $A$, $B$ and $C$ must lie in the exterior of the triangle $A'B'C'$ (see Figure 5.45).

In the triangle $A'BC$, $BC$ has minimal length, so $\angle BA'C \leq 60^\circ$, and hence $\angle B'A'C' < \angle BA'C \leq 60^\circ$, a contradiction.

It follows that the triangles $ABC$ and $A'B'C'$ have at least one common vertex. Suppose that they have exactly one common vertex, say $A = A'$. The points $B'$ and $C'$ cannot both lie in $D_A$, so one of them must lie in either $D_B$ or $D_C$. Let us assume that $B' \in D_B$ (see Figure 5.46). Then $ABCB'$ is a convex quadrilateral and we have $AB' + BC < BB' + AC$, hence $A'B' = AB' < BB'$, a contradiction.

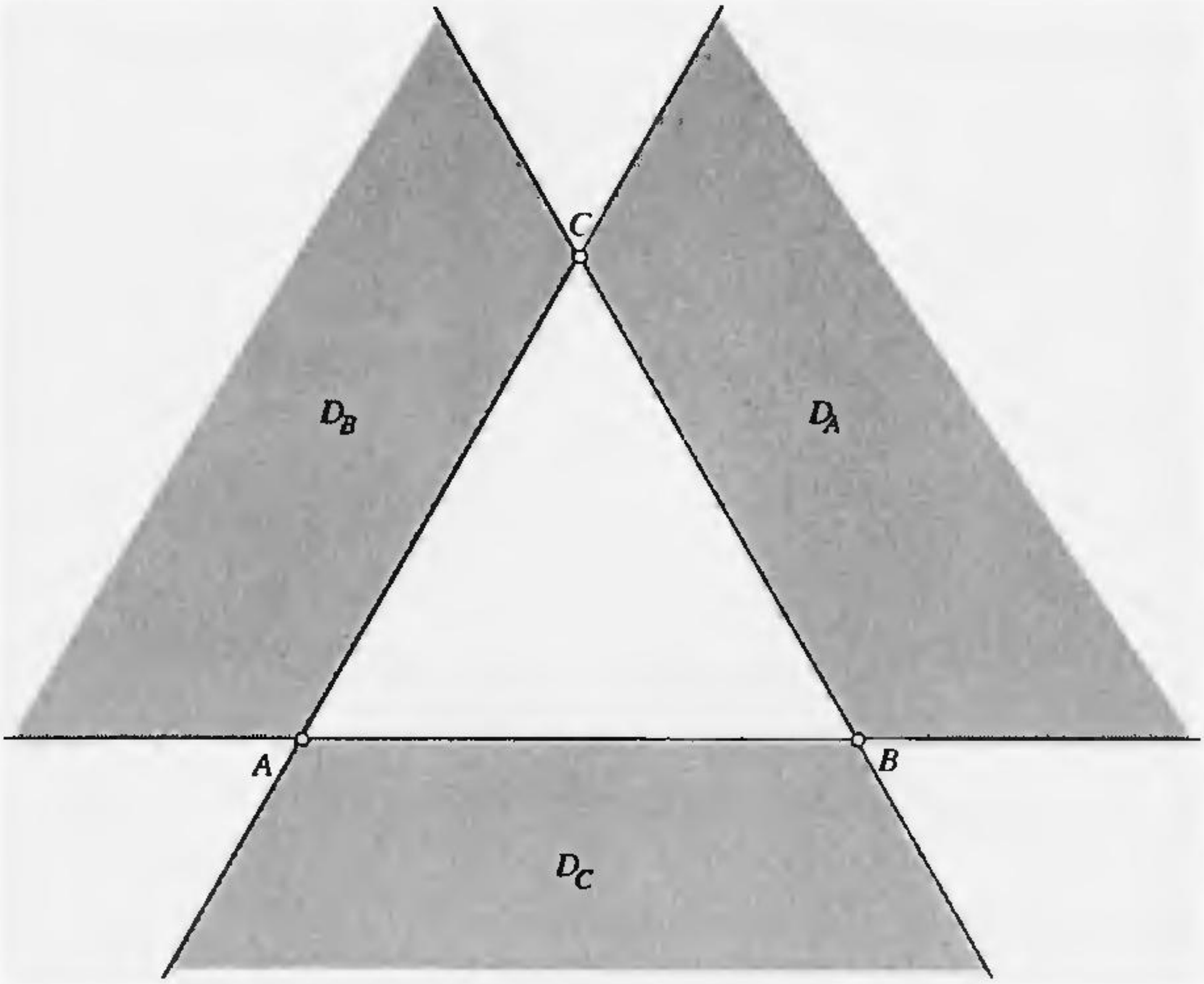


Figure 5.44

In conclusion, the triangles $ABC$ and $A'B'C'$ have at least two common vertices, so $AB = A'B'$. This implies that the distance between any two vertices of the polygon is the same. Clearly, this cannot hold unless the polygon is an equilateral triangle.

**Problem 7.** From a point on the circumcircle of an equilateral triangle $ABC$, parallels to the sides $BC, CA$, and $AB$ are drawn, intersecting the sides $CA$, $AB$, and $BC$ at the points $M$, $N$, $P$, respectively. Prove that the points $M$, $N$, and $P$ are collinear.

**Solution.** It suffices to prove that $\angle CPM = \angle BPN$ (see Figure 5.47).

Observe that the quadrilateral $CMDP$ is cyclic. Indeed, since $DP$ is parallel to $AB$, the angle $\angle DPC$ equals 120°. Hence

$$\angle DPC + \angle DMC = 120^\circ + 60^\circ = 180^\circ.$$

Consequently, $\angle CPM = \angle CDM$. In a similar way, we obtain $\angle BPN = \angle BDN$. But $\angle BDC = 120^\circ$ ($ABDC$ is cyclic) and $\angle MDN = 120^\circ$ ($AMDN$ is an isosceles trapezoid). It follows that $\angle CDM = \angle BDN$, and consequently, $\angle CPM = \angle BPN$.

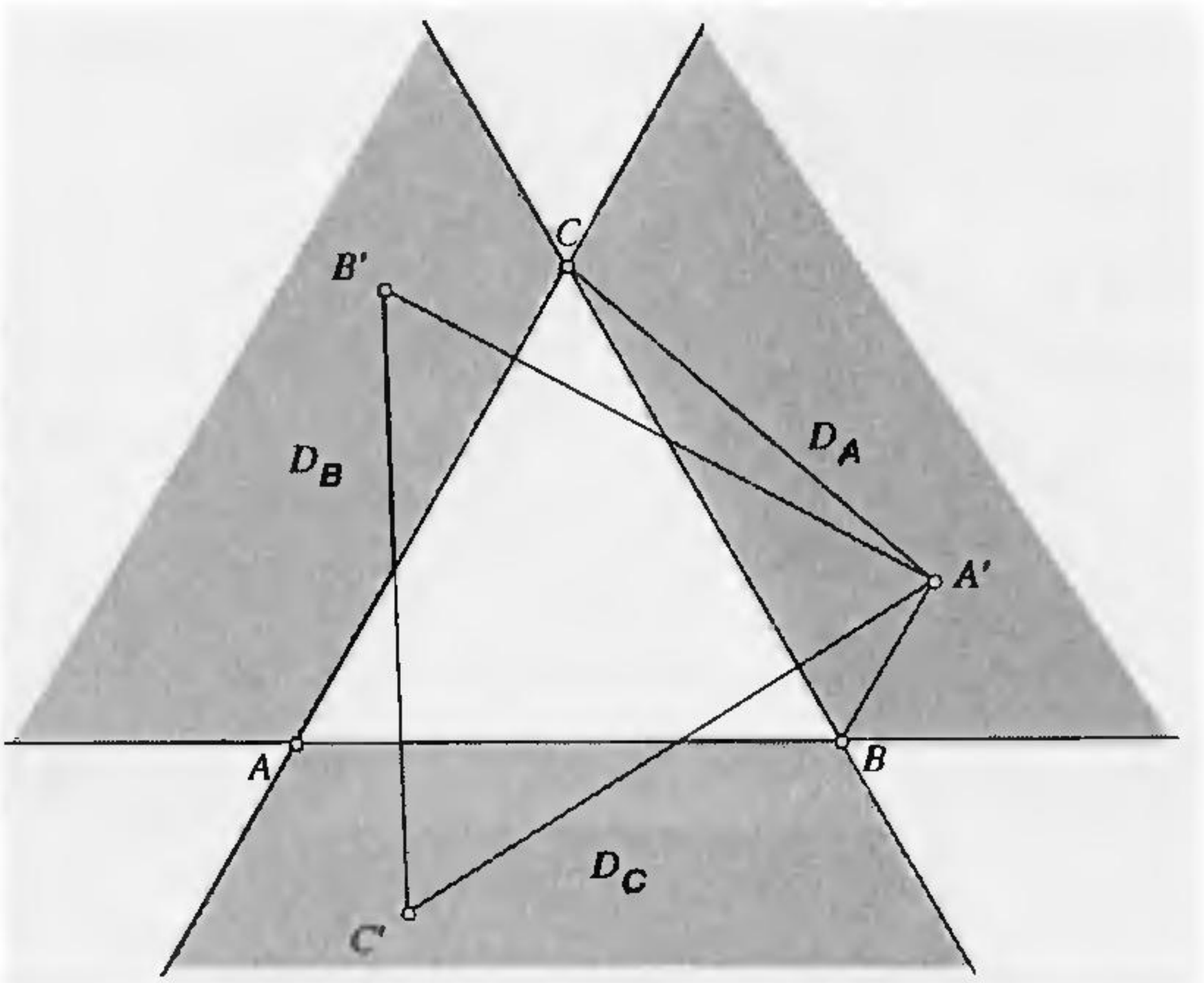


Figure 5.45

**Problem 8.** Let $P$ be a point on the circumcircle of the equilateral triangle $ABC$. Prove that the projections of any point $Q$ onto the lines $PA$, $PB$ and $PC$ are the vertices of an equilateral triangle.

**Solution.** Let $A'$, $B'$, and $C'$ be the projections of a point $Q$ onto the lines $PA$, $PB$, $PC$, respectively. If the points are located as in the Figure 5.48, then $\angle A'PB' = 120^\circ$ and $\angle A'PC' = \angle B'PC' = 60^\circ$.

From the construction, it follows that the quadrilaterals $A'PB'Q$, $A'PC'Q$, and $B'PC'Q$ are cyclic, so $\angle A'PQ = \angle A'B'Q$ and $\angle A'PQ = \angle A'C'Q$. Then $\angle A'B'Q = \angle A'C'Q$, and this shows that $A'B'C'Q$ is also cyclic. We see that

$$\angle A'B'C' = 180^\circ - \angle A'QC' = \angle A'PC' = 60^\circ,$$
$$\angle A'C'B' = \angle A'QB' = 180 - \angle A'PB' = 60^\circ,$$

and so the triangle $A'B'C'$ is equilateral.

## 5.6 The "carpets" theorem

**Problem 2.** Let $ABCD$ be a parallelogram. The points $M$, $N$, and $P$ are chosen on the segments $BD$, $BC$ and $CD$, respectively, so that $CNMP$ is a parallelogram. Let $E = AN \cap BD$ and $F = AP \cap BD$. Prove that $[AEF] = [DFP] + [BEN]$

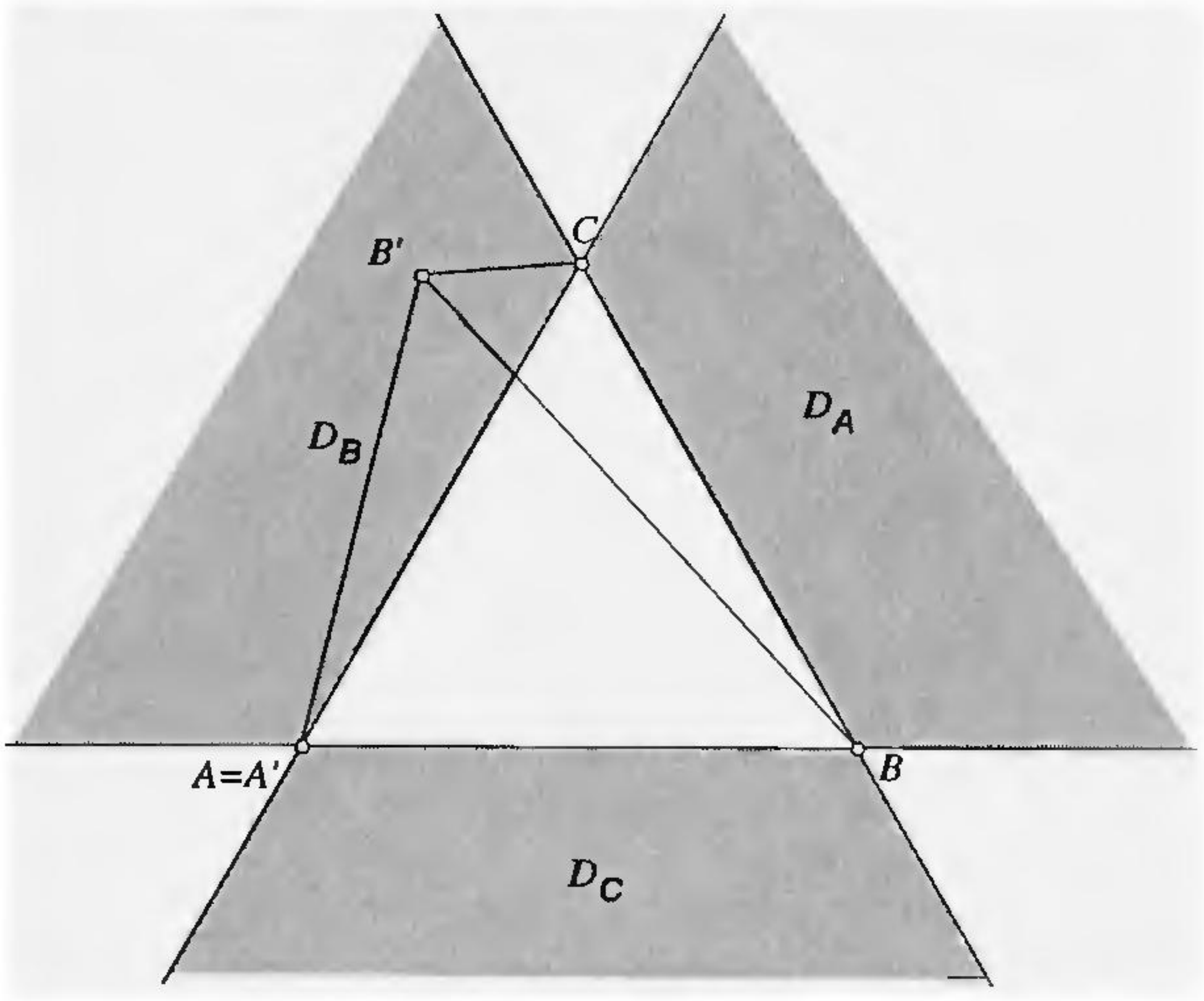


Figure 5.46

**Solution.** The carpets are the triangle $ABD$ and the quadrilateral $ANCP$. We have

$$[ABD] = \tfrac{1}{2}[ABCD]$$

and

$$[ANCP] = [ANC] + [ACP].$$

Observe that

$$\frac{[ANC]}{[ABCD]} = \frac{[ANC]}{2[ABC]} = \frac{NC}{2BC} = \frac{MD}{2BD}$$

and

$$\frac{[ACP]}{[ABCD]} = \frac{[ACP]}{2[ACD]} = \frac{CP}{2CD} = \frac{MB}{2BD}.$$

Adding these up, we obtain

$$\frac{[ANC] + [ACP]}{[ABCD]} = \frac{MD + MB}{2BD} = \tfrac{1}{2}.$$

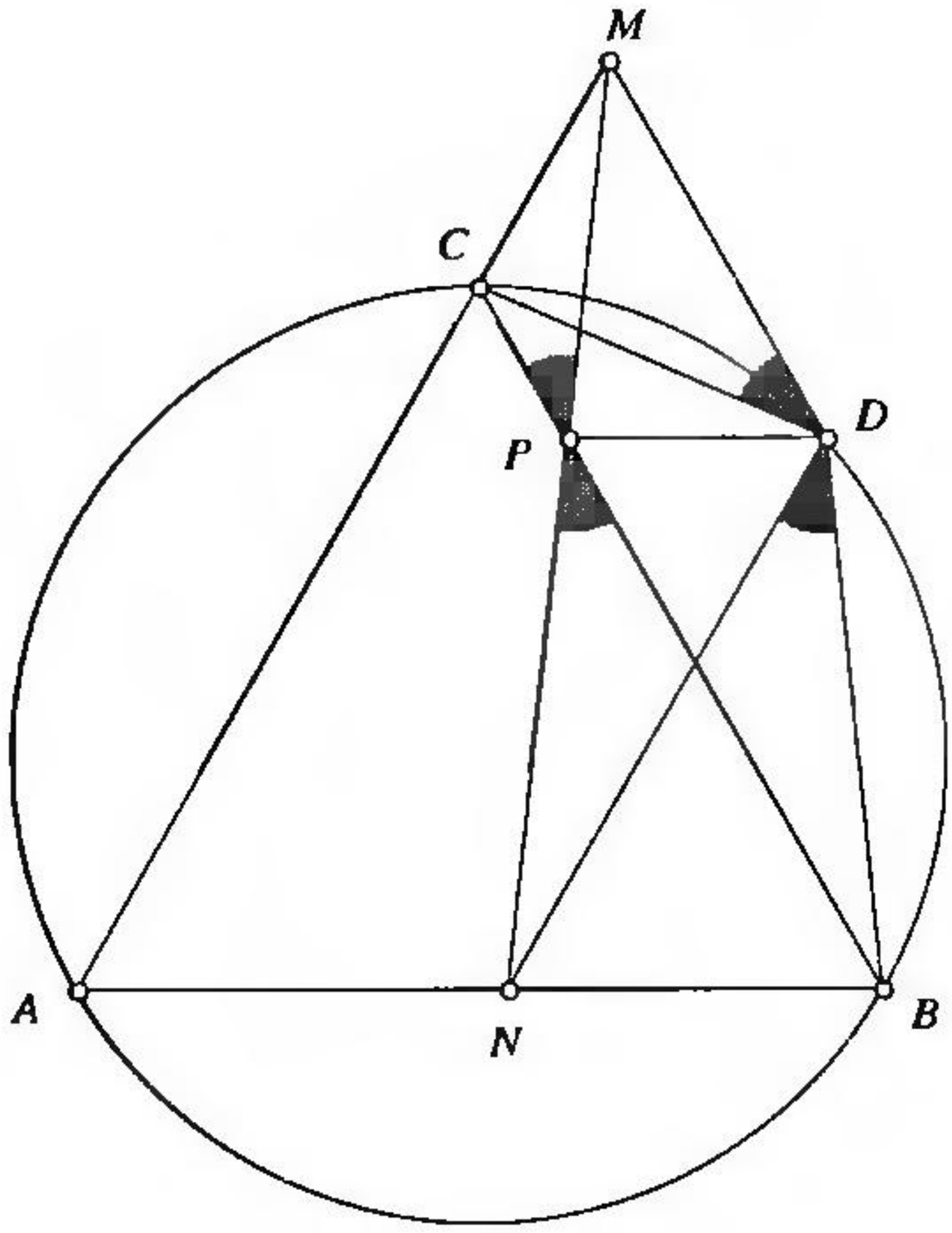


Figure 5.47

It follows that

$$[ABD] + [ANCP] = [ABCD],$$

so the area of the common part of the carpets, that is, $[AEF]$, equals the area of $ABCD$ that remains uncovered, and this is $[DFP] + [BEN]$.

*Observation.* Another proof uses a well-known property of the trapezoid. Because $[ADP] = [ADM]$, it follows that $[DFP] = [AFM]$. Similarly, $[BEN] = [AEM]$, and the result follows by adding these equalities.

**Problem 3.** Consider the quadrilateral $ABCD$. The points $M$, $N$, $P$, and $Q$ are the midpoints of the sides $AB$, $BC$, $CD$, and $DA$. Let $X = AP \cap BQ$, $Y = BQ \cap CM$, $Z = CM \cap DN$, and $T = DN \cap AP$. Prove that

$$[XYZT] = [AQX] + [BMY] + [CNZ] + [DPT].$$

**Solution.** The carpets are the quadrilaterals $AMCP$ and $BNDQ$. Because their common part is $XYZT$, all we have to prove is that

$$[AMCP] + [BNDQ] = [ABCD].$$

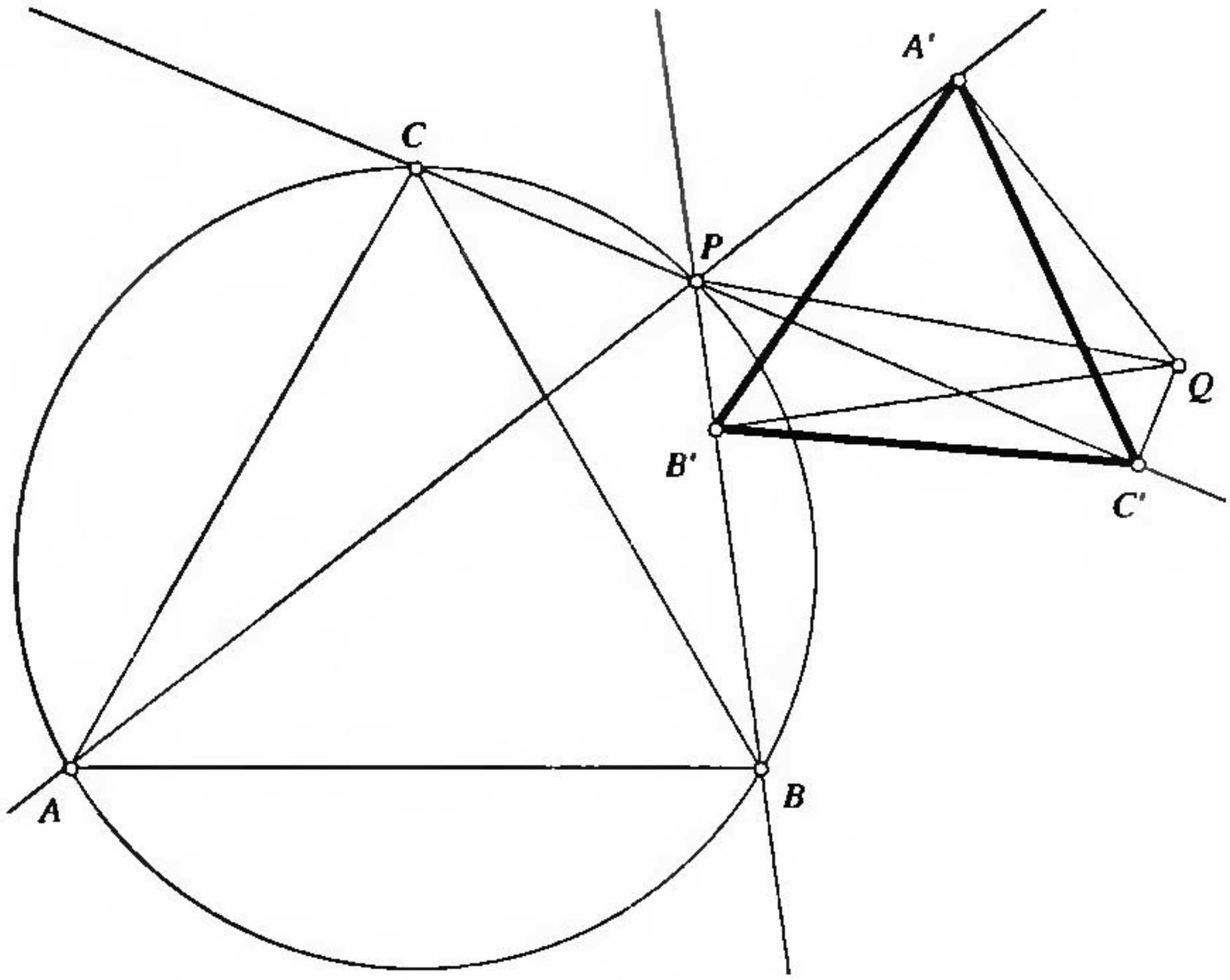


Figure 5.48

We have

$$[AMCP] = [AMC] + [ACP] = \tfrac{1}{2}[ABC] + \tfrac{1}{2}[ACD] = \tfrac{1}{2}[ABCD].$$

Similarly,

$$[BNDQ] = \tfrac{1}{2}[ABCD],$$

and hence

$$[AMCP] + [BNDQ] = [ABCD],$$

as desired.

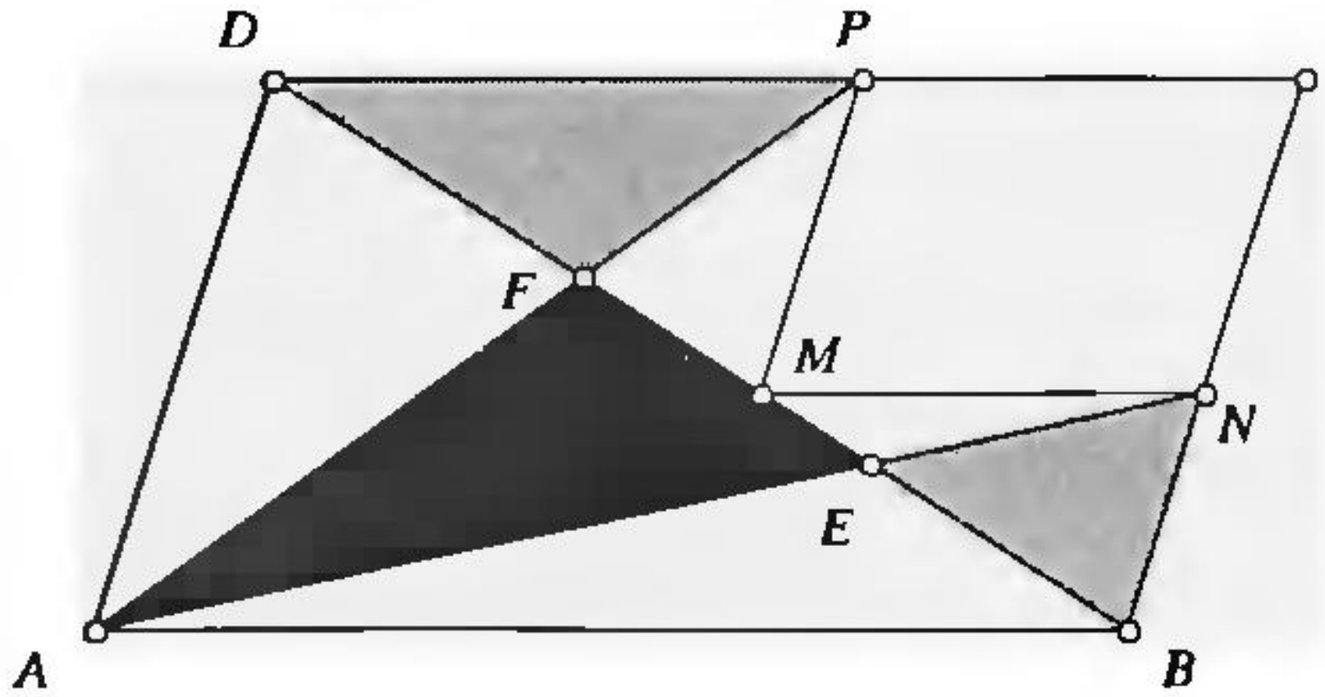


Figure 5.49

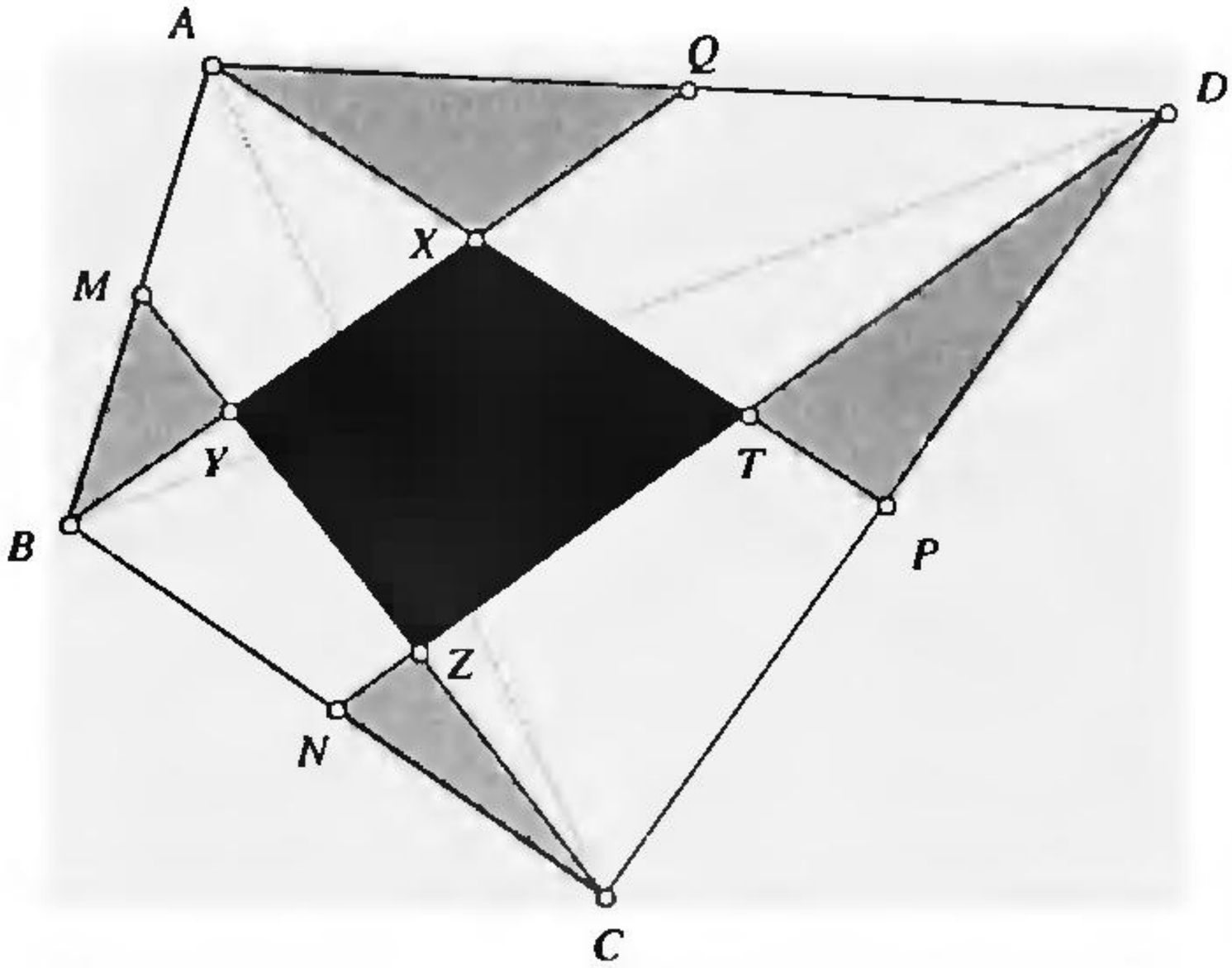


Figure 5.50

**Problem 4.** Through the vertices of the smaller base $AB$ of the trapezoid $ABCD$ two parallel lines are drawn, intersecting the segment $CD$. These lines and the trapezoid's diagonals divide it into seven triangles and a pentagon. Show that the area of the pentagon equals the sum of the areas of the three triangles that share a common side with the trapezoid.

**Solution.** Let $A'$ and $B'$ be the intersections of the parallel lines with $CD$. Take as carpets triangles $AA'C$ and $BB'D$. Their common part is the pentagon and the uncovered part of the trapezoid is the union of the three triangles sharing a common side with the trapezoid. We then have to prove that

$$[AA'C] + [BB'D] = [ABCD].$$

If we denote the trapezoid's altitude by $h$, we have $[AA'C] = \frac{1}{2}h \cdot A'C$ and $[BB'D] = \frac{1}{2}h \cdot B'D$, thus yielding

$$\begin{aligned}
[AA'C] + [BB'D] &= \tfrac{1}{2}h \cdot (A'C + B'D) \\
&= \tfrac{1}{2}h \cdot (A'B' + B'C + A'B' + A'D) \\
&= \tfrac{1}{2}h \cdot (CD + A'B') \\
&= \tfrac{1}{2}h \cdot (CD + AB) = [ABCD].
\end{aligned}$$

**Problem 5.** Let $M$ be a point in the interior of triangle $ABC$. Three lines are drawn through $M$, parallel to triangle's sides, thereby producing three trapezoids. Suppose a diagonal is drawn in each trapezoid in such a way that

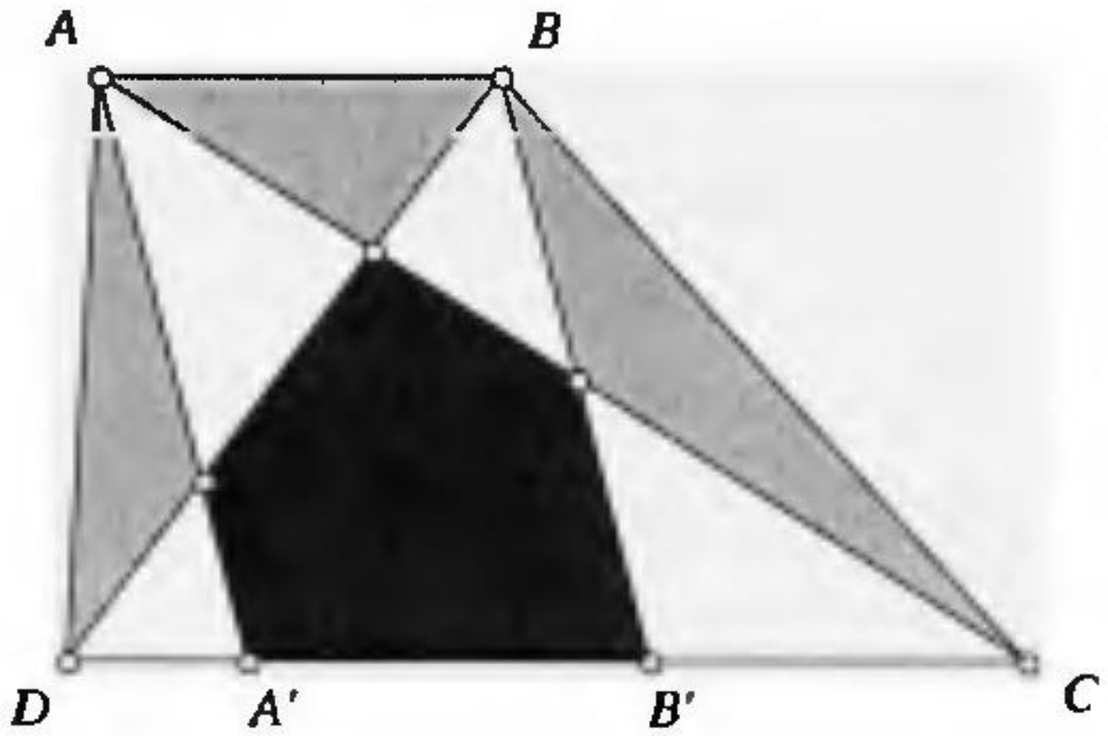


Figure 5.51

the diagonals have no common endpoints. These three diagonals divide $ABC$ into seven parts, four of them being triangles. Prove that the area of one of the four triangles equals the sum of the areas of the other three.

**Solution.** We arrange three carpets. With the notation in Figure 5.52, observe that if the carpets are $ABB'$, $BCC'$, and $CAA'$, then triangles $AB'Z$, $BC'X$, and $CA'Y$ are covered twice, while $XYZ$ remains uncovered.

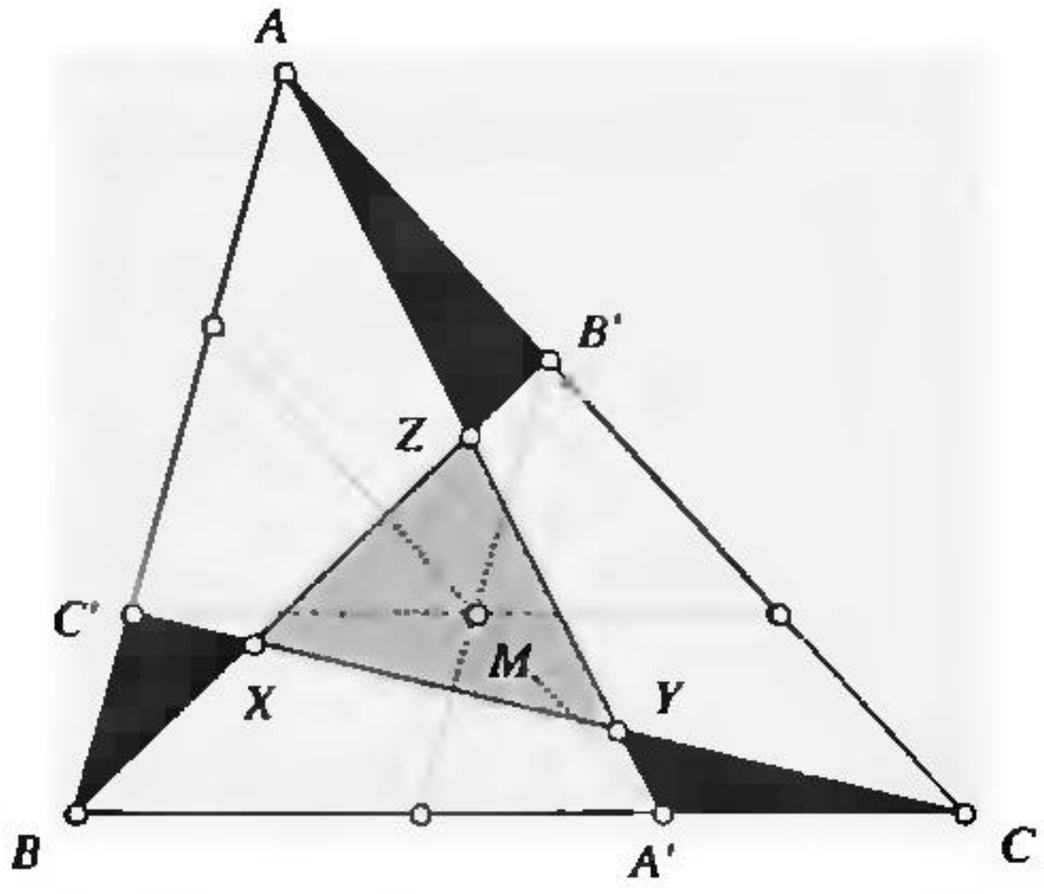


Figure 5.52

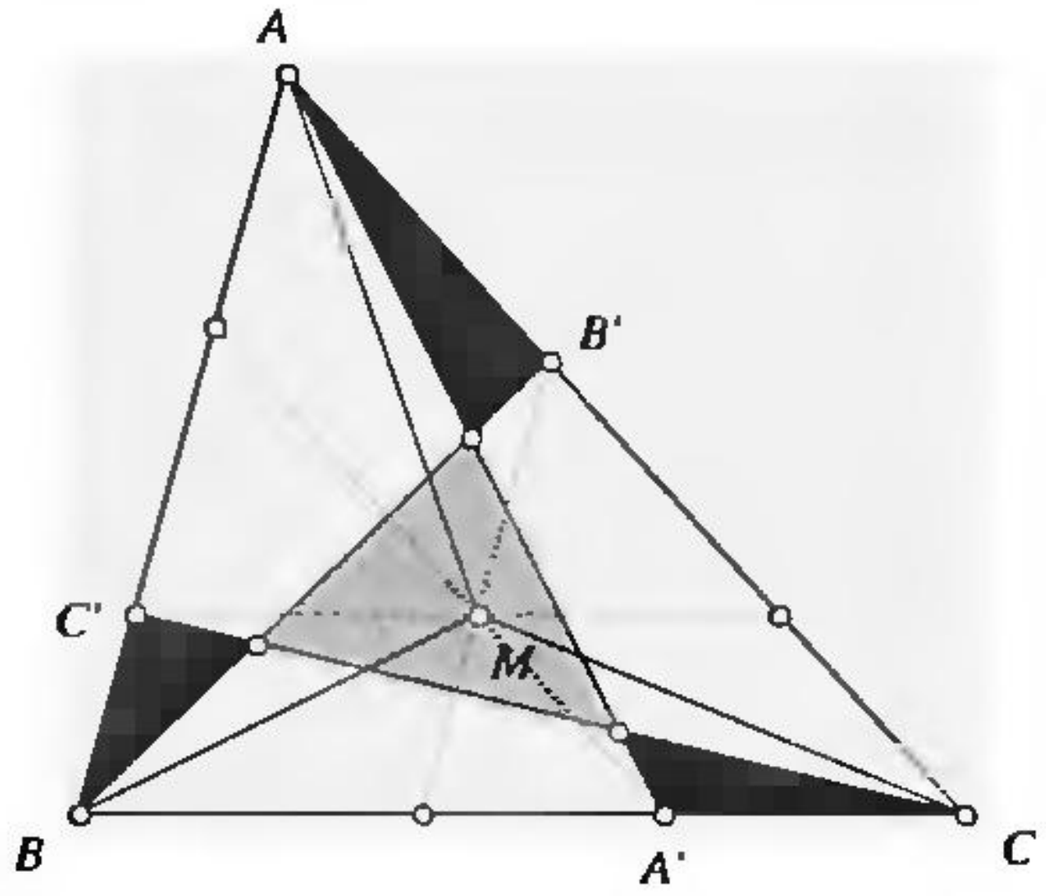


Figure 5.53

Thus, the equality

$$[AB'Z] + [BC'X] + [CA'Y] = [XYZ]$$

holds if and only if the sum of the carpets' areas equals the area of $ABC$. But this is very simple to prove if we notice that

$$[ABB'] = [ABM],$$
$$[BCC'] = [BCM]$$

and

$$[CAA'] = [CAM].$$

We obviously have

$$[ABM] + [BCM] + [CAM] = [ABC].$$

## 5.7 Quadrilaterals with an inscribed circle

**Problem 1.** Suppose that in the quadrilateral $ABCD$ we inscribe a circle with center $O$. Prove that the sum of the angles $\angle AOB$ and $\angle COD$ equals 180°. (Figure 5.54.)

**Solution.** We know that the angle bisectors of the quadrilateral intersect at $O$. Thus

$$\angle AOB = 180^\circ - \angle ABO - \angle BAO = 180^\circ - \frac{\angle A + \angle B}{2}.$$

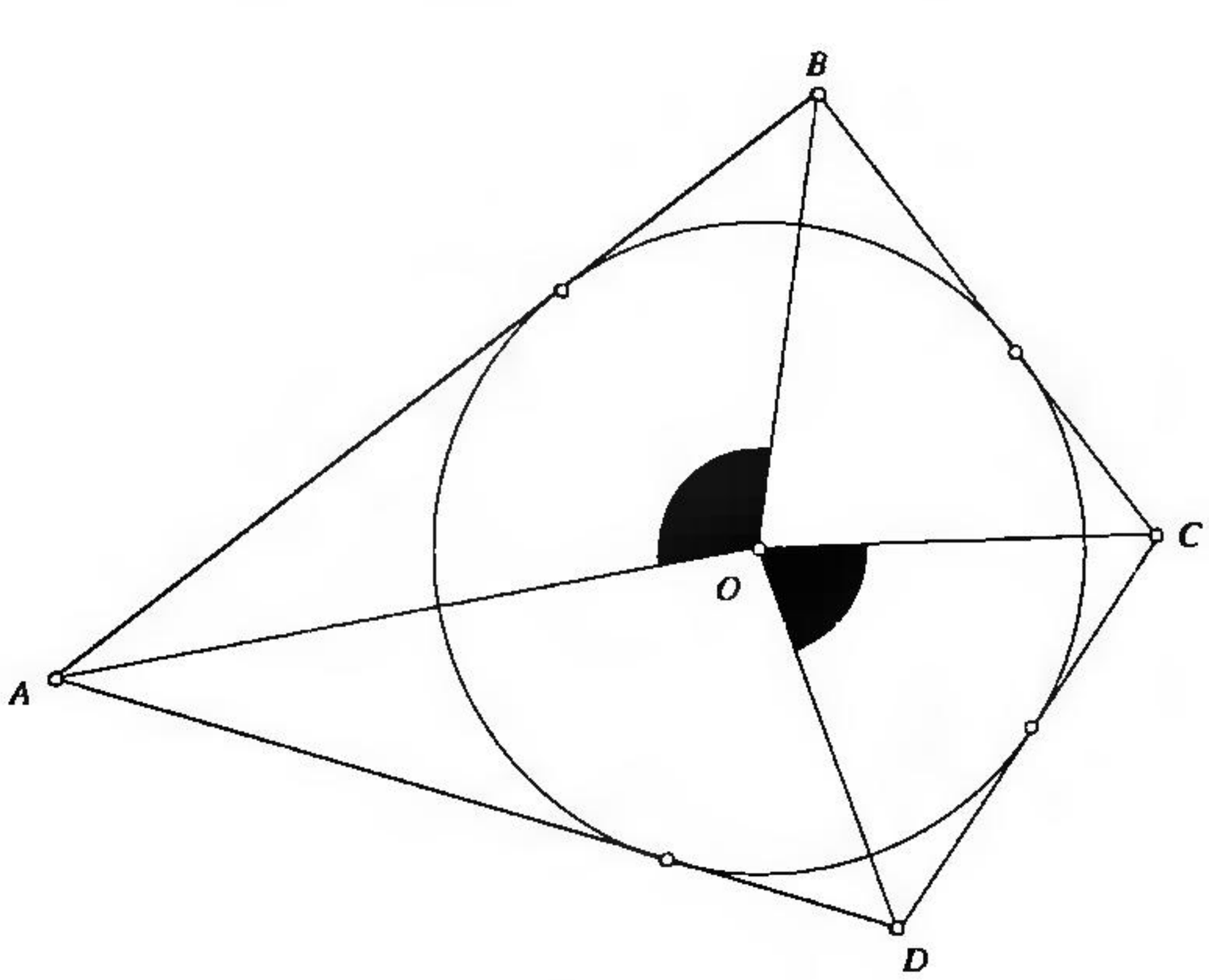


Figure 5.54

Similarly,

$$\angle COD = 180^\circ - \frac{\angle C + \angle D}{2}.$$

Adding these equalities, we obtain

$$\angle AOB + \angle COD = 360^\circ - \frac{\angle A + \angle B + \angle C + \angle D}{2} = 360^\circ - \frac{360^\circ}{2} = 180^\circ.$$

**Problem 2.** Let $ABCD$ be a quadrilateral with an inscribed circle. Prove that the circles inscribed in triangles $ABC$ and $ADC$ are tangent to each other.

**Solution.** Suppose that the circles inscribed in triangles $ABC$ and $ADC$ touch $AC$ at the points $X$ and $Y$, respectively. We have to prove that $X = Y$. With the notation as in Figure 5.55, we have: $AX = AK$ (the tangents to a circle drawn from a point are equal), $BK = BL$, $CL = CX$, $CY = CM$, $DM = DN$, $AN = AY$.

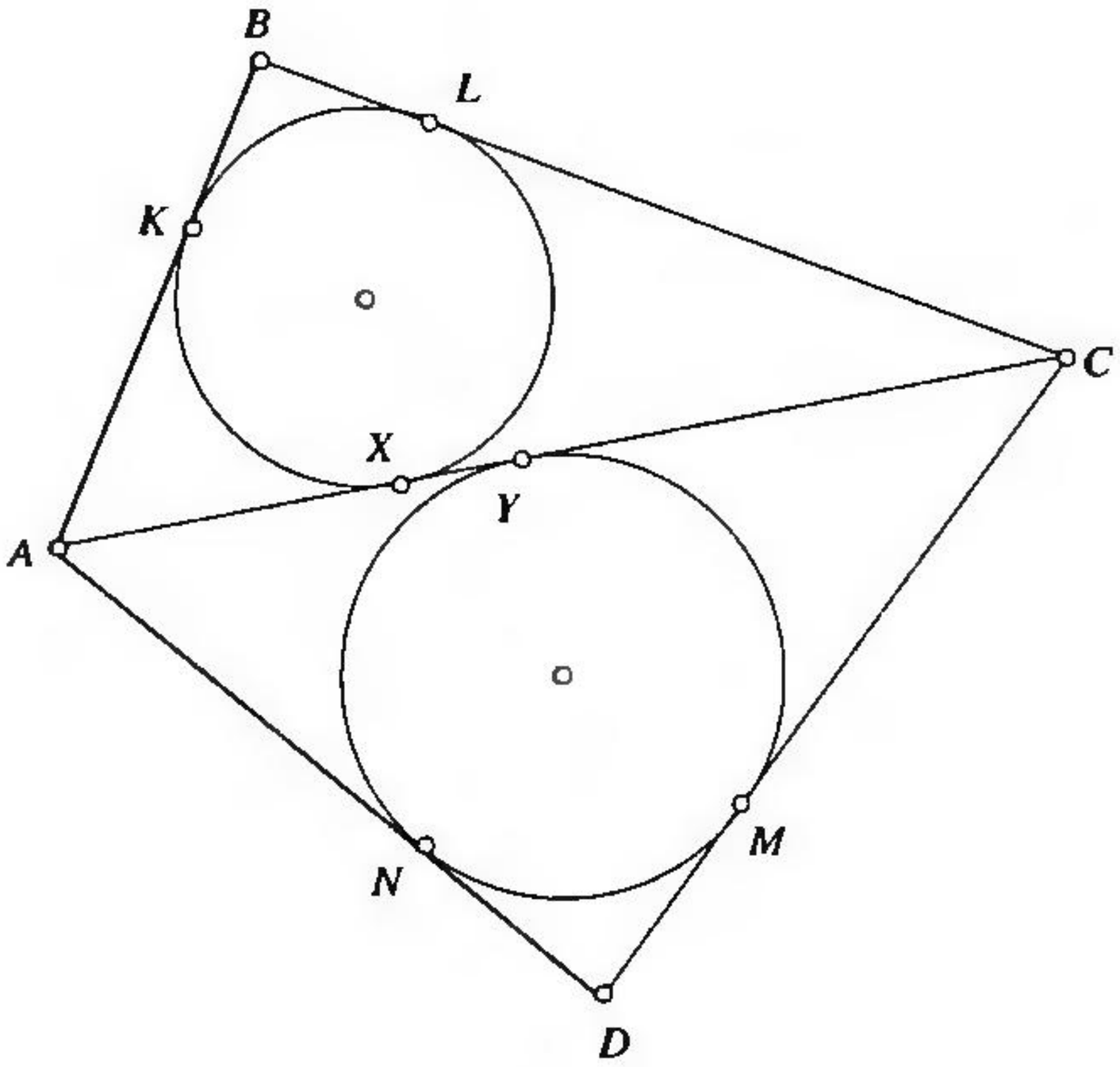


Figure 5.55

Because $ABCD$ has an inscribed circle, $AB + CD = AD + BC$, so

$$AK + BK + CM + MD = AN + DN + BL + CL.$$

Using the previous equalities, we obtain

$$AX + CY = AY + CX.$$

But $AY = AX + XY$ and $CX = CY + XY$. We deduce $2XY = 0$, and consequently $X = Y$.

If the order of the points $X$ and $Y$ on the segment $AC$ is reversed, the last two equalities are $AY = AX - XY$ and $CX = CY - XY$, but the conclusion is the same.

*Observation.* It can be proven that the points $K$, $L$, $M$, and $N$ are the vertices of a cyclic quadrilateral.

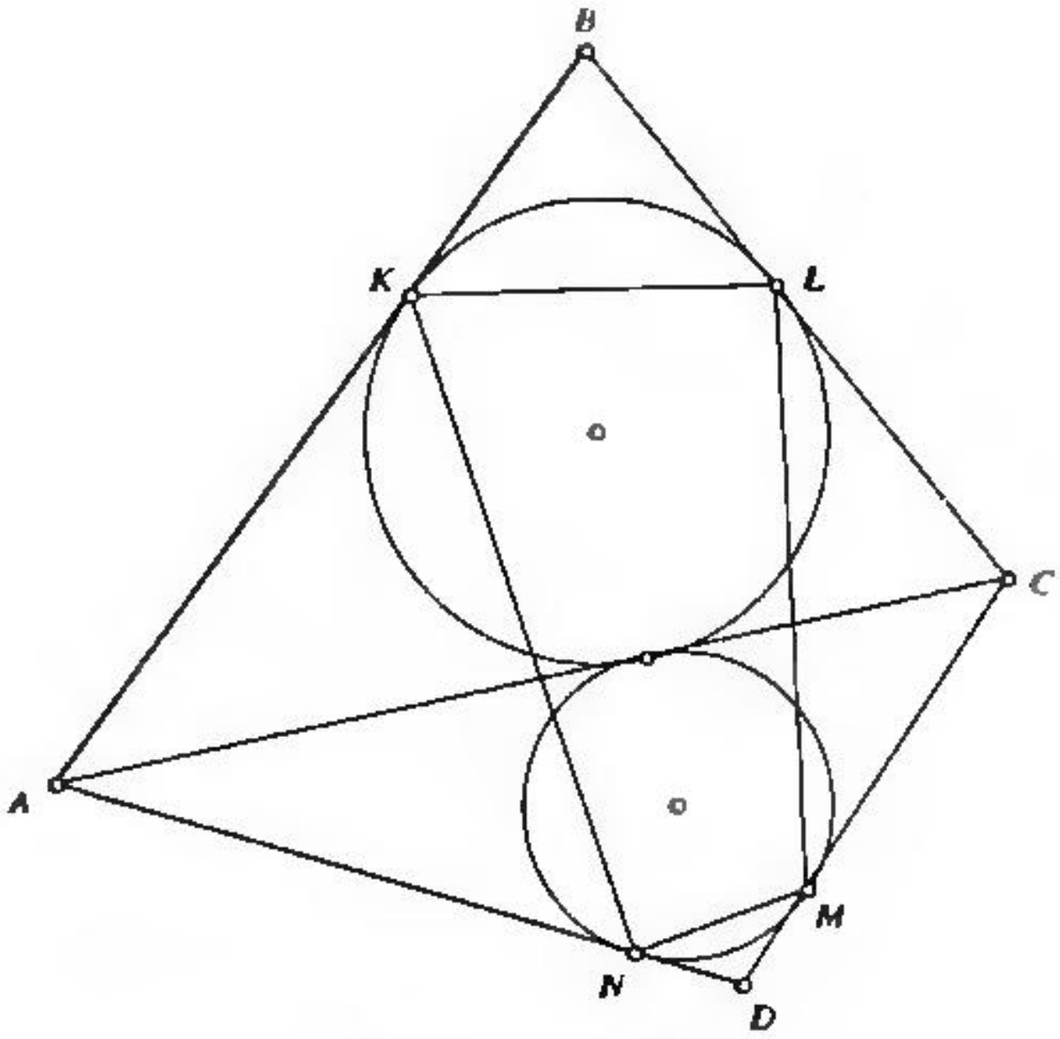


Figure 5.56

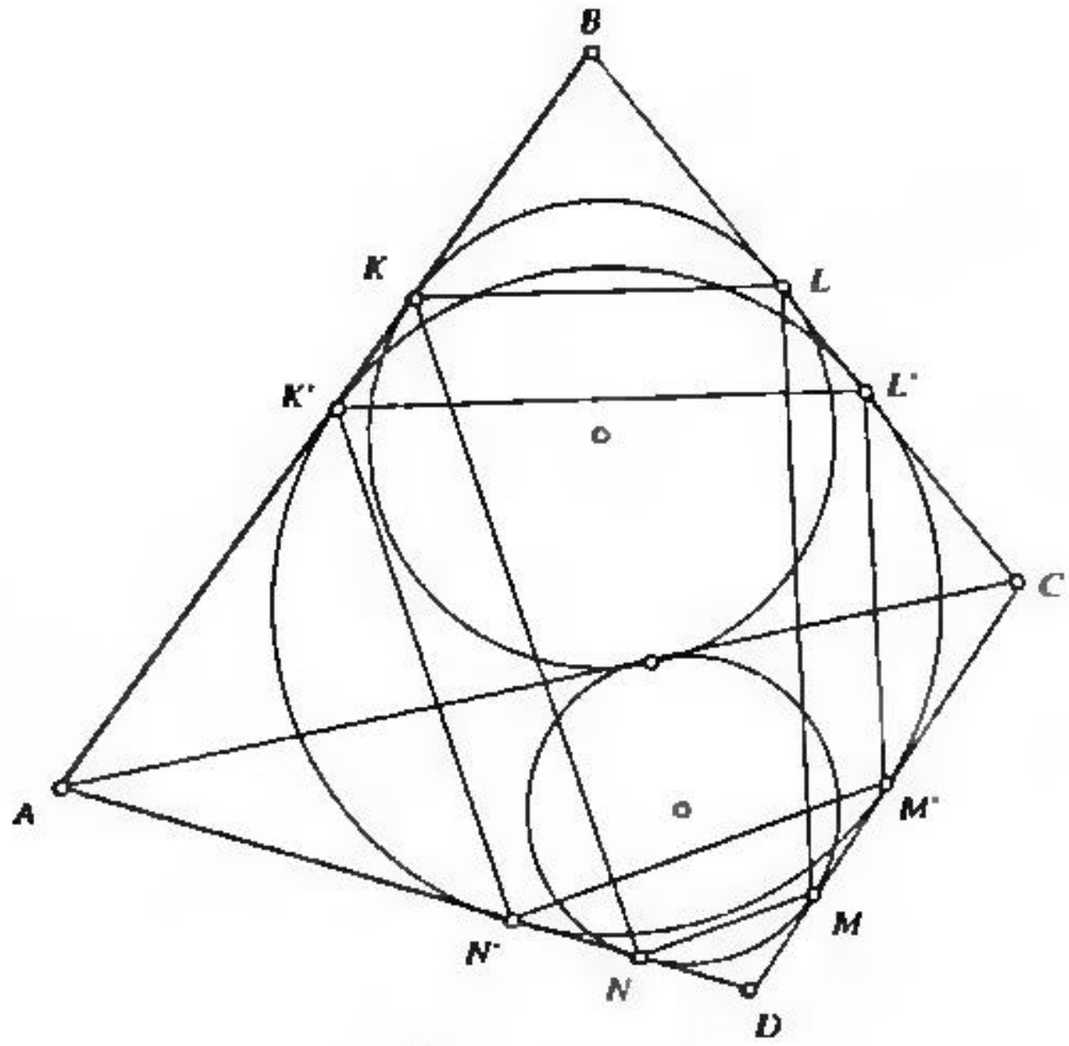


Figure 5.57

Indeed, if we draw the circle inscribed in $ABCD$ and denote by $K'$, $L'$, $M'$, and $N'$ the tangency points of the circle with the quadrilateral's sides, it is not difficult to see that the sides of $KLMN$ and $K'L'M'N'$ are parallel, and hence that their corresponding angles are equal. But $K'L'M'N'$ is cyclic; thus its opposite angles add up to 180° and the conclusion follows.

**Problem 3.** Let $ABCD$ be a convex quadrilateral. Suppose that the lines $AB$ and $CD$ intersect at $E$ and the lines $AD$ and $BC$ intersect at $F$. Prove that the following statements are equivalent:

(i) a circle is inscribed in $ABCD$;

(ii) $BE + BF = DE + DF$;

(iii) $AE - AF = CE - CF$.

**Solution.** Suppose a circle is inscribed in the quadrilateral $ABCD$ and touches the sides of the quadrilateral at the points $K$, $L$, $M$, and $N$.

Observe that

$$BE + BF = EK - BK + BL + LF = EM + NF$$
$$= EM + ND + DF = EM + MD + DF = DE + DF.$$

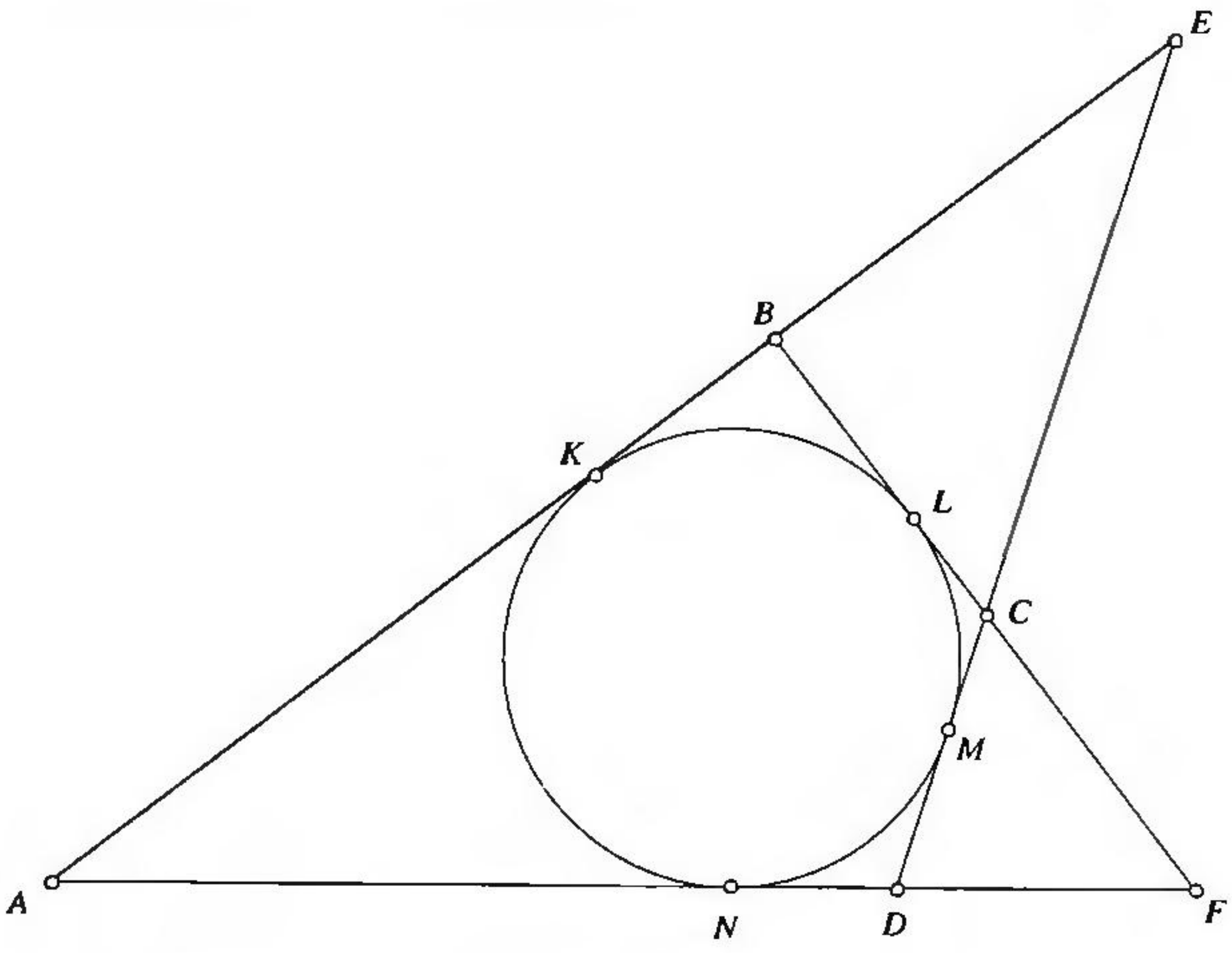


Figure 5.58

Again we used the fact that the tangents to a circle from a point are equal, so $BK = BL$, $EK = EM$, $LF = NF$, and $ND = MD$.

In a similar way, we have

$$AE - AF = AK + EK - AN - NF = EK - NF$$
$$= EM - LF = CE + CM - CL - CF = CE - CF.$$

**Problem 4.** Let $ABCD$ be a convex quadrilateral. Suppose that the lines $AB$ and $CD$ intersect at $E$, and the lines $AD$ and $BC$ intersect at $F$. Let $M$ and $N$ be two arbitrary points on the line segments $AB$ and $BC$, respectively. The line $EN$ intersects $AF$ and $MF$ at $P$ and $R$. The line $MF$ intersects $CE$ at $Q$. Prove that if the quadrilaterals $AMRP$ and $CNRQ$ have inscribed circles, then $ABCD$ has an inscribed circle.

**Solution.** Because $AMRP$ has an inscribed circle, it follows from the preceding problem that $AE + AF = RE + RF$. Analogously, since $CNRQ$ has an inscribed circle, we have $RE + RF = CE + CF$. We obtain that $AE + AF = CE + CF$, and this implies that $ABCD$ has an inscribed circle.

**Problem 5.** The points $A_1$, $A_2$, $C_1$, and $C_2$ are chosen in the interior of

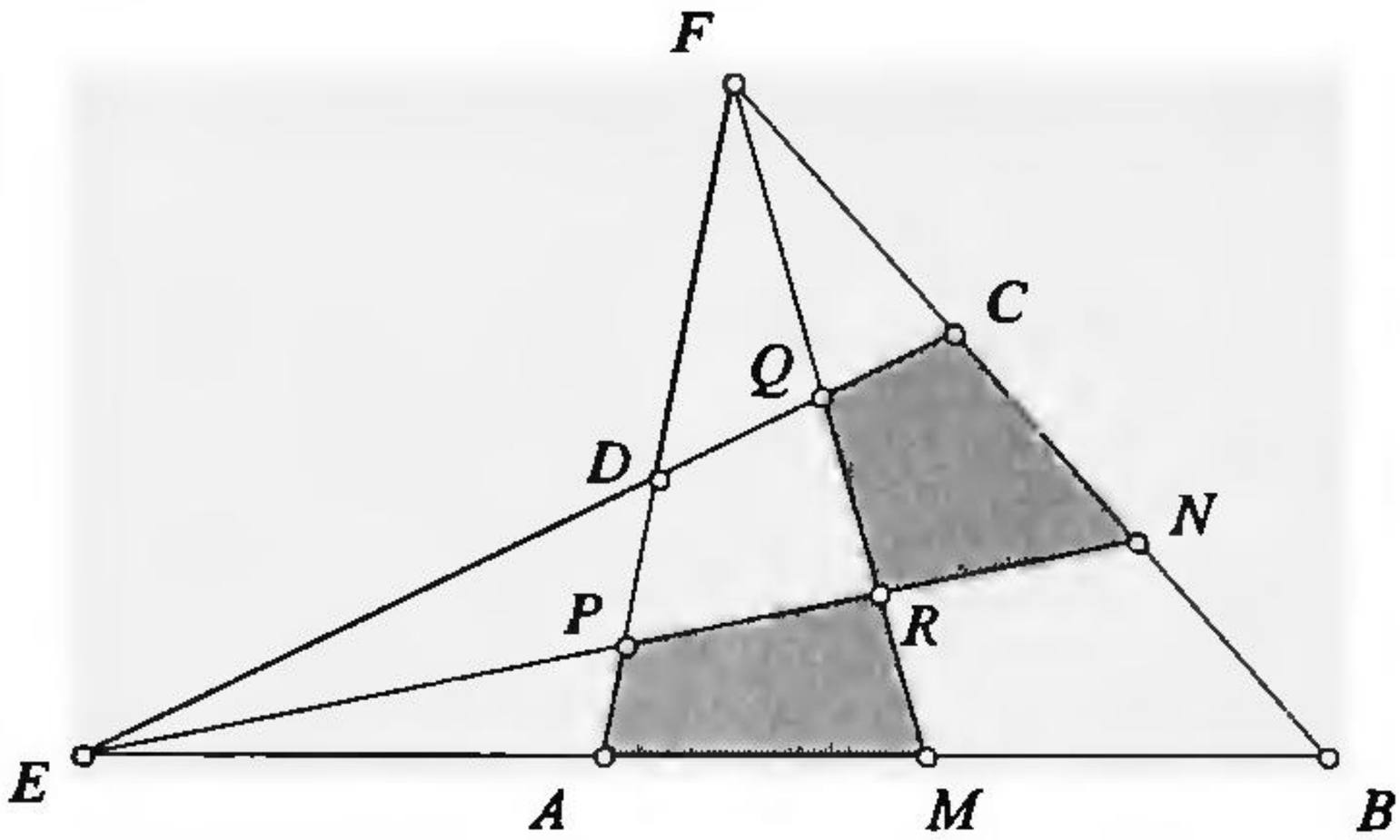


Figure 5.59

the sides $CD$, $BC$, $AB$, and $AD$ of the convex quadrilateral $ABCD$. Denote by $M$ the point of intersection of the lines $AA_2$ and $CC_1$, and denote by $N$ the point of intersection of the lines $AA_1$ and $CC_2$. Prove that if one can inscribe circles in three of the four quadrilaterals $ABCD$, $A_2BC_1M$, $AMCN$, and $A_1NC_2D$, then a circle can be also inscribed in the fourth one.

**Solution.** Let $\alpha = AB - BC - AM + CM$. From the previous problem, it follows that a circle can be inscribed in the quadrilateral $A_2BC_1M$ if and only if $\alpha = 0$.

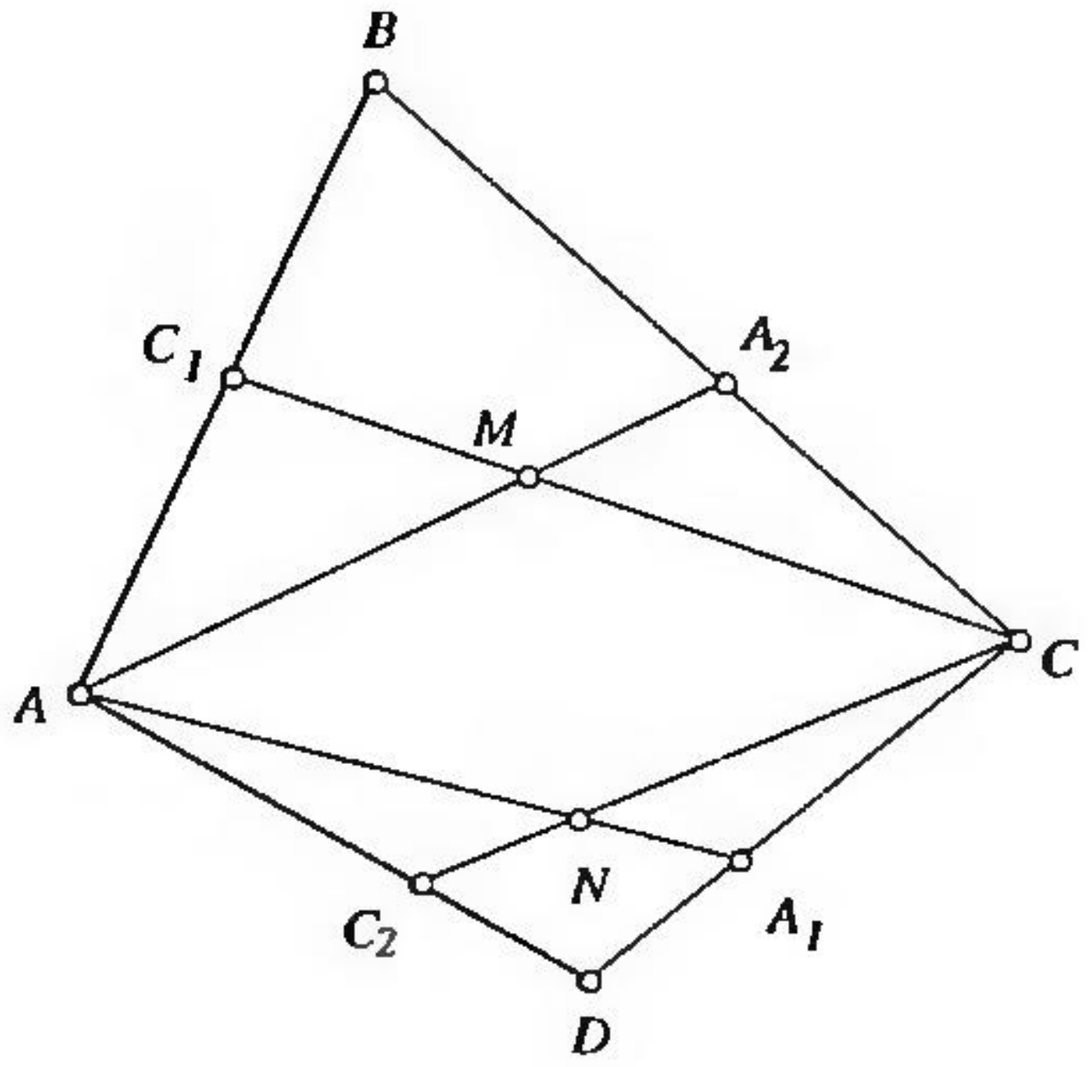


Figure 5.60

Analogously, if we set

$$\beta = CD - AD - CN + AN,$$

then $\beta = 0$ if and only if $A_1DC_2N$ has an inscribed circle. Setting

$$\gamma = AM - CM + CN - AN,$$

$$\delta = BC - AB + AD - CD,$$

from the theorem of Pithot it follows that $\gamma = 0$ and $\delta = 0$ are necessary and sufficient conditions for the existence of an inscribed circle in $AMCN$ and $ABCD$, respectively.

Now, simply observe that $\alpha + \beta + \gamma + \delta = 0$, so that if three of the four numbers are zero, then so also is the fourth one. It follows that if one can inscribe circles in three of the four quadrilaterals, a circle can be also inscribed in the fourth one.

**Problem 6.** A line cuts a quadrilateral with an inscribed circle into two polygons with equal areas and equal perimeters. Prove that the line passes through the center of the inscribed circle. (Figure 5.61.)

**Solution.** The line cuts the quadrilateral either into two quadrilaterals, or into a triangle and a pentagon, or into two triangles. The reasoning is essentially the same in all cases, so we assume that the line intersects the sides $AB$ and $CD$ at the points $X$ and $Y$. Let $O$ be the center of the inscribed circle. Because $AXYD$ and $BXYC$ have the same perimeter, it follows that

$$AX + AD + DY = BX + BC + CY.$$

Multiplying this equality with $\frac{1}{2}R$, where $R$ is the radius of the inscribed circle, yields

$$[OAX] + [OAD] + [ODY] = [OBX] + [OBC] + [OCY],$$

that is,

$$[OXADY] = [OXBCY].$$

Because

$$[OXADY] + [OXBCY] = [ABCD],$$

it follows that

$$[OXADY] = [OXBCY] = \tfrac{1}{2}[ABCD].$$

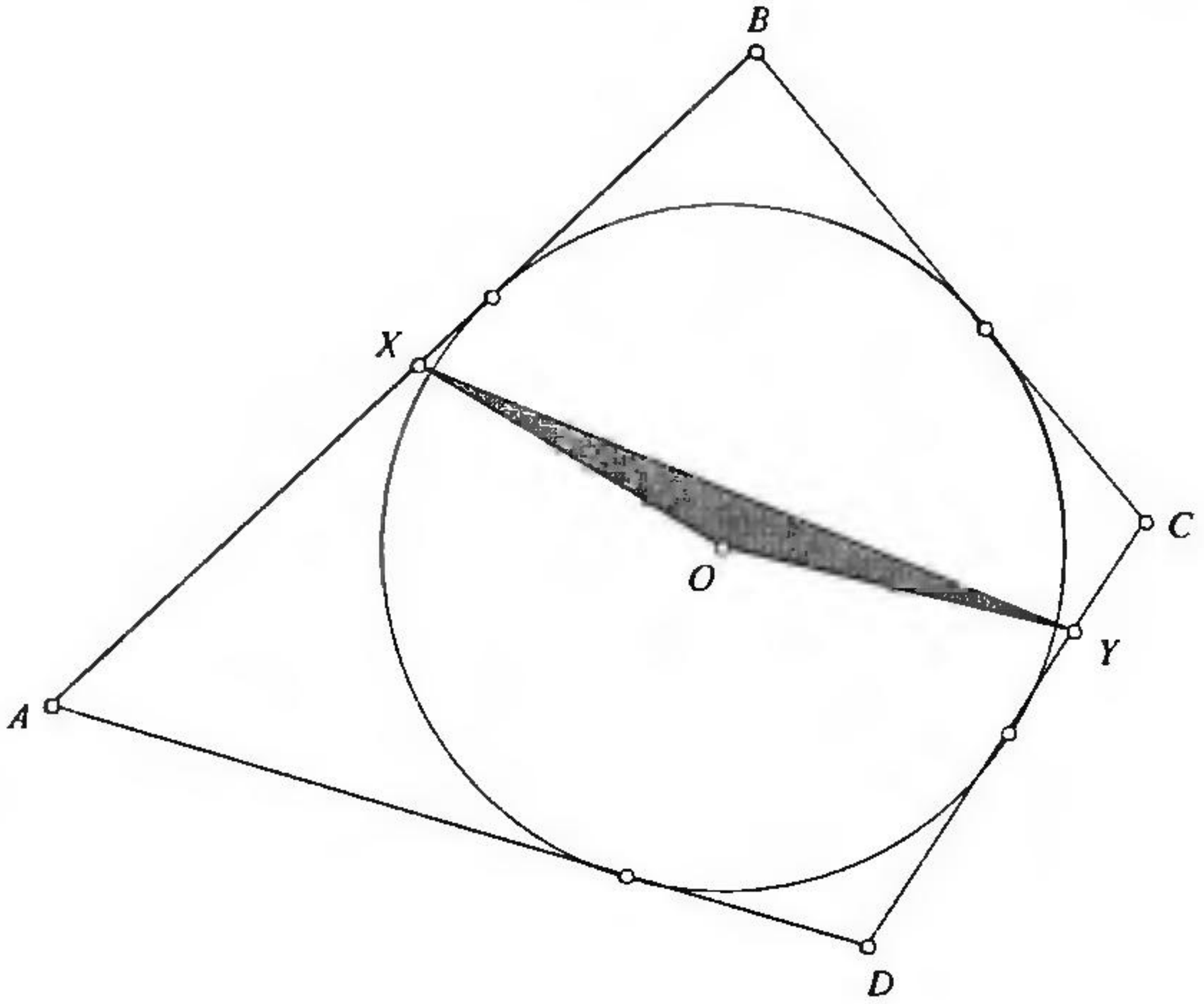


Figure 5.61

Now suppose. by way of contradiction, that the line $XY$ does not pass through $O$. Suppose $O$ lies in the interior of $AXYD$ (the case in which $O$ is in the interior of $BXYC$ is similar). We have

$$[BXYC] = \tfrac{1}{2}[ABCD] = [BXOYC],$$

but $[BXOYC] = [BXYC] + [OXY]$, and hence $[OXY] = 0$, which is a contradiction.

## 5.8 Dr. Trig learns complex numbers

**Problem 4.** Let $a, b, c$ be real numbers such that

$$\cos a + \cos b + \cos c = \sin a + \sin b + \sin c = 0.$$

Prove that

$$\cos(a+b+c) = \tfrac{1}{3}\left(\cos 3a + \cos 3b + \cos 3c\right),$$
$$\sin(a+b+c) = \tfrac{1}{3}\left(\sin 3a + \sin 3b + \sin 3c\right).$$

**Solution.** Let $x = \cos a + i \sin a$, $y = \cos b + i \sin b$, and $z = \cos c + i \sin c$. We know from the problem statement that $x + y + z = 0$. We use the identity

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - xz - yz),$$

which is valid for any numbers $x$, $y$, and $z$ (see the Algebra part of this book, section 1.1). It follows that

$$x^3 + y^3 + z^3 - 3xyz = 0,$$

and as a result,

$$xyz = \tfrac{1}{3}\left(x^3 + y^3 + z^3\right).$$

But this means that

$$\cos(a + b + c) + i \sin(a + b + c) = \tfrac{1}{3}\sum (\cos 3a + i \sin 3a),$$

and we obtain the required equalities by identifying the real parts and the nonreal parts in the above equality.

**Problem 5.** Find the value of the product of $\cos 20^\circ$, $\cos 40^\circ$, and $\cos 80^\circ$.

**Solution.** Let $z = \cos 20^\circ + i \sin 20^\circ$. Then

$$\cos 20^\circ = \tfrac{1}{2}\left(z + \frac{1}{z}\right), \cos 40^\circ = \tfrac{1}{2}\left(z^2 + \frac{1}{z^2}\right), \cos 80^\circ = \tfrac{1}{2}\left(z^4 + \frac{1}{z^4}\right).$$

Also, $z^9 = \cos 180^\circ + i \sin 180^\circ = -1$, and thus

$$\frac{1}{z} = -z^8, \frac{1}{z^2} = -z^7, \frac{1}{z^4} = -z^5.$$

The product equals

$$\begin{aligned}
&\tfrac{1}{8}\left(z - z^8\right)\left(z^2 - z^7\right)\left(z^4 - z^5\right) \\
&\quad = \tfrac{1}{8}\left(z^3 - z^8 - z^{10} + z^{15}\right)\left(z^4 - z^5\right) \\
&\quad = \tfrac{1}{8}\left(z^3 - z^8 + z - z^6\right)\left(z^4 - z^5\right) \\
&\quad = \tfrac{1}{8}\left(z^7 - z^8 - z^{12} + z^{13} + z^5 - z^6 - z^{10} + z^{11}\right) \\
&\quad = \tfrac{1}{8}\left(z - z^2 + z^3 - z^4 + z^5 - z^6 + z^7 - z^8\right) \\
&\quad = \tfrac{1}{8}\left(1 - \frac{z^9 + 1}{z + 1}\right) = \tfrac{1}{8}.
\end{aligned}$$

**Problem 6.** Prove that

$$\frac{1}{\cos 6^\circ} + \frac{1}{\sin 24^\circ} + \frac{1}{\sin 48^\circ} = \frac{1}{\sin 12^\circ}.$$

**Solution.** Let $z = \cos 6^\circ + i \sin 6^\circ$; then $z^{15} = \cos 90^\circ + i \sin 90^\circ = i$. We have

$$\cos 6^\circ = \frac{z^2+1}{2z}, \sin 12^\circ = \frac{z^4-1}{2iz^2}, \sin 24^\circ = \frac{z^8-1}{2iz^4}, \sin 48^\circ = \frac{z^{16}-1}{2iz^8}.$$

The equality to be proved becomes

$$\frac{2z}{z^2+1} - \frac{2iz}{z^4-1} + \frac{2iz^4}{z^8-1} + \frac{2iz^8}{z^{16}-1} = 0.$$

Multiplying by $z^{16}-1$, we obtain, after a short computation,

$$z^{16} - 1 - iz(z^{14}+1) = 0.$$

This is true if and only if $iz - 1 - i^2 - iz = 0$, which is obvious.

**Problem 7.** Prove that

$$\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7} + \tfrac{1}{2} = 0.$$

**Solution.** If $z = \cos\frac{2\pi}{7} + i\sin\frac{2\pi}{7}$, then $z^7 = 1$. The required equality becomes

$$\tfrac{1}{2}\left(z + \frac{1}{z}\right) + \tfrac{1}{2}\left(z^2 + \frac{1}{z^2}\right) + \tfrac{1}{2}\left(z^3 + \frac{1}{z^3}\right) + \tfrac{1}{2} = 0.$$

Multiplying by $2z^3$ and rearranging the terms, we get

$$z^6 + z^5 + z^4 + z^3 + z^2 + z + 1 = 0,$$

or

$$\frac{z^7 - 1}{z - 1} = 0,$$

which is obvious.

**Problem 8.** Prove the equality

$$\sin\frac{\pi}{2n}\sin\frac{2\pi}{2n}\cdots\sin\frac{(n-1)\pi}{2n}=\frac{\sqrt{n}}{2^{n-1}}.$$

**Solution.** Consider the polynomial $P(X)=X^{2n}-1$. Its roots are the numbers $x_k=\cos\frac{k\pi}{n}+i\sin\frac{k\pi}{n}$, with $k=0,1,\ldots,2n-1$. It follows that $P(X)=(X-x_0)(X-x_1)\cdots(X-x_{2n-1})$. We observe that, except for $x_0=1$ and $x_n=-1$, all the other roots are nonreal complex numbers and that, for $1\le k\le n-1$, $\overline{x}_k=x_{2n-k}$. Thus, we can write

$$P(X)=(X^2-1)\prod_{k=1}^{n-1}(X-x_k)(X-\overline{x}_k)$$

$$=(X^2-1)\prod_{k=1}^{n-1}\left(x^2-(x_k+\overline{x}_k)\,x+x_k\overline{x}_k\right).$$

Now, $x_k+\overline{x}_k=2\cos\frac{k\pi}{n}$ and $x_k\overline{x}_k=1$. Dividing both sides by $x^2-1$, we obtain

$$x^{2n-2}+x^{2n-4}+\cdots+x^4+x^2+1=\prod_{k=1}^{n-1}(x^2-2x\cos\frac{k\pi}{n}+1).$$

For $x=1$, the above equality becomes

$$n=2^{n-1}\prod_{k=1}^{n-1}(1-\cos\frac{k\pi}{n})=2^{n-1}\prod_{k=1}^{n-1}2\sin^2\frac{k\pi}{2n}$$

$$=\left(2^{n-1}\prod_{k=1}^{n-1}\sin\frac{k\pi}{2n}\right)^2.$$

Since $\sin\frac{k\pi}{2n}>0$ for $k=1,\ldots,n-1$, we obtain

$$\prod_{k=1}^{n-1}\sin\frac{k\pi}{2n}=\frac{\sqrt{n}}{2^{n-1}}.$$

**Problem 9.** Solve the equation

$$\sin x+\sin 2x+\sin 3x=\cos x+\cos 2x+\cos 3x.$$

**Solution.** If $z = \cos x + i \sin x$, the equation becomes

$$\frac{z^2-1}{2iz} + \frac{z^4-1}{2iz^2} + \frac{z^6-1}{2iz^3} = \frac{z^2+1}{2z} + \frac{z^4+1}{2z^2} + \frac{z^6+1}{2z^3}.$$

Multiplying by $2iz^3$ and grouping the terms, we obtain

$$z^6 + z^5 + z^4 - i(z^2 + z + 1) = 0,$$

or

$$(z^4 - i)(z^2 + z + 1) = 0.$$

If $z^2 + z + 1 = 0$, then $z = \frac{-1 \pm i\sqrt{3}}{2} = \cos \frac{\pm 2\pi}{3} + i \sin \frac{\pm 2\pi}{3}$, and therefore $x = \frac{\pm 2\pi}{3} + 2k\pi$, with $k \in \mathbf{Z}$. If $z^4 = i$, then $\cos 4x + i \sin 4x = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2}$, and it follows that $4x = \frac{\pi}{2} + 2k\pi$, so we find that $x = \frac{\pi}{8} + \frac{k\pi}{2}$, with $k \in \mathbf{Z}$.

**Problem 10.** Prove that

$$\cos \frac{\pi}{5} = \frac{1+\sqrt{5}}{4}.$$

**Solution.** If we let $z = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5}$, then $z^5 = -1$, which is equivalent to

$$(z+1)(z^4 - z^3 + z^2 - z + 1) = 0.$$

Because $z \neq -1$, we obtain $z^4 - z^3 + z^2 - z + 1 = 0$. After dividing by $z^2 \neq 0$, it follows that

$$z^2 + \frac{1}{z^2} - \left(z + \frac{1}{z}\right) + 1 = 0.$$

If $x = \cos \frac{\pi}{5}$, then

$$z + \frac{1}{z} = 2x,\ z^2 + \frac{1}{z^2} = \left(z + \frac{1}{z}\right)^2 - 2 = 4x^2 - 2.$$

It follows that $x$ is a (positive) root of the equation

$$4x^2 - 2x - 1 = 0,$$

and, as a result,

$$x = \frac{1+\sqrt{5}}{4}.$$

# Chapter 6

# Number Theory and Combinatorics

## 6.1 Arrays of numbers

**Problem 4.** Prove that the sum of any $n$ entries of the table

$$\begin{array}{ccccc}
\mathbf{1} & \frac{1}{2} & \frac{1}{3} & \cdots & \frac{1}{n} \\
\frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \cdots & \frac{1}{n+1} \\
\vdots & & & & \\
\frac{1}{n} & \frac{1}{n+1} & \frac{1}{n+2} & \cdots & \frac{1}{2n-1}
\end{array}$$

situated in different rows and different columns is not less than 1.

**Solution.** Denoting by $a_{ij}$ the entry in the $i$-th row and $j$-th column of the array, we have

$$a_{ij} = \frac{1}{i+j-1},$$

for all $i, j$, $1 \leq i, j \leq n$. Choose $n$ entries situated in different rows and different columns. It follows that from each row and each column, exactly one number is chosen. Let $a_{1j_1}, a_{2j_2}, \ldots, a_{nj_n}$ be the chosen numbers, where $j_1, j_2, \ldots, j_n$ is a permutation of indices $1, 2, \ldots, n$. We have

$$\sum_{k=1}^{n} \frac{1}{a_{kj_k}} = \sum_{k=1}^{n} (k + j_k - 1) = \sum_{k=1}^{n} k + \sum_{k=1}^{n} j_k - n.$$

But

$$\sum_{k=1}^{n} j_k = \sum_{k=1}^{n} k = \frac{n(n+1)}{2}$$

since $j_1, j_2, \ldots, j_n$ is a permutation of indices $1, 2, \ldots, n$. It follows that

$$\sum_{k=1}^{n} \frac{1}{a_{kj_k}} = n^2.$$

The Cauchy–Schwartz inequality yields

$$(x_1 + x_2 + \cdots + x_n)\left(\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}\right) \geq n^2,$$

for all positive real numbers $x_1, x_2, \ldots, x_n$. Taking $x_k = a_{kj_k}$ and using the above equality, we obtain

$$\sum_{k=1}^{n} a_{kj_k} \geq 1$$

as desired.

**Problem 5.** The entries of an $n \times n$ array of numbers are denoted by $a_{ij}$, $1 \leq i, j \leq n$. The sum of any $n$ entries situated on different rows and different columns is the same. Prove that there exist numbers $x_1, x_2, \ldots, x_n$ and $y_1, y_2, \ldots, y_n$, such that

$$a_{ij} = x_i + y_j,$$

for all $i, j$.

**Solution.** Consider $n$ entries situated on different rows and different columns $a_{ij_i}$, $i = 1, 2, \ldots, n$. Fix $k$ and $l$, $1 \leq k < l \leq n$ and replace $a_{kj_k}$ and $a_{lj_l}$ with $a_{kj_l}$ and $a_{lj_k}$, respectively. It is not difficult to see that the new $n$ entries are still situated on different rows and different columns. Because the sums of the two sets of $n$ entries are equal, it follows that

$$a_{kj_k} + a_{lj_l} = a_{kj_l} + a_{lj_k}. \qquad (*)$$

Now, denote by $x_1, x_2, \ldots, x_n$ the entries in the first column of the array and by $x_1, x_1 + y_2, x_1 + y_3, \ldots, x_1 + y_n$ the entries in the first row (in fact, we

have defined $x_k = a_{k1}$, for all $k$, $y_0 = 0$ and $y_k = a_{1k} - a_{k1}$ for all $k \geq 2$).

| $x_1$ | $x_1 + y_2$ | ... | $x_1 + y_j$ | ... | $x_1 + y_n$ |
|---|---|---|---|---|---|
| $x_2$ | | | | | |
| $\vdots$ | | | | | |
| $x_i$ | | | $a_{ij}$ | | |
| $\vdots$ | | | | | |
| $x_n$ | | | | | |

The equality

$$a_{ij} = x_i + y_j,$$

stands for all $i$, $j$ with $i = 1$ or $j = 1$. Now, consider $i$, $j > 1$. From (*) we deduce

$$a_{11} + a_{ij} = a_{1j} + a_{i1},$$

hence

$$x_1 + a_{ij} = x_i + x_1 + y_j$$

or

$$a_{ij} = x_i + y_j,$$

for all $i$, $j$, as desired.

**Problem 6.** In an $n \times n$ array of numbers, all rows are different (two rows are different if they differ in at least one entry). Prove that there is a column which can be deleted in such a way that the remaining rows are still different.

**Solution.** We prove by induction on $k$ the following statement: at least $n - k + 1$ columns can be deleted in such a way that the first $k$ rows are still different. For $k = 2$, the assertion is true. Indeed, the first two rows differ in at least one place, so we can delete the rest of the $n - 1$ columns. Suppose the assertion is true for $k$, that is, suppose we can delete $n - k + 1$ columns, in such a way that the first $k$ rows are still different. If, after the deletion of the columns, the $(k + 1)$-th row is different from all first $k$ rows, then we can put back any of the deleted columns and end up with $n - k$ deleted columns and $k + 1$ different rows. If, after the deletion, the $(k + 1)$-th row coincides with one of the first rows, then we put back the column in which the two rows differ in the original array. For $k = n$, we obtain the desired result.

**Problem 7.** The positive integers from 1 to $n^2$ are placed arbitrarily on squares of an $n \times n$ chessboard. Prove that there exist two adjacent squares (having a common vertex or a common side) such that the difference of the numbers placed on them is not less than $n + 1$.

**Solution.** Suppose the contrary: the difference of the numbers placed on any adjacent squares is less than $n + 1$. If we place a king on a square of the chessboard, it can reach any other square in at most $n - 1$ moves through adjacent squares. Place a king on the square with number 1 and move it to the square with number $n^2$ in at most $n - 1$ moves. At each move, the difference between the numbers in the adjacent squares is less than $n + 1$, and hence the difference between $n^2$ and 1 is less than $(n + 1)(n - 1) = n^2 - 1$, a contradiction.

**Problem 8.** A positive integer is written in each square of an $n^2 \times n^2$ chess board. The difference between the numbers in any two squares sharing an edge is less than or equal to $n$. Prove that at least $\left\lfloor \frac{n}{2} \right\rfloor + 1$ squares contain the same number.

**Solution.** Consider the smallest and largest numbers $a$ and $b$ on the board. They are separated by at most $n^2-1$ squares horizontally and $n^2-1$ vertically, so there is a path from one to the other with length at most $2(n^2 - 1)$. Then, since we know that any two successive squares differ by at most $n$, we have $b - a \leq 2(n^2 - 1)n$. But since all numbers on the board are integers lying between $a$ and $b$, only $2(n^2-1)n+1$ distinct numbers can exist; and because

$$n^4 > (2(n^2 - 1)n + 1)\frac{n}{2},$$

more than $\frac{n}{2}$ squares contain the same number, as needed.

**Problem 9.** The numbers $1, 2, \ldots, 100$ are arranged in the squares of a $10 \times 10$ table in the following way: the numbers $1, \ldots, 10$ are in the bottom row in increasing order, numbers $11, \ldots, 20$ are in the next row in increasing order, and so on. One can choose any number and two of its neighbors in two opposite directions (horizontal, vertical, or diagonal). Then either the number is increased by 2 and its neighbors are decreased by 1, or the number is decreased by 2 and its neighbors are increased by 1. After several such operations the table again contains all the numbers $1, 2, \ldots, 100$. Prove that they are in the original order.

**Solution.** Label the table entry in the $i$-th row and $j$-th column by $a_{ij}$, where the bottom-left corner is in the first row and first column. Let $b_{ij} = 10(i - 1) + j$ be the number originally in the $i$-th row and $j$-th column.

Observe that

$$P = \sum_{i,j=1}^{10} a_{ij} b_{ij}$$

is invariant. Indeed, every time the entries $a_{mn}$, $a_{pq}$, $a_{rs}$ are changed (with $m + r = 2p$ and $n + s = 2q$), $P$ increases or decreases by $b_{mn} - 2b_{pq} + b_{rs}$, but this equals

$$10((m-1) + (r-1) - 2(p-1)) + (n + s - 2q) = 0.$$

In the beginning,

$$P = \sum_{i,j=1}^{10} b_{ij} b_{ij};$$

at the end, the entries $a_{ij}$ equal the $b_{ij}$ in some order, so we now have

$$P = \sum_{i,j=1}^{10} a_{ij} b_{ij}.$$

By the rearrangement inequality, $\sum_{i,j=1}^{10} a_{ij} b_{ij} \leq \sum_{i,j=1}^{10} b_{ij} b_{ij}$ with equality only when each $a_{ij} = b_{ij}$. The equality does occur since P is invariant. Therefore the $a_{ij}$ do indeed equal the $b_{ij}$ in the same order, and thus the entries 1, 2, . . . , 100 appear in their original order.

## 6.2 Functions defined on sets of points

**Problem 3.** Let $D$ be a set of $n \geq 1$ concentric circles in the plane. Suppose that the function $f : D \to D$ satisfies

$$d(f(A), f(B)) \geq d(A, B)$$

for every $A, B \in D$, where $d(M, N)$ is the distance between the points $M$ and $N$.

Prove that

$$d(f(A), f(B)) = d(A, B)$$

for every $A, B \in D$.

**Solution.** Let $D_1, D_2, \ldots, D_n$ be the concentric circles with radii $r_1 < r_2 < \cdots < r_n$ and center $O$. For an arbitrary point $A \in D$, we will denote $f(A) = A'$.

We first notice that if $A, B \in D_n$ such that $AB$ is a diameter, then $A'B'$ is also a diameter of $D_n$. If $C$ is another point on $D_n$, we have

$$A'C'^2 + B'C'^2 \geq AC^2 + BC^2 = AB^2 = A'B'^2.$$

Because $OC'$ is a median of the triangle $A'B'C'$, it follows that

$$OC'^2 = \tfrac{1}{2}\left(A'C'^2 + B'C'^2\right) - \frac{1}{4}A'B'^2 = r_n^2,$$

and hence $C' \in D_n$ and $A'C' = AC$, $B'C' = BC$. We deduce that $f(D_n) \subset D_n$ and the restriction of $f$ to $D_n$ is an isometry. Now take $A, X, Y, Z$ on $D_n$ such that $AX = AY = A'Z$. It follows that $A'X' = A'Y' = A'Z$, and hence one of the points $X', Y'$ coincides with $Z$. This shows that $f(D_n) = D_n$. Since $f$ is clearly injective, it follows that $f(D_i) = D_i$, for all $i$, $1 \leq i \leq n-1$, and that all restrictions, of $f$ to $D_i$ are isometries.

Next, we prove that distances between adjacent circles, say $D_1$ and $D_2$, are preserved. Take $A, B, C, D$ on $D_1$ such that $ABCD$ is a square and let $A', B', C', D'$ be the points on $D_2$ closest to $A, B, C, D$, respectively.

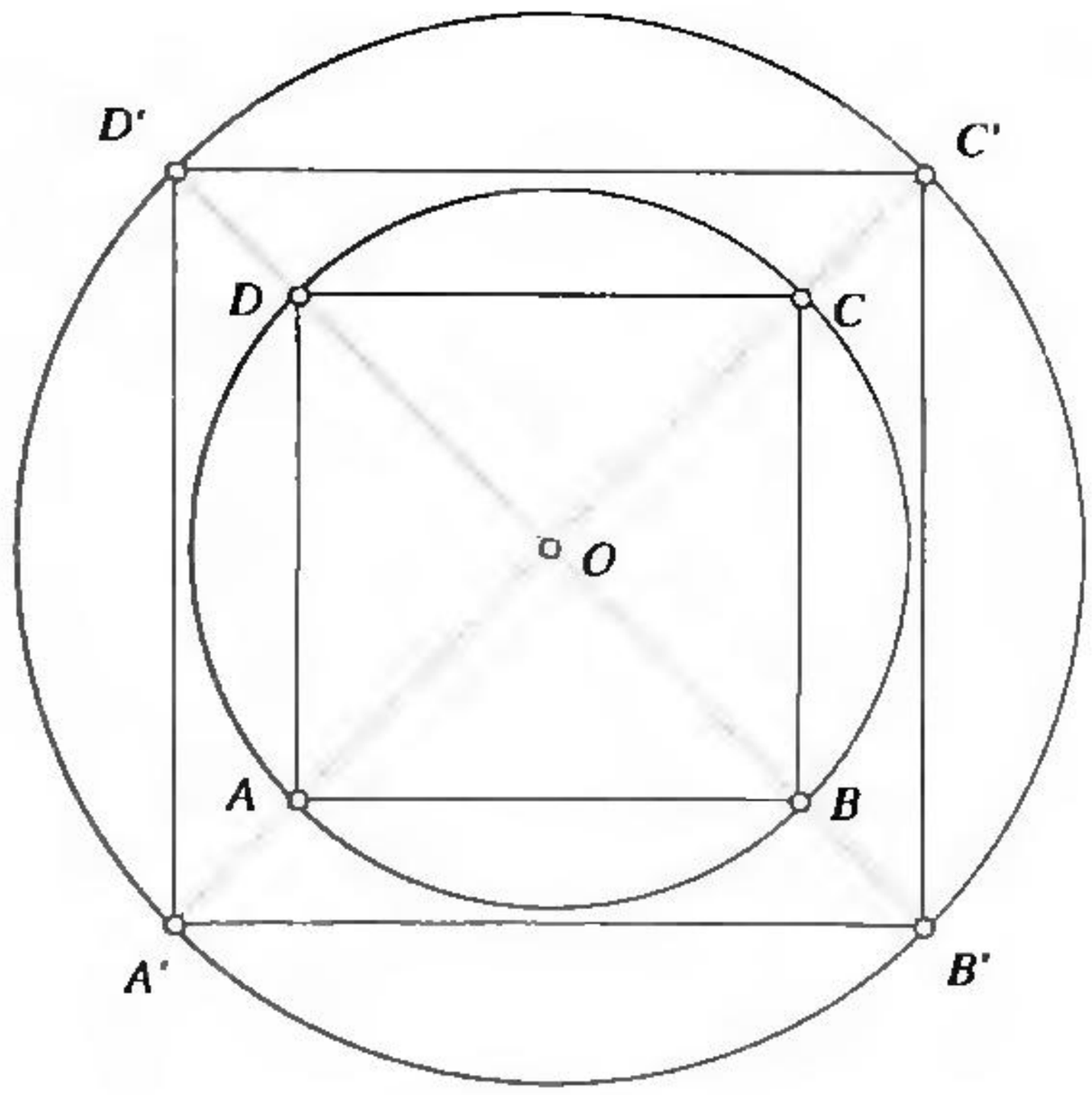


Figure 6.1

Then $A'B'C'D'$ is also a square and the distance from $A$ to $C'$ is the maximum between any point on $D_1$ and any point on $D_2$. Hence the eight points maintain their relative position under $f$ and this shows that $f$ is an isometry.

**Problem 4.** Let $S$ be a set of $n \geq 4$ points in the plane, with the property that no three of them are collinear and not all of them lie on a circle. Find all functions $f : S \to \mathbf{R}$ with the property that for any circle $C$ containing at least three points of $S$,

$$\sum_{P \in C \cap S} f(P) = 0.$$

**Solution.** For two distinct points $A$, $B$ of $S$, we denote by $C_{A,B}$ the set of circles determined by $A$, $B$ and other points of $S$. Suppose $C_{A,B}$ has $k$ elements. Since the points of $S$ do not all lie on the same circle, it follows that $k \geq 2$. Because

$$\sum_{P \in C \cap S} f(P) = 0$$

for all $C \in C_{A,B}$, we deduce that

$$\sum_{C \in C_{A,B}} \sum_{P \in C \cap S} f(P) = 0.$$

On the other hand, it is not difficult to see that

$$\sum_{C \in C_{A,B}} \sum_{P \in C \cap S} f(P) = \sum_{P \in S} f(P) + (k-1)\left(f(A) + f(B)\right).$$

Thus the sum $\sum_{P \in S} f(P)$ and $f(A) + f(B)$ have opposite signs for all $A$, $B$ in $S$. If, for instance, $\sum_{P \in S} f(P) \geq 0$, then $f(A) + f(B) \leq 0$ for all $A$, $B$ in $S$. Let $S = \{A_1, A_2, \ldots, A_n\}$. Then

$$f(A_1) + f(A_2) \leq 0,\ f(A_2) + f(A_3) \leq 0, \ldots, f(A_n) + f(A_1) \leq 0,$$

yielding

$$2\sum_{P \in M} f(P) \leq 0.$$

Hence $\sum_{P \in M} f(P) = 0$ and $f(A) + f(B) = 0$ for all distinct points $A$, $B$. This implies that $f$ is the zero function. Indeed, let $A$, $B$, $C$ be three distinct

points in $S$. It is not difficult to see that the equalities

$$f(A) + f(B) = 0,$$
$$f(B) + f(C) = 0,$$
$$f(A) + f(C) = 0$$

yield

$$f(A) = f(B) = f(C) = 0$$

and our claim is proved.

**Problem 5.** Let $P$ be the set of all points in the plane and $L$ be the set of all lines of the plane. Find, with proof, whether there exists a bijective function $f : P \to L$ such that for any three collinear points $A, B, C$, the lines $f(A)$, $f(B)$, and $f(C)$ are either parallel or concurrent.

**Solution.** Let $A_i$, $i = 1, 2, 3$ be three distinct points in the plane and $l_i = f(A_i)$. We claim that if $l_1, l_2, l_3$ are concurrent or parallel, then $A_1, A_2, A_3$ are collinear. Indeed, suppose that $l_1, l_2, l_3$ intersect at $M$ and that $A_1A_2A_3$ is a nondegenerate triangle. Then for any point $B$ in the plane, we can find points $B_2$, $B_3$ on the lines $A_1A_2$ and $A_1A_3$, respectively, such that $B$, $B_2$, $B_3$ are collinear.

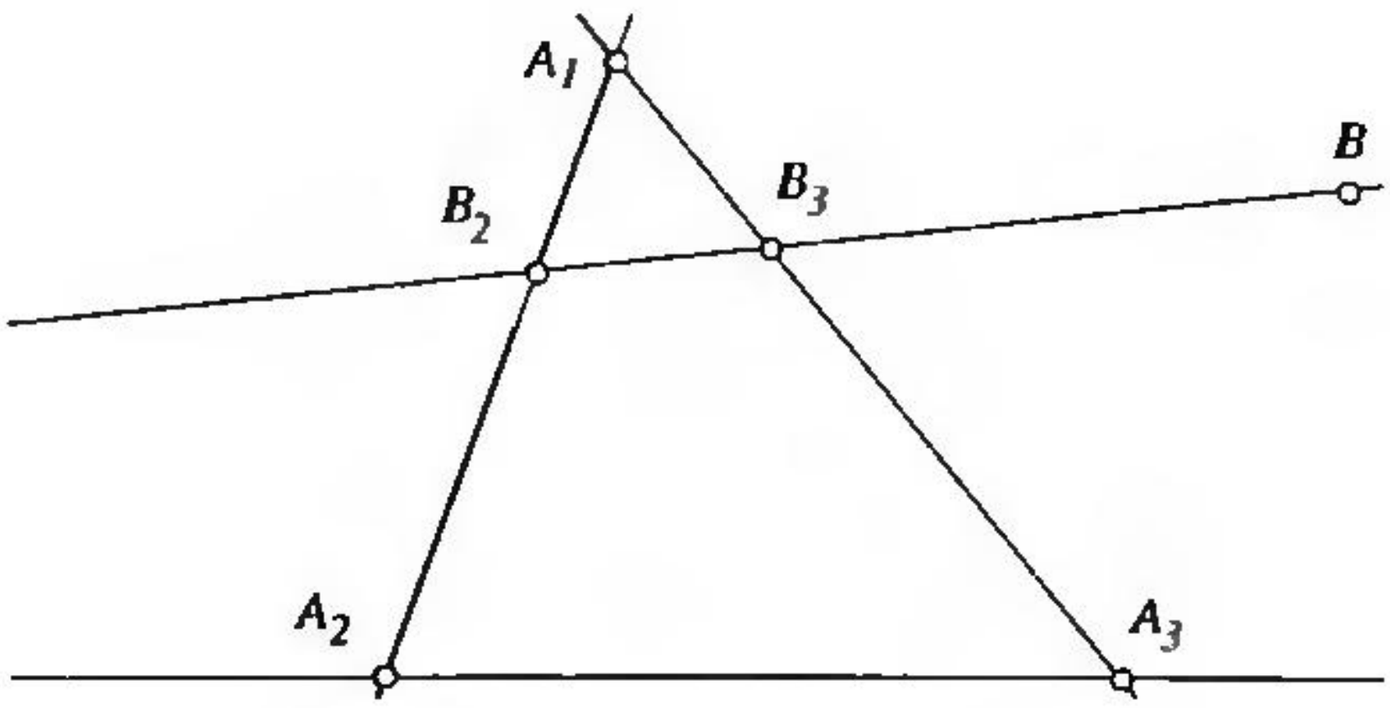


Figure 6.2

Because $A_1$, $B_2$, $A_2$ are collinear, it follows that $f(B_2)$ is a line passing through $M$. The same is true for $f(B_1)$, and thus also for $f(B)$. This contradicts the surjectivity of $f$. A similar argument can be given if $l_1, l_2, l_3$ are parallel.

We conclude that the restriction of $f$ to any line $l$ defines a bijection from $l$ to a pencil of lines (passing through a point or parallel). Consider two pencils $P_1$ and $P_2$ of parallel lines. The inverse images of $P_1$, $P_2$ are two parallel

lines $l_1, l_2$ ($P_1$ and $P_2$ have no common lines, and therefore $l_1$ and $l_2$ have no common points). Let $P_3$ be a pencil of concurrent lines whose inverse image is a line $l$, clearly not parallel to $l_1$ or $l_2$. Let $l'$ be a line parallel to $l$. Then $f(l')$ is a pencil of concurrent lines and it follows that there is a line through the points corresponding to $l$ and $l'$ whose inverse image would be a point on both $l$ and $l'$, a contradiction. Hence no such function exists.

**Problem 6.** Let $S$ be the set of interior points of a sphere and $C$ be the set of interior points of a circle. Find, with proof, whether there exists a function $f : S \to C$ such that for any points $A, B \in S$, $d(A, B) \leq d(f(A), f(B))$.

**Solution.** No such function exists. Indeed, suppose $f : S \to C$ has the stated property. Consider a cube inscribed in the sphere and assume, with no loss of generality, that its sides have length 1. Partition the cube into $n^3$ smaller cubes and let $A_1, A_2, \ldots, A_{(n+1)^3}$ be their vertices. For all $i \neq j$ we have

$$d(A_i, A_j) \geq \frac{1}{n},$$

and thus

$$d(f(A_i), f(A_j)) \geq \frac{1}{n}.$$

It follows that the disks $D_i$ with centers $f(A_i)$ and radius $\frac{1}{2n}$ are disjoint and contained in a circle $C'$ with radius $r + \frac{1}{n}$, where $r$ is the radius of $C$. The sum of the areas of these disks is then less than the area of $C'$, and consequently

$$(n+1)^3 \frac{\pi}{4n^2} \leq \pi \left(r + \frac{1}{n}\right)^2.$$

This inequality cannot hold for sufficiently large $n$, which proves our claim.

**Problem 7.** Let $S$ be the set of all polygons in the plane. Prove that there exists a function $f : S \to (0, +\infty)$ such that

1. $f(P) < 1$, for any $P \in S$;

2. If $P_1, P_2 \in S$ have disjoint interiors and $P_1 \cup P_2 \in S$, then $f(P_1 \cup P_2) = f(P_1) + f(P_2)$.

**Solution.** Consider a covering of the plane with unit squares and denote them by $U_1, U_2, \ldots, U_n, \ldots$. If $P$ is a polygon, define

$$f(P) = \sum_{k \geq 1} \frac{1}{2^k} [P \cap U_k].$$

Because $P$ intersects a finite number of unit squares, the above sum is finite, so $f$ is well defined. Moreover, $f$ satisfies the conditions of the problem. Indeed, we have $[P \cap U_k] \leq 1$ for all $k$, and if

$$N = \max\{i | P \cap U_i \neq \emptyset\},$$

then

$$f(P) = \sum_{k \geq 1}^{N} \frac{1}{2^k} [P \cap U_k] \leq \sum_{k \geq 1}^{N} \frac{1}{2^k} = 1 - \frac{1}{2^N} < 1.$$

For the second condition, observe that

$$[(P_1 \cap P_2) \cap U_k] = [P_1 \cap U_k] + [P_2 \cap U_k]$$

for all $U_k$ and for all polygons $P_1$, $P_2$ such that $P_1 \cup P_2 \in S$. Hence

$$f(P_1 \cup P_2) = f(P_1) + f(P_2).$$

## 6.3 Count twice!

**Problem 3.** Find how many committees with a chairman can be chosen from a set of $n$ persons. Derive the identity

$$\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n} = n2^{n-1}.$$

**Solution.** The number of persons in the committee may vary between 1 and $n$. Let us count the number of committees having $k$ members. The $k$ members can be chosen in $\binom{n}{k}$ ways, while the chairman can be chosen in $k$ ways, yielding a total of $k\binom{n}{k}$ committees with $k$ members. Adding up for $k = 1, 2, \ldots, n$, we obtain that the total number of committees is

$$\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n}.$$

On the other hand, we can first choose the chairman. This can be done in $n$ ways. Next, we choose the rest of the committee, which is an arbitrary subset of the remaining $n-1$ persons. Because a set with $n-1$ elements contains $2^{n-1}$ subsets, it follows that the committee can be completed in $2^{n-1}$ ways and the total number of committees is

$$n2^{n-1}.$$

*Observation.* There are many proofs of the given equality. An interesting one is the following. Consider the identity

$$(1+x)^n = 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{n-1}x^{n-1} + \binom{n}{n}x^n.$$

Differentiating both sides with respect to $x$ yields

$$n(1+x)^{n-1} = \binom{n}{1} + 2\binom{n}{2}x + \cdots + (n-1)\binom{n}{n-1}x^{n-2} + n\binom{n}{n}x^{n-1}.$$

Setting $x = 1$ gives the desired result.

**Problem 4.** In how many ways can one choose $k$ balls from a set containing $n-1$ red balls and a blue one? Derive the identity

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

**Solution.** We have to choose $k$ balls from a set containing $n$ balls, so the answer is $\binom{n}{k}$. On the other hand, the blue ball may or may not be among the selected $k$ balls. If the blue ball is selected, then, in fact, we have chosen $k-1$ red balls from $n-1$ red balls, and this can be done in $\binom{n-1}{k-1}$ ways. If the blue ball is not selected, then we have chosen $k$ red balls from the $n-1$ red ones. This can be done in $\binom{n-1}{k-1}$ which leads to a total of

$$\binom{n-1}{k} + \binom{n-1}{k-1}$$

possibilities to choose the $k$ balls.

*Observation.* Using similar arguments, we can obtain a more general identity. Let us count in how many ways one can choose $k$ balls from a set containing $n$ red and $m$ blue balls. Ignoring the color of the balls, the answer is $\binom{n+m}{k}$. Let us take into consideration the fact that the chosen $k$ balls can be

broken down as follows: all red, or $k-1$ red and 1 blue, or $k-2$ red and 2 blue, etc. We obtain the identity

$$\binom{n+m}{k} = \binom{n}{k}\binom{m}{0} + \binom{n}{k-1}\binom{m}{1} + \binom{n}{k-2}\binom{m}{2} + \cdots + \binom{n}{0}\binom{m}{k}.$$

**Problem 5.** Let $S$ be a set of $n$ persons such that:

(i) any person is acquainted with exactly $k$ other persons in $S$;

(ii) any two persons that are acquainted have exactly $l$ common acquaintances in $S$;

(iii) any two persons that are not acquainted have exactly $m$ common acquaintances in $S$.

Prove that

$$m(n-k) - k(k-l) + k - m = 0.$$

**Solution.** Let $a$ be a fixed element of $S$. Let us count the triples $(a, x, y)$ such that $a, x$ are acquainted, $x, y$ are acquainted and $a, y$ are not acquainted. Because $a$ is acquainted to exactly $k$ other persons in $S$, $x$ can be chosen in $k$ ways and for fixed $a$ and $x$, $y$ can be chosen in $k-1-l$ ways. Thus the number of such triples is $k(k-1-l)$. Let us count again, choosing $y$ first. The number of persons not acquainted to $a$ equals $n-k-1$, so $y$ can be chosen in $n-k-1$ ways. Because $x$ is a common acquaintance of $a$ and $y$, it can be chosen in $m$ ways, yielding a total of $m(n-k-1)$ triples. It is not difficult to see that the equality

$$k(k-1-l) = m(n-k-1)$$

is equivalent to the desired one.

**Problem 6.** Let $n$ be an odd integer greater than 1 and let $c_1, c_2, \ldots, c_n$ be integers. For each permutation $a = (a_1, a_2, \ldots, a_n)$ of $\{1, 2, \ldots, n\}$, define

$$S(a) = \sum_{i=1}^{n} c_i a_i.$$

Prove that there exist permutations $a \neq b$ of $\{1, 2, \ldots, n\}$ such that $n!$ is a divisor of $S(a) - S(b)$.

**Solution.** Denote by $\sum_a S(a)$ the sum of $S(a)$ over all $n!$ permutations of $\{1, 2, \ldots, n\}$. We compute $\sum_a S(a) \pmod{n!}$ in two ways. First, assuming that the conclusion is false, it follows that for each $a$, $S(a)$ has a different remainder mod $n!$, and thus these remainders are the numbers $0, 1, 2, \ldots, n! - 1$. It follows that

$$\sum_a S(a) \equiv \tfrac{1}{2}n!(n! - 1) \pmod{n!}.$$

On the other hand,

$$\sum_a S(a) = \sum_a \sum_{i=1}^{n} c_i a_i = \sum_{i=1}^{n} c_i \sum_a a_i.$$

For each $i$, in the sum $\sum_a a_i$, each of the numbers $1, 2, \ldots, n$, appears $(n-1)!$ times, hence

$$\sum_a a_i = (n-1)!(1 + 2 + \cdots + n) = \tfrac{1}{2}(n+1)!.$$

It follows that

$$\sum_a S(a) = \tfrac{1}{2}(n+1)! \sum_{i=1}^{n} c_i.$$

We deduce that

$$\tfrac{1}{2}n!(n! - 1) \equiv \tfrac{1}{2}(n+1)! \sum_{i=1}^{n} c_i \pmod{n!}.$$

Because $n > 1$ is odd, the right-hand side is congruent to 0 mod $n!$, while the left-hand side is not, a contradiction.

**Problem 7.** Let $a_1 \le a_2 \le \cdots \le a_n = m$ be positive integers. Denote by $b_k$ the number of those $a_i$ for which $a_i \ge k$. Prove that

$$a_1 + a_2 + \cdots + a_n = b_1 + b_2 + \cdots + b_m.$$

**Solution.** Let us consider an $n \times m$ array of numbers $(x_{ij})$ defined as follows: in row $i$, the first $a_i$ entries are equal to 1 and the remaining $m - a_i$

entries are equal to 0. For instance, if $n = 3$ and $a_1 = 2, a_2 = 4, a_3 = 5$, the array is

| | | | | |
|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 0 |
| 1 | 1 | 1 | 1 | 1 |

Now, if we examine column $j$, we notice that the number of 1's in that column equals the number of those $a_i$ greater than or equal to $j$, that is, $b_j$. The desired result follows by adding up in two ways the 1's in the array. The total number of 1's is $\sum_{i=1}^{n} a_i$, if counted by rows, and $\sum_{j=1}^{m} b_j$, if counted by columns.

**Problem 8.** In how many ways can one fill an $m \times n$ table with $\pm 1$ such that the product of the entries in each row and each column equals $-1$?

**Solution.** Denote by $a_{ij}$ the entries in the table. For $1 \leq i \leq m-1$ and $1 \leq j \leq n-1$, we let $a_{ij} = \pm 1$ in an arbitrary way. This can be done in $2^{(m-1)(n-1)}$ ways. The values for $a_{mj}$ with $1 \leq j \leq n-1$, and for $a_{in}$ with $1 \leq i \leq m-1$, are uniquely determined by the condition that the product of the entries in each row and each column equals $-1$. The value of $a_{mn}$ is also uniquely determined, but it is necessary that

$$\prod_{j=1}^{n-1} a_{mj} = \prod_{i=1}^{m-1} a_{in}. \qquad (*)$$

If we denote by

$$P = \prod_{i=1}^{m-1} \prod_{j=1}^{n-1} a_{ij}$$

we observe that

$$P \prod_{j=1}^{n-1} a_{mj} = (-1)^{n-1}$$

and

$$P \prod_{i=1}^{m-1} a_{in} = (-1)^{m-1}.$$

Hence (*) holds if and only if $m$ and $n$ have the same parity.

## 6.4 Sequences of integers

**Problem 4.** Prove that there exist sequences of odd integers $(x_n)_{n\geq 3}$, $(y_n)_{n\geq 3}$ such that

$$7x_n^2 + y_n^2 = 2^n$$

for all $n \geq 3$.

**Solution.** For $n = 3$, define $x_3 = y_3 = 1$. Suppose that for $n \geq 3$ there exist odd integers $x_n$ and $y_n$ such that $7x_n^2+y_n^2 = 2^n$. Observe that the integers

$$\frac{x_n + y_n}{2} \text{ and } \frac{x_n - y_n}{2}$$

cannot be both even, since their sum is odd.

If $\frac{x_n+y_n}{2}$ is odd, we define

$$x_{n+1} = \frac{x_n + y_n}{2}\,,\; y_{n+1} = \frac{7x_n - y_n}{2}$$

and the conclusion follows by noticing that

$$7x_{n+1}^2 + y_{n+1}^2 = \frac{1}{4}\left(7(x_n + y_n)^2 + (7x_n - y_n)^2\right) = 2(7x_n^2 + y_n^2) = 2^{n+1}.$$

If $\frac{x_n-y_n}{2}$ is odd, we define

$$x_{n+1} = \frac{x_n - y_n}{2}\,,\; y_{n+1} = \frac{7x_n + y_n}{2}$$

and a similar computation yields the result.

**Problem 5.** Let $x_1 = x_2 = 1$, $x_3 = 4$, and

$$x_{n+3} = 2x_{n+2} + 2x_{n+1} - x_n$$

for all $n \geq 1$. Prove that $x_n$ is a square for all $n \geq 1$.

**Solution.** We first notice that $x_1 = x_2 = 1^2$, $x_3 = 2^2$, $x_4 = 3^2$, $x_5 = 5^2$, $x_6 = 8^2$, and so forth. This leads to the assumption that $x_n = F_n^2$, where $F_n$ is the $n$-th term of the Fibonacci sequence defined by $F_1 = F_2 = 1$ and $F_{n+1} = F_n+F_{n-1}$, for all $n \geq 2$. We prove this assertion inductively. Suppose it is true for all $k \leq n + 2$. Then

$$\begin{aligned} x_{n+3} &= 2F_{n+2}^2 + 2F_{n+1}^2 - F_n^2 = 2F_{n+2}^2 + 2F_{n+1}^2 - (F_{n+2} - F_{n+1})^2 \\ &= F_{n+2}^2 + F_{n+1}^2 + 2F_{n+1}F_{n+2} = (F_{n+2} + F_{n+1})^2 = F_{n+3}^2, \end{aligned}$$

and the assertion is proved.

**Problem 6.** The sequence $(a_n)_{n\geq 0}$ is defined by $a_0 = a_1 = 1$ and

$$a_{n+1} = 14a_n - a_{n-1}$$

for all $n \geq 1$. Prove that the number $2a_n - 1$ is a square for all $n \geq 0$.

**Solution.** We have $2a_0 - 1 = 1, 2a_1 - 1 = 1, 2a_2 - 1 = 5^2, 2a_3 - 1 = 19^2$, $2a_4 - 1 = 71^2$. We observe that if we define $b_0 = -1, b_1 = 1, b_2 = 5$, $b_3 = 19, b_4 = 71$, then $b_{n+1} = 4b_n - b_{n-1}$ for $1 \leq n \leq 3$. We will prove inductively that

$$2a_n - 1 = b_n^2,$$

where $b_0 = -1, b_1 = 1$ and $b_{n+1} = 4b_n - b_{n-1}$ for all $n \geq 1$. Suppose this is true for $1, 2, \ldots, n$ and observe that

$$\begin{aligned} 2a_{n+1} - 1 &= 14(2a_n - 1) - (2a_{n-1} - 1) + 12 = 14b_n^2 - b_{n-1}^2 + 12 \\ &= 16b_n^2 - 8b_nb_{n-1} + b_{n-1}^2 - 2b_n^2 + 8b_nb_{n-1} - 2b_{n-1}^2 + 12 \\ &= (4b_n - b_{n-1})^2 - 2(b_n^2 + b_{n-1}^2 - 4b_nb_{n-1} - 6) \\ &= b_{n+1}^2 - 2(b_n^2 + b_{n-1}^2 - 4b_nb_{n-1} - 6). \end{aligned}$$

Hence it suffices to prove that $b_n^2 + b_{n-1}^2 - 4b_nb_{n-1} - 6 = 0$. This also follows by induction. The equality is true for $n = 1$ and $n = 2$. Suppose it holds for $n$ and observe that

$$\begin{aligned} b_{n+1}^2 + b_n^2 - 4b_{n+1}b_n - 6 &= (4b_n - b_{n-1})^2 + b_n^2 - 4(4b_n - b_{n-1})b_n - 6 \\ &= b_n^2 + b_{n-1}^2 - 4b_nb_{n-1} - 6 = 0, \end{aligned}$$

as desired.

**Problem 7.** The sequence $(x_n)_{n\geq 1}$ is defined by $x_1 = 0$ and

$$x_{n+1} = 5x_n + \sqrt{24x_n^2 + 1}$$

for all $n \geq 1$. Prove that all $x_n$ are positive integers.

**Solution.** We first notice that the sequence is increasing and all of its terms are positive. Next we observe that the recursive relation is equivalent to

$$x_{n-1}^2 - 10x_nx_{n-1} + x^2 - 1 = 0.$$

Replacing $n$ by $n-1$ yields

$$x_n^2 - 10x_n x_{n-1} + x_{n-1}^2 - 1 = 0.$$

Hence for $n \geq 2$ the numbers $x_{n+1}$ and $x_{n-1}$ are distinct roots of the equation

$$x^2 - 10xx_n + x_n^2 - 1 = 0.$$

Viète's relations yield $x_{n+1} + x_{n-1} = 10x_n$, or, equivalently,

$$x_{n+1} = 10x_n - x_{n-1}$$

for all $n \geq 2$. Because $x_1 = 1$ and $x_2 = 10$, it follows inductively that all $x_n$ are positive integers.

**Problem 8.** Let $(a_n)_{n\geq 1}$ be an increasing sequence of positive integers such that

1. $a_{2n} = a_n + n$ for all $n \geq 1$;
2. if $a_n$ is a prime, then $n$ is a prime.

Prove that $a_n = n$, for all $n \geq 1$.

**Solution.** Let $a_1 = c$. Then $a_2 = a_1 + 1 = c + 1$, $a_4 = a_2 + 2 = c + 3$. Since the sequence is increasing, it follows that $a_3 = c + 2$. We prove that $a_n = c + n - 1$ for all $n \geq 1$. Indeed, if $n = 2^k$ for some integer $k$, this follows by induction on $k$. Suppose that

$$a_{2^k} = c + 2^k - 1.$$

Then

$$a_{2^{k+1}} = a_{2\cdot 2^k} = a_{2^k} + 2^k = c + 2^{k+1} - 1.$$

If $2^k < n < 2^{k+1}$, then

$$c + 2^k - 1 = a_{2^k} < a_{2^k+1} < \cdots < a_n < \cdots < a_{2^{k+1}} = c + 2^{k+1} - 1,$$

and this is possible only if $a_n = c + n - 1$.

Next, we prove that $c = 1$. Suppose that $c \geq 2$. Let $p < q$ be two consecutive prime numbers greater than $c$. We have

$$a_{q-c+1} = c + q - c = q,$$

and hence $q - c + 1$ is a prime. Clearly, $q - c + 1 \leq p$. It follows that for any consecutive prime numbers $p < q$, we have

$$q - p \leq c - 1.$$

The numbers $(c+1)!+2, (c+1)!+3, \ldots, (c+1)!+c+1$ are all composite, and hence if $p$ and $q$ are the consecutive primes such that

$$p < (c+1)!+2 < (c+1)!+c+1 < q,$$

then

$$q - p > c - 1,$$

a contradiction. It follows that $c = 1$ and $a_n = n$ for all $n \geq 1$.

## 6.5 Equations with infinitely many solutions

**Problem 3.** Find all triples of integers $(x, y, z)$ such that

$$x^2 + xy = y^2 + xz.$$

**Solution.** The given equation is equivalent to

$$x(x - z) = y(y - x).$$

Let $d = \gcd(x, y)$. Then $x = da$, $y = db$, with $\gcd(a, b) = 1$. We deduce that $y - x = ka$ and $x - z = kb$ for some integer $k$. Because $\gcd(a, b-a) = \gcd(a, b) = 1$, it follows that $b - a$ divides $k$. Setting $k = m(b - a)$, we obtain $d = ma$ and hence the solutions are

$$x = ma^2,\ y = mab,\ z = m(a^2 + ab - b^2)$$

where $m, a, b$ are arbitrary integers.

**Problem 4.** Let $n$ be an integer. Prove that the equation

$$x^2 + y^2 = n + z^2$$

has infinitely many integer solutions.

**Solution.** The equation is equivalent to

$$(x - z)(x + z) + y^2 = n.$$

If we set $x - z = 1$, then we obtain

$$2x - 1 + y^2 = n$$

and

$$x = \frac{n + 1 - y^2}{2}.$$

Now, it suffices to take $y = n + m$, where $m$ is an odd integer, to insure that $x$ is an integer as well. Indeed, if $m = 2k + 1$, then

$$x = \frac{n + 1 - y^2}{2} = \frac{n + 1 - (n + 2k + 1)^2}{2}$$
$$= -\frac{n(n + 1)}{2} - 2nk - 2k^2 - 2k,$$

an integer. Thus $z = x - 1$ is an integer as well.

**Problem 5.** Let $m$ be a positive integer. Find all pairs of integers $(x, y)$ such that

$$x^2(x^2 + y) = y^{m+1}.$$

**Solution.** Multiplying the equation by 4 and adding $y^2$ to both sides yields the equivalent form

$$(2x^2 + y)^2 = y^2 + 4y^{m+1}$$

or

$$(2x^2 + y)^2 = y^2(1 + 4y^{m-1}).$$

It follows that $1 + 4y^{m-1}$ is an odd square, say $(2a + 1)^2$. We deduce that $y^{m-1} = a(a + 1)$.Because $a$ and $a + 1$ are relatively prime integers, each of them must be the $(m - 1)$-th power of some integers. Clearly, this is possible only if $m = 2$, and hence $y = a(a + 1)$. It follows that

$$2x^2 + a(a + 1) = a(a + 1)(2a + 1).$$

Thus $x^2 = a^2(a + 1)$. We deduce that $a + 1$ is a square. Setting $a + 1 = t^2$ yields $x = t^3 - t$ and $y = t^4 - t^2$.

**Problem 6.** Let $m$ be a positive integer. Find all pairs of integers $(x, y)$ such that

$$x^2(x^2 + y^2) = y^{m+1}.$$

**Solution.** The equation can be written in the equivalent form

$$(2x^2 + y)^2 = y^4 + 4y^{m+1}.$$

Observe that $y^4 + 4y^{m+1}$ cannot be a square for $m = 1$. Indeed, in this case

$$y^4 + 4y^{m+1} = y^4 + 4y^2 = y^2(y^2 + 4).$$

and no two nonzero squares differ by 4. A similar argument works for $m = 2$.

Now, for $m \geq 3$, we write the equation in the equivalent form

$$(2x^2 + y)^2 = y^4(1 + 4y^{m-3}).$$

As in the previous solution, we deduce that $y = a(a + 1)$ for some integer $a$, and then that $x = a^3(a + 1)$. It follows that $a = t^2$ for some integer $t$, so the solutions are $x = t^5 + t^3$, $y = t^4 + t^2$.

**Problem 7.** Find all nonnegative integers $a, b, c, d, n$ that satisfy

$$a^2 + b^2 + c^2 + d^2 = 7 \cdot 4^n.$$

**Solution.** For $n = 0$, we have $2^2 + 1^2 + 1^2 + 1^2 = 7$. Therefore $(a, b, c, d) = (2, 1, 1, 1)$ and all of its permutations. If $n \geq 1$, then we have that $a^2 + b^2 + c^2 + d^2 \equiv 0 (\mathrm{mod}\, 4)$, hence the numbers have the same parity. We analyze two cases.

a) The numbers $a, b, c, d$ are odd. We write $a = 2a' + 1$, etc. We obtain:

$$4a'(a' + 1) + 4b'(b' + 1) + 4c'(c' + 1) + 4d'(d' + 1) = 4(7 \cdot 4^{n-1} - 1).$$

The left-hand side of the equality is divisible by 8. This establishes that $7 \cdot 4^{n-1} - 1$ must be even, which happens only for $n = 1$. We obtain that $a^2 + b^2 + c^2 + d^2 = 28$, with the solutions (3, 3, 3, 1) and ( 1, 1, 1, 5).

b) The numbers $a, b, c, d$ are even. Write $a = 2a'$, etc. We obtain

$$a'^2 + b'^2 + c'^2 + d'^2 = 7 \cdot 4^{n-1},$$

so we proceed recursively.

Finally, we obtain the solutions $(2^{n+1}, 2^n, 2^n, 2^n)$, $(3 \cdot 2^n, 3 \cdot 2^n, 3 \cdot 2^n, 2^n)$, $(2^n, 2^n, 2^n, 5 \cdot 2^n)$, and the respective permutations of these tuples.

**Problem 8.** Show that there are infinitely many positive integers $x$, $y$, $z$, $t$, with no common divisor greater than 1, satisfying the equality

$$x^3 + y^3 + z^2 = t^4.$$

**Solution.** Consider the identity

$$(a+1)^4 - (a-1)^4 = 8a^3 + 8a.$$

Taking $a = b^3$, with $b$ an even integer gives

$$(b^3+1)^4 = (2b^3)^3 + (2b)^3 + \left((b^3-1)^2\right)^2.$$

Since $b$ is even, $b^3+1$ and $b^3-1$ are odd integers. It follows that the numbers $x = 2b^3$, $y = 2b$, $z = (b^3-1)^2$ and $t = b^3+1$ have no common divisor greater than 1.

**Problem 9.** Let $k \geq 6$ be an integer number. Prove that the system of equations

$$\begin{cases} x_1 + x_2 + \cdots + x_{k-1} = x_k, \\ x_1^3 + x_2^3 + \cdots + x_{k-1}^3 = x_k, \end{cases}$$

has infinitely many integral solutions.

**Solution.** Consider the identity

$$(m+1)^3 + (m-1)^3 + (-m)^3 + (-m)^3 = 6m.$$

If $n$ is an arbitrary integer, then $\frac{n-n^3}{6}$ is also an integer since $n - n^3 = -(n-1)\,n\,(n+1)$ and the product of three consecutive integers is divisible by 6. Setting $m = \frac{n-n^3}{6}$ in the identity above gives

$$\left(\frac{n-n^3}{6}+1\right)^3 + \left(\frac{n-n^3}{6}-1\right)^3 + \left(\frac{n^3-n}{6}\right)^3 + \left(\frac{n^3-n}{6}\right)^3 + n^3 = n.$$

On the other hand, we have

$$\left(\frac{n-n^3}{6}+1\right) + \left(\frac{n-n^3}{6}-1\right) + \frac{n^3-n}{6} + \frac{n^3-n}{6} + n = n,$$

yielding the following solution for $k = 6$ : $x_1 = \frac{n-n^3}{6} + 1$, $x_2 = \frac{n-n^3}{6} - 1$, $x_3 = x_4 = \frac{n^3-n}{6}$, $x_5 = x_6 = n$. For $k > 6$ we can take $x_1$ to $x_5$ as before, $x_6 = -n$ and $x_i = 0$ for all $i > 6$.

## 6.6 Equations with no solutions

**Problem 4.** Prove that the equation

$$4xy - x - y = z^2$$

has no positive integer solutions.

**Solution.** We write the equation in the equivalent form

$$(4x - 1)(4y - 1) = 4z^2 + 1.$$

Let $p$ be a prime divisor of $4x - 1$. Then

$$4z^2 + 1 \equiv 0 \pmod{p}$$

or

$$(2z)^2 \equiv -1 \pmod{p}.$$

On the other hand, Fermat's little theorem yields

$$(2z)^{p-1} \equiv 1 \pmod{p}$$

hence

$$(2z)^{p-1} \equiv ((2z^2)^{\frac{p-1}{2}} \equiv (-1)^{\frac{p-1}{2}} \equiv 1 \pmod{p}.$$

This implies that $p \equiv 1 \pmod 4$. It follows that all prime divisors of $4x - 1$ are congruent to 1 modulo 4, hence $4x - 1 \equiv 1 \pmod 4$, a contradiction.

**Problem 5.** Prove that the equation

$$6(6a^2 + 3b^2 + c^2) = 5d^2$$

has no solution in nonzero integers.

**Solution.** We can assume that $\gcd(a, b, c, d) = 1$, otherwise we simplify the equation with a suitable integer. Clearly, $d$ is divisible by 6, so let $d = 6m$. We obtain

$$6a^2 + 3b^2 + c^2 = 30m^2,$$

hence $c$ is divisible by 3. Replacing $c = 3n$ in the equation yields

$$2a^2 + b^2 + 3n^2 = 10m^2.$$

If $b$ and $n$ are odd, then $b^2 \equiv 1 \pmod 8$ and $3n^2 \equiv 3 \pmod 8$. Because $2a^2 \equiv 0$ or $2 \pmod 8$ and $10m^2 \equiv 0$ or $2 \pmod 8$ this leads to a contradiction. Hence $b$ and $n$ must be even and from the initial equation we deduce that $a$ is also even. This contradicts the assumption that $\gcd(a, b, c, d) = 1$.

**Problem 6.** Prove that the system of equations

$$\begin{cases} x^2 + 6y^2 = z^2, \\ 6x^2 + y^2 = t^2, \end{cases}$$

has no positive integer solutions.

**Solution.** As in the previous solution, we can assume that $\gcd(x, y, z, t) = 1$. Adding up the equations yields

$$7(x^2 + y^2) = z^2 + t^2.$$

The square residues modulo 7 are 0, 1, 2, and 4. It is not difficult to see that the only pair of residues which add up to 0 modulo 7 is (0, 0). Hence $z$ and $t$ are divisible by 7. Setting $z = 7z_1$ and $t = 7t_1$ yields

$$7(x^2 + y^2) = 49(z_1^2 + t_1^2)$$

or

$$x^2 + y^2 = 7(z_1^2 + t_1^2).$$

It follows that $x$ and $y$ are also divisible by 7, contradicting the fact that $\gcd(x, y, z, t) = 1$.

**Problem 7.** Let $k$ and $n$ be positive integers, with $n > 2$. Prove that the equation

$$x^n - y^n = 2^k$$

has no positive integer solutions.

**Solution.** Suppose the contrary and choose $n, k$ such that $x^n - y^n = 2^k$ and $n$ is minimal. If $n$ is even, say $n = 2n'$, then

$$x^n - y^n = \left(x^{n'} - y^{n'}\right)\left(x^{n'} + y^{n'}\right)$$

and it follows that $x^{n'} - y^{n'}$ is also a power of 2, contradicting thus the minimality condition. If $n$ is odd, we define the set

$$A = \left\{ p \in \mathbf{N}^* \mid \text{there exist } x, y \in N \text{ such that } x^n - y^n = 2^p \right\}.$$

Let $p'$ be the minimal element in $A$. If $x^n - y^n = 2^{p'}$, it follows that $x$ and $y$ have the same parity. Since

$$(x - y)\left(x^{n-1} + x^{n-2}y + \cdots + y^{n-1}\right) = 2^{p'},$$

we deduce that $x$ and $y$ are even. Setting $x = 2x_1$ and $y = 2y_1$ yields $x_1^n - y_1^n = 2^{p'-n}$. If $p' > n$, this contradicts the minimality of $p'$. Otherwise, $p' = n$, and it is not difficult to see that the equation $x_1^n - y_1^n = 1$ has no positive integer solutions for $n > 2$.

**Problem 8.** Prove that the equation

$$\frac{x^{2000} - 1}{x - 1} = y^2$$

has no positive integer solutions.

**Solution.** Observe that

$$\frac{x^{2000} - 1}{x - 1} = (x^{1000} + 1)(x^{500} + 1)\frac{x^{500} - 1}{x - 1}.$$

Setting

$$a = x^{1000} + 1,$$
$$b = x^{500} + 1,$$
$$c = \frac{x^{500} - 1}{x - 1},$$

we see that $b$ and $c$ divide $a - 2$ and $c$ divides $b - 2$. It follows that the greatest common divisor of any two of $a, b, c$ is at most 2. The product $abc$ is a square only if $a, b, c$ are squares or doubles of squares. It is not difficult to see that $a$ and $b$ cannot be squares. Hence they are doubles of squares.

This implies that

$$4ab = 4x^{1500} + 4x^{1000} + 4x^{500} + 4$$

is a square. But this is impossible, since a short computation shows that

$$4x^{1500} + 4x^{1000} + 4x^{500} + 4 > (2x^{750} + x^{250})^2$$

and

$$4x^{1500} + 4x^{1000} + 4x^{500} + 4 < (2x^{750} + x^{250} + 1)^2.$$

**Problem 9.** Prove that the equation

$$4\left(x_1^4 + x_2^4 + \cdots + x_{14}^4\right) = 7\left(x_1^3 + x_2^3 + \cdots + x_{14}^3\right)$$

has no solution in positive integers.

**Solution.** Suppose, by way of contradiction, that such a solution exists. Then

$$\sum_{k=1}^{14}\left(x_k^4-\tfrac{7}{4}x_k^3\right)=0.$$

Observe that

$$\begin{aligned}\sum(x_k-1)^4&=\sum\left(x_k^4-4x_k^3+6x_k^2-4x_k+1\right)\\&=\sum\left(x_k^4-\tfrac{7}{4}x_k^3-\tfrac{9}{4}x_k^3+6x_k^2-4x_k+1\right)\\&=\sum\left(-\tfrac{9}{4}x_k^3+6x_k^2-4x_k+1\right).\end{aligned}$$

Since

$$-\tfrac{9}{4}x_k^3+6x_k^2-4x_k=-x_k\left(\tfrac{3}{2}x_k-2\right)^2\le 0$$

we obtain that

$$\sum(x_k-1)^4\le 14.$$

This inequality implies that each $x_k$ is equal to either 1 or 2. Now, suppose $x$ of the numbers $x_1,\ldots,x_{14}$ are equal to 1 and $y$ of them are equal to 2. Then $x+y=14$ and the original equation gives

$$4\,(x+16y)=7\,(x+8y)\,.$$

We obtain $x=\frac{112}{11}$ and $y=\frac{42}{11}$, a contradiction.

## 6.7 Powers of 2

**Problem 3.** Let $n$ be a positive integer such that $2^n+1$ is a prime number. Prove that $n=2^k$, for some integer $k$.

**Solution.** Suppose that $n$ is not a power of 2. Then it can be written in the form $n=2^k(2p+1)$, with $k\ge 0$ and $p\ge 1$. But then we have:

$$2^n+1=\left(2^{2^k}\right)^{2p+1}+1=\left(2^{2^k}+1\right)\left(\left(2^{2^k}\right)^{2p}-\left(2^{2^k}\right)^{2p-1}+\cdots-2^{2^k}+1\right),$$

hence $2^n+1$ is not a prime number

*Observation.* The numbers $F_n = 2^{2^n} + 1$ are called Fermat numbers. Fermat conjectured that all such numbers are prime, which is true for $n \leq 4$, but Euler proved that $F_5 = 2^{32}+1$ is divisible by 641. The proof is quite short: notice that $641 = 640 + 1 = 5 \cdot 2^7 + 1$, and therefore $5 \cdot 2^7 \equiv -1 \pmod{641}$, so that $5^4 \cdot 2^{28} \equiv 1 \pmod{641}$. On the other hand, $641 = 625+16 = 5^4 + 2^4$, so $2^4 \equiv -5^4 \pmod{641}$. Multiplying these two congruencies, we obtain $5^4 \cdot 2^{32} \equiv -5^4 \pmod{641}$. Hence $2^{32} \equiv -1 \pmod{641}$.

**Problem 4.** Let $n$ be a positive integer such that $2^n - 1$ is a prime number. Prove that $n$ is a prime number.

**Solution.** If $n$ is not a prime number, then $n = ab$ for some positive integers $a, b > 1$. We obtain:

$$\begin{aligned} 2^n - 1 = 2^{ab} - 1 &= \left(2^a\right)^b - 1 \\ &= \left(2^a - 1\right)\left(\left(2^a\right)^{b-1} + \left(2^a\right)^{b-2} + \cdots + 2^a + 1\right), \end{aligned}$$

and this factorization shows that $2^n - 1$ is not a prime number.

**Problem 5.** Prove that the number $A = 2^{1992} - 1$ can be written as a product of six integers greater than $2^{248}$.

**Solution.** This again, is an exercise in factorization:

$$\begin{aligned} A = 2^{1992} - 1 = 2^{249 \cdot 8} - 1 &= \left(2^{249}\right)^8 - 1 \\ = \left(2^{249} - 1\right)\left(2^{249} + 1\right)&\left(\left(2^{249}\right)^2 + 1\right)\left(\left(2^{249}\right)^4 + 1\right). \end{aligned}$$

Now, observe that

$$\begin{aligned} \left(2^{249}\right)^2 + 1 &= \left(2^{249}\right)^2 + 2 \cdot 2^{249} + 1 - 2^{250} \\ = \left(2^{249} + 1\right)^2 - \left(2^{125}\right)^2 &= \left(2^{249} - 2^{125} + 1\right)\left(2^{249} + 2^{125} + 1\right), \end{aligned}$$

and

$$\begin{aligned} \left(2^{249}\right)^4 + 1 = 2^{996} + 1 &= \left(2^{332}\right)^3 + 1 \\ = \left(2^{332} + 1\right)&\left(2^{664} - 2^{332} + 1\right). \end{aligned}$$

Thus, the six integers are $2^{249} - 1, 2^{249} + 1, 2^{249} - 2^{125} + 1, 2^{249} + 2^{125} + 1$, $2^{332} + 1$ and $2^{664} - 2^{332} + 1$. It is not difficult to see that all six are greater than $2^{248}$.

**Problem 6.** Determine the remainder when $3^{2^n} - 1$ is divided by $2^{n+3}$.

**Solution.** Observe that after we use $n$ times the identity

$$x^2 - 1 = (x - 1)(x + 1),$$

we obtain

$$3^{2^n} - 1 = (3 - 1)(3 + 1)(3^2 + 1)\left(3^{2^2} + 1\right)\cdots\left(3^{2^{n-1}} + 1\right).$$

Now, each of the numbers $3^2 + 1, 3^{2^2} + 1, \ldots, 3^{2^{n-1}} + 1$ is divisible by 2 but not by 4. Indeed, $3 \equiv -1 \pmod 4$, so $3^{2^k} \equiv (-1)^{2^k} \equiv 1 \pmod 4$. It follows that $3^{2^k} + 1 \equiv 2 \pmod 4$. We deduce that there exists an odd integer $2m + 1$ such that

$$(3^2 + 1)\left(3^{2^2} + 1\right)\cdots\left(3^{2^{n-1}} + 1\right) = 2^{n-1}(2m + 1).$$

Then

$$3^{2^n} - 1 = 2 \cdot 4 \cdot 2^{n-1}(2m + 1) = m \cdot 2^{n+3} + 2^{n+2},$$

and this shows that the remainder is $2^{n+2}$.

**Problem 7.** Prove that for each $n$, there exists a number $A_n$, divisible by $2^n$, whose decimal representation contains $n$ digits, each of them equal to 1 or 2.

**Solution.** We prove the assertion by induction on $n$. For $n = 1$, take $A_1 = 2$; for $n = 2$, take $A_2 = 12$. Now, suppose there exists $A_n$, divisible by $2^n$, whose decimal representation contains $n$ digits, each of them equal to 1 or 2.

If $2^{n+1}$ divides $A_n$, then $A_{n+1}$ is obtained by adding the digit 2 at the beginning of $A_n$. Thus,

$$A_{n+1} = 2 \cdot 10^n + A_n = 2^{n+1} \cdot 5^n + A_n,$$

and we see that this number has $n + 1$ digits 1 or 2 and is divisible by $2^{n+1}$.

If $2^{n+1}$ does not divide $A_n$, then $A_n = 2^n k$ for some odd integer $k$. In this case, we add the digit 1 at the beginning of $A_n$. It follows that

$$A_{n+1} = 10^n + A_n = 2^n \cdot 5^n + 2^n \cdot k = 2^n(5^n + k).$$

The claim is proved by observing that since $k$ is odd, $5^n + k$ is even, and thus $A_{n+1}$ is divisible by $2^{n+1}$.

*Observation.* We can prove that the number $A_n$ is unique. Indeed, suppose $B_n$ is another number with the same properties. Then $A_n - B_n$ is divisible by $2^n$, so they have the same last digit. Let $A_{n-1}$ and $B_{n-1}$ be the numbers obtained by discarding the last digit of $A_n$ and $B_n$. Then we get that $A_n - B_n = 10(A_{n-1} - B_{n-1})$, and we deduce that $A_{n-1} - B_{n-1}$ is divisible by $2^{n-1}$. Again, they have the same last digit. Repeating this argument, we finally deduce that $A_n = B_n$. We can see that we did not use the fact that $A_n$ and $B_n$ are divisible by $2^n$, but rather only that their difference is divisible by $2^n$. This shows that the $2^n$ numbers with $n$ digits equal to 1 or 2 give different remainders when divided by $2^n$. Finally, we observe that the digits 1 and 2 from the problem statement can be replaced with any two other nonzero digits with different parities.

**Problem 8.** Using only the digits 1 and 2, one writes down numbers with $2^n$ digits such that the digits of every two of them differ in at least $2^{n-1}$ places. Prove that no more than $2^{n+1}$ such numbers exist.

**Solution.** Let us suppose that $2^{n+1} + 1$ such numbers exist. From these, at least $2^n + 1$ have the same last digit. Denote by $A$ the set of these numbers. For $a, b \in A$, let $c(a, b)$ be the number of digit coincidences. Thus, from the hypothesis it follows that $c(a, b) \leq 2^{n-1}$, for every $a$ and $b$. Let $N$ be the total number of coincidences for all numbers in $A$. Then

$$N \leq 2^{n-1} \cdot \binom{2^n + 1}{2} = 2^{3n-2} - 2^{2n-2}.$$

On the other hand, if we arrange the numbers of $A$ in a table with $2^n + 1$ rows and $2^n$ columns, we count in the last column $\binom{2^n+1}{2}$ coincidences and in each of the remaining $2^n - 1$ columns at least $2^{2(n-1)}$ coincidences. Indeed, if in some column we have $2^{n-1} - k$ digits of one type and $2^{n-1} + k + 1$ digits of the other type, then the number of coincidences equals

$$\binom{2^{n-1} - k}{2} + \binom{2^{n-1} + k + 1}{2} = 2^{2(n-1)} + k^2 + k \geq 2^{2(n-1)}.$$

We conclude that

$$N \geq \binom{2^n + 1}{2} + (2^n - 1) \cdot 2^{2(n-1)} = 2^{3n-2} - 2^{2n-2} + 2^{n-1},$$

which is a contradiction.

*Observation.* It can be shown that $2^{n+1}$ numbers with the required properties exist. We prove this inductively. For $n = 1$, take the numbers 11,

12, 21 and 22, which clearly satisfy the conditions. Suppose there exists a set $S_n$ with $2^{n+1}$ numbers having $2^n$ digits such that every two of them differ in at least $2^{n-1}$ places. We construct a set $S_{n+1}$ with $2^{n+2}$ numbers, each with $2^{n+1}$ digits. We generate from each element of $S_n$ two elements of $S_{n+1}$ as follows: if $a = \overline{a_1a_2\ldots a_n} \in S_n$, then in $S_{n+1}$ we put $b = \overline{a_1a_2\ldots a_na_1a_2\ldots a_n}$ and $c = \overline{a_1a_2\ldots a_na_1'a_2'\ldots a_n'}$, where $a_k' = 1$ if $a_k = 2$ and $a_k' = 2$ if $a_k = 1$. For instance, if $S_2 = \{11, 12, 21, 22\}$, then $S_3 = \{1111, 1212, 2121, 2222, 1122, 1221, 2112, 2211\}$. It is not difficult to check that the set thus obtained has the required properties.

**Problem 9.** Does there exist a natural number $N$ which is a power of 2 whose digits (in the decimal representation) can be permuted to form a different power of 2?

**Solution.** The answer is no. Indeed, suppose such $N$ exists and let $N'$ be another power of 2 obtained by rearranging the digits of $N$. We can assume that $N < N'$. Since $N$ and $N'$ have the same number of digits, we must have $N' = 2N$, or $N' = 4N$ or $N' = 8N$. It results that

$$N' - N \in \{N, 3N, 7N\}.$$

We get a contradiction from the following

*Lemma.* Let $n = \overline{a_ka_{k-1}\ldots a_1a_0}$ be a positive integer and

$$s(n) = a_k + a_{k-1} + \cdots + a_1 + a_0,$$

the sum of its digits. Then the difference $n - s(n)$ is divisible by 9.

*Proof.* We have

$$\begin{aligned} n - s(n) &= a_k \cdot 10^k + a_{k-1} \cdot 10^{k-1} + \cdots + a_1 \cdot 10 + a_0 \\ &\quad - a_k - a_{k-1} - \cdots - a_1 - a_0 \\ &= a_k(10^k - 1) + a_{k-1}(10^{k-1} - 1) + \cdots + a_1(10 - 1), \end{aligned}$$

and the conclusion follows by noticing that every number of the form $10^p - 1$ is divisible by 9.

Returning to our problem, we see that since $N$ and $N'$ have the same digits, $s(N) = s(N')$, thus $N' - N$ must be divisible by 9. This is impossible, because $N' - N \in \{N, 3N, 7N\}$ and none of the numbers $N$, $3N$, $7N$ is divisible by 9.

**Problem 10.** For a positive integer $N$, let $s(N)$ be the sum of its digits, in the decimal representation. Prove that there are infinitely many $n$ for which $s(2^n) > s(2^{n+1})$.

**Solution.** We use again the above lemma. Suppose, by way of contradiction, that there exist finitely many $n$ for which $s(2^n) > s(2^{n+1})$. Then there exists an integer $m$ such that for every $n > m$, $s(2^n) \leq s(2^{n+1})$. Since $s(2^n) = s(2^{n+1})$ if and only if $2^{n+1} - 2^n = 2^n$ is divisible by 9, we deduce that the sequence $s(2^n)$ is strictly increasing for $n > m$. Moreover, since the remainders of the numbers $2^n$ divided by 9 are 2,4,8,7,5,1, and these repeat periodically after six steps, we deduce that

$$s(2^{n+6}) \geq s(2^n) + 2 + 4 + 8 + 7 + 5 + 1 = s(2^n) + 27.$$

Then, inductively, we see that

$$s(2^{n+6k}) \geq s(2^n) + 27k,$$

for $n > m$ and all positive integers $k$.

On the other hand, $2^3 < 10$, so $2^{n+6k} < 10^{\frac{n}{3}+2k}$. This shows that the number $2^{n+6k}$ has at most $\frac{n}{3}+2k$ digits in its decimal representation. It follows that

$$s(2^{n+6k}) \leq 9\left(\frac{n}{3} + 2k\right) = 3n + 18k.$$

Combining the two estimates of $s(2^{n+6k})$, we deduce

$$s(2^n) + 27k \leq 3n + 18k,$$

for every positive integer $k$. This is a contradiction.

**Problem 11.** Find all integers of the form $2^n$ (where $n$ is a positive integer) with the property that, after deleting the first digit of its decimal representation, we again get a power of 2.

**Solution.** Suppose $2^m$ is obtained after deleting the first digit (equal to $a$) of the decimal representation of $2^n$. We then have $2^n = 10^k a + 2^m$, for some integer $k$. Hence $2^{n-m} - 1$ is divisible by 5. Checking the remainders of the powers of 2 when divided by 5, it results that $2^{n-m} - 1$ is divisible by 5 if and only if $n - m = 4t$, for some positive integer $t$. But then

$$\begin{aligned} 10^k a = 2^m\left(2^{4t} - 1\right) &= 2^m\left(2^{2t} + 1\right)\left(2^{2t} - 1\right) \\ &= 2^m\left(2^{2t} + 1\right)\left(2^t + 1\right)\left(2^t - 1\right). \end{aligned}$$

Observe that $2^{2^t}+1$ and $2^{2^t}-1$ are odd integers that differ by 2. Therefore they are relatively prime. The same applies to $2^t+1$ and $2^t-1$, hence $2^{2^t}+1$, $2^t+1$, and $2^t-1$ are all odd, pairwise prime integers. If $t>1$, then each of the three numbers has an odd prime divisor, but this is impossible, since $10^k a$ is divisible by 5 and at most one of the numbers 3 and 7 ( if any of them divides $a$). Consequently, $t=1$, hence $10^k a = 2^m \cdot 3 \cdot 5$, which leads to $k=1$ and $a = 2^{m-1}\cdot 3$. The only possible cases are $m=1$ and $m=2$, so the solutions to the problem are $2^5=32$ and $2^6=64$.

**Problem 12.** Let $a_0=0$, $a_1=1$ and, for $n\geq 2$, $a_n = 2a_{n-1}+a_{n-2}$. Prove that $a_n$ is divisible by $2^k$ if and only if $n$ is divisible by $2^k$.

**Solution.** Using the standard algorithm for recurrence relations, we obtain

$$a_n = \frac{1}{2\sqrt{2}}\left[\left(1+\sqrt{2}\right)^n - \left(1-\sqrt{2}\right)^n\right],$$

for every positive integer $n$. If we define

$$b_n = \tfrac{1}{2}\left[\left(1+\sqrt{2}\right)^n + \left(1-\sqrt{2}\right)^n\right],$$

we observe that $b_n$ verifies the same recurrence relation, and since $b_0 = b_1 = 1$, all $b_n$ are positive integers.

We have

$$b_n + a_n\sqrt{2} = \left(1+\sqrt{2}\right)^n$$

and

$$b_n - a_n\sqrt{2} = \left(1-\sqrt{2}\right)^n.$$

Multiplying these equalities we obtain $b_n^2 - 2a_n^2 = (-1)^n$. Therefore, $b_n$ is odd for each $n$.

Now, we prove the claim by induction on $k$. For $k=0$ we have to prove that $a_n$ is odd if and only if $n$ is odd. We have

$$2a_n^2 = b_n^2 - (-1)^n$$

and $b_n = 2m+1$, for some integer $m$. Hence

$$a_n^2 = 2(m^2+m) + \frac{1-(-1)^n}{2}.$$

It follows that $a_n$ is odd if and only if $1-(-1)^n = 2$, that is, if $n$ is odd. The inductive step follows from the equality

$$a_{2n} = 2a_n b_n,$$

which is obtained observing that

$$b_{2n} + a_{2n}\sqrt{2} = \left(1+\sqrt{2}\right)^{2n} = \left(b_n + a_n\sqrt{2}\right)^2 = b_n^2 + 2a_n^2 + 2a_n b_n\sqrt{2}.$$

**Problem 13.** If $A = \{a_1, a_2, \ldots, a_p\}$ is a set of real numbers such that $a_1 > a_2 > \cdots > a_p$, we define

$$s(A) = \sum_{k=1}^{p} (-1)^{k-1} a_k.$$

Let $M$ be a set of $n$ positive integers. Prove that $\sum_{A \subseteq M} s(A)$ is divisible by $2^{n-1}$.

**Solution.** Let $M = \{a_1, a_2, \ldots, a_n\}$. We can assume that

$$a_1 > a_2 > \cdots > a_n.$$

Let $A \subseteq M - \{a_1\}$, $A = \{a_{i_1}, a_{i_2}, \ldots, a_{i_s}\}$, with $a_{i_1} > a_{i_2} > \cdots > a_{i_s}$, and $A' = A \cup \{a_1\}$. Then

$$a_1 > a_{i_1} > a_{i_2} > \cdots > a_{i_s}$$

and we have:

$$\begin{aligned} s(A) + s(A') = {} & a_{i_1} - a_{i_2} + \cdots + (-1)^{s-1} a_{i_s} + a_1 \\ & - a_{i_1} + a_{i_2} - \cdots + (-1)^s a_{i_s} = a_1. \end{aligned}$$

Now, matching all the $2^{n-1}$ subsets $A$ of the set $M - \{a_1\}$ with the corresponding subsets $A' = A \cup \{a_1\}$, we find that

$$\sum_{A \subseteq M} s(A) = 2^{n-1} a_1,$$

and the claim is proved.

## 6.8 Progressions

**Problem 4.** Partition the set of positive integers into two subsets such that none of them contains a nonconstant arithmetical progression.

**Solution.** Partition the set $\{1, 2, 3, \dots\}$ in the following way

$$\begin{array}{cccccccccc} 1 & & & 4 & 5 & 6 & & & & \dots \\ & 2 & 3 & & & & 7 & 8 & 9 & 10 & \dots \end{array}$$

None of the sets

$$A = \{1, 4, 5, 6, 11, 12, 13, 14, 15, \dots\}$$

or

$$B = \{2, 3, 7, 8, 9, 10, 16, 17, 18, 19, 20, 21, 22, \dots\}$$

contains a nonconstant arithmetical progression. Indeed, if such a progression is contained in one of the sets, let $r$ be its common difference. But in both sets we can find a "gap" of more than $r$ consecutive integers, a contradiction.

**Problem 5.** Prove that among the terms of the progression 3,7,11, ... there are infinitely many prime numbers.

**Solution.** Suppose the contrary and let $p$ be the greatest prime number in the given progression. Consider the number

$$n = 4p! - 1.$$

It is not difficult to see that it is a term of the progression (in fact, the given progression contains all positive integers of the form $4k - 1$). Thus $n$ must be composite, since $n > p$. Observe that $n$ is not divisible by any prime of the form $4k - 1$ (all these are factors in $p!$), so all the prime factors of $n$ are of the form $4k + 1$. The product of several factors of the form $4k + 1$ is again of the form $4k + 1$, and hence $n = 4k + 1$, for some $k$. This is a contradiction.

**Problem 6.** Does there exist an (infinite) nonconstant arithmetical progression whose terms are all prime numbers?

**Solution.** The answer is no. Indeed, if such a progression exists, denote its common difference by $r$, and consider the $r$ consecutive integers

$$(r + 1)! + 2, (r + 1)! + 3, \dots, (r + 1)! + (r + 1).$$

Each of them is a composite number, but since the progression has the common difference $r$, one out of any $r$ consecutive integers must be a term of the progression. This is a contradiction.

**Problem 7.** Consider an arithmetical progression of positive integers. Prove that one can find infinitely many terms whose decimal digits add up to the same sum.

**Solution.** Let $(a_n)_{n\geq 1}$ be the progression and let $r$ be the common difference. Suppose the number $a_1$ has $d$ digits (in its decimal representation). Then for all $k > d$, the sum of the digits of the number

$$a_1 + 10^k r$$

is the same.

**Problem 8.** The set of positive integers is partitioned into $n$ arithmetical progressions, with common differences $r_1, r_2, \ldots, r_n$. Prove that

$$\frac{1}{r_1} + \frac{1}{r_2} + \cdots + \frac{1}{r_n} = 1.$$

**Solution.** Let $a_k = a_1 + (k-1)r_1$ be the progression with common difference $r_1$. Let us count how many terms of this sequence are less than or equal to some positive integer $N$. The inequality $a_k \leq N$ is equivalent to

$$a_1 + (k-1)r_1 \leq N$$

or

$$k \leq \frac{N}{r_1} - \frac{a}{r_1} + 1.$$

It follows that the number of terms of the first progression belonging to the set $\{1, 2, \ldots, N\}$ equals

$$\left\lfloor \frac{N}{r_1} - \frac{a}{r_1} + 1 \right\rfloor.$$

Similarly, we deduce that the number of terms of the progression with common difference $r_i$ belonging to the set $\{1, 2, \ldots, N\}$ equals

$$\left\lfloor \frac{N}{r_i} - \frac{a}{r_i} + 1 \right\rfloor.$$

Since the progressions form a partition of the set of positive integers, we must have

$$\sum^{n} \left| \frac{N}{r_i} - \frac{a}{r_i} + 1 \right| = N.$$

Using the inequality $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$, we obtain

$$N \leq \sum_{i=1}^{n} \left( \frac{N}{r_i} - \frac{a}{r_i} + 1 \right) < N + n.$$

Hence

$$1 \leq \sum_{i=1}^{n} \frac{1}{r_i} - \frac{1}{N} \sum_{i=1}^{n} \frac{na}{r_1} + \frac{n}{N} < 1 + \frac{n}{N}$$

and letting $n \to \infty$ yields the desired result.

**Problem 9.** Prove that for every positive integer $n$, one can find $n$ integers in arithmetical progression, all of them powers of integers, but one cannot find an infinite sequence with this property.

**Solution.** We prove the assertion by induction on $n$. For $n = 3$, we can consider the numbers 1, 25, and 49. Suppose the assertion is true for some $n$, and let $a_i = b_i^{k_1}$, $i = 1, 2, \ldots, n$ be the terms of the progression having the common difference $d$. Let $b = a_n + d$ and let $k = \text{lcm}(k_1, k_2, \ldots, k_n)$. Then the $n + 1$ numbers

$$a_1 b^k,\ a_2 b^k, \ldots,\ a_n b^k,\ b^{k+1}$$

are in progression and all are powers of some integers. Indeed, for all $i$, $1 \leq i \leq n$, there exists $d_i$ such that $k = k_i d_i$, and we obtain

$$a_i b^k = b_i^{k_i} b^k = b_i^{k_i} b^{k_i d_i} = (b_i b^{d_i})^{k_i},$$

as desired.

For the second part, we need a result from calculus: if $a, b > 0$, then

$$\lim_{n \to \infty} \sum_{k=1}^{n} \frac{1}{ak + b} = +\infty.$$

From this we deduce that if $(a_n)_{n \geq 1}$ is a progression of positive integers, then

$$\lim_{n \to \infty} \sum_{k=1}^{n} \frac{1}{a_k} = +\infty.$$

Now suppose that $(a_n)_{n \geq 1}$ is a progression in which all terms are powers of integers. Let $S$ be the set of all positive integers that are powers of integers. We will prove that

$$\sum_{a \in S} \frac{1}{a} \leq 1$$

which contradicts the previous result. Indeed, we have

$$\sum_{a\in S}\frac{1}{a} \le \sum_{n\ge 2}\sum_{k\ge 2}\frac{1}{n^k} = \sum_{n\ge 2}\frac{1}{n^2}\left(1+\frac{1}{n}+\frac{1}{n^2}+\cdots\right)$$

$$= \sum_{n\ge 2}\frac{1}{n^2}\cdot\frac{1}{1-\frac{1}{n}} = \sum_{n\ge 2}\frac{1}{n(n-1)} = \sum_{n\ge 2}\left(\frac{1}{n-1}-\frac{1}{n}\right) = 1.$$

**Problem 10.** Prove that for any integer $n$, $n \ge 3$, there exist $n$ positive integers in arithmetical progression $a_1, a_2, \ldots, a_n$ and $n$ positive integers in geometric progression $b_1, b_2, \ldots, b_n$, such that

$$b_1 < a_1 < b_2 < a_2 < \cdots < b_n < a_n.$$

**Solution.** Let $m > n^2$ be an integer. We define

$$B_k = \left(1+\frac{1}{m}\right)^k,$$

for $k = 1, 2, \ldots, n$. Observe that for $k \ge 2$,

$$B_k > 1+\frac{1}{m}.$$

For $k \le n$, we have

$$B_k = 1+\frac{k}{m}+\frac{k(k-1)}{2!m^2}+\cdots+\frac{k!}{k!m^k}$$

$$\le 1+\frac{k}{m}+\frac{n(n-1)}{2!m^2}+\cdots+\frac{n(n-1)\ldots(n-k+1)}{k!m^k}$$

$$= 1+\frac{k}{m}+\frac{1}{m}\left(\frac{n(n-1)}{2!m}+\cdots+\frac{n(n-1)\cdots(n-k+1)}{k!m^{k-1}}\right)$$

$$< 1+\frac{k}{m}+\frac{1}{m}\left(\frac{1}{2!}+\frac{1}{3!}+\cdots+\frac{1}{k!}\right) < 1+\frac{k+1}{m}.$$

We have used the fact that $m > n^2$ and the inequalities

$$\frac{1}{2!}+\frac{1}{3!}+\cdots+\frac{1}{k!} < \frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{(k-1)\cdot k} = 1-\frac{1}{k} < 1.$$

Defining

$$A_k = 1+\frac{k+1}{}$$

for $1 \leq k \leq n$, yields

$$B_1 < A_1 < B_2 < A_2 < \cdots < B_n < A_n.$$

In order to obtain progressions with integer terms, we define $a_k = m^n A_k$ and $b_k = m^n B_k$, for all $k$, $1 \leq k \leq n$.

# Glossary

**Centroid of a triangle** Point of intersection of the medians.

**Ceva's theorem and its trigonometric form** Let $AA'$, $BB'$, and $CC'$ be three cevians of triangle $ABC$. Then $AA'$, $BB'$, and $CC'$ are concurrent if and only if either

$$\frac{A'B}{A'C} \cdot \frac{B'C}{B'A} \cdot \frac{C'A}{C'B} = 1$$

or

$$\frac{\sin \angle A'AB}{\sin \angle A'AC} \cdot \frac{\sin \angle B'BC}{\sin \angle B'BA} \cdot \frac{\sin \angle C'CA}{\sin \angle C'CB} = 1.$$

**Cevian** Any segment joining the vertex of a triangle to a point on the opposite side.

**Circumcenter** Center of the circumscribed circle.

**Circumcircle** Circumscribed circle.

**Convex quadrilateral** The quadrilateral $ABCD$ is convex if the line segments $AC$ and $BD$ intersect (or, equivalently, if all its interior angles are less than 180°).

**Cyclic polygon** Polygon that can be inscribed in a circle.

**Eisenstein's criterion** Let $f = a_0 + a_1 X + \cdots + a_n X^n$ be an integer polynomial. If $p$ is a prime number such that $p$ divides $a_0, a_1, \ldots, a_{n-1}$, $p$ does not divide $a_n$ and $p^2$ does not divide $a_0$, then $f$ cannot be written as a product of two (nonconstant) integer polynomials.

**Fermat's little theorem** If $p$ is a prime number, then $a^p \equiv a \pmod{p}$ for any integer $a$.

**Homothety** A homothety of center $O$ and ratio $r$ is a transformation that maps each point $P$ in the plane to a point $P'$ such that $\overline{OP'} = r\overline{OP}$.

**Incenter** Center of the inscribed circle.

**Incircle** Inscribed circle.

**Inradius** The radius of the incircle.

**De Moivre's formula** For any angle $a$ and any integer $n$,

$$(\cos a + i \sin a)^n = \cos na + i \sin na.$$

**One-to-one function** A function $f$ such that if $x \neq y$, then $f(x) \neq f(y)$.

**Orthocenter** The point of intersection of the altitudes of a triangle.

**Pithot's theorem** A circle can be inscribed in the convex quadrilateral $ABCD$ if and only if $AB + CD = BC + DA$.

**Root of unity** Solution to the equation $z^n - 1 = 0$.

**Symmetry center** The point $O$ is a symmetry center of a figure $F$ if for any point $M \in F$, there exists $M' \in F$ such that $O$ is the midpoint of the line segment $MM'$.

# Index of notation

| | |
|---|---|
| $\mathbf{Z}$ | the set of integers |
| $\mathbf{Q}$ | the set of rational numbers |
| $\mathbf{R}$ | the set of real numbers |
| $\mathbf{C}$ | the set of complex numbers |
| $[a, b]$ | the set of real numbers $x$ such that $a \leq x \leq b$ |
| $(a, b)$ | the set of real numbers $x$ such that $a < x < b$ |
| $AB$ | the segment $AB$; also the length of the segment $AB$ |
| $\overrightarrow{AB}$ | the vector $AB$ |
| $[F]$ | the area of the figure $F$ |
| $\lfloor x \rfloor$ | the integer part of the real number $x$ |

# Index to the problems

(Chapter/Paragraph/Problem number)


ARML Power Question: 2.3.2.

Balkan Mathematical Olympiad: 1.6.7, 2.3.7, 2.4.4, 3.4.2.

British Mathematical Olympiad: 2.8.1.

Bulgarian Mathematical Olympiad: 3.4.3.

Dutch Mathematical Olympiad: 3.7.2.

Gazeta Matematica: 1.1.12, 1.6.3, 1.8.10, 2.7.5.

International Mathematical Olympiad: 1.2.6, 2.4.7, 2.4.9, 3.3.6.

Korean Mathematical Olympiad: 3.7.5.

Kvant: 2.6.4, 2.6.5.

Revista Matematica din Timisoara: 1.2.4, 2.4.5.

Romanian Mathematical Olympiad: 1.1.3, 1.2.7, 1.3.9, 1.3.10, 1.3.12, 1.5.3, 1.5.4, 1.5.7. 1.5.10, 1.5.11, 1.6.9, 1.8.9, 2.4.3, 2.5.4, 2.5.5, 2.5.6, 3.2.1, 3.2.2, 3.2.3, 3.2.4, 3.2.5, 3.2.6, 3.2.7, 3.3.5, 3.4.6, 3.4.7, 3.4.7, 3.7.6, 3.7.13.

Russian Mathematical Olympiad: 1.3.5, 1.3.13, 1.4.3, 1.4.5, 1.4.6, 1.4.7, 1.6.5, 1.7.7, 2.3.1, 2.3.3, 2.4.8, 2.5.3, 2.6.3, 3.1.1, 3.1.2, 3.1.8, 3.1.9, 3.7.7, 3.7.8, 3.7.9, 3.7.10.

Tournament of the Towns: 3.1.6, 3.1.7.

USAMO and US selection tests: 1.2.9, 1.8.3.

# Mathematical **Olympiad** Treasures

**TITU ANDREESCU AND BOGDAN ENESCU**

*Mathematical Olympiad Treasures* aims at building a bridge between ordinary high school exercises and more sophisticated, intricate and abstract concepts and problems in undergraduate mathematics. The book contains a stimulating collection of problems in the subjects of geometry and trigonometry, algebra, number theory and combinatorics. While it may be considered a sequel to *Mathematical Olympiad Challenges*, the focus of *Treasures* is on engaging a wider audience of undergraduates to think creatively in applying techniques and strategies to problems in the real world.

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